Handbook of Current and Novel Protocols for the Treatment of Infertility

Edited by
Michael H. Dahan
Human M. Fatemi
Nikolaos P. Polyzos
Juan A. Garcia-Velasco

HANDBOOK OF CURRENT AND NOVEL PROTOCOLS FOR THE TREATMENT OF INFERTILITY

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Dedication

> Success is no accident. It is hard work, perseverance, learning, studying, sacrifice and most of all, love of what you are doing. ***Pele***

When I was approached by Elsevier several years ago to edit their next textbook on infertility, I was both humbled and excited. I have been fortunate to work with wonderful co-editors, Nik, Juan, and Human, with whom I have strengthened friendships over this time. I could not have completed this without them.

I truly love being a reproductive endocrinologist, I could not imagine a more fulfilling field of medicine and there is nothing like meeting a child who I helped create. It is hard work but has always been worth it. I hope that this book helps to transfer my enjoyment and enthusiasm to readers interested in reproductive medicine, which is why I endeavored into this undertaking.

I could not have edited this textbook without the patience and love of my children, Yehuda, Daniel, Jacob, Ariella, and Rebecca. My work has always taken time from them, and I hope they understand. My wife Leah has been my greatest supporter and always is patient and understanding when I need to work at a computer and not contribute directly to my family. To quote Winston Churchill "My most brilliant achievement was my ability to be able to persuade my wife to marry me."

I dedicate this book to Leah, Yehuda, Daniel, Jacob, Ariella (Lala), and Rebecca.

My father Simon taught me to reason, which has served me so well in my research and academic career and I am eternally grateful for all he did, his love, and how he believed in me.

My mother Barbara was a pillar of support, and I will never forget her asking how I could be a doctor since I feared blood. Now I tell my patients if they ask my mother, she will tell them that they cannot do better than me (although I need to pay her to say that).

I would not be in this position without the guidance and mentoring of R. Jeffrey Chang at UCSD, where I did my REI fellowship, and my co-fellows Richard, Ketan, and Mickey who taught me a great deal about medicine and research, as did Pam our research coordinator.

Michael H. Dahan, MD, FACOG,
FRCPS of Canada

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> The knowledge of anything, since all things have causes, is not acquired or complete, unless it is known by its causes.
> Sina (Persian physician and philosopher, 980 −1037 C.E.)

I remember, I was in France during my summer vacation, when I was approached by Michael to coedit a new book on fertility treatments. Since I just had published few months before already a book on infertility treatments, initially, I was very hesitant. However, within a few seconds, Michael convinced me, and post-factum, I am very happy he did so. Furthermore, with friends and very well-respected coeditors like Michael, Juan, and Nik, I was honored to coedit this book.

One might say, there have been so many publications in the past on fertility treatments, I would reply:

> Science is dynamic and is developing fast. There is always a need for an "update"!

I dedicate this book to my wife Sanne and my daughters Charline and Arta Louise.

Leaving my country Iran and being on my own from the age of 11, due to an ongoing destructive war with Iraq, it was not evident to keep on going in the right path. Without my parent's support, I certainly wouldn't have finished my school in Germany and moving yet to different foreign countries (Belgium, Kuwait, and the UAE) to study and work. The support and encouragement of my parents were a large part of what kept me going.

My wife's love and advice kept me engaged on making steps forward. Her understanding has been more than invaluable.

My daughters Charline and Arta Louise, thank you for giving me so much joy and thank you for simply being there.

Last but not least, I would like to thank all my colleagues at ART Fertility clinics, especially, Mr. Ibrahim Khatib. Many people have failed at what you have accomplished, simply because they were busy finding problems while you were busy finding solutions. Thank you!

Human M. Fatemi, MD, PhD

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To my beloved son Pavlos, who really changed my life

Nikos P. Polyzos

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"To María, Jaime, and Carmela, my biggest supporters and strongest motivators"

Juan A. Garcia-Velasco

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Preface

Juan Garcia Velasco, Human Fatemi, Nikolaos Polyzos, and I undertook this endeavor to develop an up-to-date textbook of fertility care for both the trainee in need of instruction and the experienced clinician who wants to refresh and update their knowledge. It includes information on how to perform modern care as well as exposure to uncommon albeit contemporaneous subjects in the realm of fertility care.

We included basic chapters such as the male and female infertility evaluation, ovarian stimulation and insemination, and several chapters on managing different types of in vitro fertilization (IVF) cycles. How women with polycystic ovary syndrome (PCOS) require tailored care. We went over different types of ovarian trigger for oocyte collection. Using progestins to prevent ovulation, sperm DNA fragmentation, and its role on fertility outcomes. We discussed the mild stimulation protocol and the dual stim protocol, for which a recent prospective randomized trial demonstrated the advantage of as compared to two sequential follicular phase start cycles.

We discussed the management of recurrent pregnancy loss, and we critically reviewed care for recurrent implantation failure, endometrial receptivity, and the microbiome.

We even have chapters on how to set up an IVF and an andrology laboratory, which are not typically available in textbooks. We provide a schematic on how to troubleshoot an IVF laboratory when success rates suddenly decrease unexpectedly, knowledge which is both important for the clinicians and the embryologists responsible for the patients and the laboratory.

We made sure to include information about in vitro maturation (IVM) of human oocytes, likely the future of our field. We selected chapters to be included from both: a group which believes in stimulation and triggering for an IVM and another group which would almost never perform such procedures in their IVM cycles. This was done so that the reader could be exposed to both points of view. This concept goes back to the basis of fertility care. That there are many different ways to perform fertility care, with most believing that their way is the best, while most of these different ways work perfectly fine.

This book will address how to manage the complications from IVF and how to minimize their occurrence.

We selected international experts in the field as the authors, who are both contemporaneous and represent the diverse methods of fertility care around the world.

We were very fortunate that many of the best academic fertility experts in the field today participate by contributing chapters.

We hope that you find this book both educational and novel.

We wish to thank Elsevier for undertaking this endeavor and all the assistants who have helped this title progress.

Sincerely,
The Editors

Michael H. Dahan, Human M. Fatemi, Nikolaos P. Polyzos, and Juan A. Garcia Velasco

CHAPTER

1

The evaluation of the female infertility patient

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The World Health Organization (WHO) categorizes infertility as a disease (WHO, 1). It affects both genders and causes considerable personal suffering and disruption of family life.

When heterosexual couples try to conceive, we talk about *functional fertility*. This would translate into how easy or how difficult it is for a couple to conceive. Even in absence of a factor affecting fertility, the probability of conception declines with increasing female age. This is attributed to the decrease in ovarian reserve depicted by a quantitative and a qualitative decline in the follicular pool [1,2].

Fecundability is a term used when referring to the probability of a spontaneous conception in one menstrual cycle in couples having regular sexual intercourse. Fecundability varies from 11% to 33% in reproductive-age women (where >90% of women studied were <35 years of age and without known infertility factors being present when they initiated attempted conception) [3–5]. Cumulative pregnancy rates at 12 months of regular unprotected intercourse were traditionally reported as being about 80%–85% in the age group 25–27 years. Recently, the American Society of Reproductive Medicine reassessed cumulative pregnancy rates in young couples without infertility and determined that 85% of couples conceived after six months of regular unprotected intercourse. Of women 40–45 years of age, 55% are pregnant by one year [6]. Fecundability declines, as the female age gets older, even if there is no known infertility factor.

The time from the start of unprotected sexual intercourse to pregnancy is called "*time to pregnancy*" and is reported in months. As the female ages, *time to pregnancy* increases. The median *time to pregnancy* was reported to be three months for women under 38 years of age, four months for women 38–39, eight months for women 40–41, and longer than 12 months for women 42–44 years of age, in a population at least 30 years old, with no known infertility factors [7].

If the duration of *time to pregnancy* is long enough that it passes a predetermined threshold, then the fertility investigation for both partners is recommended. The term "*infertility*" is

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historically described as not conceiving for one year despite regular unprotected sexual intercourse, two to three times a week, in a heterosexual relationship. The infertility investigation is often recommended to start earlier, after six months of subfertility, if the female age is older than 35 years or if there is an obvious problem like menstrual irregularity, a previous diagnosis of endometriosis, or a known moderate male factor [8]. The reason to set the age cut off at 35 years for initiation of an evaluation at six months of infertility is that the quantitative and qualitative follicular pool depletion accelerates with age, the decline in monthly fecundity rate is faster, and the time to pregnancy becomes longer after age 35. For women, over 40 years of age, the investigations need to be completed more immediately, and the treatment needs to be expedited.

The WHO published a technical report in 1992, titled "Recent Advances in Medically Assisted Conception" following the WHO Scientific Group Meeting in April 1990, in Geneva [9]. In that report, the prevalence of female factors was reported as higher than the prevalence of male factors. The most common female infertility causes were ovulatory disorders and tubal factors even though the prevalence of causes varied according to the various geographical regions (Tables 1.1 and 1.2).

Female fertility often needs to be evaluated in women and nonbinary individuals (with female karyotype) who do not fit the classical definition of failure to conceive in 6–12 consecutive months despite regular unprotected sexual intercourse where a male partner is contributing. These are women and sometimes nonbinary individuals planning oocyte preservation, women in homosexual relationships planning reciprocal in vitro fertilization (IVF), and women who desire conception using donor sperm with or without a partner. Transmales may also need evaluation for both conception and fertility preservation with testosterone suplementation having no effect on outcomes in IVF cycles.

Having both male and female evaluated concomitantly is recommended in all cases where applicable, since 20%–35% of the time, there are factors affecting fertility in both partners in couples. Further testing may be planned for a woman where the man's test results are found to be in the normal range. Or, for instance, tubal evaluation may be deferred if a man is known to have severe sperm factor, and the couple is recommended to undergo IVF where tubal patency is not needed to be tested. In these cases, an evaluation of the uterine cavity may be indicated to rule out polyps, fibroids or Asherman's syndrome.

TABLE 1.1 Etiology of infertility and distribution according to sex.

Etiology of infertility	Developed countries	Sub-Saharan Africa
Female factor only	37%	31%
Male factor only	8%	22%
Both male and female factor	35%	21%
No cause found	5%	14%
Became pregnant during investigation	15%	12%

TABLE 1.2 Etiology of female infertility.

Female etiology	Developed countries	Sub-Saharan Africa
Ovulatory disorders	24%	17.9%
Pelvic adhesions	12.4%	11.1%
Acquired tubal abnormalities	11.2%	10.3%
Bilateral tubal occlusion	10.6%	41.9%
Hyperprolactinemia	6.7%	4.3%
Endometriosis	5.7%	0.9%
No demonstrable cause	28.6%	13.7%

History assessment

The fertility investigation starts with a comprehensive history (ideally to be complemented with a thorough physical exam in a setup where an in-person clinic visit takes place). History and through the physical exam alone will reveal vast information for planning further tests and treatment options and will also guide the preconceptional counseling. The significant parameters to ask in history are listed and explained in Table 1.3.

Physical exam

Height and weight measurements, BMI calculation, and appearance

Extremes of body mass index (BMI) on both ends and fat distribution need to be noted. Very low BMI is also associated with reduced fertility. Eating disorders leading to very low BMI are associated with hypothalamic amenorrhea (WHO group 1) by decreasing hypothalamic gonadotropin-releasing hormone secretion [12].

Together with high BMI, central obesity may suggest insulin resistance; abdominal circumference should be measured. As these women are often given dietary and exercise advice, and sometimes prescribed metformin, the resultant change in weight, BMI, abdominal circumference or fat distribution, and change in menstrual cycle length need to be noted during care. Patients with high or low BMI should ideally be referred to a nutritionist.

Not only the BMI but also the patient's weight alone is also important, particularly in the selection of gonadotropin dose in ovulation induction and controlled ovarian stimulation for IVF. Higher doses are usually required above 170 lbs (77 kg) even if the ovarian reserve is good.

A short stature (less than 5 feet in height, 152.4 cm) with or without primary amenorrhea may suggest Turner syndrome (TS). Other phenotypic features, such as the webbed neck, low-set ears, shield chest, pigmented nevi, cubitus valgus, and low hairline, may be recognized when the patient is seen in the clinic. Pure 45,X cases are mostly diagnosed during childhood whether due to phenotypic appearance or due to nondevelopment of secondary sexual

TABLE 1.3 Key parameters in history taking and their relevance.

Parameter	Relevance
Age	The most important determinant of fecundability, time to pregnancy, and treatment outcome.
Previous obstetric history	A complete obstetric history should be sought including pregnancy dates, time-to-pregnancy for each conception, if previous conceptions were with the current partner, how each pregnancy ended-up, gestational age at delivery, delivery method, indication for a cesarean section (breech presentation may indicate uterine anomalies—septum or bicornuate), birthweight of the offspring (a large baby for gestational age may suggest gestational diabetes), obstetrical complications (i.e., gestational diabetes, preeclampsia, chorioamnionitis), and concomitant diagnoses with pregnancy and treatments received (i.e., thyroid disease). Whether a curettage was performed postpartum or postmiscarriage is associated with intrauterine synechia risk (Asherman's syndrome). If hypomenorrhea/amenorrhea occurred after an intrauterine intervention or postpartum fever, Asherman's syndrome may need to be ruled out. A proven recent spontaneous pregnancy, even if it is miscarried shortly after confirmation of a gestational sac by ultrasound or a positive serum hCG test, suggests at least one patent Fallopian tube and possibly functional sperm parameters of the contributing partner. It should be considered that a pregnancy that is believed to have miscarried without confirmation of it being intrauterine by ultrasound could have been ectopic.
Duration of current infertility	The longer the duration of infertility, the lesser the chance of spontaneous conception. Previous contraception methods—effective or not? The duration of infertility may be longer than reported.
Menstrual history	Age at menarche Cycle interval, duration of menstrual bleed (days) Oligomenorrhea may be due to polycystic ovarian syndrome (PCOS). Current menometrorrhagia in women with previous oligo-amenorrhea, particularly in women with high BMI, may indicate endometrial pathology (hyperplasia, endometrial intraepithelial neoplasia, or rarely invasive endometrial adenocarcinoma) even in women less than 35 years of age. Recent longer cycle interval may sometimes be associated with PCOS, particularly if there is a recent history of weight gain and/or discontinuation of mmetformin, as well as developing hyperprolactinemia (may cause amenorrhea, too) or thyroid dysfunction. Recent shortening cycle length may be associated with declining ovarian reserve [10]. The gradual development of hypermenorrhea, menorrhagia, and/or history of anemia may be a sign for growing uterine fibroid(s). History of dysmenorrhea—particularly secondary dysmenorrhea may suggest endometriosis.
Sexual intercourse	Coital frequency and timing regarding menstrual cycle/ovulation History of sexual dysfunction on either party Dyspareunia
Galactorrhea	May suggest pituitary adenoma or hyperprolactinemia possibly affecting ovulation.
Hirsutism and acnea	Long-term hirsutism with oligo/amenorrhea may suggest PCOS or nonclassical congenital adrenal hyperplasia (NCCAH). Hirsutism varies according to ethnic background. It is rare in Asian women with PCOS; both hyperandrogenism and hyperandrogenemia are more

TABLE 1.3 Key parameters in history taking and their relevance.—cont'd

Parameter	Relevance
	common in women of Mediterranean origin and those from the Indian Subcontinent. Acute appearing hirsutism rather suggests an androgen-producing tumor most commonly of the ovaries or the adrenal glands, or hyperthecosis (rare in reproductive-age women, seen mostly postmenopausal). In acute hirsutism in a reproductive-age woman, exogenous testosterone use needs to be kept in mind.
Sexually transmitted disease	History of chlamydia or gonorrhea positivity Pelvic inflammatory disease (PID), history of tuboovarian abscess, or any hospital admission for a pelvic infection raise the possibility of tubal blockage or dysfunction (and increased risk of ectopic pregnancy).
History of malignancy	Previous chemotherapy or pelvic radiotherapy? Chemotherapy, depending on agent and dose, has varying levels of gonadotoxicity. In the case of previous pelvic irradiation, not only the ovarian reserve (if ovarian transposition or ovarian shielding was not done) but also uterine function (endometrial thickness or potential) and the vagina may be affected. Women who received pelvic irradiation have an increased risk of miscarriage, preterm labor, low birth weight, and placental implantation anomalies [11].
Systemic diseases present	Autoimmune diseases may have an association to decreased ovarian reserve. Drugs used for autoimmune disease, not infrequently, may affect ovarian reserve and be teratogenic (i.e., methotrexate). Systemic disease and drugs used for them, past and current.
Medication and supplement use	Current and past long-term medication use. Anabolic steroid use needs to be sought; in men may cause reversible sperm parameter deterioration. Maternal anabolic steroid use may cause virilization of the baby; conception while on antiandrogens may cause feminization. Isotretinoin and other vitamin A derivatives commonly used to treat acne may be teratogenic.
Surgical history	Particularly abdominopelvic surgeries. The surgical and pathology reports should be evaluated for the presence of intraabdominal adhesions and endometriosis, which can also be detected in appendectomy specimens, signs of previous PID (Fitz-Hugh-Curtis syndrome) may suggest Fallopian tube dysfunction and increased risk of ectopic pregnancy. Appendectomy—was it perforated? Was any abdominal surgery complicated by peritonitis? This is a risk for future intraabdominal adhesions. Ovarian surgeries including oophorectomies and cystectomies may contribute to diminished ovarian reserve and may have also caused peritubal adhesions. Endometriosis is particularly important, as not only the surgery but also the existence of endometriosis may cause a decline in the ovarian reserve and/or tubal dysfunction. Thyroid, pituitary, and adrenal gland surgeries indicate relevant bloodwork for these organs.
Lifestyle, habits, and exposures	Occupation and living environment; environmental exposure to hazardous material: lead and other chemicals, excessive heat, continuous vibration, or radiation may impact pregnancy potential. Eating habits: high consumption of swordfish and tuna may be associated with the ingestion of mercury. Morbid obesity may decrease pregnancy rates in women (and sperm quality in males) and also increases miscarriage rates.

(*Continued*)

TABLE 1.3 Key parameters in history taking and their relevance.—cont'd

Parameter	Relevance
	Ayurvedic or traditional Chinese medications, older paints may contain lead. Ventilation conditions of the work and living environment. Smoking habits including the use of noncigarette nicotine delivery systems (vaping, shisha) may cause decline in ovarian reserve. The amount of coffee or caffeine consumption. Amount of alcohol consumption. Illicit drug use. Exercise history: excessive exercise through a decrease in GnRH secretion due to excessive secretion of cortisol may cause hypothalamic amenorrhea [12,13]
Family history	Maternal diethylstilbestrol (DES) use could be questioned although affected women are less common (discontinued in the United States in 1971 but continued until the early 1980s in other countries). History of premature menopause history in the woman's family warrants ovarian reserve evaluation in woman. Any family member of the woman with multiple miscarriages. Birth defects in either partner's family or developmental delays.
Previous fertility treatment	Treatment types, response to medication prescribed, procedures performed, and outcome of previous treatments may help in investigation and in planning of the new treatment.

characteristics. Before pregnancy, all women with classic or mosaic Turner's syndrome should undergo evaluation of aortic root dilation and congenital urinary system abnormalities. TS may be mosaic with varying degrees of 46XX cell lines present, as well. In this case, the women may have menses and are at risk for premature ovarian failure. In the case of TS mosaic with a Y chromosome present (45X/46XY or other combinations), the gonads need to be removed due to future malignity risk.

In the presence of primary amenorrhea, secondary sex characters need to be observed. Underdeveloped secondary sex characteristics may refer to hypothalamic causes (hypogonadotropic hypogonadism—low or normal serum follicle-stimulating hormone [FSH] and luteinizing hormone [LH]) or gonadal failure where hypothalamo–pituitary axis works well (high serum FSH an LH) [14]. A karyotype test for these patients will likely be needed. Other genetic and sex development disorders, Swyer syndrome (46XY), which is pure gonadal dysgenesis, androgen insensitivity syndrome (testicular feminization, 46, XY), and Leydig cell agenesis commonly have a female phenotype with primary amenorrhea. There is no uterus and no Fallopian tubes in androgen insensitivity syndrome and complete Leydig cell agenesis; however, there may exist a uterus in Swyer syndrome, and gonads are underdeveloped streak gonads. All gonads in phenotypic females with Y chromosomes need surgical removal to prevent malignancy.

Mullerian agenesis, complete transverse vaginal septum, or imperforate hymen are other causes of primary amenorrhea with normal 46, XX karyotype.

A recent concept, idiopathic hirsutism, may be seen in women who are hirsute but having regular cycles. Circulating androgens are also found to be in normal levels. However, this does not guarantee a normal ovulation; up to 40% of these women may be anovulatory [15].

Body hair distribution is another finding that needs to be examined. Hirsutism refers to the presence of male-like pattern of terminal hairs in women. This may indicate elevated serum androgens. It should be differentiated from hypertrichosis as hypertrichosis is excessive hair anywhere on the body, while hirsutism is only on the androgen-sensitive areas, that is, chin, upper lip, chest, upper back, etc. Hirsutism is traditionally evaluated according to Ferriman–Gallwey (FG) scoring system [16], and this system was modified by multiple investigators and is called the modified FG system [17–20]. A score of 8 or higher is considered hirsutism. However, today, visual assessment of terminal body hair on the androgen-sensitive areas being >5 mm in length has become the most widely used method to evaluate hirsutism [21].

Thyroid and breast

The thyroid gland needs to be examined for goiter or thyroid nodules in all patients. A breast exam may uniformly be performed in all patients or only in the presence of a history of oligomenorrhea to investigate galactorrhea, when breast secretion is reported by women and when there are concerns about underdeveloped secondary sex characteristics (and anyone with suspicion of a breast mass). While evaluating galactorrhea, current medication use and recent pregnancy/delivery history need to be kept in mind. Consideration should be given to the testing of serum thyroid-stimulating hormone (TSH) and prolactin levels in all patients with ovulatory problems and irregular menstruations. Hyper- or hypothyroidism may cause menstrual abnormalities, and subclinical levels should likely be corrected prior to pregnancy to prevent any complications, particularly in the presence of antithyroid antibodies [22–24].

Pelvic exam

The pelvic exam revealing a fixed uterus brings about questions regarding adhesions caused by endometriosis or pelvic adhesions due to previous PID, peritonitis, tuberculosis, or previous surgery. It may also reveal vaginismus preventing sexual intercourse. Detection of a large uterus may suggest uterine fibroids, or a bulky uterus may suggest the presence of adenomyosis. Again, pain and/or rectovaginal sensitivity or nodules during pelvic exam may suggest endometriosis.

Pelvic ultrasound

The pelvic ultrasound examination when integrated with history and physical exam provides valuable further information and/or clarification to the fertility evaluation. Although an examination by a transvaginal transducer often provides sharper images of the uterus and ovaries in most cases, a transabdominal ultrasound exam is also performed when image optimization is not conceivable by the transvaginal route due to patient habitus (often high BMI), large uterus and/or fibroids, ovaries located high in the pelvis, and when the structure of interest falls out of the depth of penetration of the transvaginal transducer. A transabdominal ultrasound exam is particularly instrumental in obtaining a clear image in selected cases where the ovaries are anteriorly located and far from the vaginal vault. The ultrasound exam can provide accurate information about ovarian morphology, antral follicle count (AFC), presence of uterine fibroid(s) and their locations, uterine anomalies, and adenomyosis.

Pelvic ultrasound examination provides information not only on pelvic anatomy but also on ovarian reserve through ovarian volume and AFC.

The quality of the pelvic ultrasound exam is highly machine-, operator-, and subject-dependent. Therefore, information obtained from poor-quality ultrasound images needs to be interpreted cautiously, and if the AFC estimation is not believed to be accurate, ovarian reserve evaluation needs to be completed by blood tests.

Evaluation of ovarian reserve

Declining fecundability rates with increasing female age are well documented. Female age is an important parameter of the ovarian reserve evaluation. It is often used as a *post hoc* test as the ovarian reserve of younger women is expected to be higher compared to older females and with oocytes from younger women having the greatest potential for euploidy [25]. The most often used ovarian reserve tests (ORTs) are the AFC, serum anti-Mullerian hormone (AMH), and early follicular phase serum FSH (to be interpreted with the concomitant serum estradiol level). Another biomarker for ovarian reserve, more rarely used as compared to AFC, AMH, and FSH, is serum inhibin B levels.

Ovarian reserve evaluation not only helps diagnosing the underlying pathology in cases of menstrual irregularity/amenorrhea but also helps determine the FSH dose to be administered in ovulation induction for timed intercourse or insemination and ovarian stimulation for IVF treatment.

AFC: The presence of 12–20 or more follicles, 2–9 mm in diameter, in an ovary, and/or ovarian volume >10 mL qualifies the ovary as polycystic ovary (PCO) [26]. These women may or may not have regular menstrual cycles, may or may not have polycystic ovarian syndrome (PCOS). However, they are expected to respond well or overrespond to the ovarian stimulation during IVF treatment. On the other hand, women with total AFC of 12 or less are at risk for decreased ovarian reserve and suboptimal cumulative pregnancy rates at IVF. An AFC less than six is indicative of poor ovarian reserve [25].

AMH: AMH is a glycoprotein produced by antral and preantral follicles. It can be measured anytime throughout the menstrual cycle as it is predominantly FSH-independent and it has low intra- and intermenstrual cycle variability [27,28]. It is arguably the most accurate predictor of ovarian reserve and treatment outcome [29–32]. In the past, there was no consensus regarding the AMH levels referring to normal ovarian reserve due to poor assay reproducibility [33,34]. However, the Beckman–Coulter assay for AMH has provided high levels of reproducibility. With other assays, levels between 1 and 4 ng/mL may be considered as cut offs for normal ovarian reserve [35]. Lower values refer to poorer ovarian reserve. With the Beckman–Coulter assay, serum levels of less than 1.2 ng/mL are considered a poor prognosis group.

FSH: Basal (cycle day 2–5 of a spontaneous or induced menstrual cycle) serum FSH and estradiol levels are used in ovarian reserve evaluation. In women with oligomenorrhea or amenorrhea, these tests (FSH and estradiol) can be performed at random and may detect diminished ovarian reserve (serum FSH levels above 12–15 IU/L and normal serum estradiol levels), menopause (high FSH typically above 40 IU/L and very low estradiol levels), or hypothalamic amenorrhea (FSH low [30% of subjects, <2.0 IU/L] or normal [70% or subjects, 2–10 IU/L] and low or undetectable serum estradiol levels). This occurs because the assay often measures immunizable and not bioactive FSH that is released in hypothalamic amenorrhea. Rarely serum LH levels can help evaluate primary amenorrhea.

Suboptimal cumulative pregnancy rates are reported in women with lower ovarian reserve in IVF treatments. However, in women without infertility and low ovarian reserve, spontaneous pregnancy rates may be similar to women with normal ovarian reserve. For example, infertile patients had similar AMH levels compared with controls with no history of infertility in an age-adjusted regression analysis [36]. In multiple other studies, the biomarkers (not age) of ovarian reserve are not convincingly shown to have correlation neither with chances of spontaneous pregnancy nor with miscarriage rates [37–40]. Women with poorer ovarian reserve, compared with same age group women with normal ovarian reserve, have similar spontaneous pregnancy chances and similar miscarriage rates. Therefore, interpretation of test results and patient counseling should be done with caution.

Inhibin-B: Inhibin-B declines with dwindling ovarian reserve in most studies [41,42], however not in all of them [43,44]. It is mostly used in research settings rather than being commonly employed as an ovarian reserve biomarker. It is argued that there may be a delay in decline of inhibin-B compared to the actual decline in ovarian reserve [44].

Ovulation assessment

Ovulation status needs to be assessed in all women evaluated for infertility.

The majority of women with a history of menstrual cycles between 24 and 35 days in duration have ovulatory cycles [45,46], especially if they have molimina symptoms. Molimina symptoms are feelings in women associated with progesterone production or withdrawals which include breast tenderness, sleep difficulties, a sense of bloating, mood changes and irritability, headache, fatigue, acnea, increased appetite in the luteal phase, etc. However, in a moderate-sized study, 89% of women with regular cycles and molimina were ovulating according to progesterone measurements [47].

In addition to menstruation cycle history, women may present their basal body temperature records as a tool of *retrospective* identificator of their ovulatory status. In combination with menstrual history, the presence of molimina symptoms and basal body temperature charts can help identify the ovulatory women without further testing [48]. For prospective identification of ovulation, urinary LH kits (ovulation predictors) can reliably be used particularly in women with regular cycles [49]. However, it must be kept in mind that women in menopause, perimenopause, or having PCOS syndrome may have elevated LH levels above the cut off of 20 IU/L and not be ovulating.

It's reported that over one-third of women with regular menstrual cycles may not have ovulation [50]. Therefore, a midluteal serum progesterone measurement can be used as a reliable test; >5 ng/mL serum progesterone concentration suggests a specificity of 98.4 (95% CI 96.0–99.5) and with a sensitivity of 89.6 (95% CI 85.2–92.9) for ovulation [51].

Women with menstrual cycle intervals longer than 35 days are considered anovulatory; however, among them up to 50% may be ovulating where the study group included women with cycles 36–45 days [52]. Therefore, oligo-ovulatory may be a more accurate term for them rather than anovulatory. Ovulation can be assessed with serial pelvic ultrasound exams and urinary ovulation predictors (urinary LH test kits) as in this study, and it may further be combined with serum progesterone measurement one week after the ovulation for a more accurate assessment. However, this is neither practical nor cost-effective for a general use. Yet, when intended, to clarify ovulation status, weekly serum progesterone measurement starting from cycle day 21 or weekly ultrasound monitoring to observe follicular development and presence of a corpus luteum may be used to clarify the situation in these women when necessary.

For women who do not have regular cycles, whether amenorrheic or oligomenorrheic, and/or where serum progesterone level indicates anovulation, serum FSH and estradiol levels together with or without other parameters (AFC and AMH) of ovarian reserve are used to establish the differential diagnosis of anovulation (Table 1.4). Serum prolactin and TSH assays need to be included in the tests. In routine, daily practice these tests; in case of oligo- or amenorrhea, these are mostly requested together instead of a stepwise approach. When PCOS is diagnosed, a metabolic screening is also warranted, especially in women with high BMI. Testing serum lipid levels, renal function (blood urea nitrogen and creatinine), and diabetes via hemoglobin HbA1c, ideally less than 6% or 42 mmol/mol is desired prior to treatment, must be less than 7% to conceive in diabetic women; otherwise, risks of fetal anomalies increase. Diabetes is suspected when HbA1c is >6.5% or >48 mmol/mol. Liver function tests should be considered where HbA1c is high.

Hyperprolactinemia prevents ovulation through LH suppression. It may or may not cause galactorrhea and therefore needs to be tested in women with anovulation. Mild hyperprolactinemia (20–50 mIU/L) may impair fertility without causing amenorrhea by causing inadequate progesterone secretion.

To keep in mind, if the amenorrhea (investigated as anovulation) follows a delivery, curettage, myomectomy, and rarely PID, it may be mistaken for anovulation while the etiology is mechanic (Asherman's syndrome or cervical stenosis).

TABLE 1.4 Differential diagnosis of anovulation with respect to WHO classification.

	FSH	Estradiol	AFC	Endometrium	AMH
WHO Group 1	Low–Normal	Low	Low–Normal–High	Thin	Like Normal controls [53]
WHO Group 2	Normal	Normal	Normal–PCO	Normal—thick	High
WHO Group 3	High	Low	Low	Thin	Low
Hyperprolactinemia	Low–Normal	Normal	Low–Normal–High	Thin–Normal	

Tubal patency evaluation

Tubal patency should be evaluated in all women unless ovulatory dysfunction is present. However, it may be deferred in a severe male factor infertility necessitating IVF where the woman does not have any history of prior sexually transmissible disease. Similarly, women without history of infertility planning donor sperm insemination can defer the tubal evaluation and try insemination for an initial three to six cycles. In anovulatory infertility, failure to conceive within three to six months of ovulation induction should then result in evaluations of tubal patency.

The gold standard of tubal function evaluation is exploratory laparoscopy and chromotubation. However, this is not recommended for routine use for women who do not have endometriosis symptoms, prior pelvic surgery, and unsatisfactory hysterosalpingography. An exploratory laparoscopy would enable diagnoses of peritubal adhesions, which could result in a patent but nonfunctional Fallopian tube.

Both hysterosalpingogram (HSG) and hysterosalpingo-contrast sonography (HyCoSy) are common methods to evaluate fallopian tube patency. In the presence of a single patent fallopian tube, pregnancy rates are like those of women with both tubes patent per the current metaanalyses. The fallopian tubes are being evaluated using saline instillation sonography (SIS), ultrasound visible foam solutions, or HyCoSy. SIS provides an excellent view of the uterine cavity for both congenital anomalies (especially when combined with 3D ultrasound) and acquired pathologies (polyp, submucosal fibroid, and synechiae). However, it is difficult to diagnose a hydrosalpinx with SIS as compared to HSG. Tubal patency may also be present and not seen with SIS. With HyCoSy, using 3D ultrasonography and intrauterine foam installation, fallopian tube patency evaluations are comparable to outcomes seen with HSG and laparoscopy but with less discomfort.

Other investigations

Serum testosterone (free or total), DHEAS (dehydroepiandrosterone sulfate), and early morning fasting 17-hydroxyprogesterone levels can aid in evaluating women with hirsutism. (17-hydroxyprogesterone should be done in the early or mid-follicular phases in ovulatory women).

Tests for hepatitis B, hepatitis C, syphilis, and human immunodeficiency virus are helpful since gametes and sperm will be handled.

Rubella and varicella immunity can be evaluated prepregnancy, and vaccination is offered to women who are nonimmune given the potential to protect the fetus.

Serum prolactin levels should be tested in women with galactorrhea and menstrual disturbances.

A hemoglobin hematocrit and platelet count should be obtained before any surgical procedures including IVF.

Genetic screening for carrier status is often offered before conception to couples with large panels available today based on ethnic origin. Although this is not required in many countries, it is prudent to have documents offering it to the couple before conception.

Screening for thalassemia and sickle cell carrier status in women or couples of Asian, Mediterranean, and African descents may also be indicated preconception.

Many laboratories offer comprehensive genetic carrier screening that can be tailored to ethnic origin and may be helpful for certain couples to prevent rare recessive disease. The payment for such may or may not be covered by insurance.

Conclusion

All women need a thorough history and physical exam. They should undergo transvaginal ultrasonography, ovulation assessment, ORT, and tubal patency evaluation in most cases. Documentation of ovulation by blood test may not be required in all cases. A tailored screening before initiating fertility care for each patient is recommended regarding their needs.

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CHAPTER

2

The fertility evaluation of the male partner

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Overview

Infertility is a global health issue that affects millions of people worldwide, and it is defined as the inability to conceive after 12 months of attempted conception. Furthermore, it has been shown that each gender contributes equally to infertility [1]. Hence, it is recommended to evaluate both male and female partners concurrently. The causes of male infertility are mainly involved in either defect of sperm production or defects in sperm transport and emission; however, in some patients, the exact reason for infertility is unidentifiable and termed idiopathic or unexplained (summarized in Fig. 2.1) [2]. Thus, it is imperative to acquire a proper detailed evaluation as the treatment approach can vary significantly. The recommended time for initiating fertility evaluation is typically after 12 months of failed attempts of conception. However, a study of fertile couples showed that after six months of regular conception attempts, the cumulative pregnancy rate was 88%, increasing to 98% after 12 months [3]. Therefore, as a minority of patients achieve conception after the six months mark, it is reasonable to initiate the fertility evaluation after six months in the following circumstances: in men with abnormal reproductive history (e.g., bilateral cryptorchidism, history of pelvic radiotherapy, and history of receiving cytotoxic chemotherapy) or if a man questions his fertility potential in the absence of a female partner, or when female partner's age is over 35 years or in the presence of abnormal female reproductive history [4], with the understanding that conception may occur in some couples shortly after.

Another indication for male fertility evaluation is recurrent pregnancy loss (RPL), defined as two or more failed pregnancies [5]. Evaluation may help identify the exact cause for RPL in nearly 50% of couples. Female factors include genetic causes, anatomic abnormalities in the female genital tract such as uterine septum, submucosal uterine fibroids, and adhesions, as

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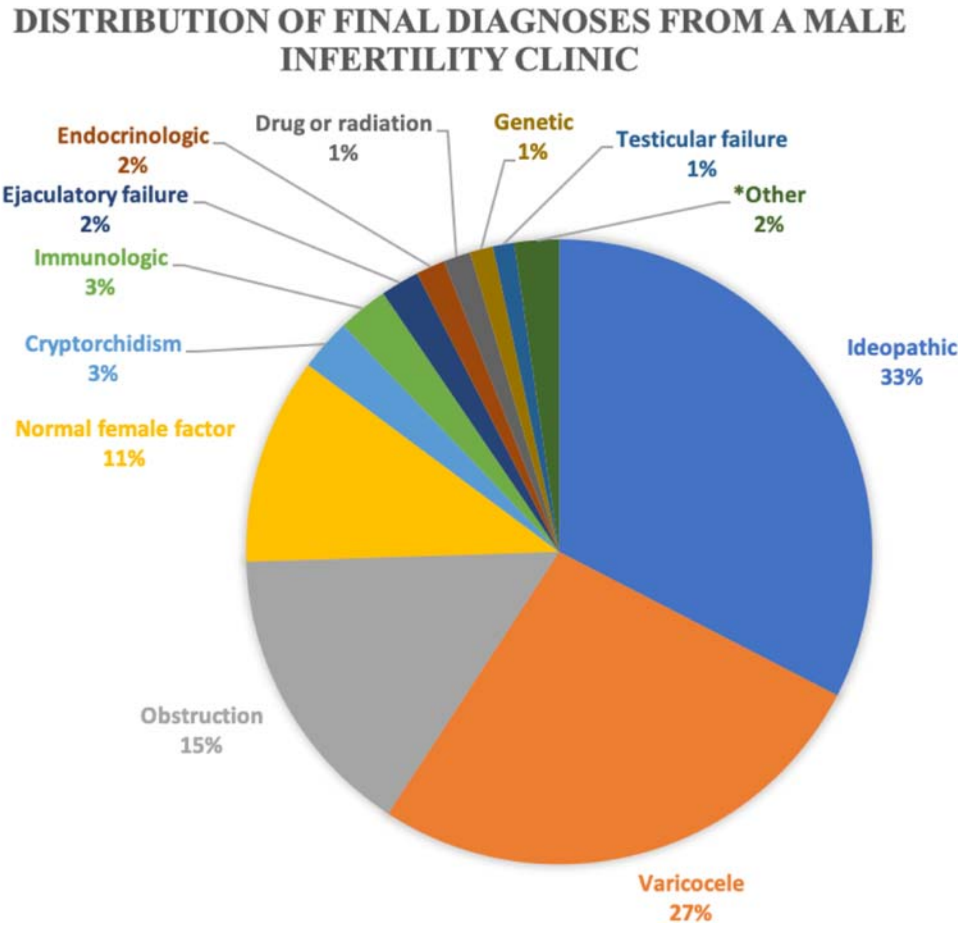

FIGURE 2.1 * Other diagnoses include (sexual dysfunction, pyospermia, cancer, systemic disease, infection, torsion, and ultrastructural). Data from Sigman, M., Lipshultz, L., & Howards, S. Office evaluation of the subfertile male. In: Lipshultz, L., Howards, S., & Niederberger, C. (Eds.), Infertility in the Male. Cambridge: Cambridge University Press; 2009. pp. 153–176. http://doi.org/10.1017/CBO9780511635656.011

well as endocrine disorders, infections, and hematologic and immunologic disorders. Male factors mainly involve genetic anomalies, such as karyotypic abnormalities and abnormal sperm DNA fragmentation [4,6,7].

Clinical history

Multiple key factors must be determined when acquiring a clinical history. The first factors to acquire are the male and female age, the duration of attempted conception, coital frequency and timing, and whether couples have primary infertility (never conceived in the past) or secondary infertility (prior conception with the same partner or different partner). The importance of age stems from the finding that this factor has a significant impact on female fecundity, particularly, with age over 35 [8]. It is also the most important predictor for fertility outcome in couples undergoing therapy [9–11]. The effect of paternal age on reproductive potential is less clearly defined. Although studies have shown that aging in men can negatively impact bulk semen parameters [12–15], the decline is modest. However, paternal aging can induce chromosomal anomalies, genetic mutations, and epigenetic changes. Disorders such as trisomy 21, achondroplasias, hemophilia A, and neurodevelopment disorders (schizophrenia, bipolar disease, and autism) have been linked to advanced paternal age [16–18]. One possible explanation for these associations is that the spermatogonia of older men have undergone numerous cell divisions with a consequent increase in the rate of DNA transcription errors and defects in chromosomal integrity [19]. Therefore, age may impact clinical decisions when counseling infertile couples as the infertility specialist may recommend more invasive reproductive technologies (i.e., in vitro fertilization [IVF], intracytoplasmic sperm injection [ICSI]) as the optimal treatment.

The next step is to determine the possible etiologic factor for infertility. Common nonmonic used is TICS which stands for (T-spermatotoxicity, I-infection, C-childhood diseases, S-sexual history) [20].

Several mechanisms can induce spermatotoxicity. One mechanism involves hormonal dysregulation by medications such as antiandrogens, spironolactone, retroviral protease inhibitors, reverse transcriptase inhibitors [21], exogenous steroids, and testosterone [22]. Another mechanism that may adversely impact spermatogenesis is thermal toxicity. The testes are temperature-sensitive organs, and scrotal temperature is maintained at 2°C–4°C below core body temperature. Behaviors that elevate scrotal temperature include the use of hot baths/sauna, use of laptop computers, and prolonged sitting [23]. Important environmental factors that can cause testicular dysfunction include exposure to pesticides and heavy metals [4]. Radiation and cytotoxic chemotherapy may also induce toxic injury to the testes. Agents such as cyclophosphamide, cisplatin, etoposide, and bleomycin are associated with spermatotoxicity in a dose- and time-dependent manner [24].

Infections of the genitourinary tract can cause a significant impact on fertility potential by either impaired spermatogenesis or anatomic obstruction from scarring of the genital tract [25]. Functional impairment entails either a direct effect of the infectious organism on spermatozoa or through induction of an immunologic response to male reproductive organs leading to secondary sperm dysfunction [26].

The history of childhood diseases is an integral part of fertility evaluation. History components include past surgical history, specifically inguinoscrotal surgeries such as hydrocele, inguinal hernia repair, and correction of cryptorchidism. Although rare, possible complications are vasal obstruction or testicular atrophy [27,28]. Furthermore, surgeries involving the urethra and bladder neck can cause strictures or retrograde ejaculation [2,29]. Moreover, history of testicular torsion can impact fertility potential. Studies have shown that contralateral testicular biopsy findings are abnormal in 57%–88% of males when torsion occurs [30]. Furthermore, approximately half of men with torsion develop adverse spermatogenic effects [30]. Overall, 36%–39% of men will have sperm concentration below 20 million/mL [30].

Sexual history should include any disorders of erectile function or ejaculation. Furthermore, the timing of intercourse, coital frequency, and use of contraceptives should be noted.

Physical examination

Physical examination of the male partner should include a general physical exam and a focused genital exam. The goal of the general physical examination is to look for any stigmata of genetic and syndromic conditions such as Klinefelter syndrome (classical features of an eunuchoid appearance, tall stature for age, and gynecomastia) [31]. Furthermore, obesity should be noted as it has been associated with male reproductive dysfunction. Obese men generally have elevated estradiol due to peripheral conversion from testosterone by the enzyme aromatase produced by the abundant adipose cells [32].

The male genital exam should include examination of the penis, scrotum, testis, epididymis, and cord structures. Penile exam should exclude abnormalities that interfere with the semen emission such as phimosis, meatal displacement in hypospadias or epispadias, meatal stenosis, and significant penile curvature [33]. In scrotal exam, the clinician should note the testicular position, size (by either Prader orchidometer or Caliper orchidometer, Fig. 2.2), consistency, and the presence of masses or hydrocele. Furthermore, the epididymis should be examined to rule out engorgement (suggestive of obstruction), and the presence of the vas deferens must be confirmed. Finally, one must examine the spermatic cord to rule out the presence of varicocele. The modern evidence-based clinical grading varicocele system is summarized in Table 2.1 [34,35].

Semen analysis

The joint American Urological Association (AUA) and American Society for Reproductive Medicine (ASRM) guidelines on male infertility recommend one or more semen analyses for couples presenting for fertility evaluation [36]. The semen analysis provides important information on the functionality of the seminiferous tubules, sperm production, transport, and emission. The semen analysis should include semen volume, pH, sperm concentration and count, sperm motility, and morphology [37]. It is generally advised that semen should be collected after a minimum of two days of ejaculatory abstinence. One of the drawbacks of the semen analysis is that it is a crude indicator of male fertility potential and should be

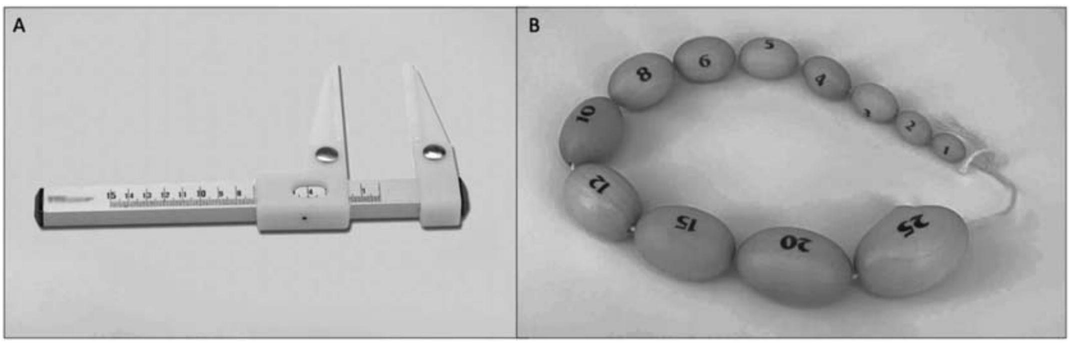

FIGURE 2.2 (A): Caliper orchidometer and (B): Prader orchidometer.

TABLE 2.1 Varicocele grading system [34,35].

Varicocele grade	
Subclinical	Only detected by radiographic imaging such as Doppler ultrasound
I	Palpable only with the patient standing and performing a Valsalva maneuver
II	Palpable but not visible by examination
III	Both palpable and visible by examination

interpreted with caution. Moreover, the reference range for semen parameters published by the World Health Organization (WHO) was set in a population of fertile couples who achieved natural conception [37]. However, Guzick et al. [38] examined the semen parameters in cohorts of fertile and infertile men and demonstrated that the presence of multiple abnormalities in semen parameters progressively increased the odds of infertility. Certain anomalies in the semen parameters can preclude a man's ability for natural conception, such as azoospermia, complete globozoospermia (round headed spermatozoa without acrosomes), necrozoospermia (dead spermatozoa), or complete asthenozoospermia (lack of sperm motility) [36]. Therefore, semen analysis remains an essential tool in the evaluation of infertile men. The latest (sixth edition) WHO 5th centile semen parameter reference limits (with CI) are summarized in Table 2.2 [39].

TABLE 2.2 2020 WHO sixth edition semen parameter reference limits (lower 5th centile) [39].

	5th centile	(95% CI)
Sperm concentration (10^6 per ml)	16	(15–18)
Total sperm number (10^6 per ejaculate)	39	(35–40)
Total motility (PR + NP, %)	42	(40–43)
Progressive motility (PR, %)	30	(29–31)
Nonprogressive motility (NP, %)	1	(1–1)
Immotile spermatozoa (IM, %)	20	(19–20)
Vitality (%)	54	(50–56)
Normal forms (%)	4	(3.9–4)

Hormonal evaluation

The importance of hormonal evaluation stems from the fact that spermatogenesis is highly dependent on intratesticular testosterone synthesis and hypoandrogenism is highly associated with male infertility [40]. A study has shown that approximately 45% of men with azoospermia caused by spermatogenic dysfunction are observed to have testosterone levels below 300 ng/dL [41]. Therefore, the AUA [36] and ASRM [42] recommend endocrine evaluation with follicle-stimulating hormone (FSH) and testosterone for infertile men with impaired libido, erectile dysfunction, oligozoospermia or azoospermia, atrophic testes, or evidence of hormonal abnormality on physical evaluation. Furthermore, in men who have low morning testosterone (<300 ng/dL), it is recommended to repeat the measurement of total and free testosterone, as well as serum LH, estradiol, and prolactin measurements [36]. Moreover, hormonal profile, semen volume, and physical exam could aid in differentiating genital tract obstruction from impaired sperm production. Schoor et al. [43]. showed that 96% of men with obstructive azoospermia had FSH assay values of 7.6 IU/L or less and testis long axis greater than 4.6 cm, whereas 89% of men with azoospermia caused by spermatogenic dysfunction had FSH values greater than 7.6 IU/L and testis long axis 4.6 cm or less [4]. In men with suspected obstructive azoospermia, a confirmatory diagnostic testicular biopsy is recommended to establish the diagnosis of obstruction and sperm cryopreservation from the biopsy specimen should be considered if ART is contemplated.

Genetic studies

Genetic disorders are diagnosed in nearly 1% of men presenting for fertility evaluation [2]. The most common abnormalities are numerical karyotype abnormalities and less frequently structural anomalies such as deletions, duplications, and inversions of a region of autosomal or sex chromosomes [44]. An important example of a karyotype abnormality is Klinefelter syndrome (47, XXY). Most men with Klinefelter syndrome are azoospermic at presentation. However, men should be counseled that few nonmosiac XXY men will have sperm in the ejaculate but have a very low probability of natural conception [45–47]. As such, patient with Klinefelter syndrome could be offered sperm retrieval using conventional testicular sperm extraction (TESE) or microscopic testicular sperm extraction (micro-TESE) with subsequent assisted reproductive technologies (such as ICSI). Meta-analysis showed that mean sperm retrieval rates in patients with Klinefelter syndrome were 43% for TESE and 45% for micro-TESE with overall live birth rates of 43% [48]. The second most common genetic cause of male infertility is Y chromosome microdeletion, specifically, the azoospermia factor (AZF) region microdeletion. The AZF region consists of three areas encoding genes involved in spermatogenesis, namely AZFa, AZFb, and AZFc. Knowledge of the affected area is clinically important and can aid in decision-making as most individuals with AZFc deletion exhibit active spermatogenesis [49]. Individuals with partial microdeletion of the AZFa and AZFb may also harbor areas of spermatogenesis. In contrast, a complete deletion of AZFa and AZFb is associated with absent spermatogenesis and affected individuals should not be offered TESE for ART but rather consider other pathways for parenthood [49,50]. According to the AUA guidelines, karyotype and Y-chromosome microdeletion analysis should be recommended for men with primary infertility and azoospermia or severe oligozoospermia (<5 million sperm/mL) with elevated FSH or testicular atrophy or a presumed diagnosis of impaired sperm production as the cause of azoospermia [36].

Histopathology

Testicular biopsy is usually indicated to confirm the diagnosis of obstructive azoospermia by showing a normal spermatogenesis. In contrast, the role of biopsy in men with nonobstructive azoospermia is limited. However, studies have shown that testicular histology could provide an insight into the chance of successful sperm retrieval [43,51]. Studies have shown that men with Sertoli cell-only syndrome (SCO) have a significantly lower rate of sperm retrieval when compared with men with predominantly maturational arrest or hypospermatogenesis (HS) (16.3% vs. 62.5% and 89.2%, respectively) [52]. Despite this promising predictive value, the performance of a biopsy in men with nonobstructive azoospermia is not recommended as it is an invasive procedure with potential complications such as postprocedural pain, infection, and hematoma formation. Furthermore. it may not change management as men with Sertoli-only cell may still proceed for sperm retrieval methods to achieve conception [53].

Conditions with special considerations

Congenital bilateral absence of the vas deferens

Congenital bilateral absence of the vas deferens (CBAVD) occurs in about 1%–2% of infertile men [54]. The clinical presentation of CBAVD involves normal to slightly small testicles with nonpalpable vas deferens, normal hormonal profile (specifically FSH), and low volume azoospermia with acidic pH and undetectable or low fructose concentration [55]. CBAVD is commonly associated with cystic fibrosis, which is an autosomal recessive disorder that arises from mutation in cystic fibrosis transmembrane conductance regulator (CFTR gene) [56]. The gene encodes for a cyclic adenosine monophosphate (cAMP)–dependent chloride channel. The mutation results in defect in anion transport and fluid secretion that ultimately leads to obstruction and/or atrophy of the epididymis and vas deferens during embryogenesis [57,58]. Moreover, studies have identified more than 1600 CFTR mutations with variable disease phenotype presentations [59–62]. The most common severe mutation is ΔF508 [60]. This mutation consists of a three nucleotide deletion in codon 508, resulting in loss of phenylalanine amino acid, which subsequently prevents the protein migration to the plasma membrane [63]. Although the most common genitourinary manifestation of CFTR mutation is CBAVD, other manifestations include congenital unilateral absence of the vas and idiopathic epididymal obstruction [56,64]. Given the hereditary nature of the CFTR mutation and its significant clinical implication to offspring, the AUA guidelines recommend genetic evaluation of the female partner of men with CBAVD or a CFTR mutation. Formal genetic counseling includes discussing carrier status, genetic hereditability, and preimplantation genetic diagnosis for couples who test positive for a mutation [36].

The AUA also recommends renal ultrasound for patients with vasal agenesis to evaluate for renal abnormalities [4]. During embryogenesis, the development of the mesonephric duct which gives rise to the epididymis, vas deferens, and seminal vesicle is highly associated with the primitive kidney. Therefore, abnormalities in the mesonephric duct can lead to renal anomalies. Studies have found that 26%–75% of men with unilateral absence of the vas deferens will have ipsilateral renal anomalies including agenesis [65,66]. In men with bilateral vasal agenesis, the prevalence of renal anomalies is lower at 10% [67]. Renal anomalies may also be present in men with CBAVD and CFTR mutation [68,69]. Therefore, it is recommended to offer abdominal imaging to all men with vasal agenesis regardless of the CFTR status [4].

Ejaculatory duct obstruction

Ejaculatory duct obstruction is rare and diagnosed in 1%–5% of infertile men [70]. The etiology of the obstruction may be congenital or due to acquired disorders such as seminal vesicle calculi, postinflammatory scar formation at the prostate level, and trauma or previous transurethral surgery [2]. Main presenting complaints include infertility, ejaculatory and testicular pain, perineal discomfort, and hematospermia [71]. Rarely, it can present with low back pain, urinary obstruction, dysuria, and difficulty in defecation [70]. The diagnosis should be suspected in the presence of low volume azoospermia, fructose negative semen with normal hormonal profile, and genetic screening [72]. The diagnostic modality of choice is radiological in the form of either transrectal ultrasound or pelvic magnetic resonance imaging [73].

Globozoospermia

Globozoospermia is a structural sperm defect, characterized by the presence of rounded head spermatozoa along with lack of acrosome and cytoskeleton defects around the nucleus [74]. This condition is classified into two types. Type I (total or classic globozoospermia, or true round-head only syndrome) is defined by the finding that all spermatozoa present in the semen harbor a small, rounded, and acrosome-free heads leading to primary infertility as the spermatozoa cannot penetrate the zona pellucida due to lack of acrosome. Alternatively, type II globozoospermia (partial globozoospermia) is characterized by the presence of both normal and round-headed spermatozoa, with an estimate of 20%–90% of spermatozoa have no acrosome [75,76]. Men with type I globozoospermia cannot achieve natural conception and require ICSI for fertilization. However, ICSI fertilization rate is reported to be lower in globozoospermic spermatozoa compared with normal spermatozoa due to its reduced ability to activate oocyte, as well as their poor chromatin condensation and high sperm DNA fragmentation, with reported overall fertilization rate below 50% [77–79].

Recurrent pregnancy loss

RPL is defined as two or more failed pregnancies [5]. Evaluation may identify the exact cause for RPL in nearly 50% of couples. In men, causes are usually related to genetic anomalies, such as numerical or structural karyotypic abnormalities and abnormal sperm DNA fragmentation [4,6,7]. Therefore, the AUA recommends evaluation of the male with karyotype and sperm DNA fragmentation. Genetic assessment may also include sperm aneuploidy testing [4].

Surgical sperm extraction techniques

Several methods have been described and utilized to harvest spermatozoa in infertile men.

Testicular sperm aspiration (TESA) involves percutaneous needle aspiration of the testis using a 16 to 22-guage needle [80]. TESA with subsequent IVF or ICSI is most useful in men with obstructive azoospermia such as men with congenital absence of the vas deferens or men with failed or surgically unreconstructible obstruction or in men who do not desire for reconstruction [81]. In contrast, TESA is not recommended in men with NOA as it is inefficient (sperm retrieval rate of 22%–30%) [82,83] and could be traumatic [80]. Another method involves testicular sperm extraction through open biopsy (TESE). This method offers a higher sperm retrieval rate than TESA in men with NOA [84]. However, TESE involves extraction of a large amount of testicular tissue and interruption of testicular vessels [85,86] resulting in testicular damage through tissue loss and formation of hematoma with subsequent pressure atrophy and inflammation that may limit patients to one retrieval attempt [87]. In 1999, Schlegel et al. described a new technique aiming to reduce testicular tissue loss and damage. It involves microdissection testicular sperm extraction (micro-TESE). Based on the concept of identifying spermatogenically active regions of the testis through direct examination of the individual seminiferous tubules by high power (20-30X) microscope, seminiferous tubules containing large number of developing germ cells are likely to be larger and more opaque than tubules containing fewer germ cells or Sertoli cells only [88]. Studies have shown that in men with NOA, micro-TESE offers sperm retrieval rates in the range of 40%–60% compared to 16.7%–45% for conventional TESE [89] while improving the frequency of successful sperm retrieval due to significantly less tissue loss and testicular damage [88,90–94]. Furthermore, for men with cryptozoospermia, micro-TESE resulted in sperm recovery of up to 96% [95]. Complications are rare and most are managed conservatively such as contained hematoma, wound infection, and de novo testosterone deficiency; more serious complication includes testicular abscess requiring surgical drainage [83].

Infertility and general health

Male infertility with abnormal semen parameters may be a sign of a serious underlying disorder. Studies have shown that 1%–6% of men have undiagnosed medical disease during fertility evaluation [96,97]. Furthermore, infertile subjects were found to have more comorbidities compared to fertile controls. Malignancies are diagnosed at a higher rate in infertile men [98]. Studies have shown that men with abnormal semen parameters or those requiring ART for conception have a higher rate of testicular and prostate cancer [99–102]. Glazer et al. have shown that mortality rates are positively associated with abnormal semen analysis [103]. Therefore, it is recommended to counsel infertile male about the associated health conditions with infertility [4] such as testosterone deficiency in men with Klinefelter syndrome [104,105], pulmonary, pancreatic disorders, and dental carries in men with cystic fibrosis [106], and testicular cancer in men with a history of cryptorchidism [107,108].

Summary

A male factor contributes to couple infertility in up to 50% of infertile couples. Therefore, it is recommended to evaluate both male and female partners concurrently when an abnormality is detected on semen analysis. The etiology of male infertility is multifactorial and is generally caused by a defect in sperm production and/or maturation or in sperm transport and emission. However, in some patients, it is idiopathic or unexplained. A complete history and physical examination are essential in the evaluation of the infertile male. Additional evaluation including hormonal profile, genetic testing, and imaging studies are required depending on the initial findings. Infertile men with abnormal semen parameters should be counseled on the health risks associated with their infertility.

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CHAPTER

3

Sperm DNA fragmentation and male infertility: a comprehensive review for the clinicians

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Sperm genetics

Sperm chromatin and DNA packaging

In the study of human male reproduction, chromatin structural integrity and DNA assessment represents a focal point in the current research, which may offer a better and more comprehensive understanding of male fertility disorders. With improvements in the profiling assays, the current biological evidence observed a paternal age-related and transgenerational effects on the various male reproductive disorders [1]. Basic semen parameters, though being widely used to assess men's fertility potential, still lack efficiency in the evaluation of male fertility, unlike sperm DNA and chromatin integrity, which appear more efficient in gauging realistic male reproductive potential. Also, it is not uncommon that a sperm with normal morphology may harbor damaged DNA, and hence are functionally inadequate, an observation that could help resolve the ambiguity behind many cases of unexplained male infertility [2,3]. A healthy and functional human DNA is expected to lack single- or double-stranded nicks or breaks, in addition to the absence of debilitating chemical DNA modifications. DNA repair system exists in the

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pre- and postreplication mechanisms, but this is limited to small breaks, while larger ones are not amenable for repair. Double-stranded DNA breaks are considered extremely mutagenic, which happens due to the absence of repair system during the pronucleus stage where the genomes are being separated [2]. Paternal contribution to embryonic development and subsequent offspring's health and viability mandates the existence of appropriate DNA integrity through a rigorous state of methylation, which is a fundamental feature of the germ cell line developmental process [1,4]. Sperm maturation occurs during the spermiogenesis stage, after the early mitotic and meiotic divisions. This last step of spermatogenesis involves sperm chromatin packaging [1,2]. Eventually, the haploid DNA is packaged into the sperm head which is relatively small [5]. The structural and molecular changes undergone by the sperm genetic material result in a compacted and insoluble chromatin, which protects the sperm genetic material during its transport [2]. This results in organization of the sperm DNA in the form of loops, which are subsequently attached in the nuclear matrix to specific attachment sites [6]. Chromatin remodeling and compact packaging reduce the sperm nuclear size, and hence facilitating transport while adding further protection to the sperm genome [2,7]. In mammals, histones are initially replaced by a transition protein synthesized and released by the differentiating spermatids, which binds to sperm DNA prior to the protamines [8–10]. DNA is predominantly bound to protamines (85%–90%) and to lesser extent histones (10%–15%) and could be measured using DNA Raman spectroscopy [3]. Protamines serve to condense the genetic material resulting in transcriptional inactivation [7,11]. The process of protamination, which begins in the testicles, is continued during the epididymal transit phase with the formation of the disulfides bonds, which add further compaction to the DNA [2]. Protamines compris arginine-rich structural elements acting as DNA anchoring sites. They exist in two forms, protamine 1 and 2, which get temporary phosphorylated after their formation until they bind to the DNA, after which the phosphate group is released, concomitantly with oxidation of the cysteine resides, ultimately creating disulfide bridges binding protamines molecules [3,12]. The loss of the disulfide bonds would result in a 10-fold volume increase due to nuclear swelling, that is, chromatin decondensation [13].

Sperm contribution to the developing embryo

The recognition of the degree of sperm contribution toward the developing embryo is important, particularly in the context of assisted reproductive technology (ART). The sperm harbors crucial genetic and epigenetic material that is transferred to the offspring. The sperm contribution is not limited to the genomic transmission, which includes the DNA and chromatin structures, but also similarly important are the epigenetic entities including DNA methylations, histone modifications, noncoding RNAs, oocyte activating factor, and the centrioles [14–16]. During spermiogenesis, 90%–95% of histones to protamines transition take place, which add more chromatin compaction and added genetic preservations [17]. Furthermore, centrioles are considered significant contributions, and though their paternal inheritance remains controversial, they are involved in assembling the sperm aster, involved in microtubule organization and facilitate pronuclear migration, which are important steps in the early embryonic development process [18,19]. Additionally, small noncoding RNAs (nonprotein coding) are forms of sperm epigenomic attributions, which reflect possible transgenerational inheritance of the different paternal exposures [16,20]. This type of RNA is implicated in gene silencing, in addition to the regulation of genetic expression, transcription, and is involved in what is so-called epigenetic memory. Small noncoding RNAs are transmitted to the zygote, regulating the early phases of development [15,21]. Paternal impact on the developing embryo may include failure of fertilization, altered syngamy, and defective early cleavage, in addition to the delayed effects including implantation failure, miscarriage, and developmental aberrations [14].

Sperm DNA repair by oocyte: what we know and what we don't?

Protection of the sperm genome in the male reproductive system is offered by protamine-related DNA dense packaging [22]. Zinc plays an important role in the sperm genetic protection, being bond strongly with chromatin, resulting in further DNA stability and safeguarding from oxidative damage. Upon fertilization, paternal zinc gets eliminated leading to dissociation of the DNA strands [22,23]. The process of DNA repair appears limited to the early phases of sperm development, after which, the stored mature sperm would rely mostly on oocyte and early embryonic DNA repair systems [24–26]. The significance of DNA repair stems from the fact that damaged sperm DNA may transmit abnormal genetic structure to the offspring [27]. Effective oocyte repair system can help in sperm genetic recovery during and after fertilization with evidence suggesting the presence of increased expression of DNA repair genes in the oocyte, which appears to decline with advanced maternal age [28]. Sperm with damaged DNA may prompt the initiation of apoptosis or be offered a chance for repair. It is, however, not uncommon that the oocyte fails to repair paternal damaged sperm DNA, which may eventually result in miscarriage [26,28]. Recurrent ART failure despite the absence of obvious zygotic or cleavage abnormalities may be linked to elevated sperm DNA fragmentation (SDF) rates [26]. Even though oocyte fertilization by a dysfunctional sperm accommodating fragmented DNA is still possible, yet deleterious impact would be expected to manifest after the paternal genome is involved, which happens to be the third day of cleavage and afterward. This negative impact is shown to involve embryonic fragmentation with evidence of developmental retardation, and consequently failure of implantation, if not miscarriage. Conversely, oocytes from healthy and young donors may prevent transmission of DNA paternal aberrations to the embryonic DNA [22,24,27,28].

How do sperm genetics become dysfunctional?

Impairment of sperm genetics can refer to many different molecular abnormalities that damage the sperm genome and can lead to clinical consequences. These include nuclear SDF, mitochondrial DNA damage, epigenetic abnormalities, Y-chromosome microdeletions, and telomere attrition, as well as base modifications or cross-linking that can cause sperm genome damage [29]. Three pathophysiological mechanisms have been described as leading to SDF including defective sperm chromatin packaging, abortive apoptosis, and oxidative damage.

Defective sperm chromatin packaging

Defects in the aforementioned process of sperm chromatin packaging are associated with increased SDF. Immature sperm chromatin, which contains a larger proportion of histones due to incomplete exchange with protamine, has been found in significantly larger proportions among sperm with SDF compared with those without SDF, as demonstrated by aniline blue staining which selectively binds histones [30]. Defective compaction, or decondensed chromatin, renders sperm DNA more susceptible to damage by various insults [31]. Furthermore, during the compaction process, downstream torsional stress in DNA is relieved by topoisomerase enzymes, which create nicks in the DNA strands and subsequent failure to repair these nicks leads to persistent breaks in the sperm DNA strands [32].

Abortive apoptosis

Apoptosis or programmed cell death occurs physiologically in the testes to maintain an appropriate population of germ cells and to remove any abnormal germ cells [33]. Apoptosis can occur via intrinsic pathways, involving mitochondria, and extrinsic pathways. The intrinsic pathway involves certain stimuli such as radiation or free radicals, which lead to changes in mitochondrial permeability and the release of mitochondrial proteins bringing about apoptosis. The extrinsic pathway involves cell–surface receptor interactions, such as the Fas, which is a death receptor that when binds to the Fas ligand leads to downstream cell signaling ending in apoptosis. Both of these pathways lead to the activation of caspases followed by endonucleases, which cause DNA fragmentation as part of the cell death process [34]. Failure to complete the process of apoptosis in spermatozoa, that is, abortive apoptosis, leads to the release of sperms with fragmented DNA [32]. Sperm Fas expression is found to be higher among men with abnormal semen parameters, indicating abortive apoptosis [35]. Activity of caspases, as a marker of apoptosis was found to be significantly higher among the fragmented sperm population when compared with nonfragmented sperm [30]. Finally, magnetic-associated cell sorting (MACS), which removes apoptotic spermatozoa by identifying surface phosphatidylserine (a marker of apoptosis) via binding of annexin V beads, was found to significantly reduce SDF [36].

Oxidative damage

Reactive oxygen species (ROS) are present in low physiological amounts in the male reproductive organs and semen, serving several functions that are essential for normal spermatogenesis, sperm maturation, and fertilization [37]. Oxidative stress occurs when there is imbalance between ROS and antioxidants and which has been extensively studied as a cause of impaired male fertility and increased SDF [38]. ROS are produced by endogenous sources, such as immature spermatozoa and white blood cells, and exogenous sources such as radiation, toxins, and smoking [38,39]. High ROS can damage sperms via lipid peroxidation of polyunsaturated fatty acids in their cell-surface membrane, forming cytotoxic aldehydes such as malondialdehyde (MDA) [39]. Levels of MDA were found to be positively correlated to the amount of SDF and were significantly higher in infertile men compared with fertile men [40]. Additionally, ROS can also directly damage sperm DNA and activate the intrinsic process of apoptosis leading to SDF [39,41]. 8-hydroxydeoxyguanosine (8-OHdG) is a known indicator of oxidative sperm DNA damage and has long been associated with infertility and abnormal semen parameters [42]. The amount of 8-OHdG in sperm was also found to be significantly correlated to SDF ($r = 0.67$; $P < .001$) [43].

Single- or double-strand breaks: does it really matter?

SDF can involve a single strand or both strands of the DNA double helix. This is called single-strand breakages (SSBs) or double-strand breakages (DSBs) [44]. Differentiating between single- and double-strand breaks can be of twofold interest. The first interest is the identification of the mechanism of occurrence of these breaks. Double-strand breaks would correspond to a defective repair of breaks occurring in the elongated spermatid during chromatin remodeling, while SSBs are thought to be produced under the influence of oxidative stress [45,46]. The second interest in distinguishing SSBs and DSBs would lie in the fact that their impact on the development of the future embryo could vary depending on the type, site, and complexity of the breaks [44,46]. The loss of continuity of the DNA molecule induced serious problems. Indeed, if the DNA molecule is not or incorrectly repaired, it can lead to the formation of chromosomal aberrations and the death of the cell [34,47]. After fertilization, it has been shown that the oocyte is able to repair certain breaks, especially if they are not too numerous [44,48]. Though SSBs seem less severe, yet if numerous, they can lead to the same consequences. Indeed, the more numerous they are, the greater the risk of faulty repair, possibly due to limited capability of the oocyte repair system to implement its regenerative process beyond certain limit [44,49–51]. Hence, DSBs and SSBs could have negative impact on reproductive outcomes [51]. The DSBs can easily lead to nonviable cells and absence of pregnancy [46]. As the male and female pronuclei are separated in the early embryo [52], there is no complementary strand to allow repair. On the other hand, the presence of SSBs could possibly be associated with a greater probability of embryonic survival [46]. Finally, certain DSBs located in specific regions of the genome as in the histone-rich matrix attachment regions could lead to recurrent miscarriages, embryo implantation failures following intracytoplasmic sperm injection (ICSI), and abnormalities in post-ICSI embryonic kinetics [46].

In any case, since most of SDF tests are unable to differentiate between SSBs or DSBs [46], this makes the distinction between these two types of sperm DNA breaks difficult. The only technique to differentiate SSBs from DSBs and to locate breakpoints is the comet technique [46,53]. However, because the comet assay involves multiple methodological steps, demands a high level of expertise for interpretation of the results, and has an important level of interlaboratory variation, its use may not be appropriate for all laboratories [54]. In conclusion, even if the DNA strand breaks can have different origins and different impacts on the embryo development, the interest in differentiating the two types of breaks (SSBs and DSBs) remains limited in clinical practice.

Which test to assess sperm chromatin integrity and SDF?

Terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate-nick end labeling (TUNEL) assay

The Terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate-nick end labeling (TUNEL) assay is a direct method used in the assessment of SSBs and DSBs using flow cytometry and being the method of choice as it can rapidly evaluate and measure up to 10,000 cells [55]. In a metaanalysis studying the utility of the different DNA and chromatin integrity measuring assays, the TUNEL assay was most predictive in estimating miscarriage [56]. A threshold of 20% is used to differentiate fertile from their infertile counterparts [57,58].

Sperm chromatin structure assay (SCSA)

The Sperm chromatin structure assay (SCSA) assay utilizes flow cytometry, indirectly assessing DNA integrity. It assesses the sperm DNA susceptibility toward the in situ acid-induced denaturation, which is further stained using the fluorescent acridine orange (AO) stain, and then examined under flow cytometer for the emission of green or red fluorescence, signifying the binding with double- or single-strand DNA breaks, respectively [59]. The SCSA has the capability of analyzing between 5000 and 10,000 cells in few seconds with the use of flow cytometer. It is characterized by a low interlaboratory variability being governed by standardized and strict protocols [60]. The SCSA measures single-stranded DNA motifs and is considered an expensive test, same as with TUNEL assay, being flow cytometry-dependent [61]. The SDF threshold used for SCSA is 18.9%−20% [58,62,63].

Sperm chromatin dispersion (SCD) test

The Sperm chromatin dispersion (SCD) technique assesses DNA integrity by evaluating the emission of the typical DNA halos in sperm with nonfragmented DNA after immersion in a layer of agarose and then a lysing medium to remove the nuclear proteins [64]. In a study by Chohan et al. SDF levels measured by SCD, SCSA, and TUNEL assay had significant correlation when used to compare DNA integrity in infertile versus fertile counterparts [65]. In patients undergoing intrauterine insemination (IUI), SCD was not found to correlate with pregnancy outcome [66].

In situ nick translation (NT) assay

In situ nick translation (NT) assay uses template-dependent enzyme (DNA polymerase) to assess DNA single-stranded breaks with lower sensitivity when compared to commonly used assays [67].

Comet assay

The general principle of the comet assay, also known as the single cell gel electrophoresis assay, is based on migration of the broken strands of the DNA toward the anode, during electrophoresis, generating the characteristic dispersion pattern that resembles a comet tail, while the intact DNA constitutes the comet's head [68]. The use of automated high throughput comet helps visualize a patient's sperm chromatin damage profile by plotting the percentage of DNA in the tail against tail length [69]. Cortés-Gutiérrez et al. developed a two-dimensional two-tailed comet assay that can differentiate between SSBs and DSBs in the same comets in sperm [70].

Acridine orange test

AO test is used with flow cytometry or fluorescent microscopy. The AO test like the SCSA test measures the susceptibility of sperm nuclear DNA to acid-induced denaturation in situ by quantifying the metachromatic shift of AO fluorescence from green (native DNA) to red (denatured DNA) [71]. Fluorescence after AO staining was neither related to basic sperm parameters nor with sperm functional capacity as evaluated by sperm−mucus interaction tests in vitro and in vivo [72]. However, when the predictive potential of AO was tested for ART outcome, no pregnancy was found when the proportion of green fluorescent spermatozoa was <50% even though an average of 26% of the oocytes were fertilized [73]. In general, results of AO test are limited by the subjective interpretation of the fluorescent color shades [74].

High-performance liquid chromatography

The High-performance liquid chromatography is the method used for quantitative assessment of the level of 8-OHdG, which is a by-product of oxidative DNA damage [75]. The level of 8-OHdG was found to be significantly higher in infertile men as compared to healthy controls and significantly decreased following antioxidant treatment [76].

Less commonly used tests for SDF assessment

Other cytochemical techniques have been described for evaluation of the sperm chromatin status and maturity including acidic aniline blue (AAB), Toluidine Blue (TB), and Chromomycin A3 (CMA3) tests. Results of AAB assay have been correlated with abnormal sperm chromatin and male infertility [77]. Additionally, chromatin condensation visualized by AAB has been shown as a good predictor for in vitro fertilization (IVF) outcome [78]. Strong positive correlation was found between the proportion of sperm heads with abnormal chromatin structure (violet heads) as determined by the TB assay and the counts of red–orange sperm heads as revealed by the AO method [79], and SCSA and TUNEL test [80]. A threshold of 45% TB violet cells was found to be highly predictive for infertility with a specificity of 92%, but with poor predictive potential for fertility (42% sensitivity) [81]. Semen samples with high CMA3 positivity (>30%) were found to have significantly lower fertilization rates if used for ICSI [82].

Etiologies and risk factors for SDF

Varicocele

Varicocele can have a negative effect on the developing sperm, resulting in a wide range of aberrations including SDF. In a metaanalysis, higher degree of SDF was observed among patients with varicocele compared to control groups with a mean difference calculated as 9.84% ($P < .00,001$). Additionally, the latter study showed that varicocele repair is associated with significant reduction of SDF with a calculated mean difference reaching −3.37% ($P < .00,001$) [83]. Another systematic review and metaanalysis including 1070 infertile men with clinical varicocele indicated significant reduction of SDF rates following varicocele repair, with a pronounced effect among the patients presenting initially with elevated SDF compared to those with normal values [84]. Moreover, the impact of subclinical varicocele on SDF was examined in a study by García-Peiró et al., which demonstrated a statistically significant reduction in SDF following repair of clinical varicocele but not in the subclinical varicocele counterpart [85]. In a different study by Ni et al. SDF and the oxidative stress marker, MDA levels were considerably higher in the clinical varicocele group compared to controls, which was not the case in the subclinical varicocele group [86].

Genital tract infection

It has been proposed that 15% of male infertility could be attributed to genitourinary infections [87]. Contamination of the semen by infectious pathogens may arise from urinary tract or from sexual transmissions [88]. Genital tract infection is associated leukocytospermia with resultant rise in ROS levels, causing oxidative stress and subsequently SDF [89]. In a study by Pagliuca et al. it was found that in 53 semen samples obtained from patients presenting for fertility evaluation, 70% of patients with seminal abnormalities had a positive infectious status, with six times increase in the risk of SDF among those with altered semen parameters and associated infection [90].

Aging

Controversial evidence exists on the negative impact of advanced paternal age, not only on conventional semen parameters but also on chromatin maturity. A study has examined the differences between ejaculated sperm from 1124 men aged above and below 40 years. Results showed that the odds ratio of acquiring SDF in advanced male age is twice compared to that in younger counterparts, with evidence of more deterioration in chromatin integrity [91]. Another study analyzing 2178 semen samples from men attending infertility clinics showed that advanced male age is associated with significant deterioration in sperm DNA integrity, in addition to altered mitochondrial function [92].

Smoking and alcohol

Smoking is recognized as a risk factor for SDF [93]. Smokers have been shown to have higher rates of SDF than nonsmokers [94–97]. Smoking is thought to induce decondensation of sperm chromatin, which can lead to SDF [98,99]. These effects would be proportional to the number of cigarettes smoked per day and the duration of exposure to tobacco [98]. Regarding the mechanism of action, smoking could alter mRNA expression levels of protamines and transition proteins [99], inducing a susceptibility of the chromatin to the oxidative stress. Markers of oxidative stress were also shown to be increased in smokers compared with nonsmokers [97,100,101], and an unbalanced antioxidant/oxidation ratio could explain sperm DNA damage and particularly SDF in smokers [97,102].

Alcohol consumption alone has been found to be a risk factor for SDF [102]. For other authors, the combined consumption of alcohol and tobacco would increase SDF but not the consumption of alcohol alone [95]. In this latter study, drinkers consumed more than one drink per day, but actual consumption was not recorded. Since alcohol and tobacco use may be jointly present in the same patient, another study investigated the effects of this cointoxication showing that heavy drinkers (>2 drinks per day) had higher levels of SDF compared to isolated smokers or isolated alcohol users [103]. This was not true for moderate drinkers (less than two drinks per day) [103].

Obesity and overweight

Obesity has been shown to be associated with increased SDF. For men with a body mass index (BMI) greater than 28 kg/m^2 [104], a 2.5-fold increase in SDF rate was shown compared with men with a BMI between 18.5 and 28 [104]. On the other hand, men with less pronounced overweight (BMI between 24 and 28) did not show an increase in SDF compared with men with normal BMI [104]. These trends have been found in other studies that used a classical BMI threshold of 30 to describe obesity [105–108]. On the other hand, other studies did not show a significant difference in SDF rates according to BMI [109,110]. These two latter studies that included more than 2000 patients stated that the association between obesity, overweight, and elevated SDF is still controversial after adjusting for the confounding factors [109,110]. Two metaanalyses published in 2015 [107] and 2020 [108] including four and three studies, respectively, found higher rates of DNA fragmentation in obese men (BMI 30–34.99) compared with men with normal BMI (<25). However, the authors of the latter study [108] stated that data were inconsistent for an association between BMI and SDF, arguing that only obesity (and not overweight) could be linked to increased SDF. It is probably for this reason that obesity was not retained as SDF risk factor in the recent EAU guidelines [93]. Hence, additional studies are still needed to address the relationship between SDF and BMI.

Occupational and environmental exposures

Several occupational and environmental exposures have been linked to increased SDF levels [111]. At the patient level, this may be due to exposure to scrotal heat [112], radiotherapy, or chemotherapy [111,113–115]. At the environmental level, links between SDF and air pollution [116], radioactivity [113], or exposure to chemicals and pesticides [117] have been demonstrated.

The impact of SDF on male reproductive outcomes

SDF has been associated with male infertility and can reduce the ability of a couple to conceive naturally as well as having significant harmful effects on the outcomes of ART. Moreover, the repercussions of SDF can be carried forward and cause harm in offspring. These detrimental effects of SDF are discussed in detail in the following section.

Implications of SDF on natural pregnancy

The adverse effects of defective sperm DNA on male fertility potential have been extensively investigated over the last two decades. Evenson et al. used the SCSA to reflect abnormalities in sperm DNA [71]. They followed couples who were presumed fertile and were trying to conceive over one year and reported significantly higher ($P < .001$) SCSA values for couples who were not able to conceive. Furthermore, when comparing the SCSA values from men who were presumed infertile to the fertile counterparts, significant differences were also reported.

Sperm DNA damage is well correlated with the inability to achieve natural pregnancy, as demonstrated by Zini et al. who conducted a metaanalysis consisting of 616 couples and reported significantly higher failure rates of natural conception with higher SCSA values ($OR = 7.01$, $P < .001$) [118]. Another group studied the ability of SDF, measured by TUNEL assay, in predicting success of natural conception [119]. They reported that SDF levels ≤25%, an optimal threshold determined by ROC analysis, was associated with significantly higher odds for conception (OR = 5.22, $P < .05$) among infertile men. They also reported that median SDF values among infertile men who did not conceive over one year (31%) were significantly higher than fertile controls (18%, $P = .008$).

The SDF not only affects the ability to achieve a pregnancy but also to maintain it. In this regard, a metaanalysis that included 13 studies reported a significantly higher SDF by an average difference of 11.98% ($P < .001$) among men whose partners experienced recurrent pregnancy loss (RPL), defined as two or successive miscarriages, compared to controls who had natural conception [120]. Another metaanalysis included 579 men with RPL and 434 controls and demonstrated a similar association between SDF and RPL with a mean difference of 10.7% ($P < .001$) between the two groups [121].

Implications of SDF on ART outcomes

Elevated SDF can reduce pregnancy rates after IUI as demonstrated in a metaanalysis that included 10 studies with 2839 IUI cycles and reported significantly lower pregnancy rates with high SDF levels (RR = 0.34, $P < .001$) [122]. This same metaanalysis also reported significantly lower live birth rates with higher SDF. A recent prospective study from Belgium that included 209 IUI cycles found that with SDF levels >26%, no clinical pregnancies were reported [123]. This cut-off level was determined using ROC analysis to predict success of IUI in terms of clinical pregnancy with 100% sensitivity and 36.3% specificity.

However, not all studies have reported consistent findings [124]. An analysis of 1185 IUI cycles reported no significant differences in clinical pregnancy rates among groups with high, medium, or low DFI, but did report significantly higher miscarriage rates among the groups with high and medium DFI compared with low DFI [125]. We can conclude that elevated SDF may impair pregnancy rates after IUI and is associated with higher pregnancy loss and lower live birth rates.

The different reproductive outcomes after IVF or ICSI have been correlated to SDF levels. Fertilization rates in conventional IVF are reduced with elevated SDF [126,127]. The same, however, was not reported for ICSI [126]. This may be because the normal process of fertilization is bypassed in ICSI when the spermatozoon is directly injected into the oocyte. The SDF can also affect embryo quality after IVF and ICSI, as demonstrated by a metaanalysis that included 17,879 embryos and observed a lower rate of good quality embryos with higher DFI (RR = 0.65, $P < .01$) [128].

A metaanalysis that included 56 studies with 8086 IVF or ICSI cycles reported a strong and significant negative impact of elevated SDF on clinical pregnancy rates (OR = 1.84, 95% CI = 1.5−2.27, $P < .0001$) [129]. When analyzing IVF alone, metaanalyses have consistently reported significantly reduced pregnancy rates with elevated SDF [118,128,130]. For ICSI alone, however, the results of most metaanalyses have found nonsignificant associations between SDF and clinical pregnancy rates following ICSI [118,128,130].

SDF can also increase the risk of miscarriage after IVF or ICSI with relative risk of 2.16 among the high SDF group compared with the low SDF group ($P < .00,001$) [56]. Other metaanalyses have reported similar results in terms of increased miscarriage with high

SDF levels after both IVF and ICSI [128,131]. Finally, live birth rates have also been reported to be impacted by SDF [132]. We can conclude that SDF can affect fertilization and pregnancy rates after IVF but not ICSI; however, SDF may impair embryo quality and exert delayed impacts in terms of increasing risk of miscarriage and reducing live birth rates after both IVF and ICSI.

Impact of SDF on offspring

Sperm DNA damage and fragmentation can cause segments of the paternal chromosome to be randomly and unequally distributed between cells during the divisions of early embryonic development, leading to chaotic mosaicism and a wide array of chromosomal abnormalities [133]. If these embryos survive and pregnancy is carried to term, this can lead to a variety of congenital anomalies in the offspring.

Although high-quality studies on the impact of SDF on human embryos and offspring do not exist mainly due to ethical concerns, it has been described that abnormalities in the sperm genome can lead to malformations, birth defects, genetic abnormalities and disorders, and even infertility in the offspring [134–136]. It is important to keep in mind that infertile men with elevated SDF who were eventually able to father children with ART procedures such as ICSI can carry the abnormal DNA to their offspring with its many consequences. This can explain the findings of a metaanalysis in 2005 that concluded a significantly increased risk of birth defects after ART when compared to natural pregnancy [137].

Finally, an important sequela of paternal SDF may be an increased risk of childhood cancers in offspring. This was demonstrated by Kumar et al. who reported significantly higher levels of SDF, seminal ROS, and sperm DNA oxidative damage among fathers of children with nonfamilial sporadic heritable retinoblastoma compared to controls who father healthy children [101]. Moreover, these parameters were significantly higher among tobacco users, implying an increased risk of childhood cancers among children of men who use tobacco, which may be attributed to elevated SDF levels among fathers who smoke.

How to manage patients with high SDF?

Addressing patient risk factors

When deciding on a management plan and treatment strategy for patients with elevated SDF, it is important to treat any underlying cause or possible etiology. Men with clinical varicocele should be managed accordingly and may be offered varicocele repair. A recent metaanalysis including 11 studies with a total of 394 patients who underwent varicocelectomy demonstrated a significant reduction in SDF by 5.79% after the procedure [138]. The positive reproductive outcomes of varicocele repair have also been demonstrated by another metaanalysis which compared ICSI outcomes in patients with clinical varicocele who underwent varicocele repair to those who did not, and reported higher clinical pregnancy rates (OR = 1.59, $P = .002$) and higher live birth rates (OR = 2.17, $P < .00,001$) among patients who underwent varicocele repair [139]. Men with genital tract infections should be treated with antibiotics as this has been demonstrated to reduce SDF and improve reproductive outcomes [140].

The link between lifestyle modification and reduction of SDF is weak. Considering the impact of tobacco and alcohol consumption on SDF, reducing or stopping the consumption of these toxins could improve the integrity of sperm DNA, but no study has proven this to date [58,141]. If obesity seems to be linked to an increased SDF, weight reduction would improve it as reported by a prospective study in which obese men followed a healthy diet and exercise program for 12 weeks significantly decreased their mean SDF after the program [142]. A diet favoring fruits, vegetables, fish, and white meats, and reducing sugars, red meat, fat, and alcohol consumption are linked to lower levels of SDF [143,144]. Combined with physical exercise and an antioxidant treatment for three months, diet could reduce SDF in infertile patients who have failed ICSI [145]. Adequate glycemic control for men with diabetes is also crucial, as men with diabetes were found to have significantly higher SDF compared to nondiabetic controls [146,147]. In addition, diabetes was found to adversely impact the outcomes of ICSI with reduced blastocyst formation rates (38.1% vs. 55.5%, $P < .001$), reduced clinical pregnancy rates (28.6% vs. 46.3%, $P < .001$), and increased miscarriage rates (50% vs. 24.6%, $P < .001$) when compared to nondiabetic men [147]. Thus, recent guidelines acknowledge the managing of lifestyle factors among patients with high SDF (Grade C) [58,141,148].

Clinicians can also recommend frequent ejaculation as a conservative means to reduce SDF. Shorter ejaculatory abstinence was found to decrease SDF levels [149,150]. This is translated into improved reproductive outcomes, as significantly improved pregnancy rates after IUI were reported with an abstinence period of two days or less compared to more than two days [151]. Another study compared ICSI outcomes between two groups, an experimental group with recurrent ejaculation and reduced abstinence (12 h) before the final ejaculation for ICSI and a control group with 3–4 days of abstinence before the final ejaculate, and reported significantly higher pregnancy rates in the experimental group (56.4% vs. 43.3%, $P = .03$) [152].

Use of antioxidants

Given the role of ROS in the pathogenesis of male infertility and elevated SDF, many studies have investigated the use of different antioxidants for the management of infertile men. Antioxidant supplementation in 112 men for three months resulted in significantly reduced SDF after treatment compared to pretreatment values (34.3% vs. 43.5%, $P = .0017$) [153]. A significant reduction in SDF was also demonstrated by another study which supplemented infertile men with n-acetyl-cysteine for three months with values decreasing from 19.3% before treatment to 15.1% after treatment ($P = .001$) [154]. A study comparing the effect of different antioxidant regimens showed that all were able to significantly reduce SDF with no significant differences between the different regimens [155].

Antioxidant supplementation has also been studied in men with known risk factors. For example, giving vitamin C to men with occupational exposure to lead showed significant reductions in SDF as measured by comet assay [156]. Additionally, supplementation of vitamin C and vitamin E in men who have failed ICSI led to more than 10% reduction of SDF in 76.3% of the men [157]. The authors of the latter study indicated significantly improved ICSI outcomes in the second attempt after supplementation compared to the first trial before supplementation, with increased pregnancy rates (48.3% vs. 6.9%, $P < .05$) and implantation rates (19.6% vs. 2.2%, $P < .01$).

Use of ART

Given the lack of negative impact of SDF on fertilization and clinical pregnancy rates after ICSI when compared to IUI, IVF, or natural conception, couples with failure to achieve pregnancy due to SDF may be directed toward ICSI. Although pregnancy rates are unaffected, couples undergoing ICSI due to elevated SDF should be aware of their increased risk of pregnancy loss after the procedure. The different ART modalities can be further enhanced using sperm selection techniques, which select spermatozoa with less SDF and use them to fertilize the oocytes, leading to improved outcomes.

Swim-up (SU) and density gradient centrifugation (DGC) are both commonly used sperm processing modalities that select more motile and functional spermatozoa and have been found to significantly reduce SDF and seminal ROS [158]. MACS, which removes apoptotic spermatozoa and reduces SDF as discussed earlier, has been investigated in infertile men with SDF >20% undergoing ICSI, where 366 ICSI cycles were performed after DGC followed by MACS and compared to 358 ICSI cycles after DGC alone [159]. No significant difference in fertilization rate was reported between the two groups; however, the MACS group was associated with significantly higher pregnancy rates, lower miscarriage rates, and improved live birth rates.

Intracytoplasmic morphologically selected sperm injection (IMSI) selects morphologically normal and motile spermatozoa without vacuoles in their head under high magnification for use in ICSI. These selected spermatozoa have been reported to have significantly less SDF [160]. When compared to standard ICSI, IMSI was associated with significantly higher implantation rates, as well as higher clinical pregnancy and live birth rates, although the latter two did not reach statistical significance [161].

Physiologic ICSI (PICSI) is another sperm selection technique, which selects mature spermatozoa via binding to hyaluronic acid (HA), as mature sperm have a higher density of HA receptors, and this was found to be significantly associated with lower SDF [162]. Microfluidic sperm sorters use fluid dynamics and guide spermatozoa through microchannels, leading to the selection of motile spermatozoa which have been found to contain significantly less SDF compared to sperm selected by SU or centrifugation [163]. Finally, testicular sperm has been found to contain less SDF compared to ejaculated sperm and can be used for ICSI with improved outcomes [164]. However, it should be noted that the aforementioned conclusion is based on poor-quality evidence and that SDF testing for testicular sperm is not standardized [165]. Clinicians should also keep in mind that testicular sperm retrieval is an invasive procedure that is not without risks and complications, such as hematoma formation, fibrosis, pain, bleeding, or infection [166]. Furthermore, inconsistent findings were reported by a recent study that showed no significant difference in terms of clinical pregnancy, miscarriage, and live birth rates after ICSI using testicular sperm compared to ejaculated sperm [167].

Clinical practice guidelines and recommendations on SDF testing

The SDF test has been included in the latest version of the WHO manual (WHO 2021) in the chapter "extended analyses." Since SDF is only partially related to semen quality [54], it could indeed represent an important addition in the work-up of male infertility. The WHO manual provides a detailed description of different tests used in the evaluation of SDF [54] and highlights the need for laboratories to establish their reference thresholds. However, in this edition, no clinical description of the possible indications for SDF testing is given, leaving each user unclear as to the conditions that entitles prescribing this test.

The indications governing the clinical utility of SDF testing has been established in the recent EAU guidelines, which recommends SDF testing in the assessment of [93]:

- couples with RPL (from natural conception and ARTs) (strong recommendation)
- men with unexplained infertility (strong recommendation)

Based on metaanalyses and high-quality articles, two research groups have published recent guidelines on the indication of SDF testing [141,148]. These two reports have been evaluated, and their convergences and divergences were recently synthesized [58]. In these two reports, the indications for SDF testing are RPL (Grade B–C) and unexplained male infertility (Grade B–C). These two reports also recommend testing in cases of ART (before or after failure), varicocele (Grade C), and the presence of risk factors (as described above) (Grade C). Only the latter group [148] recommends testing in the case of sperm freezing (grade D recommendation).

Limitations of sperm SDF testing in clinical practice?

A negative association has been suggested between SDF and time to pregnancy [168], and hence upon counseling infertile men, SDF testing could be offered to selected patients where the decision relies on the DNA integrity, as with patients with clinical varicocele and males from couples with RPL or ART failure [169,170]. However, many challenges hinder the consideration of SDF testing as a tool used in the routine management of male infertility. A commentary by Tadros et al. described the main concerns behind poor SDF incorporation in daily clinical practice, including the inability of the test to discern the couples who will get pregnant, naturally or in the ART context, from those who will not, variability of the available techniques with standardization concerns, and the associated financial burden. There have been significant efforts to resolve many of these limitations [170]. Recently, trials of standardization and validation to SCSA and TUNEL assays have been performed [171,172].

Updates in the recent research on SDF (2019–21)

Given the growing interest in SDF as a marker for male infertility as well as the potential for its application in the reproductive field, research is constantly evolving with many new studies investigating pathophysiologic aspects of SDF, potential associated risk factors and conditions, as well as prospective therapeutic options. Continuous efforts to understand the molecular aspects and mechanisms behind SDF are underway, such as the effect of methylation and silencing of genes involved in repair of DNA breaks and their association with SDF. For example, Kabartan et al. recently studied BRCA1 and BRCA2 promoter methylation and SDF but found no significant association [173].

A new pathway for SDF generation that involves cell-free DNA (cfDNA) was recently described [174]. cfDNA refers to extracellular-free double-stranded DNA fragments that can be found in a variety of body fluids and increases in inflammatory states, cancer, and cellular stress [175]. In vitro exposure of spermatozoa to cfDNA significantly increased SDF with no effect on conventional parameters such as motility and vitality and no relation to oxidative stress as determined by 8-OHdG [174]. To further support this, concomitant addition of DNase to cfDNA resulted in no increase in SDF. This was applied to men with spinal cord injury and elevated SDF who were found to have 2.8 times higher mean cfDNA in their seminal plasma compared to fertile normozoospermic controls [176].

With many different conditions, contributing to SDF, there is always room for investigating whether certain chronic diseases, infections, exposures, or risks may lead to SDF. For example, new studies have found significantly higher SDF levels in men with systemic lupus erythematosus [177] and in men who were positive for human papilloma virus [178]. Environmental factors on SDF are also being investigated, such as studying the seasonal variation and its impact on SDF in horses, where SDF levels were found to be significantly higher in summer compared to spring, and this was associated with significantly reduced pregnancy rates [179]. New studies also suggest a diurnal variation in SDF levels, with values being lowest at 11 AM in humans [180].

Finally, the effects of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), which is responsible for the recent pandemic, on male reproductive function is also being investigated. SARS-CoV-2 infection has been postulated to increase SDF via several factors including oxidative stress due to local inflammation and psychological stress, use of antiviral drugs, and acute hypogonadism causing alteration in hormonal balance and fever [181,182]. A recent study conducted on men who were infected with SARS-CoV-2 reported improved sperm DNA integrity and significantly reduced markers of oxidative stress 120 days compared with 14 days after the infection [183].

Clinical scenarios on SDF management

To perform SDF testing, and necessitate counseling the patients as to the therapeutic options that are available to treat a subset of infertile men with elevated SDF, herein, we describe few examples of these clinical case scenarios:

Case 1

Scenario: A 35-year-old male patient presents with a history of primary infertility of four-year duration. The reproductive history of the female partner is unremarkable. The mean values of semen parameters on the last two occasions reveal the following results: semen volume: 3.5 mL, sperm concentration: 10×10^6/mL, total sperm motility: 25%, progressive sperm motility: 5%, vitality: 40%, normal sperm forms: 2%, and seminal leukocytes: 0.3×106/mL. How will you manage this patient?

Solution: The sixth edition of the WHO manual for semen analysis acknowledges SDF assessment as an extended test of sperm that can be requested under certain circumstances

[54]. Also, two recent metaanalyses independently recommended SDF testing in infertile couple with idiopathic male infertility (IMI) with the level of evidence graded as C [141] or B−C [148]. So, this patient can be offered SDF testing. Additionally, patients with IMI and elevated SDF are to be counseled regarding the significance of changes of their lifestyle and dietary habits, as well as various exposures.

Case 2

Scenario: A 37-year-old male patient presents with a history of secondary infertility of five-year duration. Genital examination reveals bilateral grade 3 varicocele that was confirmed by scrotal color Doppler ultrasound. The reproductive history of the female partner is unremarkable. The mean values of semen parameters on the last two occasions reveal the following results: semen volume: 2.5 mL, sperm concentration: 18×10^6/mL, total sperm motility: 30%, progressive motility: 5%, vitality: 45%, normal sperm forms: 5%, and seminal leukocytes: 0.5×106/mL. How will you manage this case?

Solution: The finding of clinical varicocele in infertile men is listed among the possible indications of SDF testing in two recent metaanalyses with both studies grading the level of evidence as C [141,148]. So, this patient can be offered SDF test. Also, varicocele repair can be offered to patients with high SDF (level of evidence graded as B−C) [141]. The results of a metaanalysis indicated significant reduction of postoperative SDF rates following varicocele repair (weighted mean difference −7.23%; 95% CI: 8.86 to −5.59; I2 = 91%) [84]. Additionally, the treatment effect was greater in men with elevated versus normal preoperative SDF levels.

Case 3

Scenario: A couple presents with a history of primary infertility of seven-year duration and three failed ICSI trials (one negative pregnancy and two miscarriages). The female partner is 33 years old with no remarkable abnormalities in her infertility work up. The husband's semen analysis reveals the following: semen volume: 3.5 mL, sperm concentration: 20×10^6/mL, total sperm motility: 25%, progressive motility: 0.0%, vitality: 30%, normal sperm forms: 6%, and seminal leukocytes: 0.3×10^6/mL. How will you manage this couple?

Solution: The SDF test has been recommended in infertile couples that suffer repeated ICSI failure with the level of evidence graded as B−C [141,148]. So, SDF test can be ordered for this couple before additional ICSI trials. In cases proved to have elevated SDF, the outcome of subsequent ICSI can be enhanced by using IMSI [161], PICSI [162], microfluidic sperm sorting [163], or the use of testicular sperm [164].

Conclusion

In recent years, numerous studies have highlighted the role of spermatozoa on embryo development and the negative impact of SDF on natural and medically assisted pregnancy rates. The pathophysiological mechanisms that cause SDF are explained. These mechanisms help in our understanding of how these single- and double-strand breaks occur in the male gamete and why some sperm are more susceptible to DNA damage than others. When it comes to DNA repair, it was shown that oocyte and early embryonic repair systems are crucial entities in the DNA reparative process preventing transmission of DNA aberrations to the offspring. Furthermore, SDF assessment is progressing with trials of standardization taking place, enabling laboratories and clinicians to establish more rigid thresholds to aid in the clinical utility of the different assays. The indications of these tests are summarized and illustrated in the form of clinical scenarios that are based on recent guidelines. Finally, the management of patients with SDF is based on prevention, correction of etiological factors (such as smoking cessation, repair of varicocele, etc.), and supplementation of antioxidants before considering ART. Future implications using this valuable tool in the assessment of male fertility are ongoing, with continuous efforts to understand the molecular aspects and mechanisms behind SDF being underway including new pathways for SDF generation like those involving cfDNA.

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CHAPTER

4

The evaluation and management of recurrent pregnancy loss

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Introduction

Recurrent pregnancy loss (RPL) affects approximately 1%–4% of fertile couples [1–3] and places immense strain on a couple's life. It is one of the most difficult conditions for clinicians to diagnose and manage [4]. "Pregnancy loss" is the spontaneous termination of a pregnancy before the fetus reaches viability [5]. RPL is defined by the European Society of Human Reproduction and Embryology (ESHRE) as the loss of two or more pregnancies; this includes all pregnancy losses from conception to 24 weeks of gestation, but not including ectopic or molar pregnancies [5]. In contrast, the American Society for Reproductive Medicine (ASRM) defines RPL as the loss of two or more clinically identifiable conceptions [6].

Nonvisualized pregnancy losses, while not included in the ASRM definition of RPL, appear to have the same unfavorable impact on future pregnancy outcomes as clinical pregnancy loss [7].

Recurrent miscarriage can be classified as primary (no previous live birth) or secondary (following a previous live birth) [8]. RPL may be caused by chromosomal abnormalities in the developing embryo, structural uterine anomalies, endocrine disturbances, or autoimmune disorders, but nearly half of RPL cases are unexplained [9]. A recently published series of articles on miscarriage recommended a global effort to increase support and care for women

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and their partners following one miscarriage. The authors in this series advocated a tiered model of care in which women would have their health needs assessed and be given information and assistance following one miscarriage. After a second miscarriage, their care should be assigned to a miscarriage clinic for initial investigations, additional care, and monitoring of future pregnancies. After three miscarriages, they would be offered a full series of evidence-based investigations and care [10–13].

When evaluating a couple with RPL, investigations following a detailed history should include parental karyotyping, uterine cavity assessment, laboratory testing for certain endocrinopathies, and testing for antiphospholipid antibody syndrome (APS) [6]. Psychological support is an important component of care for RPL patients [4]. Couples with RPL can be reassured that the odds of having a live birth are good even with no interventions [14], and having an adverse pregnancy outcome is unlikely [15].

Incidence and prevalence

Spontaneous miscarriage occurs in up to 15%–20% of couples, and 1%–4% of these will develop recurrent miscarriages [1–3]. The prevalence may be underestimated because cultural and social attitudes in some groups may prevent women from discussing their pregnancy loss freely [9]. Regardless of medical intervention, most women with RPL have a high rate of subsequent live birth [14]. The incidence of successful subsequent pregnancy correlates strongly with maternal age and the number of prior losses [3,16]. In a population-based study, the risk of miscarriage in women aged 25–29 years was 9.8%, rising to 53.6% in women aged ≥45 years [1]. The increased risk in women of advancing maternal age can be attributed to increased rates of aneuploidy in oocytes [17]. Irrespective of maternal age, the number of prior miscarriages is another risk factor for RPL; several population-based studies have shown that the age-adjusted odds ratio for miscarriage increases after each pregnancy loss [18–20]. The ESHRE supports determining a woman's prognosis based on her age and her detailed pregnancy history, including the number of previous miscarriages, live births, and their order [5]. The relationship between age, number of losses, and the likelihood of subsequent miscarriage highlights the significance of stratifying patients based on these factors to allow for better data analysis in clinical research.

Etiology and risk factors

The widely accepted causes of RPL are parental chromosomal abnormalities, APS, some endocrinal disorders and uterine abnormalities. Early embryonic losses are most often due to genetic causes [21], but second trimester losses are more often explained by immunologic or anatomical disorders [22].

Genetic causes

Clinical findings suggest that couples with RPL have a greater incidence of karyotype anomalies than the general population, and their female first-degree relatives are six times more likely to experience RPL [23]. Embryos from couples with RPL have higher rates of aneuploidy than those from couples without RPL [24]. Both numerical and structural chromosomal abnormalities in either partner can lead to recurrent miscarriage [25]. Aneuploidies are quite variable in incidence among women who experience recurrent miscarriages [26]. Aneuploidy most commonly arises from random nondisjunction meiotic errors in the oocyte [24]. Autosomal trisomy, monosomy X, and polyploidy are the most prevalent karyotypic abnormalities [27]. As most cases are due to random de novo errors, aneuploidy is unlikely to recur in a future pregnancy. Therefore, the prevalence of chromosomal abnormalities is lower in couples with RPL than in those with sporadic pregnancy loss [3]. Advancing maternal age is the greatest risk factor for embryonic aneuploidy, with clear increased risk in women aged 40–45 years. There is also a small increase in risk in women aged less than 20 years [28]. Most structural chromosomal abnormalities are balanced translocations, either reciprocal or Robertsonian. Paracentric and pericentric inversions are uncommon, but also linked to an increased risk of RPL [29,30]. The highest risk for miscarriage is associated with reciprocal translocations [31]. Balanced translocations are observed in 2%–5% of couples with RPL and result in unbalanced translocation in the embryo [32]. These translocations may remain undetected for generations if the associated pregnancies are euploid, either diploid or balanced abnormally [23]. Copy number variation (CNV) occurs when regions of the genome are replicated [33]. Large CNVs (with high numbers of replications) are believed to cause miscarriage, especially if they are de novo and contain clinically relevant genes [34]. Single gene disorders have also been associated with RPL, although these disorders are relatively uncommon [23].

TABLE 4.1 Summary of studies done to evaluate the reproductive outcome in couples with structural chromosomal abnormalities.

Study author and date	Study design	Intervention	Reproductive outcome
Carp et al. (2004) [35]	Retrospective comparative cohort study	916 RPL patients including 99 patients with chromosomal aberration were followed up for one year without any intervention	33/73 pregnancies (45.2%) in patients with chromosomal aberration and 325/588 pregnancies (55.3%) resulted in live births
Kyu Lim et al. 2004 [36]	Prospective cohort study	PGD using FISH was performed in 43 couples with reciprocal translocations and 6 couples with Robertsonian translocations	16 healthy babies were delivered in 12 single and 2 twin pregnancies and one case of preterm twin delivery occurred
Franssen et al. 2006 [30]	Index control study	278 carrier and 427 noncarrier RPL couples. Couples were followed up for at least 2 years after the chromosomal analysis	83% of carrier couples and 84% of noncarrier couples had at least one healthy child
Stephenson and Sierra 2006 [37]	Prospective study	Reproductive outcomes for 1893 RPL couples including 51 carriers of a structural chromosome rearrangement were followed up	- The live birth rate was 15% before evaluation, and 71% after evaluation and treatment of concomitant factors

(*Continued*)

TABLE 4.1 Summary of studies done to evaluate the reproductive outcome in couples with structural chromosomal abnormalities.—cont'd

Study author and date	Study design	Intervention	Reproductive outcome
Sugiura-Ogasawara et al. 2008 [38]	Prospective comparative study	Determination of chromosome abnormalities prevalence in 2382 RPL couples and comparison of subsequent success rates between cases with normal and abnormal karyotyping	- 29/46 (63.0%) who became pregnant with reciprocal translocations had a live birth with spontaneous conception - 950/1207 (78.7%) with normal karyotype had live births
Fischer et al 2010 [39]	Retrospective data review	Evaluating the reproductive outcomes for 192 RPL couples with chromosomal translocations after PGD	Pregnancy success rate was 87%
Keymolen et al 2012 [40]	Retrospective cohort study	Evaluating the reproductive outcomes after PGD in 69 male and 73 female carriers of reciprocal translocations	Live birth rate 12.8%/oocyte retrieval and 26.7%/embryo transfer
Kochhar and Ghosh 2013 [41]	Prospective study	Pregnancy outcomes were recorded over 2 years in patients with chromosomal rearrangements among 788 RPL couples	Over 2 years follow-up: 2/3 of 49 pregnancies resulted in a normal live birth
Flyn et al (2014) [42]	Retrospective study	Comparison of the reproductive outcome, between couples with chromosomal abnormalities and those with unexplained repeated miscarriages in 795 RPL couples	36 live births/159 pregnancies (22.6%) among the carriers of chromosomal abnormality
Ikuma et al (2015) [43]	Retrospective study	Comparing live birth rates in patients with RPL associated with translocation underwent PGD (37 patients) or natural conception (52 patients)	Cumulative live birth rates: - PGD 67.6% - Natural conception 65.4%
Bedaiwy et al. 2016 [44]	Retrospective chart review	23 out of 2321 RPL couples found to have parental chromosome rearrangement of which 13 couples underwent PGD	Live birth rates: - PGD: 6 cases (66.6%) (an incidence of 1/5.63 person-years) - Clinical management: 24 cases (53.3%) (an incidence of 1/4.09 person-years).
Kato et al 2016 [45]	Retrospective cohort	Assessment the reproductive outcomes following PGD using FISH in 52 RPL couples with translocations	Average cumulative live birth rate 76.90%
Pundir et al 2016 [46]	Retrospective cohort study	Evaluation the reproductive outcomes after PGD using FISH in 91 RPL couples carrying reciprocal translocations	- Live birth rate/cycle 19% - Cumulative live birth rate/couple 32%
Awartani and Al Shabibi 2018 [47]	Retrospective cohort study	77/1074 RPL couples with chromosomal anomalies were followed up throughout pregnancy	67% of the following pregnancies ended in live births

Studies focused on reproductive outcomes of those who carry structural chromosomal rearrangements are shown in Table 4.1.

Anatomic causes

A normal uterine cavity is critical for embryo implantation and development [48]. Disruptions of endometrial homeostasis increase the probability of pregnancy loss [49,50]. Anatomical factors implicated in RPL can be congenital (most commonly a uterine septum) or acquired (most commonly uterine fibroids) [51].

Congenital uterine anomalies

Congenital uterine abnormalities appear to be more common than previously assumed [52]. They usually develop from disturbances in the formation, fusion, or differentiation of the mesonephric ducts, or in the regression of the septum [53]. The most common anomalies are the septate uterus, unicornuate uterus, bicornuate uterus, and uterus didelphys [54]. These defects are generally associated with late first trimester or second trimester miscarriage and other adverse outcomes such as preterm labor or fetal malpresentations [55]. In women with congenital uterine anomalies, the risk of miscarriage is greatest in those with a uterine septum, followed by those with a bicornuate uterus [56].

MRI images for different Mullerian anomalies that may be implicated in RPL are shown in Fig. 4.1.

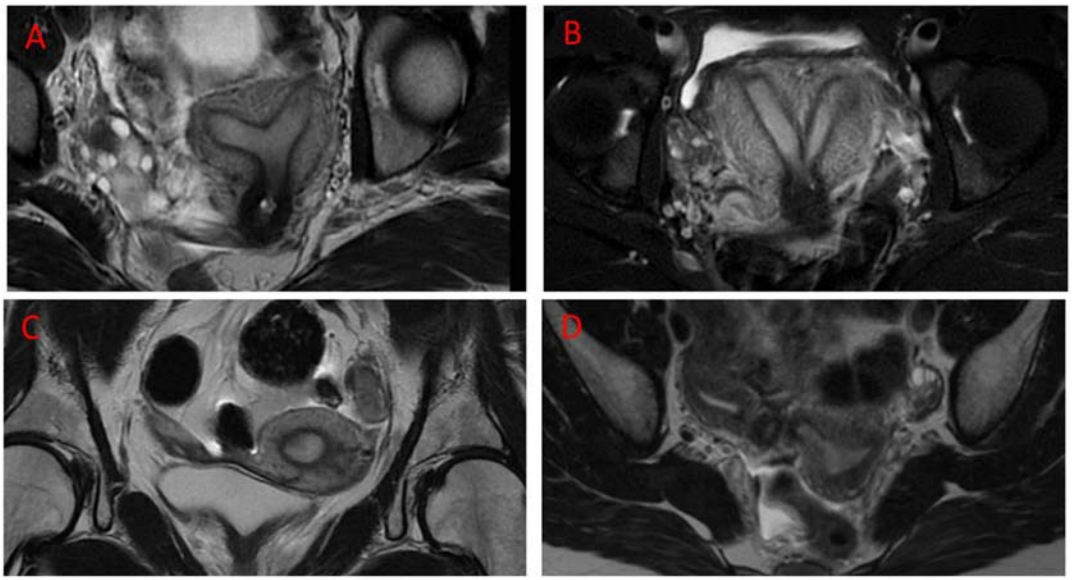

FIGURE 4.1 Shows MRI images for different mullerian anomalies that may be implicated in RPL. (A) MRI image shows partial uterine septum. (B) MRI image shows complete uterine septum. (C) MRI image shows unicornuate uterus. (D) MRI image shows uterus didelphys.

Acquired uterine abnormalities

Common acquired uterine anomalies related to RPL include intrauterine adhesions (IUAs), uterine fibroids, or endometrial polyps [57].

IUAs (Asherman syndrome, intrauterine synechiae) frequently result from the destruction of the endometrium by curettage or ablative techniques [58] and ultimately lead to complications such as menstrual abnormalities, infertility, and increased miscarriage risk [59]. It has been hypothesized that they provoke miscarriage by decreasing the amount of normal endometrial tissue, reducing the vascularity of the remaining endometrial tissue, and constricting the uterine cavity, despite a lack of supporting evidence [60]. The risk of IUAs increases with rising numbers of previous curettage procedures and miscarriages [61].

Uterine fibroids and their association with RPL have been extensively studied [62]. Submucous fibroids in particular can be associated with RPL as they may hinder implantation and reduce endometrial receptivity by increasing cytokine production [63]. However, a large metaanalysis of a general obstetric population showed no increase in risk of miscarriage in women with fibroids. It was concluded that the presence of confounding factors may have accounted for the widespread clinical perception that fibroids are a cause of pregnancy loss [64]. In a recent retrospective analysis of women undergoing in vitro fertilization (IVF), women with intramural fibroids of more than 2 cm diameter reaching the endometrial cavity had decreased rates of implantation, clinical pregnancy, and live birth [65].

The impact of endometrial polyps on sporadic or RPL has not been specifically addressed [60].

Cervical insufficiency (incompetent cervix) results mostly from congenital mesenchymal weakness or trauma due to surgery or birth [66]. Previous cervical cone biopsy or a sonographically short cervix is associated with an increased risk of second trimester loss [67].

Endocrine causes

Endocrine disorders play a role in approximately 8%−12% of RPL cases [68]. These include thyroid disorders, luteal phase deficiency (LPD), polycystic ovary syndrome (PCOS), insulin resistance, and hyperprolactinemia.

Thyroid dysfunction

Women with hyperthyroidism are at a higher risk of miscarriage and fetal death [69]. Miscarriage and poor fetal neurocognitive development are established consequences of overt hypothyroidism [70]. Subclinical hypothyroidism may [71,72] or may not [73] increase the risk of sporadic or recurrent miscarriage. Hashimoto thyroiditis is associated with the development of thyroid antibodies [74]. The association between thyroid antibodies (such as thyroid peroxidase (TPO) and thyroglobulin (Tg) antibodies) and RPL has been a popular area of research, with contradictory findings [75,76]. The majority of women with TPO or Tg antibodies are euthyroid, while those with overt hypothyroidism usually have greater antibody levels [77]. Some authors claim that thyroid antibodies are associated with increased risk of pregnancy loss even in euthyroid women [77−79].

Hyperprolactinemia

Hyperprolactinemia may cause infertility and miscarriage through anovulation, impaired oocyte maturation, or LPD. Hyperprolactinemia is usually secondary to medication use or primary hypothyroidism [80]. Maintaining a normal level of prolactin may be useful in women with RPL [81].

Polycystic ovary syndrome

PCOS has been linked to increased rates of pregnancy loss, fetal death, gestational diabetes mellitus, and preeclampsia [82]. Although there is no conclusive evidence for PCOS being a specific risk factor for RPL [83], PCOS may contribute to miscarriage [84]. Insulin resistance, hyperinsulinemia, and abnormal glucose tolerance in women with PCOS have recently been linked to a higher risk of miscarriage [85].

Insulin resistance

Insulin resistance is more common in women with RPL than in fertile controls [86], but the reason for this is unknown.

Luteal phase deficiency

LPD (reduced luteal progesterone production) is widely believed to be a cause of pregnancy loss [87]; however, evidence that LPD is associated with RPL is lacking [88].

Immune causes

The immune system plays a significant role in RPL, acting through autoimmune or alloimmune mechanisms.

Autoimmune causes

Pregnancy loss has long been associated with systemic lupus erythematosus (SLE), and many women with SLE have antiphospholipid antibodies [89]. Women with these antibodies who also have poor pregnancy outcomes or vascular thrombosis are said to have APS. This may cause RPL, intrauterine fetal death, preterm labor, placental insufficiency leading to intrauterine growth restriction, and early-onset and severe preeclampsia [90]. APS occurs in 15%–20% of women with RPL, is the most common treatable cause of RPL, and is the only autoimmune condition clearly associated with miscarriage [3]. How APS causes poor obstetric outcomes is not fully understood [91]. It was initially assumed that thrombosis in the placental vasculature was the underlying cause, but subsequently an inflammatory effect of antiphospholipid antibodies on developing trophoblast and placental cells has become a more accepted explanation [92]. The gestational age at miscarriage due to APS is usually after 10 weeks [93].

Alloimmune causes

The endometrium is the first point of contact between the maternal immune system and the trophoblast [49]. The critical role of immunological tolerance to a semiallogeneic fetus in pregnancy has prompted decades of investigation into immune-related etiologies and therapeutic options for RPL [49]. Many explanations for the immunological tolerance of fetal tissues have been proposed. These include the production of blocking factors that hinder paternally derived foreign fetal antigens from being rejected by the mother, similarities in human leukocyte antigens (HLAs), the role of uterine natural killer (NK) cells in trophoblastic invasion, angiogenesis, and the local maternal immune response to pathogens [94]. Depletion of CD4+ CD25+ regulatory T (Treg) cells causes miscarriage in allogeneic but not syngeneic pregnancies, emphasizing the role of Treg cells in supporting pregnancy [49]. Higher rates of RPL are linked to higher incidences of identical HLA-A and HLA-B alleles [95]. However, the evidence suggesting that specific HLA alleles and HLA sharing can cause recurrent miscarriage is limited [96]. Women with RPL have higher peripheral NK cell percentages and numbers than controls [97], although these do not appear to mirror endometrial levels [98]. Moreover, they do not appear to predict future miscarriage in RPL patients [99]. In carefully timed endometrial biopsies, women with RPL had a considerable increase in uterine NK cell density [100], but the predictive significance of measuring uterine NK cells is unknown [101]. Changes in the amount and efficiency of uterus-resident lymphocytes may result in a proinflammatory endometrium that predisposes to miscarriage [102]. Depletion of tolerogenic dendritic cells has been implicated in pregnancy loss [103], and an imbalance of T helper 1 (Th1) and T helper 2 (Th2) cytokines has been linked to poor pregnancy outcomes including RPL [104].

Inherited thrombophilia

Thrombophilias are caused by mutations in the genes encoding factor V Leiden, prothrombin, antithrombin, protein C, and protein S [105]. Inherited thrombophilias are potential causes of systemic thrombosis. Thrombosis of spiral arteries within the placenta may affect perfusion and ultimately lead to late fetal loss in the second or third trimester, restriction of intrauterine growth, preeclampsia, or placental abruption. Although RPL may be linked with hereditary thrombophilia, the association has not been validated [106]. Hereditary thrombophilias have a greater and more persistent correlation with second-trimester miscarriages than with first-trimester losses [107,108].

Infection

No pathogens have been proven to cause RPL, since the majority of uterine infections are isolated events or elicit protective maternal antibodies [109]. Chronic endometritis is associated with gonorrhea or chlamydia and nonsexually transmitted infections including *Escherichia coli*, *Streptococcus*, *Staphylococcus*, *Enterococcus faecalis*, and yeast. However, the causative organism is often not known [110]. Chronic endometritis has been claimed to cause miscarriage, but there is no proof that enhanced endometrial immune cell invasion in chronic endometritis causes RPL [3]. Future molecular studies of the uterine microbiome should provide further information on the role of infections in RPL [111]. Bacterial vaginosis has been linked to late miscarriage and premature labor [112], but there is little information regarding how it links to RPL [113].

Obesity/overweight/underweight

Obesity (BMI $\geq$30 kg/m^2) is an independent risk factor for RPL [5]. However, there is no clear link between increased BMI alone and RPL [114]. Women who are underweight (BMI $<$19 kg/m2) or overweight (BMI $>$25 kg/m^2) have an increased miscarriage risk [115].

Male factor

Men whose partners have RPL have more abnormal sperm DNA parameters such as sperm DNA fragmentation, but the data are heterogeneous [116]. Y chromosome microdeletion and shortened telomere length in sperm warrant further investigation [117]. Advancing paternal age is related to increased rates of miscarriage independent of maternal age and other variables [118]. Paternal age also adversely affects pregnancy outcomes following intrauterine insemination (IUI) and IVF [119]. However, there is still limited evidence for a paternal contribution to RPL.

Psychological factors

Multiple studies have confirmed greater depression and anxiety in patients following RPL [120]. Psychological support and dedicated care from clinicians have been shown to benefit pregnancy outcomes [121]; thus, both parents should be afforded psychological support and care [5,6].

Lifestyle and environmental factors

Smoking cigarettes may influence trophoblastic function and lead to an increased risk of miscarriage [122]. Other adverse factors include the use of alcohol ($\geq$3–5 drinks per week), cocaine, and caffeine ($>$3 cups of coffee per day) [123,124]. Air pollution and endocrine-disrupting agents such as phthalates and bisphenol A may increase the risk of miscarriage, [125,126] but not necessarily RPL.

Dietary factors

Vitamin D deficiency may be associated with pregnancy-induced hypertension, preterm birth, gestational diabetes, and intrauterine growth restriction [127], but its relationship with RPL is unclear [128]. It may increase levels of antiphospholipid and antithyroid antibodies [129]. Other dietary variables have not shown conclusive links with RPL [130,131].

Unexplained recurrent miscarriage

In 50% of cases of RPL, no cause is identified [9]. Recent studies have focused on endometrial factors in the regulation of implantation [132]. Integrating a genetic analysis of products of conception (POC) and an evidence-based evaluation of RPL into an algorithm is expected to reduce the proportion of unexplained RPL cases to 10% and to reduce associated costs of management by half [133].

Evaluation of RPL

Genetic evaluation of RPL

Parental karyotyping

Genetic irregularities are found in only 3%–5% of couples with RPL [134], but identifying the 2%–5% of couples with a balanced reciprocal or Robertsonian translocation justifies parental karyotyping [32]. Furthermore, parental karyotyping provides predictive information for future pregnancies, and routine parental karyotyping is recommended by the ASRM [6]. However, the ESHRE guidelines recommend parental karyotyping only if a couple has an increased risk of aneuploidy (couples with a child with congenital anomalies, a family history of unbalanced chromosomal translocations, or detection of a genetic translocation in POC) [5].

Cytogenetic testing of products of conception

The ASRM recommends genetic testing of POC only where there is ongoing therapy for RPL or where an explanation for a miscarriage might confer psychological benefit [6]. The ESHRE recommends that genetic analysis of POC should only be performed to provide an explanation, noting that other treatable risk factors may be missed if an abnormal karyotype is found [5]. Karyotyping is expensive and may misdirect management in cases of placental mosaicism [135] or if there is contamination from maternal tissues [136]. Routine karyotypic analysis, G-banded karyotyping, cell culture, and fluorescence in situ hybridization are all hampered by high rates of maternal cell contamination and cell culture failure [137]. Chromosomal microarray testing is currently considered the optimal method for genetic testing of POC because of its ability to detect aneuploidy in patients with recognized reciprocal chromosomal translocations and to reduce the probability of maternal contamination [138]. It is more sensitive than traditional cytogenetic testing [139]. Next-generation sequencing including DNA sequencing of the whole genome can recognize particular genes responsible for disease, but it is still not widely used in reproductive medicine [140]. Of the options for screening, cell-free DNA testing appears to be the preferred choice for couples who want to avoid additional risk of fetal loss, especially when the test covers a panel of microdeletion and duplication disorders [141].

Anatomical evaluation

All women with RPL should have evaluation of the uterine cavity to exclude congenital or acquired uterine abnormalities, despite the uncertain relevance of these in RPL [5,6]. Three-dimensional (3D) ultrasound is the preferred first diagnostic method for congenital anomalies and MRI can be used where 3D ultrasound is not available [5]. Hysteroscopy can be helpful in evaluating the uterine cavity in RPL patients [142]. Sonohysterography with saline infusion has many advantages in evaluating women with RPL, since it is essentially noninvasive and cost-effective [142]. Hysterosalpingography (HSG) is used to diagnose IUAs and will detect (but not evaluate) congenital uterine anomalies [143]. 2D ultrasound could be performed for all RPL patients to exclude adenomyosis [5].

Hysteroscopic images for different uterine pathologies that can attribute to RPL are shown in Fig. 4.2.

Endocrinological evaluation

Thyroid evaluation

The ASRM recommends assay of serum TSH but not anti-TPO in women with RPL [6]. The ESHRE recommends assays of both TSH and anti-TPO and Thyroxine (T4) measurement should be done if abnormal TSH levels detected [5].

Insulin resistance evaluation

Elevated hemoglobin A1C levels (especially >8%) have been linked to an increased risk of miscarriage and congenital malformations [144]. The ASRM recommends measuring HbA1C in women with RPL to assess the glycemic status [6]. However, the ESHRE guidelines do not recommend routine fasting insulin or fasting glucose measurement in RPL pateints [5].

Hyperprolactinemia evaluation

The ASRM recommends measuring serum prolactin in women with RPL [6], but the ESHRE recommends doing so only where there are symptoms of hyperprolactinemia (galactorrhea and oligo- or amenorrhea) [5].

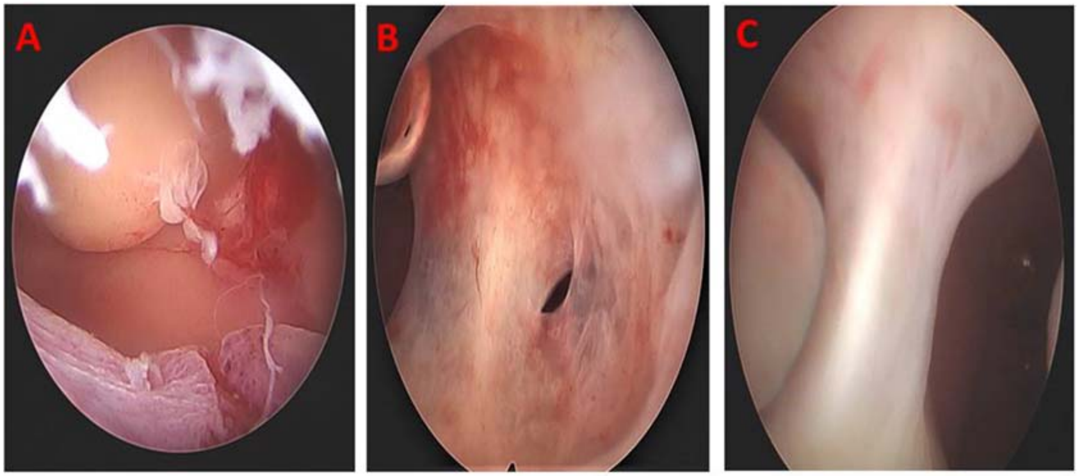

FIGURE 4.2 Shows hysteroscopic images for different uterine pathologies that can attribute to RPL. (A) Hysteroscopic image for uterine fibroid and polyp. (B) Hysteroscopic image for Asherman syndrome. (C) Hysteroscopic image for uterine septum and polyp.

Luteal phase deficiency evaluation

Luteal deficiency evaluation is not recommended in women with RPL [5,6].

Polycystic ovary syndrome evaluation

Testing for PCOS, insulin resistance, and ovarian reserve is not recommended in women with RPL [5,6].

Immunological testing

Antiphospholipid antibody testing

Women with RPL should be screened for antiphospholipid antibodies [6]. The ASRM recommends testing for lupus anticoagulant, anticardiolipin IgG and IgM antibodies, and β 2 glycoprotein I IgG and IgM antibodies. Whereas ESHRE recommends testing only for lupus anticoagulant and anticardiolipin IgG and IgM antibodies; however, screening for β 2 glycoprotein I antibodies can be considered after two pregnancy losses [5].

Alloimmunity testing

Neither the ASRM nor the ESHRE recommends routine testing for immune disorders in couples with RPL [5,6]. However, an exception could be investigation of class II HLA in Scandinavian women with secondary RPL after a delivery of a boy [5].

Inherited thrombophilias evaluation

Screening for inherited thrombophilias in women with RPL is not recommended except in the context of research or women with increased risk such as a personal history of thrombosis in a nonrisk setting (i.e., not associated with surgery or trauma) or a strong family history [5,6].

Infections testing

Routine screening for infection is not recommended in couples with RPL [6].

Male factor assessment

Routine sperm DNA fragmentation screening in males with RPL is not recommended according to the ASRM [6]. The ESHRE indicates that measuring sperm DNA fragmentation could be considered based on the mounting evidence demonstrating a substantial correlation between RPL and sperm DNA damage [5]. Lifestyle in the male partner including age, exercise, body weight, smoking, and alcohol use should be assessed [5].

Prevention

Lifestyle modifications such as weight loss, stress reduction, quitting smoking, limiting alcohol consumption, and avoiding drug use including cannabis and cocaine may all reduce the risk of RPL [5].

Treatment

Couples with structural chromosomal abnormalities

Couples with abnormal fetal or parental karyotypes should be offered genetic counseling [5]. The outcome of future pregnancies is determined by the chromosome(s) implicated and the pattern of rearrangement [6]. The available therapeutic options include attempting further conception with close monitoring during the first trimester and treatment of the concomitant morbidities, assisted reproduction (IVF/ICSI) with preimplantation genetic diagnosis (PGD) or preimplantation genetic testing (PGT), or the use of donor gametes [6].

The use of PGD-SR (also known as preimplantation genetic testing for structural rearrangements) after IVF/ICSI in couples with RPL is increasingly used to select embryos not affected by parental chromosomal rearrangement. However, there is little evidence that it improves the live birth rate in couples with RPL who have structural chromosomal abnormalities [145]. In a systematic review, natural conception was comparable to PGD with regard to rates of live birth, time to conception, and miscarriage [146], but in a comparative study, PGD was associated with a lower miscarriage rate, despite similar cumulative live birth rates and substantial cost [43]. IVF/ICSI followed by preimplantation genetic testing for aneuploidy (PGT-A) is another therapeutic option, but a multicenter RCT found no significant differences in ongoing pregnancy or live birth rates after transferring embryos screened by PGT-A or embryos selected using conventional morphological evaluation [147]. Routine PGT in couples with RPL cannot be justified [5,6].

Anatomical factor correction

Major uterine cavity anomalies should be surgically corrected, despite the lack of RCT evidence of value [6]. IVF with embryo transfer to a screened gestational carrier may be an option in women with irreparable uterine anomalies [6].

Congenital anatomical factors

Hysteroscopic resection of a uterine septum is currently standard management of women with a complete uterine septum and RPL [51]. Many of observational studies showed an association between uterine septum and RPL. In addition, hysteroscopic uterine septum resection has been shown to improve subsequent pregnancy outcomes [3]. However, the randomized uterine septum transsection trial (TRUST), the first registered multicenter randomized controlled trial (RCT) of hysteroscopic septum excision, showed that hysteroscopic resection of the septum does not enhance live birth rates or any other reproductive outcomes in women with a septate uterus [148]. Thus, the ESHRE guidelines formulated a conditional recommendation based on the results of this RCT with limited sample size showing no benefits of hysteroscopic septum resection [5]. Metroplasty is not recommended for women with bicornuate uterus, uterus didelphys or unicornuate uterus, and RPL [5]. However, surgical removal of a rudimentary uterine horn is recommended to reduce dysmenorrhea and other menstrual complications and to reduce the risk of ectopic pregnancy [149].

Acquired anatomical factors

There is no evidence that surgical treatment reduces the risk of pregnancy loss in women with acquired uterine anomalies who have lost a pregnancy [6]. The reproductive gains should be balanced against the surgical risks, which may include new adhesion formation [150]. Counseling in these women should be individualized.

Uterine fibroids

The management of women with RPL and uterine fibroids depends on the location, number, and size of the fibroids and the woman's plans for future reproduction [6]. Despite the lack of high-quality evidence, most authorities support excision of submucosal and intracavitary fibroids to enhance live birth rates, although subsequent IUAs may affect future pregnancies [151,152]. The ESHRE does not recommend hysteroscopic removal of submucous fibroids or removal of fibroids distorting the uterine cavity in women with RPL. Removal of intramural fibroids is also not recommended [5].

Intrauterine adhesions

Standard management of IUA includes hysteroscopy, dissection using cold scissors, and postoperative insertion of an intrauterine balloon [59]. Precautions must be taken after hysteroscopic division of IUAs in women with RPL to avoid recurrence, despite insufficient evidence of benefit [5].

Uterine polyp

There is no reliable evidence supporting hysteroscopic removal of endometrial polyps in women with RPL [5].

Cervical insufficiency

Serial cervical sonographic surveillance should be offered to women with a history of second-trimester pregnancy losses and suspected cervical weakening [5]. Cerclage may be considered in women who have a singleton pregnancy and a history of recurrent second-trimester miscarriage due to cervical weakening, although this does not appear to enhance perinatal survival [5].

Management of endocrine factors

Treatment of thyroid dysfunction associated with RPL

Levothyroxine should be prescribed to women with RPL who manifest hypothyroidism prior to pregnancy or in early pregnancy. However, there is varying evidence of its value in women with subclinical hypothyroidism. The American Thyroid Association recommends thyroxine treatment in women with RPL who have serum TSH > 4mIU/L and positive antithyroid antibodies. It is less strongly recommended in women with a TSH level between 2.5 and 4.0 mIU/L and positive antithyroid antibodies, or in women with TSH > 4mIU/L and negative antithyroid antibodies [153,154]. The TABLET trial compared live birth rates among euthyroid females with TPO antibodies and a history of miscarriage treated with either levothyroxine or placebo, and found no difference between the two groups, including women with RPL [155]. Therefore, the ESHRE recommends against levothyroxine treatment in euthyroid RPL patients with thyroid antibodies [5]. TSH levels should be measured in women with RPL and subclinical hypothyroidism or thyroid autoantibodies at 7–9 weeks' gestation and levothyroxine given if hypothyroidism is diagnosed [5].

Treatment of luteal phase deficiency and RPL

The lack of consensus on diagnostic criteria for LPD has made treating it controversial in women with RPL. The ESHRE states that there is insufficient evidence to recommend the use of progesterone or hCG treatment to improve live birth rates in women with RPL and LPD. However, vaginal progesterone may enhance live birth in patients with $\geq$ three losses and vaginal bleeding [5]. The ASRM concluded that progesterone is ineffective when given to women who have sporadic miscarriages, but that empiric progestogen treatment may be beneficial in women who have had $\geq$ three successive losses prior to their current pregnancy [6].

Treatment of polycystic ovary syndrome and insulin resistance in women with RPL

The initial management of women with PCOS and RPL is to recommend diet and lifestyle modification. Weight loss can improve hormonal profiles and induce regular ovulation [156]. Insulin-sensitizing agents taken before and during pregnancy can reduce the risk of miscarriage in women with PCOS who have a history of RPL and an abnormal glucose tolerance test [157,158]. However, a systematic analysis has concluded that metformin therapy does not reduce the risk of miscarriage [159]. ESHRE does not currently recommend metformin therapy for women with PCOS and RPL who have abnormal glucose metabolism [5].

Treatment of hyperprolactinemia and RPL

Women with RPL and hyperprolactinemia may benefit from treatment with a dopamine agonist [5].

Treatment of immunological disorders with RPL

Treatment of antiphospholipid antibody syndrome with RPL

Many treatment options have been suggested for the management of APS, including low-dose acetylsalicylic acid (ASA) (81 mg/day), unfractionated heparin, low-molecular weight heparin (LMWH), low-dose or high-dose prednisone, and intravenous immunoglobulin (IVIG), used as single agents or combination therapy [160,161]. Many studies have evaluated the effect of different treatments on live-birth rate in the RPL population, and the combination of unfractionated heparin with low-dose ASA appears to provide the optimal results [162–166]. The combination of twice daily unfractionated heparin and low-dose ASA is significantly beneficial in pregnancies with APS and otherwise unexplained RPL [6]. It is important to continue with prophylactic doses of LMWH in the postpartum period to guard against maternal deep vein thrombosis [167].

Treatment of inherited thrombophilia in women with RPL

The decision to use antithrombotic prophylaxis in women with inherited thrombophilia should be individualized according to the perceived risk of thrombosis [5].

Treatment of infections with RPL

The use of antibiotics to treat infection in patients with RPL is not supported by evidence [6]. The association between endometritis and RPL is postulated but not proven, and the effectiveness of antibiotic treatment has not been evaluated in RCTs.

Treatment of male factor in RPL couples

There is no evidence supporting physiological intracytoplasmic sperm injection for sperm selection in RPL couples [5]. Antioxidant supplements did not improve the chance of a live birth [5]. Cessation of smoking, maintaining a normal body weight, limiting alcohol use, reducing pollutant exposure, and maintaining a regular exercise routine are all advised for the male partner [5].

Psychological support in RPL patients

Psychosocial care for couples with RPL is a cornerstone in their management [168]. This care should be offered in a clinic specializing in early pregnancy loss [4]. Counseling should address the couple's bereavement, with supportive follow-up calls, provision of comprehensive data about causes of pregnancy loss, and including family members or friends if desired [3,169].

Couples with unexplained recurrent pregnancy loss

When the underlying etiology of RPL is unknown, clinicians are sometimes urged to provide nonevidence-based therapies. This is particularly common when patients insist that some medications must be prescribed to them in order to avoid a miscarriage.

Role of progesterone in unexplained recurrent miscarriage

Progesterone supplementation in the first trimester of patients with RPL is of questionable value. While a Cochrane review suggested that its use significantly lowered the risk of miscarriages compared to patients who received either a placebo or no treatment, the quality of the trials was poor [170]. The ASRM states that empiric progestogen administration may be of some potential benefit [6]. The ESHRE reports that vaginal progesterone may enhance live birth in patients with $\geq$ three losses and vaginal bleeding in a subsequent pregnancy [5]. According to the results of the PROMISE trial, in asymptomatic women with unexplained RPL, routine progesterone supplementation should be used with caution [171].

Role of immunotherapy in unexplained recurrent miscarriage

The ESHRE has recommended against lymphocyte immunization therapy, glucocorticoids, granulocyte colony-stimulating factor, and intralipid infusion in treatment of unexplained RPL. However, it was reported that the use of repeated and high doses of IVIG very early in pregnancy may improve live birth rate in women with 4 or more unexplained RPL [5].

ASA and low-molecular weight heparin in unexplained recurrent pregnancy loss

Treatment with ASA and/or LMWH does not increase the live birth rate in women with unexplained RPL who do not fulfill the requirements for APS. Moreover, they are associated with adverse effects [172], and the ESHRE has not recommended their use in women with RPL [5]. The ASRM concluded that a combination of twice daily unfractionated heparin and low-dose ASA is significantly effective in women with APS and otherwise unexplained RPL [6].

Preimplantation genetic testing for aneuploidy in patients with unexplained RPL

There is no clear benefit from PGT-A in couples with RPL, according to the ESHRE [5].

Endometrial scratch in women with RPL

The most recent Cochrane review evaluating women undergoing endometrial scratch/injury before IUI or sexual intercourse found no evidence that it enhanced live birth or ongoing pregnancy rates [173], but there have been no studies of its effect in women with RPL. Endometrial scratch should not be recommended for women with unexplained RPL [5].

Future directions

Future efforts should be directed toward reaching agreement on definitions and terminology for RPL that are evidence-based and physiologically sensible; the definition of RPL should include nonvisualized pregnancy losses because they share the same concerns and risks as visualized losses. It is also important to determine the true global prevalence of RPL, rather than the prevalence in developed countries with specialized maternity care, and to identify factors contributing to a higher rate of miscarriage in different ethnicities. This will require the development of multicenter and worldwide collaborations in large cohorts, especially for controversial issues such as the value of IVF in couples with chromosomal rearrangements, the effect of uterine septum resection on reproductive outcomes, the role of inherited thrombophilia, the effect of acquired uterine anomalies and their treatment, the effect of TPO antibodies and treatment with levothyroxine, the significance of antibeta 2 glycoprotein 1 antibodies, and the prognostic value of sperm DNA fragmentation.

Conclusion

RPL is a devastating condition. The psychological and physical toll on patients is often intolerable, and this burden is exacerbated with each loss experienced. Clinicians should provide psychological support and try to define the underlying etiology of the condition. Due to the strict clinical definition of RPL according to certain guidelines, couples having fewer miscarriages may be denied access to medical care. On the other hand, it may be claimed that a more precise definition allows for better allocation of resources and may also result in more relevant study findings. Despite steady improvements in diagnostic modalities, approximately 50% of cases of RPL remain unexplained. Many empirical therapies without evidence of effectiveness are used in couples with RPL, but all current guidelines recommend against such therapies. Instead, couples with unexplained RPL should be offered counseling and supportive care. The prognosis for RPL couples is generally good and depends mainly on maternal age and the number of the prior losses. Well-designed RCTs are needed to explore the causes and management of RPL and their effects on reproductive outcomes.

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CHAPTER

5

The role of surgery in infertility, an evidence-based approach

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Introduction

Congenital or acquired abnormalities of female reproductive tract can affect fertility. While some abnormalities like bilateral tubal obstruction clearly prevent pregnancy, the effects of most anatomic abnormalities and their surgical treatment on fertility are not as clear. We will briefly review these below.

Congenital uterine abnormalities

Uterine septum

While arcuate uterus is accepted as a variant of normal without an impact on fertility, the distinction between arcuate and septate uteri is controversial [1–3]. Available evidence suggest that septum is not associated with decreased fecundity or assisted reproductive technology (ART) pregnancy rates, but with significantly increased pregnancy loss rates (>2-fold) [4,5]. While many women with untreated septum have good obstetric outcomes, septum is associated with increased risk of several obstetric problems including preterm delivery (>2-fold) and malpresentation (>4-fold) [4,5]. The mere presence of the septum can be considered as the cause of the abovementioned problems, through decreased and distorted space for the fetus in the uterus, and septum resection, a relatively straight forward procedure with low complication rates, was considered the standard treatment. Until recently, the medical community did not seem to think equipoise existed to justify a randomized controlled trial (RCT) to test these assumptions. Rikken et al. conducted the first RCT, which randomized women to septum resection (n = 40) and to control (n = 40) groups [6]. More than 90% of women had incomplete septa (not reaching the internal ostium) without a mention of size, and 97% had a successful

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procedure. Overall, pregnancy loss (28 vs. 13%), preterm birth (13 vs. 10%), live birth (31 vs. 35%), cephalic presentation (85 vs. 54%), and cesarean section (69% vs. 36%) rates were statistically similar in resection and control groups, respectively. Small sample size undermines the validity of the results; yet, the observations are in keeping with another retrospective cohort study, which has been severely criticized on multiple grounds [7–9]. Overall, septum is a congenital uterine anomaly that is possibly associated with subfertility and is associated with pregnancy loss, preterm delivery, malpresentation, and cesarean delivery. While high-quality evidence supporting reproductive benefit of hysteroscopic resection is absent, evidence against its effectiveness is also of low quality.

The authors inform their patients about the uncertainty and try to empower them for an informed choice. Hysteroscopic resection is possible even for compete uterine septum extending into the cervix and vagina, and there is no need for an abdominal approach. We mostly use an office hysteroscope with bipolar energy, postoperatively administer hyaluronic acid (HA) gel to prevent adhesions if the resultant opposing raw surfaces are large, and put the patient on 6 mg/day estradiol valerate for four weeks.

Dysmorphic uterus

Dysmorphic uterus, also named T-shaped uterus, is defined with a narrow uterine cavity due to thickened lateral walls, with proportions of 2/3 uterine corpus and 1/3 cervix by ESHRE/ESGE Classification of Mullerian Anomalies [1]. Yet, it is not included in the most recent American Society of Reproductive Medicine Müllerian Anomalies Classification [2]. That alone is the proof of the lack of a universally accepted and reproducible definition of dysmorphic uterus. The etiology, impact on fertility, and effectiveness of hysteroscopic metroplasty to correct a possible effect on fertility are unknown [10]. Recently quantitative diagnostic criteria, based on 3D ultrasound (US) imaging have been proposed [10,11]. Almost all studies lack a control group, and many have been conducted before the publication of the abovementioned diagnostic criteria [12]. As such, even if causal relationships with infertility, ART failure or pregnancy loss, and a benefit of surgical correction exist, they were likely to be overestimated in these studies due to their methodological limitations. Studies investigating effectiveness of hysteroscopic metroplasty are crippled by their before—after design, which inevitably shows some effectiveness for any intervention and lead to unjustified procedures [13].

We inform our patients about the diagnosis of dysmorphic uterus, based on CUME criteria [10], its possible association with reproductive problems, and reserve surgical correction for those with otherwise unexplained problems, for example, long-standing unexplained infertility, recurrent implantation failure, or recurrent pregnancy loss. Available data do not suggest a difference between incising only the lateral walls and HOME-DU technique which involves also incising the anterior and posterior uterine walls; we prefer the former and administer HA gel to prevent adhesions on incision sites followed by four weeks long estradiol valerate at a dose of 6 mg/day [12,14].

Acquired uterine pathology

Endometrial polyps

Endometrial polyp is a mostly benign overgrowth of glands and stroma around a vascular peduncle [15]. Polyps appear as hyperechogenic round structures within the endometrial cavity with transvaginal US; one should be cautious to distinguish between a genuine polyp and a thick, proliferative endometrium presenting a fold-like image particularly on either lateral wall. A saline infusion sonography (SIS) can be done to confirm to diagnosis [16].

Whether polyps affect fertility and surgery is beneficial is controversial. High-quality evidence supporting the routine practice of surgical removal of incidentally diagnosed endometrial polyps in subfertile women is lacking [17]. Notably, a retrospective study evaluating the importance of the location of polyps (n = 230) reported that excision of polyps that were located at the uterotubal junction significantly improved the pregnancy rate [18]. The sole RCT that evaluates the benefit of polypectomy in otherwise unexplained infertility included 215 couples [19]. After up to four cycles of IUI, women in the polypectomy group had a higher chance of clinical pregnancy after polypectomy, with a relative risk of 2.1 (95% CI: 1.5−2.9). Of interest, pregnancies in the study group were obtained before the first intrauterine insemination (IUI) in 65% of cases [19]. When a metaregression analysis was performed including the abovementioned RCT and two additional retrospective studies [20,21], odds ratio (OR) of clinical pregnancy was 3.24 (95% CI: 2.20−4.79) when compared with no intervention in IUI cycles [22]. Unfortunately, data are not available for live birth rate in IUI cycles.

In patients starting an ART cycle, a metaregression analysis of five studies suggests no benefit of polypectomy on either clinical pregnancy (OR: 1.36; 95% CI: 0.89−2.07) or live birth (OR: 1.37; 95% CI: 0.90−2.09) rates when compared with no intervention [22]. It should be noted that the maximum diameter of polyps was <20 mm and polypectomy was performed before the commencement of ovulation induction except in one study [23]. In the latter [23], polypectomy was performed during the ART cycle (n = 43) without any improvement in implantation or live birth rates when compared with controls without polypectomy (n = 43). The authors reported that none of 10 patients who underwent polypectomy within five days before embryo transfer achieved a pregnancy [23].

While the single RCT suggests a benefit of polypectomy prior to IUI, overall data do not support a routine recommendation of polypectomy for ART success. On the other hand, spontaneous resolution of polyps is rare and unpredictable, and the procedure is minimally invasive and has a low risk of intrauterine adhesion (IUA) [24,25].

We recommend polypectomy before infertility treatment in the presence of symptoms such as spotting and heavy bleeding. In asymptomatic patients, polypectomy can be done particularly when visualized in the uterotubal junction. We also discuss the option of polypectomy when the diameter exceeds 1.5 cm.

Intrauterine adhesions

IUAs consist of fibrous tissue that obliterate endometrial cavity partially or completely. IUAs can be associated with infertility, pregnancy loss, menstrual abnormalities, and pelvic pain based on the severity and location of the entity.

A thin endometrium and/or endometrial irregularity with hyperechoic regions, difficulty to visualize endometrial lining, translucent "cyst-like" areas suggest IUAs [26]. Contrast filling defects with homogeneous opacity surrounded by sharp edges or even a completely distorted and narrow uterine cavity occluding the uterine ostia are findings of IUAs in hysterosalpingography (HSG) [27,28]. Three-dimensional SIS appears to be relatively novel technology that should be further evaluated for the efficacy of the method in the diagnosis of IUAs [29].

Hysteroscopic lysis of visible lesions is the current standard treatment of IUAs [27]. Real-time pelvic US guidance during hysteroscopic adhesiolysis is useful to guide the procedure if landmarks such as tubal ostia are not visible and to prevent uterine perforation. US guidance is better than laparoscopic-guided adhesiolysis, with lower cost, lack of an additional invasive procedure, and lower risk of complication [30]. Filmy adhesions can be opened by uterine distension pressure and the tip of the hysteroscope alone, or scissors might be used even without sedation. In severe cases of IUAs, still, it might be reasonable to avoid thermal energy due to potential risk of thermal damage to the adjacent residual endometrial tissue. The use of scissors would be also safer in case of uterine rupture during adhesiolysis, and further intervention to check the organ injury may not be required; however, electric energy is much faster for dense adhesions.

Given the reported recurrence risk of 3.1%–23.5% [31,32], it is reasonable to employ a preventive method especially in severe cases after adhesiolysis. While former studies with intrauterine devices (IUDs) reported encouraging results [33,34], they were not corroborated by later studies [35]. Temporary placement of intrauterine balloons [36] or pediatric Foley catheters [37] might be also employed. In one study, Foley catheter provided higher rates of menstrual restoration (81.4% vs. 62.7%) and conception (33.9% vs. 22.5%) after adhesiolysis than IUD [38]. An RCT suggests that keeping an intrauterine balloon for 28 days instead of the more common 7 or 14 days after hysteroscopic adhesiolysis might result in a greater reduction of recurrence rate [39].

A network metaanalysis of various antiadhesive methods reported that HA gel with or without insertion of IUD demonstrated the greatest change in mean adhesions scores and highest pregnancy rates [17]. It is our distinct opinion that HA is easy to administer, easier for the patient, and conforms to the shape of the cavity, and is the most effective adhesion prevention strategy. While cost may be a concern in some settings, it is unlikely to be higher than repeat surgery that could be required for recurrence.

We always require an HSG before hysteroscopic adhesiolysis for mapping adhesions and planning surgical strategy. We perform hysteroscopy under real-time US guidance; when possible, we first try to locate landmarks as tubal ostia to guide the procedure. When restroation of the cavity is completed, we apply HA gel and four weeks long estrogen treatment after surgery. We evaluate the endometrial cavity with 3D US and 3D SIS, if required, toward the end of four weeks, while the patient is still on estrogen, A progestin is prescribed for 10 days to induce withdrawal bleeding. Patients are always informed about possible need for reoperations in advance.

Fibroids

Fibroids, or leiomyomas, are the most common gynecological tumors in women of reproductive age. Up to 27% women presenting with infertility are diagnosed with fibroids [40], and incidence of fibroids in infertile women without any significant etiology is estimated to be between 1% and 2.4% [41].

Despite the high prevalence of fibroids and advances in diagnostic and therapeutic options, a number of issues prevent production of high-quality evidence regarding the effects of fibroids and their surgical treatment on fertility.

First of all, risk factors for infertility and fibroid development overlap, for example, age and nulliparity [42,43]. Second, natural history of fibroids is unpredictable, while some keep growing and others do not [44]. More importantly, several methodological problems limit the validity of studies on fibroids and fertility. Fibroids are found in various locations, sizes, and numbers in numerous combinations. This heterogeneity prevents uniform definitions or classifications in studies. Management varies greatly, limiting internal and external validity of the findings. Owing to the practical impossibility of performing an adequately sized clinical trial with a long enough follow-up period in a relatively homogenous population, we may never have the definitive answer about the relationship between fertility and fibroids, or the effect of surgery on fertility. Thus, available evidence is mostly from observational studies, which are prone to several biases.

A metaanalysis of 23 studies on fibroids and infertility reported that clinical pregnancy rates were 63% lower in women with submucous (SM) fibroids (FIGO types 0-1-2 [45]) compared to those with no fibroids [46]. Moreover, spontaneous miscarriage rates were increased by 68%. Ongoing pregnancy/live birth rates were significantly lower in the SM group as well.

To the contrary, it has been widely accepted that subserosal (SS) fibroids have no impact on fertility or live birth rates [46]. Indeed, all research on the effect of SS fibroids in fertility has been practically abandoned in the last two decades. Most recent metaanalysis reporting on SS fibroids was performed in 2007, and OR for clinical pregnancy and live birth was 1.2 (95% CI 0.8−1.7) and 1.0 (95% CI 0.7−1.5), respectively [47].

Evidence on effect of fibroids on spontaneous pregnancy is scarce, and the subject seems to be underinvestigated [43]. A 2020 metaanalysis on the impact of noncavity-disturbing IM fibroids on ART outcomes reported that women with noncavity-distorting fibroids had 44% lower odds of live birth as compared with those without any fibroids [48]. A similar and significant association was found when women with only IM fibroids and "IM and SS" fibroids were analyzed separately. Clinical pregnancy rates were reported by 15 studies, and likewise, women with noncavity-distorting fibroids were found to have significantly lower odds of clinical pregnancy when compared those without any fibroids (OR 0.68; 95% CI 0.56−0.83). It should be noted that when the analyses were limited to only prospective studies ($n = 5$), difference in clinical pregnancy rates were not statistically significant ($OR = 0.78$; 95% $CI = 0.58-1.06$). Finally, according to data from 10 studies, miscarriage rates were marginally similar between women with IM fibroid and without ($OR = 1.38$, 95% $CI = 0.98-1.95$). The authors concluded that noncavity-disturbing IM fibroids seem to adversely affect live birth and clinical pregnancy rates after ART, but the results should be interpreted with caution due to heterogeneity and most studies being retrospective.

Our impression is that IM fibroids affect fertility to some extent, but even if that is true, whether their surgical removal result in restoration of fertility is another question. Besides the common risks of surgery such as hemorrhage and infection, myomectomy may also result in adhesions, scar tissue, uterine perforation, or increased risk of uterine rupture, if pregnancy is achieved [49]. Moreover, regenerated tissue after myomectomy may lack some, if not all, functions of a healthy myometrium. Therefore, potential benefits, along with the risks and costs of surgery, should be taken into consideration before offering myomectomy.

Although the concept that removal of SM fibroids increases fertility and its practice is widely accepted, there is only one very low-quality RCT on the subject [50]. The Cochrane review on the subject included only one RCT that fit their criteria [51]. They combined data from both SM and "SM and IM" groups, and reported that OR for clinical pregnancy and miscarriage were 2.44 (95% CI 0.97−6.17) and 1.54 (95% CI 0.47−5.00), when surgery was compared with expectant management [51]. "Very low-quality" evidence rendered the benefit of hysteroscopy in SM and "SM and IM" uncertain. Still, surgical removal of SM fibroids and IM fibroids that distort endometrial cavity is generally recommended for women seeking spontaneous or assisted fertility. ASRM and Society of Obstetricans and

Gynaecologists of Canada (SOGC) declare that there is fair amount of evidence to offer myomectomy to these women [52,53].

The hottest debate is on the role of myomectomy for noncavity-distorting fibroids. The results from relatively small-sized observational studies are conflicting. The choice of controls, that is, women with no fibroids or with fibroids in situ, is heterogenous as well.

In a retrospective cohort study, 63 infertile women with IM fibroids (19 of which underwent myomectomy) were compared with 100 age-matched controls undergoing ART [54]. Clinical pregnancy rates were similar at 36%, 29%, and 36% ($P = .25$) in the myomectomy, fibroid in situ, and control groups, respectively. Another study compared 47 women who underwent myomectomy with 11 women with fibroids left in situ prior to ART [55]. Clinical pregnancy rates were comparable at 16.9% in the myomectomy versus 20.8% in the control group. It should be noted that the study was not limited with IM fibroids, and 10% of women who underwent myomectomy had SM and 50% had SS fibroids. Thus, the small size and the heterogeneity of the fibroid locations decrease the quality of evidence considerably.

In a relatively larger observational study, Bulletti et al. included 168 infertile women with at least one fibroid of at least 5 cm in diameter and no SM component [40]. Half of these women underwent laparoscopic myomectomy prior to ART, while the other half had no surgery prior to ART. Cumulative clinical pregnancy rates were 34% versus 15% ($P < .05$), and live birth rates were 25% versus 12% ($P < .05$) in the surgery and no-surgery groups, respectively. While the results are encouraging, age of the control group was not reported, and other possible confounding factors were not accounted for.

Despite the lack of robust evidence, given the possibility that noncavity-distorting IM fibroids may hinder fertility, especially when they are large, some authors suggest the removal of noncavity-distorting fibroids if they are larger than 4–5 cm in diameter. They also suggest considering myomectomy for smaller fibroids if the woman has a history of failed ART attempts [56]. SOGC states that there is fair evidence to recommend against surgery for noncavity-distorting fibroids, regardless of their size. However, for women with no other options, after discussing the potential benefits and risks, myomectomy can be offered on an individual basis [53]. Similarly, ASRM declares that myomectomy is generally not advised for women with asymptomatic fibroids that do not distort the cavity but may be offered if fibroids make oocyte retrieval complicated or impossible. Still, they add that there is fair evidence that myomectomy does not impair ART outcomes [52].

Last of all, since SS fibroids are not thought to interfere with fertility, they are not removed unless there are other indications for surgery. There is scarce evidence on the effect of their removal on fertility. The study by Casini et al. expectantly observed 11 women with SS fibroids and did not randomize women to undergo surgery in this subgroup [50]. At the end of one year, pregnancy rate was 63.6%. Removal of SS solely for fertility purposes is not recommended [43,46,52,53].

Studies on the effect of different surgical approaches on future fertility are less than a handful. A prospective observation of 1095 reproductive-aged women undergoing surgery for symptomatic fibroids over 36 months reported that the route of myomectomy does not have substantial effect on the probability of achieving pregnancy and live birth [57]. Limitations such as the study population not being limited to women with infertility as the primary symptom, tendency to prefer abdominal route for more severe cases, and lack of information on the use of infertility treatments should be noted. A 2020 Cochrane review included only two RCTs comparing laparoscopic myomectomy with myomectomy by laparotomy or mini-laparotomy reported similar live birth rates [58]. Abdominal, laparoscopic, or hysteroscopic approaches are declared acceptable by ASRM [52]. However, it is sensible and common practice to perform hysteroscopic myomectomy for SM fibroids or IM fibroids that distort endometrial cavity [52,53,56]. The choice of hysteroscopy for large cavity-distorting fibroids depends on available equipment and surgeon's expertise.

In 2020, a systematic review of 68 studies including 1575 pregnancies after myomectomy, 424 after uterine artery embolization, and 420 after fibroid ablation reported that embolization was associated with significantly higher miscarriage and lower live birth rates compared to other methods [59]. It seems wise to avoid embolization for women who may conceive in the future.

In our fertility practice, we recommend the removal of SM and IM fibroids that distort the uterine cavity. SS fibroids are left in situ in the absence of another indication than infertility. Decision on noncavity-distorting IM fibroids depends on the presence of other indications, patient age and reproductive history. In the absence of other indications, we recommend against surgery prior to initial ART attempt in young women, with good ovarian reserve. Women with otherwise unexplained recurrent pregnancy loss or recurrent implantation failure, based on our recent age-specific definition, are offered myomectomy before a new attempt at pregnancy/embryo transfer [60]. Older women, who would not reach the threshold required for unexplained recurrent implantation failure due to increased risk of aneuploidy with age, are given the option of undergoing myomectomy before embryo transfer after pooling a few, preferentially euploid, blastocysts prior to surgery.

While we prefer hysteroscopic surgery for all SM fibroids and some type 3 fibroids depending on size and the distance between fibroid capsule and serosa, the choice of surgical approach is individualized for large type 3 or transmural fibroids with SM component (FIGO type 2–5). Choice of laparoscopy or laparotomy for IM fibroids is a joint decision with the patient, considering number and size of fibroids, prior surgical history, possible risk of malignancy, and cost issues.

Isthmocele

A myometrial defect, a.k.a. niche, in the lower segment of uterus following uterine surgery, most often cesarean section, is called an isthmocele. It is observed as a fluid filled, anechoic, triangular defect close to isthmus (Fig. 5.1) [61]. A prospective study reported 22% incidence six months after a cesarean delivery [62]. While the association of postmenstrual spotting and isthmocele is understandable, whether it is causally related to subfertility is less clear. Retention of menstrual debris and/or intracavity mucus within the defect can affect sperm motility, fluid reflux into endometrial cavity can decrease implantation probability and increase the risk of miscarriage like hydrosalpinx, or inflammatory condition around the defect, potential iron toxicity from the breakdown of hemoglobin can also affect embryo implantation [61]. At the moment, these are theories that require proof. To the contrary, two retrospective studies reported similar live birth/ongoing delivery rates in women undergoing frozen embryo transfer with and without isthmocele [63,64]. While embryo transfer was postponed to another cycle in the presence of intracavitary fluid collection (IFC) (see Fig. 5.1 for IFC) in the study by Lawrenz et al., whether IFC was detected during frozen embryo transfer cycles is not mentioned [63,64]. These two studies lend credit to reflux theory

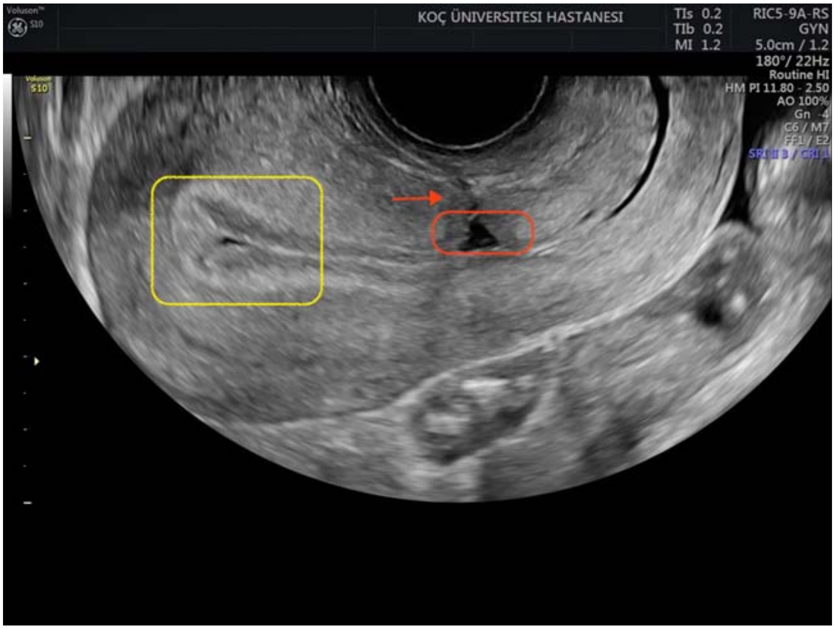

FIGURE 5.1 Sagittal view of isthmocele. *Red* box shows isthmocele cavity, *red* arrow points residual myometrial thickness, and *yellow* box shows intracavitary fluid collection.

and provide some evidence against the inflammation and iron toxicity theories. IFC occurred in 12%–40% of women during ovarian stimulation in the fore-mentioned studies [63,64]. Women with ICF had significantly more cesarean deliveries and larger isthmoceles [63]. So far, the most effective way to prevent the occurrence of isthmocele seems limiting the rate of first cesareans [65]. Double layer closure of myometrium does not seem to decrease the incidence of isthmocele, but increase residual myometrial thickness (RMT), which can be relevant for the choice of surgical technique if surgery is indicated [66,67].

A 2020 systematic review found that surgical treatment was 80% effective in treating AUB related with isthmocele, but there was no evidence for a benefit of surgical treatment for fertility or obstetric outcomes [68]. Yet, a 2021 systematic review suggests hysteroscopic treatment can be considered for secondary infertility, based on one small-sized RCT [61]. We think the currently available evidence is inadequate to support routine surgical treatment of isthmocele in infertile women. In our practice, we selectively offer surgery only to women who have visible ICF even during the secretory phase, that is, planned day of embryo transfer. Our choice is almost always hysteroscopy, which involves shaving the distal (closer to cervical os) base of the defect to increase drainage of any collection toward the cervix and to cauterize the dome of the defect to make it shrink to some extent (Fig. 5.2). The patient is informed in advance that the defect will still be visible with US, and we only aim to prevent ICF. Hysteroscopy may not be the optimal approach when RMT is low, for example, <2.5 mm, an arbitrary choice in some publications [61]. It is possible to repair isthmocele with laparoscopy or through the vagina [69]. Whether one method is better than another is unknown now.

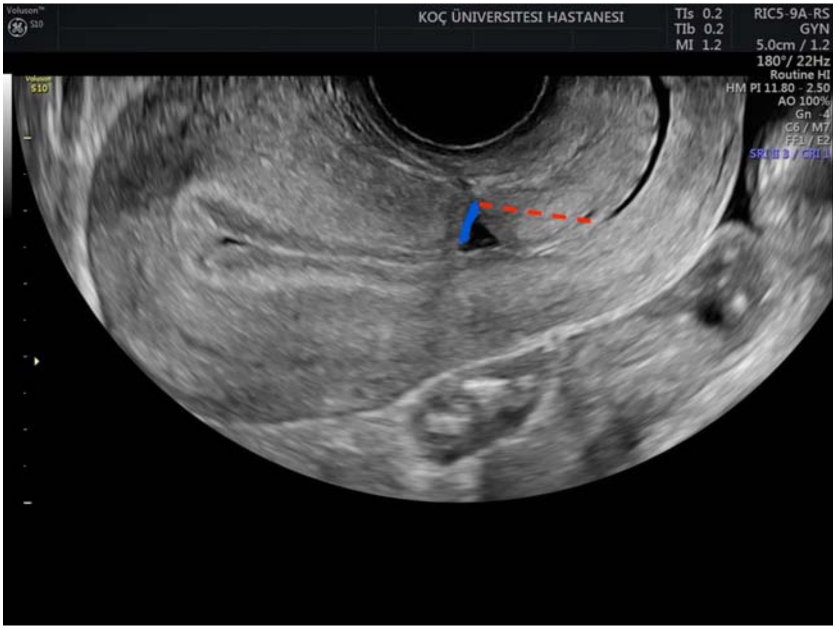

FIGURE 5.2 *Red* dashed line shows the line of hysteroscopic resection, and *blue line* shows the surface to be cauterized in the dome of the defect.

Tubal pathology

ART is the treatment of choice for most tubal factor infertility, particularly in the presence of other infertility factors or severe tubal disease, because ART success rates often exceed those of tubal surgery. Interpretation of surgical success is difficult, since generalizability of results is limited and the denominator includes varying periods, for example, 6 mo.s, to two or more years [70]. In the absence of another indication for ART, younger women, with normal/high ovarian reserve, proven fertility, and desiring multiple children may be considered candidates for surgical repair. Tubal anastomosis can be considered for women desiring sterilization reversal. Laparoscopic tubal surgery abiding microsurgical principles can be performed with or without the use of a robot; since the success rates are similar, there is no need for laparotomy. Surgeon must have adequate training and expertise on tubal surgery.

Distal tubal obstruction

While increased ART success rates have limited the indications for reconstructive tubal surgery, yet distal tubal obstruction still comprises an indication for surgery because hydrosalpinges adversely affect ART outcome. Pregnancy, implantation, and delivery rates are decreased by approximately 50%, and miscarriage rates are increased in the presence of hydrosalpinges [71]. Laparoscopic salpingectomy or tubal occlusion improves ART pregnancy rates two- to fourfold in women with hydrosalpinges, even when only one tube is affected [72]. Indeed, disconnection of a communicating unilateral hydrosalpinx by salpingectomy or obstruction improves chances of spontaneous pregnancy through the contralateral tube [73–75].

While earlier publications suggested an impact of salpingectomy on ovarian reserve, latter studies suggest that salpingectomy prior to ART, opportunistic salpingectomy during hysterectomy, bilateral salpingectomy for permanent sterilization during cesarean delivery, or for management of ectopic pregnancy does not significantly decrease ovarian reserve [72,76–78]. Ovarian response to stimulation and IVF outcome seems to be maintained after salpingectomy [79]. Response to ovarian stimulation, implantation, and clinical pregnancy rates is similar in women who underwent salpingectomy or laparoscopic proximal tubal occlusion [80].

US-guided aspiration of hydrosalpinx at the time of oocyte retrieval and hysteroscopic tubal occlusion with microinserts (Essure) are alternatives to salpingectomy [81,82]. However, ART outcome is not as good as with salpingectomy or proximal tubal occlusion [72,81]. They can be regarded as second-line interventions preferred for women who are at risk of complications with pelvic surgery.

Proximal tubal obstruction

Most proximal tubal occlusions observed with HSG are artifacts due to mucus plugs, cellular debris, or uterotubal spasm [83]. If proximal occlusion is not due to salpingitis isthmica nodosa, tubal cannulation using hysteroscopic or fluoroscopic methods is a proven alternative to traditional microsurgical repair. Where available, immediate proximal tubal cannulation at the time of HSG is the ideal method to confirm and treat true proximal obstruction. Otherwise, a repeat HSG as short as a month later would decrease the number of false-positive tests of tubal patency [84,85]. Clinical pregnancy rate following tubal catheterization for unilateral or bilateral proximal tubal occlusion was 27% (95% CI, 25%–30%). The pooled live birth and ectopic pregnancy rates were 22% (95% CI, 18%–26%) and 4% (95% CI 3%–5%), respectively. However, reocclusion rate of the opened tubes was approximately 30% [83,86].

Microsurgical segmental tubal resection and anastomosis is a proven treatment for true proximal tubal obstruction. Experienced surgeons can achieve pregnancy rates ranging between 50% and 60%, but results vary with the cause of the obstruction; reocclusion rates are relatively high with causes other than salpingitis isthmica nodosa [83,87]. In general, success rates achieved with surgery have been extremely poor and IVF represents the best treatment option [70,87].

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CHAPTER

6

Ovarian stimulation and intrauterine insemination

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Introduction

Assisted reproduction technologies (ARTs) may be classified according to complexity, in either low or high. Intrauterine insemination (IUI) belongs to the first group and can be mainly characterized as a set of measures aiming to increase the likelihood of a timely in vivo fertilization of the male and female gametes. Therefore, due to its conservative nature and that it is otherwise intended to mimic what is expected to occur in spontaneous conception, the effectiveness of each IUI attempt, even under optimal clinical conditions, is inherently limited by the narrowed reproductive performance of the human species. This reality has led to diverging views regarding the role IUI should play in the context of infertility treatments among reproductive medicine specialists [1–3], a disagreement rooted also by ethical and logistic dilemmas that attempt to answer as best as possible the question *what is better for the patient?*

a. To either start infertility treatment by undergoing a less invasive and less expensive approach (whenever the proper clinical conditions are available), in spite of its lower efficacy per attempt;
b. Or to advance directly to a more effective treatment (e.g., in vitro fertilization), in spite of its higher cost and invasiveness.

While expanding further into this controversy is out of the scope of this chapter, it must be made clear that IUI is widely considered in the medical literature as a first-line infertility therapy and still frequently used in clinical practice, especially for couples without longwithstanding infertility or advanced reproductive female age [4,5]. Therefore, understanding its rationale, the factors that influence its efficacy, the variations in methodology, and the risks associated with its performance are relevant for an adequate medical care to be offered to patients undergoing IUI.

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Prior to entering into all these aspects, there is one point that must be emphasized: in our perspective, a shift is currently taking place regarding what are considered to be the optimal conditions that must be obtained in order to carry out an IUI treatment cycle. Most importantly, the ongoing discussion whether ovarian stimulation with the achievement of more than one mature follicle is desired to improve IUI success should be finally answered. Given the current acknowledgment of how undesirable multiple gestations are, due to their significantly increased risk of serious complications for both the fetuses and the patient, any circumstance that may be associated with such occurrence should be avoided.

Definition and indications

IUI attempts to establish an analogous setting to that of a fertile couple having intercourse in the ovulatory period and to, ideally, optimize it. In order to promote such a scenario, the treatment can be typically divided into the following phases:

1. Stimulation of follicular growth;
2. Ovulation induction, once follicular maturation is documented;
3. Processing of the sperm sample;
4. Insemination of the processed sample into the uterine cavity, through the cervical canal catheterization;
5. Luteal phase support.

While variations in the clinical steps aforementioned, their advantages and risks will be discussed in this chapter, the laboratorial processing of the sperm sample is not in our purview.

Under optimal clinical conditions, the effectiveness of the IUI approach is typically reported to be around 15%–20% per cycle. As expected, the figures become lower if the clinical conditions are suboptimal. Therefore, for a precise characterization of the patients' context and prognosis, the assessment of the following clinical parameters should be considered:

1. The number of spermatozoa with progressive motility present in the ejaculate;
2. Female's age;
3. Tubal patency;
4. Infertility diagnosis;
5. Duration of infertility.

Optimal conditions for IUI are those in which the couple's evaluation confirms that all the premises for the occurrence of a pregnancy in such a context are present: a good sperm count and motility, a female patient without advanced reproductive age, bilateral tubal patency with no sign of pelvic adhesions, no diagnosis of endometriosis (although this latter premise lacks consensus) [6], and a short duration of infertility.

Postwash total motile sperm count (TMSC) is associated with IUI pregnancy outcome, with counts above 9–15 million providing optimal results [7,8]. Below these levels, there seems to be a progressive decline in pregnancy rates in an approximately linear fashion. That said, a specific threshold below which IUI should no longer be performed is difficult to establish, given that pregnancy outcomes especially in the 0.8 to 5 million range vary considerably [9]. Hence, while TMSC might allow to counsel patients regarding outcome expectations, it may be of limited value to guide the actual choice of treatment.

Given that female age is a major determinant of assisted reproduction success, age cut-offs (e.g., 40 years old) to determine the indication for IUI are frequently proposed by physicians. However, other researchers have proposed a more permissive approach in women performing donor insemination, given the higher pregnancy rates in such cases [10,11] (Fig. 6.1).

The approach toward the assessment of tubal patency testing prior to IUI has fluctuated substantially over time. Although laparoscopy with dye instillation is generally regarded as the gold standard to evaluate tubal function, it is usually reserved only for women with other concomitant indications for invasive abdominal surgery, owing not only to the costs and risk of this procedure but also to the availability of a multitude of more readily available and safer alternatives. The classical approach has been to perform a hysterosalpingogram, since it is relatively easy to execute and uses equipment which is widely available [12]. However, it is important to acknowledge that this exam frequently causes substantial pain and that its accuracy may be lower in some patient populations [13], which is important to consider when guiding patients, especially following an abnormal result. Such is especially important to account for given that tubal obstruction, in low-risk women (i.e., nonsmokers with no history pelvic inflammatory disease or abdominal surgery), is unlikely [14,15], raising many physicians to question whether these women should undergo this procedure routinely or not [16], with some advocating against its routine use prior to donor insemination [17]. Moreover, the use of oil-soluble solutions, that used to be the mainstream contrast media, has been associated with complications such as granuloma formation and embolism, which has progressively led to its replacement with water-based alternatives. More recently, less painful approaches have been increasingly used, including hysterosalpingography with foam-based ultrasound contrast media. This strategy has been shown to have high accuracy, while being substantially

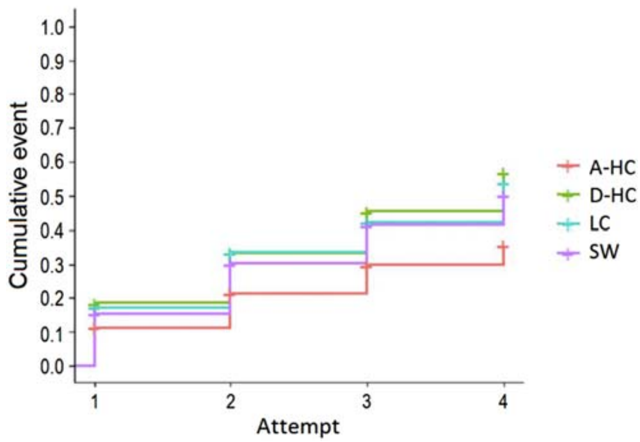

FIGURE 6.1 Kaplan–Meier estimate of cumulative live birth rate (LBR) for each successive IUI attempt for heterosexual couples undergoing autologous IUI (A-IUI) and single women (SW), lesbian couples (LC), and heterosexual couples (D-HC) undergoing donor IUI, in cases of female patients at an age range of good prognosis (≤37 years). *Fig. 6.2 from Soares SR, Cruz M, Vergara V, Requena A, Garcia-Velasco JA. Donor IUI is equally effective for heterosexual couples, single women and lesbians, but autologous IUI does worse.* Human Reprod. *2019;34:2184–2192.*

less painful. However, despite this potential benefit in terms of pain reported by the patients, oil-based media have recently regained further attention following a recent study showing a potential added therapeutic value. Specifically, post hoc analysis of a large randomized controlled trial concluded that oil-based hysterosalpingographies may be associated with higher cumulative live birth rates at least in the first nine months after the procedure [18], a benefit that seems to be most notable in women who experienced most pain during the procedure [19]. These results have been justified by some as evidence that oil-based solutions may contribute to the elimination of debris that could hinder tubal function (Table 6.1).

Regarding the duration of infertility, its relevance depends on the main cause of infertility: although a longer duration may be related to a lower probability of success with IUI [20], such assumption depends on whether:

a. we are facing a couple in whom the failure to conceive is due to a potentially restorable factor that may dramatically alter the prognosis following correction (for instance, chronic anovulation), or;
b. we are facing a couple with a long-duration infertility of unexplained cause, in which the odds of tubal dysfunction and/or gametes of reduced quality are higher, ultimately impairing the eventual effectiveness of an IUI attempt.

All these factors considered, the indications for IUI would be:

1. Mild male factor infertility;
2. Short duration infertility of unexplained cause;
3. Anovulation;
4. Difficulty or impossibility of carrying out intercourse, due to either malformation, disability, or psychological factors;
5. Serodiscordant couples for viral infections such as HIV or HCV (with the latter being less consensual).

TABLE 6.1 Estimated hazard ratio for ongoing pregnancy of oil versus water hysterosalpingogram at different time points after the exam.

Follow-up time	HR (95% CI) for ongoing pregnancy using log time
Immediately after HSG	1.71 (1.27–2.31)
1 month	1.57 (1.24–1.99)
3 months	1.36 (1.17–1.59)
6 months	1.25 (1.09–1.44)
9 months	1.19 (1.03–1.38)
1 year	1.15 (0.98–1.35)
2 years	1.06 (0.86–1.3)
3 years	1.00 (0.79–1.28)

CI, confidence interval; *HR*, hazard ratio; *HSG*, hysterosalpingography.
Table 6.1 from van Welie N, Rosielle K, Dreyer K, van Rijswijk J, Lambalk CB, van Geloven N. et al., 2020. How long does the fertility-enhancing effect of hysterosalpingography with oil-based contrast last? Reprod Biomed online *41, 1038–1044.*

Ovarian stimulation protocols

Natural cycle IUI

The avoidance of the use of medication to induce follicular growth and ovulation in regularly cycling women is proposed by part of the medical community and patients [21,22]. Given that a monofollicular cycle is considered by many as the ideal scenario for IUI, the natural cycle seems to be, at a first glance, an excellent choice. However, in practice, the efficacy of this approach is predominantly shown to be lower than that of cycles with ovarian stimulation, at least in autologous IUI [5,23]. Such an improvement, that may be related to the pharmacological correction of minor follicular and/or endometrial maturation dysfunctionalities, seems to occur even when one restricts the comparison to cycles in which only one mature follicle is present [23]. The scarce data available for donor-IUI indicate that the natural cycle may perform better in these cases than in autologous IUI [24,25]. As expected, significantly reduced multiple pregnancy rates are observed when a natural IUI is performed.

The logistics of natural IUI cycle entails the following:

- Since the natural luteinizing hormone (LH) surge occurs approximately 16 days before the following menstruation, in patients with regular cycles, the first transvaginal ultrasound scan (TVUS) scan is usually scheduled for two—three days before the expected date of the surge (e.g., in a 28-day menstrual cycle, it may be scheduled for Day 9 or 10). In a patient with irregular cycles, the most conservative approach is to schedule the first TVUS scan for two—three days before the expected day of the LH surge of her shortest possible cycle. In these cases, the need of more than one follicular tracking TVUS should be anticipated.
- Identifying the LH surge: Once a follicle with mean diameter $\geq$13 mm and endometrial proliferation are documented, daily measurements of serum LH are initiated. The LH surge is detected when an LH rise of 180% above the latest serum value is observed, and its absolute value is above 18 IU/L. Alternatively, a urinary LH kit may be used daily, albeit with less precision on the timing of the hormonal surge.

Stimulated cycle IUI

Our practice is to schedule a TVUS scan during or just prior to the (predicted) beginning (first three days) of the menstrual cycle, to rule out factors that contraindicate the initiation of treatment, such as the presence of a functional ovarian cyst. Drugs used for ovarian stimulation for IUI may include gonadotropins, clomiphene citrate (CC), or the aromatase inhibitor letrozole.

- Gonadotropins:

Daily doses of recombinant FSH or human menopausal gonadotropins are initiated between Days 2 and 5 of the cycle. Starting dose in a first attempt should range between 37.5 and 75 IU/day and is defined based on patient's ovarian reserve and body weight. Higher doses should be avoided as they increase the risk of multiple pregnancy without increasing pregnancy rates [26]. Response to a previous stimulation, when such has occurred, is also a valuable information. During stimulation, the dose is adjusted according to the ovarian response monitored by serial TVUS carried out every one–three days from Day 5 or 6 of stimulation. Typically, final oocyte maturation and ovulation are triggered by the administration of human chorionic gonadotropin (hCG), when one mature follicle (mean diameter ≥17 mm) is seen in the vaginal scan.

- CC and aromatase inhibitors (letrozole):

Standard protocols for CC and letrozole are similar, with 1 or 2 pills being administered orally between Days 2 and 5 of the menstrual cycle. A first TVUS scan can be done on Day 10 or 11 of the cycle. When using these drugs, it is advisable to administer hCG only when the largest follicle mean diameter is at least above 20 mm [27]. The reason for the benefit of a larger follicle, in comparison to when gonadotropins are used, is not fully clarified, although some postulate that it may be related to a delayed endometrial maturation (Fig. 6.2).

Combined protocols

Combined protocols are those in which CC or letrozole use is followed by the initiation of gonadotropins, either FSH or hMG, three to five days after the last day of oral medication. From this point onwards, the follow-up is the same as that of a gonadotropin stimulation cycle.

The use of a GnRH-antagonist in the last days of ovarian stimulation

There are two possible motivations to add a daily dose of a GnRH-antagonist in the final days of a gonadotropin stimulation cycle:

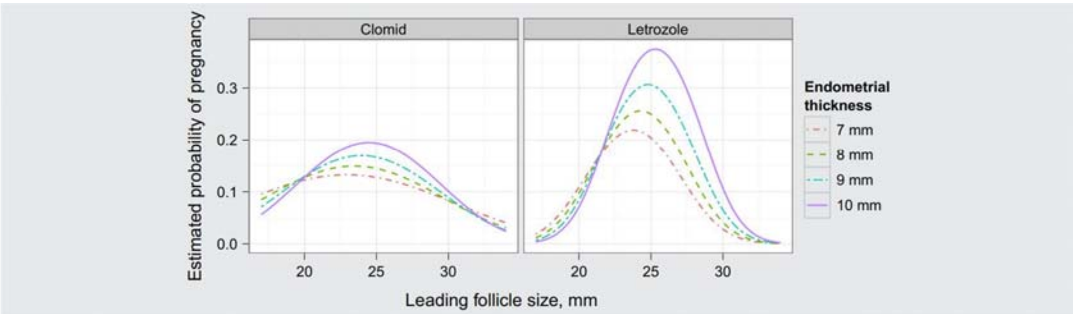

FIGURE 6.2 The estimated effect of the leading follicle size on the probability of pregnancy depending on endometrial thickness and cycle type (clomiphene citrate or letrozole ovarian stimulation). The figure is adjusted for other covariates via logistic regression. Reference levels (mean for continuous and mode for discrete variables): age=34 years; primary diagnosis=unexplained. *Supplemental Fig. 6.1 from Ref. Palatnik A, Strawn E, Szabo A, Robb P. What is the optimal follicular size before triggering ovulation in intrauterine insemination cycles with clomiphene citrate or letrozole? an analysis of 988 cycles.* Fertil Steril *2012;97:1089–1094.e3: Welie N, Rosielle K, Dreyer K, van Rijswijk J, Lambalk CB, van Geloven N et al. How long does the fertility-enhancing effect of hysterosalpingography with oil-based contrast last?* Reprod Biomed Online *2020;41:1038–1044.*

- Avoiding the risk of spontaneous ovulation as we wait for the dominant follicle to reach at least 17 mm of mean diameter, impairing the adequate synchronization between ovulation and insemination and often leading to cycle cancellation. The GnRH-antagonist administration, in this case, should be initiated as soon as the spontaneous LH surge becomes a possibility, with a leading follicle mean diameter around 14–15 mm, and may be maintained until the day of hCG administration;
- In order to minimize the risk of spontaneous ovulation when prolonging stimulation for a few more days due to scheduling constraints (e.g., in order to avoid the need to perform IUI during the weekends).

Efficacy and safety of the different protocols

In spite of the abundance of data generated from autologous IUI, controversy persists regarding the superiority of gonadotropins over oral medication in terms of clinical outcome [23,28,29]. Noteworthy, findings suggest that the improvement in pregnancy rate in some occasions observed with gonadotropins is undetachable from the increased risk of multiple gestations, and that the efficacy per follicle is not superior to that obtained with oral medication, especially with letrozole [5,23,29]. The same applies to combined protocols [5].

Considering the above and, importantly, the fact that the use of gonadotropins is more complex and expensive, there seems to be little reason to use such drugs as first-line therapy for IUI.

Cancellation criteria during stimulated IUI

The use of exogenous ovarian stimulation, employed primarily as a strategy to increase IUI pregnancy outcome, will result in the development of 3 or more mature follicles in up to a third of all cases [30]. In women below 40 years of age, such a response is associated with a substantial increase in the risk of multiple pregnancy. Meanwhile, in older women, having up to four mature follicles almost tripled the odds of pregnancy while still maintaining the risk of multiple gestation below 13%.

Hence, careful consideration should be given to cycle cancellation in cases in which the risk of multiple pregnancies increases disproportionately to the pregnancy rates [31]. Another alternative which may be proposed is excess follicle aspiration, which offers a possibility of continuing with treatment without hindering live birth rates and greatly reducing the risk of multiple pregnancy [32]. This latter approach, while increasingly used, still requires further investigation.

Final oocyte maturation and ovulation triggering and timing of insemination

There is a substantial debate on what is the optimal strategy to trigger ovulation prior to IUI. The most classical approach involves the administration of exogenous hCG 36 h prior to IUI, although others have proposed serial endocrine monitoring until an LH surge is detected. One metaanalysis [33] and a subsequent randomized controlled trial [34] from the same research group alluded to the potential benefit of opting to wait for spontaneous LH ovulation triggering. However, subsequent studies have shown discrepant results [26,35,36], including a

Cochrane review which failed to show any benefit of either strategy [37]. A possible explanation for the difference in results could be a variation in insemination timings among the studies. Specifically, another clinical trial demonstrated that insemination the day after an LH peak could provide a nearly twofold increase in pregnancy rate when compared to performing the IUI in the following day [38]. The rationale is that timing is crucial, owing to the limited period during which capacitated spermatozoa survive in the female genital tract to fertilize the oocyte(s) between 12 and 16 h following ovulation. Conversely, in hCG-triggered cycles, scheduling the IUI for 24 h after hCG administration resulted in a significant reduction of pregnancy rates (vs. 34–36 h), although this difference was no longer significant when restricting the analysis to studies reporting live birth [37]. These results, however, should be considered with caution, owing to the limited number of studies available.

Bed rest

Many centers advocate women should remain immobilized for approximately 10–15 min following IUI. The rationale behind this recommendation is that sperm requires such time to reach the fallopian tubes after insemination, thus avoiding "flow back." Most of the first studies on this matter seemed to confirm a benefit of post-IUI immobilization, including two randomized controlled trials [2,39] which were the basis for the World Health Organization (WHO) moderate quality recommendation for bed rest [26]. However, subsequent studies have questioned whether this recommendation is indeed warranted, including another large clinical trial [40] and a subsequent metaanalysis [41]. Finally, others have proposed the use of devices to promote a slow-release and intrauterine sperm containment over a period of four hours [42]. While promising, the clinical usefulness of such devices requires further validation.

Number of inseminations during treatment

Most centers currently perform a single IUI per treatment attempt, in line with what is recommended by the WHO [26]. However, more recent evidence has shed new light on this issue, questioning whether double IUI may further improve treatment outcome. The first study worth mentioning includes a large cohort of over 20,000 cycles and demonstrated a twofold increase in pregnancy rates in heterosexual couples using both autologous and donor sperm. Conversely, in lesbian and single patients, this apparent benefit was no longer evident, where only donor sperm was used [43]. In 2021, a Cochrane metaanalysis also posited that double IUI could indeed enhance pregnancy rates, but cautioned that the evidence was of low quality, requiring further investigation [44].

Luteal phase support

The known benefit of luteal phase support in ART is frequently extrapolated to IUI. However, despite the wide use of this strategy, there is a surprisingly low number of studies of this topic. One of the most recent clinical trials, from 2016 and including 393 women, posited that luteal phase support could be beneficial in gonadotropin-stimulated IUI cycles, although the study itself was underpowered to substantiate such a claim [45]. Shortly after, a systematic review and metaanalysis also concluded that, specifically in the group of women performing stimulation with gonadotropins, luteal phase support with exogenous progesterone increased live birth rates [46]. The authors of this latter study postulated that differences in drug mechanisms of action, namely the hypothalamic targeting of CC, may justify why the luteal phase in such cycles may be endogenously enhanced, thus lacking the need for exogenous luteal support. A Cochrane systematic review is also currently underway to reply this same scientific question [47].

Final remarks

The effectiveness of IUI is inherently limited by the narrowed reproductive performance of the human species. However, due to its low invasiveness and cost, it is still considered as a first-line treatment for cases of mild male factor infertility, short duration infertility of unexplained cause, anovulation, difficulty or impossibility of carrying out intercourse (due to either malformation, disability or psychological factors), serodiscordant couples for viral infections such as HIV, and single women and lesbian couples, mainly for female patients without advanced reproductive age.

Strategies to improve its success rate that are associated with an increased risk of multiple gestation should be avoided. The scarce data available for donor-IUI indicate that the natural cycle may be a good option in these cases, while the stimulated cycle is associated with better results in autologous IUI. For such cases, oral medication can be considered the first-line therapy.

There is no definitive evidence of improved results with any specific final oocyte maturation approach (natural LH surge vs. hCG administration) and number of inseminations per cycle (single or double). The same is true for bed rest after the procedure. Luteal phase support is associated with increased live birth rate in gonadotropin stimulated cycles, while such evidence is lacking in cycles in which stimulation is carried out with oral medication.

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CHAPTER

7

Infertility care of the PCOS patient

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Introduction

In this chapter, our goal is to provide a framework for the fertility care of women with polycystic ovarian syndrome (PCOS) ranging from preconception to evaluation to treatment. We will utilize the recent evidence-based guideline for the assessment and management of PCOS developed by the Australian National Health and Medical Research Council of Australia (NHMRC) with support and endorsement by the European Society for Human Reproduction and Endocrinology (ESHRE) and the American Society for Reproductive Medicine (ASRM) [1]. However, we acknowledge that in many instances, there were insufficient data to form an evidence-based recommendation, and in such cases, clinical expertise or extrapolation from other populations was utilized to make an "educated guess." In such areas, with apologies to the reader, we confess our own biases will weigh heavily on our written words.

Preconception risk factors

It is well known that women with PCOS are plagued with both reproductive and metabolic abnormalities that may adversely affect maternal and neonatal outcomes, if left untreated. Such factors, including obesity, impaired glucose tolerance, and anxiety and depression should be optimized prior to fertility treatment.

The international NHMRC evidence-based guideline recommends several health, lifestyle, and behavioral targets to optimize prior to attempting fertility interventions. Clinicians and patients should focus on optimizing weight, blood glucose, blood pressure, sleep habits, and mental, emotional, and sexual health [1]. Preconception counseling should focus on lifestyle modifications such as weight loss, maintaining healthy diet and exercise habits, smoking cessation, and limiting alcohol intake. A prenatal vitamin with adequate folic acid should

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be instituted prior to conception. For all women with PCOS seeking pregnancy, a 75-gram oral glucose tolerance test should be offered to screen for preexisting diabetes [1]. It is important to keep in mind the increased prevalence of risk factors for cardiovascular disease (CVD) among patients with PCOS. Preconception screening should assess for the presence of CVD risk factors such as obesity, dyslipidemia, cigarette smoking, sedentary lifestyle, hypertension, and impaired glucose tolerance. The existence of impaired glucose tolerance prior to pregnancy highly predicts the development of gestation diabetes during pregnancy [2]. Further, it has been shown that impaired glucose tolerance is significantly associated with decreased fecundity in women with PCOS undergoing in vitro fertilization (IVF) [3]. It should be noted that the risk of diabetes and CVD varies across ethnicities among women with PCOS [1].

Another important consideration prior to initiating treatment for anovulatory infertility is screening for other infertility factors such as male-factor infertility, uterine factors, or fallopian tube disease [4]. In one prospective study of women with PCOS and anovulatory infertility, 10% of males were found to have significant sperm abnormalities and 5% of patients had bilateral tubal occlusion [5]. Such findings necessitated further evaluation or treatment beyond ovulation induction alone. For couples with infertility due to anovulation alone and a normal semen analysis, evaluation for tubal patency should be pursued on a case-by-case basis when there is suspicion for tubal occlusion [1]. Finally, for those patients who achieve pregnancy, the NHMRC/ESHRE/ASRM guidelines remind us that monitoring in pregnancy is an important consideration in this population to reduce the likelihood of pregnancy complications. Indeed, women with PCOS are at increased risk of pregnancy-induced hypertension, preeclampsia, gestational diabetes, and premature delivery [6].

Weight loss and obesity

Among obese women with PCOS, weight loss should be primary consideration when crafting a treatment strategy. Weight loss as little as 5%−10% of total body weight is associated with improvements in ovulation, menstruation, and spontaneous pregnancy, not to mention the health and psychological benefits associated with loss of this excess weight [7−10]. Further weight loss prior to conception likely increases the likelihood of responding to oral ovulation induction drugs [11]. Lifestyle changes are first line with a focus on healthy eating and exercise habits. International guidelines note that an energy deficit of 30% (or 500−750 kcal/day) is a suitable target goal, though flexible, individual approaches should be employed [1]. No energy-equivalent diet has been shown to be superior to another, and a nutritionally balanced diet is recommended. The same guidelines recommend a minimum of 150 min per week of moderate-intensity exercise or 75 min per week of high-intensity exercise (or a combination of the two), with strength-enhancing exercise at least twice weekly [1]. Unfortunately, a paucity of evidence exists to validate these recommendations in clinical practice. A 2019 Cochrane review examined 15 studies comparing dietary, exercise, or behavioral interventions to minimal interventions. No studies assessed menstruation, live birth, or miscarriage. The review found that lifestyle intervention may improve body mass index

(BMI), weight, and free androgen index, but ultimately the majority of the studies were deemed low quality due to high risk of bias and high heterogeneity [12]. It was unclear whether glucose tolerance improved with lifestyle interventions. Ultimately, several challenges such as adherence to prescribed regimens, high subject drop-out rates, and the labor-intensive, multidisciplinary nature of lifestyle intervention programs make it difficult to translate regimens to clinical practice [13–15].

Weight-loss medications have been investigated as a means of weight loss prior to ovulation induction in women with PCOS; however, concerns remain about potential fetal harms. Certain antiobesity drugs have known teratogenicity, such as topiramate, while others have unknown effects in pregnancy [16]. Thus, international guidelines emphasize that pregnancy should be avoided while taking these medications. If weight-loss medications are used for the purposes of improving fertility, they should be considered experimental therapy as the risk–benefit ratios are currently uncertain [1]. Orlistat is one Food and Drug Administration–approved over-the-counter weight loss drug which appears to be relatively safe [16]. It inhibits intestinal lipase which reduces fat absorption. Side effects include steatorrhea, flatulence, and rarely hepatic injury [14]. Supplementation of fat-soluble vitamins, such as vitamin D, should be performed while utilizing this drug, and its use should be immediately stopped with pregnancy.

Studies which combine multiple tools to achieve weight loss including caloric restriction, brief counseling sessions, medications, and increased physical activity have been shown to have additive effects on weight loss [17]. When adapted as a 16-week preconception intervention in overweight or obese women with PCOS, subjects lost between 6% and 7% weight from baseline and had higher ovulation rates than the group of women who maintained weight over this time period (Fig. 7.1) [11].

Bariatric surgery is another approach to weight loss that is increasingly performed on women with PCOS. Typically performed laparoscopically, the most common procedures are Roux-en-Y gastric bypass, adjustable gastric banding, and sleeve gastrectomy [18]. Candidates include patients with a BMI greater than 35 kg/m^2 who have attempted lifestyle changes for at least one year without success [19]. A growing body of evidence demonstrates the benefits of bariatric surgery on overall health among morbidly obese PCOS patients. Several studies, including a large 2021 metaanalysis, have demonstrated improvements in sustained weight loss, restoration of menstrual cycles, decreased circulating androgens, reduction of hirsutism, hypertension, diabetes, and even depression [20–25]. A 2020 cohort study investigated fertility rates following bariatric surgery among women with PCOS. It found pregnancy rates of 95.2% in women seeking fertility and live birth rates of 81%, compared with pregnancy rates and live birth rates of 76.9% and 69.2%, respectively, among controls [26].

Further research is needed with high-quality prospective randomized trials to fully determine the effect of bariatric surgery on live birth rates in morbidly obese women. It should be noted that there are concerns of adverse fetal effects including shorter gestation, small for gestational age infants and potentially increased infant mortality in pregnancies following bariatric surgery [27]. Ultimately, international guidelines recommend avoidance of pregnancy with appropriate contraception during periods of rapid weight loss and for at least 1 year following bariatric surgery [1].

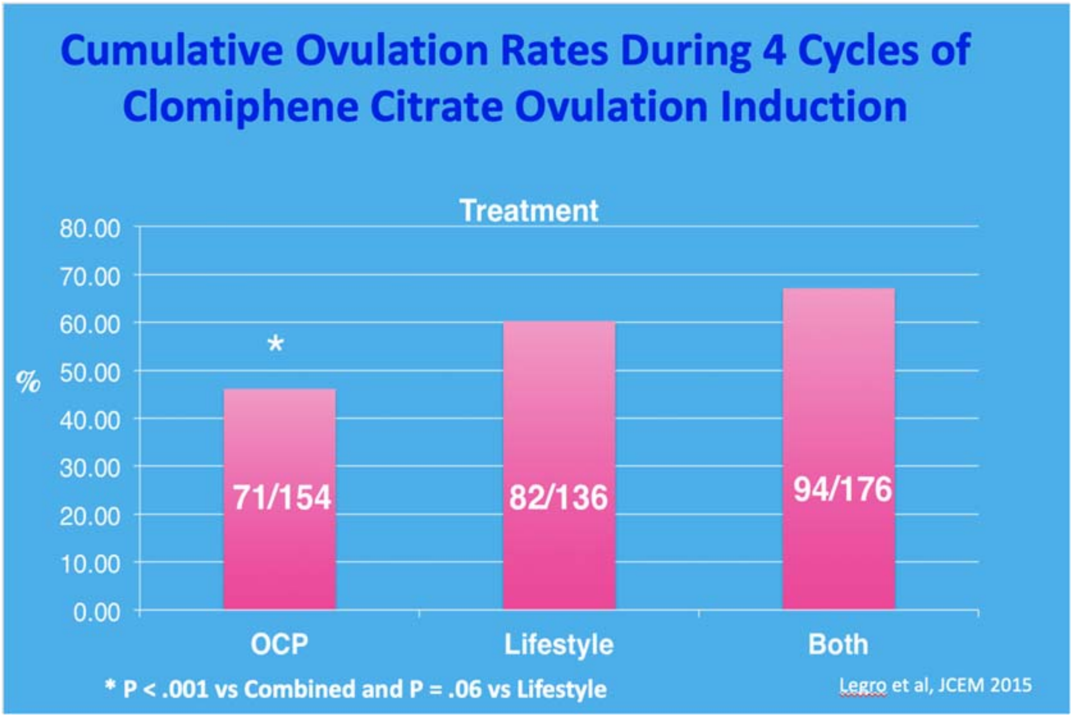

FIGURE 7.1 Randomized controlled trial comparing ovulation rates with clomiphene citrate following 16-week preconception interventions.

Pharmacologic treatment strategies

Metformin

Women with PCOS are at a significantly increased risk of glucose intolerance and type 2 diabetes mellitus compared to their counterparts without PCOS. Metformin is an antihyperglycemic agent which lowers basal and postprandial plasma glucose through several mechanisms. It decreases hepatic glucose production, reduces intestinal absorption of glucose, and improves peripheral glucose uptake and utilization to increase insulin sensitivity [28]. In addition to lifestyle interventions, the use of metformin is recommended by the international evidence-based guideline for PCOS, particularly in patients who are obese or who have other risk factors for impaired glucose tolerance [1]. Similarly, the Endocrine Society has recommended the use of metformin in women with PCOS and type II diabetes or impaired glucose tolerance who have failed lifestyle modification alone [29].

There are several metabolic benefits of metformin use in patients with PCOS. As a combination therapy along with diet and exercise, metformin modestly aids in weight loss. A recent metaanalysis of 12 randomized controlled trials (RCTs) found that among overweight women with PCOS, metformin resulted in a statistically significant reduction in BMI with a weighted mean difference of $-1.25\ \text{kg/m}^2$ compared to control subjects [30]. Similarly, the study found a significant reduction in waist diameter. Other studies have reported similar findings [31–33]. Metformin may also aid in preventing progression to diabetes mellitus. Among women with PCOS, metformin use results in maintenance or improvement in glucose tolerance [34]. It should be noted there are currently no large studies demonstrating protection against type II diabetes among women with PCOS, and a Cochrane review evaluating metformin versus oral contraceptive pill (OCP) use in PCOS patients reported insufficient data to make conclusions on the subject [35].

Metformin also appears to work in multiple ways to improve ovulation, fertility, and the functioning of the hypothalamic–pituitary–ovarian (HPO) axis. By reducing levels of circulating insulin acting on the ovaries, metformin indirectly acts to lower androgen production by theca cells as well as reducing the premature arrest of follicle growth [36,37]. Metformin is also thought to directly act on the ovary itself to lower androgen production, though the specific mechanisms are not yet fully elucidated [38]. Overall, in addition to reducing androgens and increasing ovulation, metformin improves menstrual frequency [38].

Metformin has been extensively studied for use as an ovulation induction agent. When compared to placebo, metformin is associated with increased ovulation rates and live birth, as detailed in a 2019 Cochrane review [39]. That said, metformin is inferior to clomiphene citrate and letrozole as a monotherapy, and its use as a first-line therapy is discouraged [1,31,40,41] (Fig. 7.2). There does appear to be a role for metformin as adjuvant treatment. In clomiphene-resistant PCOS patients, metformin can be combined with clomiphene to improve ovulation and clinical pregnancy rates, though it is unclear whether this improves live birth rates [1,31,40,41]. Thus, in clomiphene-resistant patients, the international PCOS guidelines recommend considering gonadotropins when available, with or without metformin, over clomiphene and metformin assuming proper monitoring with strict cancellation guidelines to prevent complications [1]. RCTs have suggested that metformin with clomiphene, or metformin as a pretreatment prior to ovulation induction, may specifically benefit obese women with PCOS [31,42]. However, the 2019 Cochrane review was unable to support these findings [39].

Metformin when used in conjunction with IVF reduces the risk of ovarian hyperstimulation syndrome (OHSS) in patients with PCOS [1,43,44]. This is true with both GnRH agonist and antagonist protocols [1]. However, its use with IVF remains controversial as it appears it may lower the number of retrieved oocytes and live birth rates, though studies are mixed [43–46]. Regardless of method of ovulation induction, metformin does not seem to affect the rate of miscarriage [40,47]. Similarly, studies have not found differences in pregnancy complications such as gestational diabetes or hypertension/preeclampsia with metformin use in pregnancy [47,48].

Adverse effects are predominately GI disturbances and are dose-dependent and self-limiting. These include abdominal discomfort, diarrhea, flatulence, nausea, and vomiting which may lead to anorexia [40]. Lactic acidosis is a rare complication which is almost exclusively found in patients with renal dysfunction [40,49]. Patients can be easily screened for the possibility of this complication with serum renal function tests. Appropriate dosing can minimize the risks of these side effects. Typical dosage is 1500–2500 mg per day in divided doses (two to three times daily); starting with small doses and gradually up-titrating minimizes side effects [4].

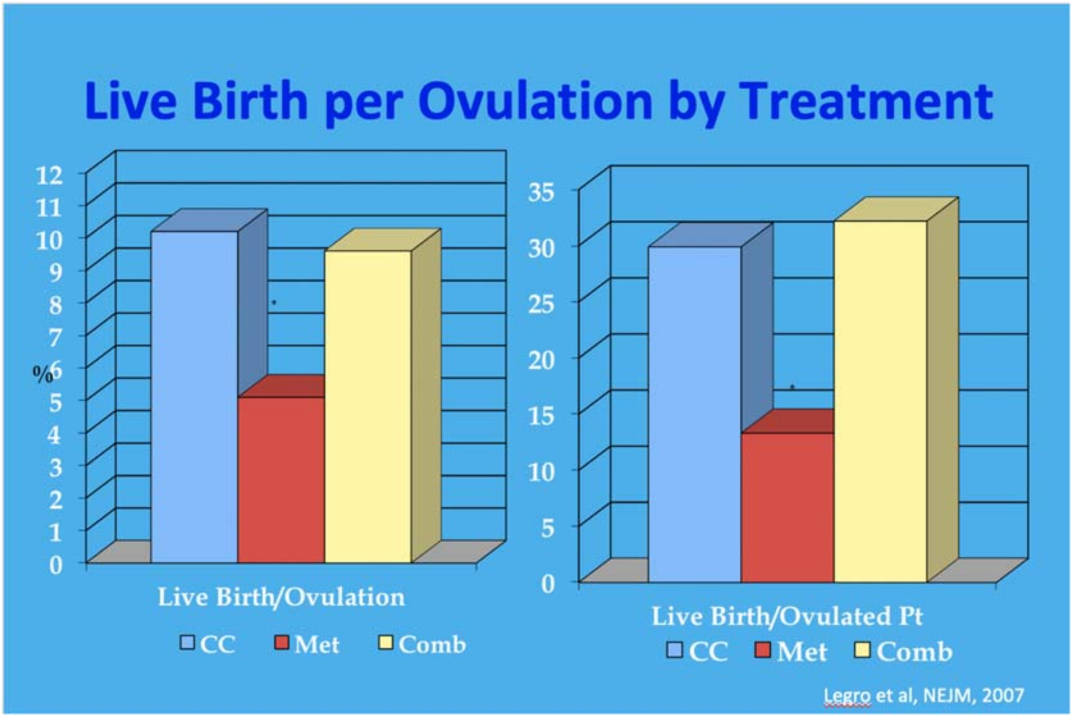

FIGURE 7.2 Metformin alone led to fewer live births per ovulation than clomiphene citrate or combined metformin/clomiphene citrate.

Letrozole

Letrozole is a member of the aromatase inhibitor family and is now considered a first-line treatment for ovulation induction in women with PCOS [1]. Originally developed for the treatment of hormone-sensitive breast cancers, this class of drugs blocks the aromatization step of estrogen synthesis, resulting in lower levels of circulating estrogens. In women with PCOS, particularly those with obesity and large stores of peripheral adipose tissue, aromatase inhibitors reduce the peripheral conversion of androgens to estrone [50]. It is thought that this reduction in estrone and other circulating estrogens restores HPO axis function by reducing inappropriate negative feedback resulting from excess estrogen. This leads to increased secretion of follicle-stimulating hormone (FSH) from the pituitary gland and follicular development [51]. One advantage of letrozole compared with clomiphene citrate is that letrozole does not affect endometrial estrogen receptors and thus does not reduce endometrial thickness or affect cervical mucous production [52]. Questions were initially raised regarding the possibility of teratogenic effects of letrozole when used for ovulation induction; however, increased risks of fetal anomalies have not been demonstrated in the literature [7]. In fact, rates of birth defects with letrozole use appear to be less than 5%, on par with clomiphene rates and the expected population rates in subfertile women undergoing ovulation induction or IVF [16,53]. This makes sense logically, as the short half-life of letrozole means it should be cleared from circulation before implantation occurs [54].

Compared with clomiphene, an established first-line treatment for anovulatory women with PCOS, letrozole has been shown to increase ovulation and live birth rates (Fig. 7.3) [55]. A 2018 Cochrane review involving 42 RCTs and 7935 women with PCOS further supports these findings [54]. The review found that letrozole led to higher pregnancy and live birth rates compared to clomiphene. Furthermore, high-quality evidence demonstrated no statistically significant difference in rates of OHSS, miscarriage rates, or multiple pregnancy rates [54], though other reviews have demonstrated lower multiple pregnancy rates [41]. One theory as to why letrozole leads to higher birth rates than clomiphene is that, compared with other ovulation induction methods, letrozole leads to decreased midluteal estradiol levels and increased midluteal progesterone levels, creating an environment for implantation more similar to that of natural conception [16,55].

Side effects of letrozole include headaches and cramping, with increased fatigue and dizziness compared to clomiphene but reduced hot flashes [16]. Typical administration and dosing for ovulation induction is 2.5–5 mg for 5 days (with a maximum of 7.5 mg daily), beginning in the early follicular phase (cycle days 2–5) [51]. It should be noted that anovulatory PCOS patients, however, perpetually exist in the follicular phase [16]. Thus, some clinicians will induce a progesterone withdrawal bleed prior to ovulation induction; this does not appear

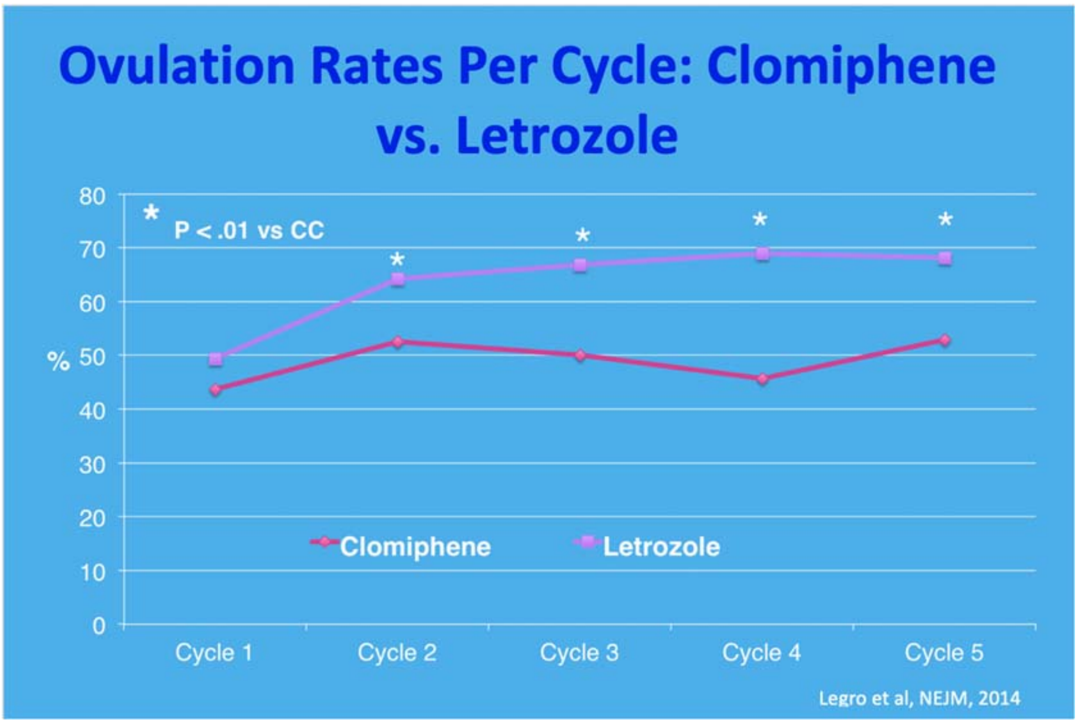

FIGURE 7.3 Ovulation rates in randomized controlled trial comparing letrozole and clomiphene citrate.

to improve pregnancy rates, and preconception progestin use has been associated with lower pregnancy rates in multiple studies, including after use of oral agents and IVF [56,57]. It also prolongs ovulation induction therapy and introduces additional cost and concern if there is no or insufficient withdrawal bleed. Ovulation induction regimens eliminating progestin withdrawal and utilizing ultrasound and/or serum assays to detect follicular development, such as the "stair-step" protocol, have become popular in clinical practice [58]. With the stair-step protocol, a transvaginal ultrasound is performed 5–7 days after administration of the lowest dose of a given agent (e.g., 5–7 days after the fifth dose of letrozole 2.5 mg daily) [59,60]. If there are no follicles greater than 10 mm in size, the patient is immediately administered a 5-day course of the "step-up" dose (e.g., letrozole 5 mg daily). Ultrasonography is repeated 1 week after the initial ultrasound and follicles are again observed for response. The cycle is continued until development of follicles greater than 10 mm in size or maximum dosing is reached.

Selective estrogen receptor modulators

Selective estrogen receptor modulators (SERMs), like aromatase inhibitors, were originally developed for the treatment of hormone-sensitive breast cancers. Clomiphene citrate, the most studied SERM for use in ovulation induction, is a racemic mixture of two stereoisomers: en-clomiphene and zu-clomiphene [51]. Of the two, en-clomiphene is the more biologically active compound. Clomiphene binds to and inhibits estrogen receptors at the hypothalamus [61]. This leads to release from the negative feedback exerted by excess circulating estrogens, similar to letrozole, and increased production of gonadotropins by the pituitary gland. Clomiphene can act as an estrogen-receptor antagonist elsewhere in the body as well. As mentioned above, this may lead to reduced endometrial development and production of less favorable cervical mucous, though the clinical significance of these effects is uncertain [52]. Clomiphene exerts estrogen-agonist properties in addition to the above antagonistic effects [62]. The half-life of clomiphene is longer than letrozole at 5–7 days, and the less active stereoisomer zu-clomiphene can persist for much longer still [61].

Currently, clomiphene is still considered by some to be a first-line treatment for ovulation induction; however, the international guidelines for the assessment and management of PCOS favor letrozole, where available, as initial treatment [1]. Side effects include hot flashes and visual disturbances, though these are usually self-limiting [51]. It should be noted that in rare cases, these visual changes, which include blurry vision, scotomata, and double vision, are irreversible [63–65]. Thus, the development of sudden visual changes should prompt discontinuation of therapy. Rates of multiple pregnancies are 5%–8%, and these are usually twin gestations [16]. Typical starting dosing for clomiphene is 50 mg daily for 5 days; this can be increased to 150 mg daily if no ovulation occurs [7]. Similar to letrozole, this would occur in the early follicular phase if following a menstrual cycle. Otherwise, in anovulatory women, it is given with or without a preceding progesterone withdrawal bleed. Generally, clomiphene use is limited to 12 cycles due to concerns raised by a 1994 cohort study associating prolonged clomiphene use with borderline or invasive ovarian tumors [66]. However, this has not been clearly demonstrated in subsequent studies, and a Cochrane found no evidence that clomiphene increases ovarian cancer risk [67]. Other selective estrogen receptor modulators including tamoxifen have been studied for their role in ovulation induction with similar outcomes to clomiphene, though their use in clinical practice is not widespread [41,68–70].

Gonadotropins

Gonadotropins should be considered a second-line treatment in women with PCOS and may be used for ovulation induction when first-line agents have failed [1]. FSH, in urinary-derived or recombinant form, is often preferred for women with PCOS, though human menopausal gonadotropin (HMG), a mixture of luteinizing hormone (LH), FSH, and small amounts of HCG, is also used [16]. A 2019 Cochrane review was unable to determine whether differences in outcomes existed between FSH and HMG use; thus it was recommended to consider costs and convenience when choosing one over the other [71]. Ultimately, gonadotropin use for ovulation induction in expert hands appears to result in higher live birth rates and ongoing pregnancy than clomiphene alone or clomiphene combined with metformin [70,72,73]. At present, it is unclear whether a difference in outcomes exists between letrozole and gonadotropins for ovulation induction in women with PCOS [7]. The risks associated with gonadotropin use in women with PCOS center around the increased risk of OHSS in this patient population as well as the increased risk of multiple pregnancies owing to the high number of antral follicles [16]. Thus, when gonadotropins are utilized, ultrasound monitoring is mandatory [1].

The typical starting dosage is 37.5–75 IU daily for two weeks; if follicle development occurs, then this dose is maintained until a lead follicle size of 18–20 mm is reached, at which point HCG or LH is administered to trigger follicular maturation [51]. If lack of follicle development necessitates increased dosage, increases should be small and only as-needed. Use of this low-dose step-up regimen is generally considered best practice to limit the risk of OHSS and multiple pregnancies in PCOS patients [74]. Dose reductions for over-responders should be a consideration to further minimize risks. The step-down protocol, which begins with a larger dose which is reduced over time, is inferior to the step-up approach [75].

Considerations for administration of ovulation induction agents

Prior to initiation of ovulation induction, a urine pregnancy test should be performed to avoid exposure of an early pregnancy to induction agents. As occasional spontaneous ovulation may occur, it is worth considering obtaining serum progesterone and a baseline ultrasound before beginning an induction cycle [16]. The necessity of ultrasound monitoring of cycles remains a topic of debate, though monitoring does allow for utilization of step-up protocols to sequentially increase dosing in the absence of ovulation [16,19,59,76].

While the optimal number of treatment cycles for clomiphene and letrozole is not fully established, patients are considered to have clomiphene citrate failure if pregnancy is not achieved after six cycles [7]. RCTs of first-line therapy with ovulation induction agents including metformin, clomiphene, and letrozole show stable pregnancy rates per cycle up to six cycles [31,55]. It should be noted clomiphene resistance is diagnosed if ovulation does not occur with administration of a clomiphene dose of 150 mg.

In vitro fertilization/Intracytoplasmic sperm injection

The international guidelines on the evaluation and treatment of PCOS concluded that unless there is an absolute indication for IVF, women with PCOS and anovulation can be offered IVF, with or without intracytoplasmic sperm injection (ICSI), as a third-line treatment when first- or second-line therapies have been unsuccessful [1,7]. At the time of publication of these guidelines, no RCTs were identified that compared IVF with ovulation induction treatments or placebos in women with PCOS [7]. It appears no RCTs have been published since. Further research comparing IVF and ovulation induction treatments in this patient population would be useful to further bolster recommendations on treatment strategies.

As with gonadotropin use for ovulation induction, use of IVF in patients with PCOS is associated with a higher risk of developing OHSS than in those without PCOS. It appears this risk is reduced, though still present, when using a GnRH antagonist protocol compared with an agonist protocol [51,77]. Compared with a GnRH long agonist protocol, an antagonist protocol leads to lower gonadotropin dose exposure as well as a shorter duration of stimulation, and its use is preferred when utilizing IVF in this patient population [1]. Furthermore, an antagonist protocol allows for use of a GnRH agonist trigger, avoiding HCG administration to further minimize the risk of OHSS. This is particularly useful with "freeze-all" cycles as pregnancy rates appear to be lower when embryo transfer is performed in the same cycle [78,79]. With regards to the risk of multiple pregnancies, this can be reduced with the use of single embryo transfer. Another consideration is whether to employ fresh or frozen embryo transfer (FET). A large RCT from China evaluated fresh transfer versus FET among PCOS women undergoing a GnRH antagonist protocol [80]. The study found a statistically significant increased rate of live birth after the first transfer with FET compared with fresh transfer (49.3% versus 42.0%). Rates of pregnancy loss were lower in the FET group than fresh transfer group (22% versus 32.7%, respectively), as were rates of OHSS (1.3% versus 7.1%, respectively). Interestingly, a higher rate of preeclampsia was noted with FET (4.4% versus 1.4%).

In vitro maturation

In vitro maturation, or IVM, has been advocated as an alternative to traditional IVF for patients at a high risk of OHSS, including PCOS patients. The purpose of IVM is to collect immature oocytes and mature them outside of the body in an attempt to avoid, or at least limit, controlled ovarian stimulation. In contrast to conventional IVF, where oocytes are collected after gonadotropin-induced maturation of a cohort of follicles, in IVM, cumulus-oocyte complexes (COCs) are collected from antral follicles and further developed once harvested [1,81,82]. IVM in the strictest sense involves maturation of these COCs from the prophase I (germinal vesicle) stage up to metaphase II (MII) after they have been harvested from patients not exposed to a preovulatory trigger [82,83]. However, in clinical practice, patients are often treated with one of several precollection protocols in which exogenous FSH, HCG, or both are administered with the goal optimizing the proportion of oocytes able to successfully mature in vitro [82–84]. The use of these protocols, however, leads to substantially less exposure to exogenous gonadotropins than conventional IVF. Thus, in patients with PCOS and other predicted high responders, use of IVM over IVF may dramatically lower the risk of OHSS.

Multiple studies have evaluated IVM against IVF in PCOS patients and those with PCO-like features [85–87]. A recent large RCT was performed to assess whether IVM was noninferior to conventional IVF [88]. This study randomized 546 women with a high antral follicle count (at least 24 follicles among both ovaries) to receive IVM or IVF. Of the 273 women who received one cycle of IVM, live birth after the first embryo transfer occurred in 96 patients (35.2%) compared with 118 (43.2%) of the 273 who received one cycle of conventional IVF. At 12 months, cumulative ongoing pregnancy rates were 44% in the IVM group versus 62.6% in the IVF group [88]. However, no cases of OHSS occurred in the IVM group, compared with two cases in the IVF group [88]. Overall, while this study was unable to demonstrate noninferiority of IVM compared to IVF, the results indicate that IVM may be a legitimate alternative to IVF in some patients with a high AFC.

In summary, the advantages of IVM over IVF include decreased need for medications, reduced monitoring, and virtual elimination of the risk of OHSS due to the need for little to no gonadotropin use [89]. However, with current protocols, IVM fertility rates lag behind IVF, and more research is needed to continue developing and refining this alternative "mild ART" strategy.

Inositol

Inositols, such as myo-inositol (MI) and D-chiro-inositol (DCI), may be of clinical interest to practitioners and PCOS patients as a "pretreatment" before IVF or for use in ovulation induction. MI, through its role as an intracellular second messenger, is involved with regulation of hormones such as insulin, FSH, and TSH [90]. Several formulations of inositol-containing products have been introduced to the market for treatment of PCOS. An initial study of DCI showing a large treatment effect at inducing ovulation in women with PCOS [91] was never replicated in subsequent studies, but nevertheless triggered an ongoing interest in inositols as a treatment for PCOS. These products vary widely in composition and may include MI, DCI, both MI and DCI, and various other additives [92]. Roseff et al. performed an exhaustive sampling of available products and found that a dizzying array of combinations exist; perhaps, more often than not these formulations were not guided by any scientific basis or therapeutic rationale. Ratios of MI/DCI varied wildly, from 0.4:1 to 104:1, while additives were extremely varied and included micro- and macroelements, vitamins, and several other molecules (even including N-acetyl cysteine). For a variety of reasons, including optimization of blastocyst quality, attenuation of intestinal absorption of MI, and the revelation that DCI is an aromatase inhibitor, Roseff et al. argue that the optimal MI/DCI ratio for the treatment of anovulation in PCOS patients is 40:1. This ratio is supported by animal models and clinical trials [93,94]. Unfortunately, most products on the market utilize seemingly arbitrary combinations of these molecules along with questionable additives. Importantly, a Cochrane review was performed in 2018 to evaluate the efficacy of inositol in treating subfertility in PCOS patients [95]. The review included 13 trials with 1472 subjects with PCOS and subfertility; 11 studies examined MI as a pretreatment to IVF, while two studies investigated its use in ovulation induction. Ultimately, due to poor-quality evidence, the review was unable to determine whether the usage of MI improved reproductive outcomes such as clinical pregnancy rate or live birth rate when used as a pretreatment for IVF. Furthermore, the review was unable to assess its performance as an ovulation induction agent as no pooled evidence was available. Inositol should be considered an experimental therapy in PCOS according to the international evidence-based guideline for the assessment and management of polycystic ovary syndrome [1].

Laparoscopic ovarian drilling

Surgical treatment of anovulation can be used as a second-line therapy for those women who do not respond to clomiphene or letrozole and who cannot or do not wish to use gonadotropins. Ovarian wedge resection, first described in the mid-to-late 1930s by Stein and Leventhal, was an initial treatment of choice for PCOS-related infertility [96]. However, due to high rates of pelvic adhesions, the approach was all but discarded with the advent of pharmacologic approaches to ovulation induction.

Laparoscopic ovarian electrocautery or laser vaporization, often known as ovarian “drilling,” is a modern surgical approach that attenuates some of the drawbacks of the classic wedge resection. Initially described by Gjönnaess in 1984, the procedure allows for a minimally invasive approach to reducing ovarian tissue. Multiple surgical techniques have been described in the literature, and no standardized method has been established. Generally, multiple perforations are created at each ovary using either diathermy or various forms of laser vaporization (such as argon, CO_2, or neodymium-doped yttrium aluminum garnet crystal lasers) [97]. These perforations penetrate the ovarian surface to involve the stroma. This local destruction of ovarian tissue is believed to reduce ovarian androgens, lowering both intraovarian androgen concentrations and the peripheral conversion of androgens to estrogens [98]. This decline in androgen concentrations, along with decreased inhibin levels, are thought to lead to a rise in FSH levels and fall in LH levels [97–100]. Overall, it appears that ovulation occurs in 30%–90% of women after laparoscopic ovarian drilling, while pregnancy rates range from 13% to 88% [98]. A Cochrane review indicates that compared with medical ovulation induction with agents such as gonadotropins, ovarian drilling may decrease live birth slightly, though there may not be a difference in the chances of clinical pregnancy [97]. The analysis also suggests decreased multiple pregnancy rates compared with medical ovulation induction, while the relative risk of miscarriage between the two remained unclear. There may be a decreased risk of OHSS; however, the infrequency of the complication limits conclusions.

In addition to the risks associated with general anesthesia, laparoscopic ovarian drilling carries the risk of postoperative adhesion formation, which in theory could cause mechanical infertility [101]. While adhesion formation after ovarian wedge resection was near universal, rates of adhesion formation after ovarian drilling are hard to know due to a dearth of large second-look laparoscopic studies. Studies with small sample sizes have found adhesion formation in anywhere from 20% to 100% of patients; however, a randomized trial of 96 women performed by Mercorio et al. found adhesions in 60% of subjects, and 46% of ovaries treated with ovarian drilling [102]. Utilizing second-look minilaparoscopy, the researchers found that number of punctures did not seem to affect the likelihood of adhesion formation; however, the left ovary was more likely to develop severe adhesions than the right.

Another potential risk of ovarian drilling is the possibility of iatrogenic premature ovarian insufficiency (POI). As measured by postoperative antimullerian hormone (AMH) concentrations, ovarian drilling would appear to lower ovarian reserve. Indeed, the procedure has been demonstrated to consistently, dramatically lower concentrations of this marker [103–105]. However, this must be interpreted with caution as it may in fact reflect a normalization of high preoperative AMH levels! To date, there has been no evidence of increased rates of POI, early menopause, or other clinical manifestations of diminished ovarian reserve.

Acupuncture, herbal remedies, and traditional Chinese medicine

Treatment modalities beyond the scope of western medicine, including acupuncture and herbal therapies, have been investigated for their potential use in PCOS fertility treatment. Chinese herbal medicine (CHM) has been applied to PCOS for decades. A June 2021 Cochrane review investigating the effectiveness of CHM included eight RCTs and evaluated CHM against clomiphene, follicle aspiration, ovarian drilling, and placebo [106]. Ultimately, the reviewers found that the overall quality of the evidence was low with no studies reporting on live birth rate. There was insufficient evidence to support its use in women with PCOS and infertility, with no consistent evidence that it affected fertility outcomes. The authors noted limited, low-certainty evidence that CHM combined with clomiphene might improve fertility rates; however, there were little data on adverse events to determine whether CHM is safe for this purpose. Further studies including high-powered RCTs are needed to determine whether CHM is safe and effective for the purpose of treating infertility in patients with PCOS.

Likewise, acupuncture has been investigated for its potential role in infertility treatment among PCOS patients. Unlike CHM, high-quality studies have been conducted, but have not noted a benefit to acupuncture, alone or in combination with clomiphene, for women with PCOS [107]. Reviews have also found little evidence to evaluate the safety and efficacy of acupuncture for this purpose, and it is unclear whether reproductive outcomes are affected [108,109]. A 2019 Cochrane review evaluated acupuncture against sham acupuncture, relaxation, clomiphene, exercise, or no treatment [110]. The review included eight RCTs involving 1546 women and found no evidence of statistically significant differences with respect to live birth rate, multiple pregnancy rate, ovulation rate, clinical pregnancies, or rates of miscarriage. The review was hampered by the limited number of RCTs and evidence deemed very low quality, with wide confidence intervals. Similar to CHM, increased rigorous studies are needed to fully evaluate acupuncture as a treatment modality for PCOS.

Conclusion

In this chapter, we have reviewed evaluation and treatment of infertility for patients with PCOS. From preconception health optimization to pharmacologic modalities, IVF, and surgical options, there are many facets of the overall care of the infertile PCOS patient to consider when developing an individualized treatment plan. It is our sincere hope this chapter, guided by the latest international guidelines, serves as a framework when approaching this complex syndrome.

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CHAPTER

8

The GnRH antagonist protocol

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Introduction

In 1984, gonadotrophin-releasing hormone (GnRH) agonists were introduced in ovarian stimulation for in vitro fertilization (IVF), as a means to increase the number of oocytes retrieved [1] by suppressing premature luteinizing hormone (LH) surge. Premature LH surge was a frequent problem in ovarian stimulation in the preanalog era, leading to cycle cancellation in approximately 20% of cases [2]. Although by the mid-80s GnRH antagonists were also available and capable of suppressing LH surge [3], due to the presence of allergic reactions associated with their use [4], GnRH agonists dominated ovarian stimulation until the early 2000s.

At that time point, the third generation of GnRH antagonists, free of allergic reactions, was introduced in ovarian stimulation. This was eagerly awaited since the use of GnRH agonists was characterized by several disadvantages, such as an initial unwelcome increase in the secretion of gonadotrophins (flare-up effect), a lengthy treatment period until hypophysis desensitization occurred, and a frequent occurrence of ovarian hyperstimulation syndrome (OHSS) [5].

By competitively binding to the pituitary GnRH receptor [6], GnRH antagonists result in an immediate, reversible [7], and dose-dependent inhibition of gonadotrophin release [8] without the flare-up effect observed when using GnRH agonists. Discontinuation of GnRH antagonist administration leads to a rapid recovery of the pituitary−gonadal axis.

GnRH antagonists simplified ovarian stimulation and led to comparable live birth rates to GnRH agonists (RR: 0.91, 95% CI: 0.79−1.04) [9] in the general population as well as in poor responders (RR: 0.89, 95% CI: 0.56−1.41) [9]. In addition, they were associated with a substantial reduction in the incidence of OHSS (RR: 0.63, 95% CI: 0.50−0.81) in patients undergoing assisted reproductive technologies (ARTs) [9]. The ovarian stimulation guideline published

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by the European Society of Human Reproduction and Embryology (ESHRE) in 2020 recommends GnRH antagonists over GnRH agonists, given their comparable efficacy and higher safety in the general IVF population [10].

Since the introduction of GnRH antagonists, several antagonist protocols have been described [11], reflecting the active research for optimizing ovarian stimulation using this advantageous treatment modality.

GnRH antagonist protocols

The majority of GnRH antagonist cycles performed today follow the daily dose scheme [12], although in the past, a single-dose antagonist scheme has been used [13]. Two GnRH antagonist analogs are available for ovarian stimulation in ART: Cetrorelix and Ganirelix, the administration of which can be implemented through either a fixed or a flexible protocol. Administration of GnRH antagonists in either the fixed or the flexible protocol is continued until the day of triggering final oocyte maturation. Both morning and evening administration is feasible, although the morning administration allows adequate LH suppression on the day of triggering, which usually occurs in the evening. Based on dose-finding studies, the optimal dose for the daily dose GnRH antagonist scheme is 0.25 mg for both Cetrorelix and Ganirelix [14,15].

Cetrorelix versus ganirelix

Both Cetrorelix and Ganirelix appear to be safe and equipotent in the suppression of FSH and LH, which is fully effective within four−eight hours after subcutaneous administration [16,17]. The incidence of premature LH surge in patients undergoing ovarian stimulation was similar between the two antagonistic analogs in phase III trials; however, no conclusive direct comparison between Ganirelix and Cetrorelix for ovarian stimulation for IVF has so far been performed [18].

GnRH antagonist fixed protocol

In the fixed protocol, antagonist is administered on a certain day of stimulation, on which the risk of LH rise is thought to be present in most patients. In the introductory trials, GnRH antagonist was started on Day 6 of follicle-stimulating hormone (FSH) administration [19], while in subsequent studies, this was also performed on Day 5 of stimulation [20].

GnRH antagonist flexible protocol

In the flexible GnRH antagonist protocol, antagonist is administered only after certain sonographic and/or endocrine criteria, indicating a risk for LH rise, are met. A large variation in the criteria used in the flexible protocol is present in the literature (Table 8.1).

TABLE 8.1 Criteria used for GnRH antagonist initiation in the flexible protocol.

Ultrasound criteria	Ultrasound and/or hormonal criteria
Follicle $\geq$12 mm [21,22]	Follicle $\geq$13 mm or E2 400 pg/mL [23]
Follicle $\geq$13 mm [24–26]	Follicle $\geq$14 mm or E2 $\geq$ 300 pg/mL [27]
Follicle $\geq$14 mm [28–46]	Follicle $\geq$14 mm and E2 > 600 pg/mL [47,48]
Two follicles $\geq$14 mm [49]	Follicle >12 mm and/or LH > 10 IU/L and/or E2 > 150 pg/mL [50]
Follicle $\geq$15 mm [51]	Follicle >12 mm and/or LH > 10 IU/L and/or E2 > 600 pg/mL [52]
Follicle $\geq$12–14 mm [53–55]	Follicle $\geq$14 mm or LH $\geq$ 10 IU/L or E2 level $\geq$200 pg/mL [53]

Fixed versus flexible GnRH antagonist protocol

Apparently, fixed GnRH antagonist initiation is a simpler protocol that requires less intense monitoring compared to the flexible one. On the other hand, flexible GnRH antagonist administration might avoid unnecessary GnRH antagonist administration, in those patients who are not at risk for an LH rise on Day 5 or Day 6 of stimulation, due to the absence of follicular development.

A metaanalysis of four randomized controlled trials (RCTs) [24,50,56,57] comparing flexible versus fixed GnRH antagonist administration did not show a significant difference in clinical pregnancy rate between the two protocols (OR: 0.85, 95% CI: 0.65–1.11) [58]. Following that metaanalysis, three additional RCTs comparing the fixed versus the flexible GnRH antagonist protocol have been performed in normo-ovulatory patients [59], in patients with polycystic ovary syndrome (PCOS) [28] and in patients with predicted high ovarian response [52]. More specifically, similar pregnancy rates were observed in 213 normo-ovulatory patients randomized to a fixed Day 6 or a flexible protocol (antagonist initiation in the presence of a follicle $\geq$14 mm, or E2 level $\geq$200 pg/mL per dominant follicle or LH $\geq$ 10 IU/L) (35.8% vs. 29.2% respectively, $P = .34$) [59]. In addition, ongoing pregnancy rates were not significantly different in 100 PCOS patients randomized to a fixed Day 6 or a flexible protocol (antagonist initiation by a follicle of 14 mm) (36% vs. 20%, respectively, $P = .12$) [28]. Similarly, in 201 patients with predicted high ovarian response, randomized to a fixed Day 5 or a flexible protocol (antagonist initiation when LH was >10 IU/L, and/or a follicle with mean diameter >12 mm was present, and/or serum E2 was >600 pg/mL), no difference in the probability of ongoing pregnancy was observed (43.5% vs. 36.2%, respectively, $P = .47$) [52].

Currently, no evidence is present to suggest a statistically significant difference in the probability of pregnancy between the fixed and the flexible protocol. It is important to note that in the comparative studies on fixed versus flexible GnRH-antagonist protocol, only in a fraction of the patients randomized to the flexible approach, GnRH-antagonist was administered later compared to the fixed protocol. This was due to the fact that the criteria used in the flexible protocol had already been met in a proportion of patients by Day 5 or Day 6 of stimulation. Accordingly, the true effect of delayed GnRH antagonist initiation has not been precisely determined in these trials.

Pretreatment with oral contraceptive pill in a GnRH antagonist cycle

Pretreatment with OCP in GnRH antagonist cycles has been used to improve the scheduling of an IVF center, by adjusting the onset of patients' menstruation [60], as well as to improve follicular synchrony by preventing endogenous FSH rise [61].

A systematic review and metaanalysis pooling results from six RCTs, including 1335 patients showed a decreased probability of live birth in patients pretreated with OCP prior to the initiation of gonadotrophin stimulation (OR 0.74, 95% CI: 0.58−0.95) [62]. According to the ovarian stimulation guideline published by ESHRE in 2020, OCP pretreatment (12−28 days) is not recommended in the GnRH antagonist protocol because of reduced efficacy [10]. However, it is still not clear whether the type of OCP pretreatment regarding the estrogen and progestogen components, the duration of OCP administration, or the starting day of gonadotrophin stimulation following OCP cessation [63] may alter the above recommendation [10].

Elevated progesterone at initiation of stimulation in a GnRH antagonist cycle

Elevated progesterone levels at menstruation are present in a proportion of patients undergoing IVF. The proportion of patients with progesterone levels >1.6 ng/mL on cycle Day 2 is reported to be 4.9% (95% CI 3.2−7.4) [64], while that with progesterone levels >1.5 ng/mL ranges from 6.2% (95% CI 4−9) [65] to 13.3% (95% CI 8−20) [66].

A recent metaanalysis combining three prospective cohort studies (1052 patients) reported that elevated progesterone level (>1.5−1.6 ng/mL) on cycle Day 2 prior to the initiation of stimulation is associated with a decreased probability of ongoing pregnancy in patients treated by gonadotrophins and GnRH antagonist for IVF (risk difference −15% 95% CI −23% to −7%) [66]. Thus, the evaluation of progesterone on Day 2 of the cycle prior to the initiation of stimulation appears to be justified. On the contrary, since progesterone levels >1.6 ng/mL are encountered only in 0.3% of patients (95% CI 0.01−1.15) on Day 3 of the cycle [67], their evaluation on that day appears to be unnecessary.

FSH dose in GnRH antagonist protocols

The optimal FSH starting dose in ovarian stimulation for IVF is usually selected based on patient's body mass index, age, ovarian reserve (as assessed by antral follicle count and/or anti-Mullerian hormone) [68], and previous response to stimulation. However, efforts have also been made to determine it objectively [69,70]. A starting dose of 150−200 IU is generally considered appropriate for a typical patient [10].

Initiation of FSH dose in GnRH antagonist protocols

In GnRH antagonist cycles, FSH stimulation can be started either on Day 2 or Day 3 of the cycle [71], without affecting the chance of pregnancy [72]. A later initiation of FSH stimulation on Day 5 has also been performed in the so-called "mild stimulation protocols" [29], the target of which is the avoidance of OHSS and decreased gonadotrophin consumption [73,74]. The fact, however, that OHSS can nowadays be eliminated with the replacement of human chorionic gonadotrophin (hCG) by GnRH agonist [75] and a lower probability of pregnancy associated with mild stimulation has tempered the initial enthusiasm regarding these protocols [76].

Administration of antagonist in the early follicular phase

Early initiation of GnRH antagonist has also been evaluated [77,78]. This results in an endocrine milieu that is more stable and closer to the normal cycle conditions, with lower levels of LH and E2 during the mid and late follicular phases [79].

Day 2 GnRH antagonist initiation did not improve the probability of ongoing pregnancy compared with Day 6 antagonist initiation in an RCT including 593 patients [80]. Similarly, the administration of GnRH antagonist for three days prior to the initiation of FSH stimulation did not improve the probability of pregnancy in three RCTs, including 69 [81], 53 [82], and 136 patients [83], treated by a fixed Day 6 GnRH [81] or a flexible [82,83] antagonist protocol.

Early antagonist initiation has also been examined in patients with elevated progesterone on Day 2 of the cycle, prior to FSH initiation. Administration of GnRH antagonist for three consecutive days has been shown to result in acceptable pregnancy rates compared to those achieved in patients with normal progesterone levels prior to gonadotrophin initiation, in a prospective cohort study including 484 patients (13.3% vs. 24.7%, respectively, $P = .16$) [65].

Delayed start GnRH antagonist protocol

In this protocol, ovarian stimulation is preceded by pretreatment with oral estradiol for 7–10 days in the late luteal phase, followed by seven days of GnRH antagonists. In this way, it is expected that the suppression of endogenous FSH, prior to the initiation of ovarian stimulation, will result in a more synchronous follicular growth. This protocol has been applied so far in poor responder patients and has been evaluated in two systematic reviews and metaanalyses [84,85]. The latest one, published in 2021 [85], pooled the results of five RCTs, including 514 patients comparing the delayed start GnRH antagonist protocol with a flexible GnRH antagonist protocol. Delayed start GnRH antagonist protocol was associated with an increased probability of clinical pregnancy (RR = 2.30, 95% CI: 1.38–3.82, $P = .001$). No data on the probability of ongoing pregnancy or live birth have been provided by the included RCTs in that metaanalysis, while no difference was observed in the number of embryos obtained or transferred between the delayed GnRH antagonist protocol and the conventional protocol. Thus, more data are necessary to arrive at solid conclusions regarding the usefulness of this protocol in the treatment of poor responders.

Luteal GnRH antagonist administration

Luteal GnRH antagonist administration has been suggested as a means to decrease the incidence of OHSS or the severity of its manifestations. In patients at high risk for OHSS, administration of GnRH antagonist in the luteal phase has been attempted as a means to avoid the occurrence of the syndrome [86]. In a prospective cohort study, 105 patients at high risk for OHSS undergoing cryopreservation of all embryos received 0.25 mg of Cetrorelix from days 3–5 postoocyte retrieval or no additional medication. The incidence of moderate/severe OHSS was significantly lower in patients who received Cetrorelix in the luteal phase (18.03% vs. 37.14%, respectively; $P = .037$) [87]. Moreover, in an observational study including 21 patients at high risk of OHSS, in whom triggering of final oocyte maturation was performed with GnRH agonist, a second dose of GnRH agonist in combination with luteal GnRH antagonist was associated with OHSS elimination [88].

In patients with established severe early OHSS who underwent embryo transfer (n = 22), low-dose luteal GnRH antagonist administration resulted in similar live birth rates compared to high-risk patients who did not develop severe early OHSS and did not receive GnRH antagonist in the luteal phase (40.9 vs. 43.6%, respectively) [89]. None of the 22 patients with severe early OHSS required hospitalization following luteal antagonist administration. Similar clinical pregnancy rates have also been reported in a retrospective study including 94 patients at high risk for OHSS who received or not luteal antagonist administration (28.8% vs. 18.5%, respectively) [90]. Regressions of OHSS in patients who received GnRH antagonist in the luteal phase appear to be associated with a decrease in vascular endothelial growth factor (VEGF) levels [87,91].

Criteria for triggering final oocyte maturation

Three RCTs have so far examined the association between the criteria used for triggering final oocyte maturation and the probability of pregnancy in antagonist cycles [92–94]. In the largest one, 413 patients undergoing IVF, with recombinant FSH, in a Day 6 fixed antagonist protocol, were randomized to receive 10,000 IU of hCG either as soon as at least three follicles were ≥17 mm on ultrasound (early-hCG group, 208 patients) or two days later after this criterion was met (late-hCG group, 205 patients). Patients were assessed by ultrasound on Day 6 of stimulation and thereafter as necessary to ensure that hCG was injected on the first day that the patient had ≥3 follicles ≥17 mm (early-hCG group) or two days later (late-hCG group). For that purpose, a follicular growth of 2 mm per day was assumed. A significant decrease in the probability of ongoing pregnancy was observed in the late-hCG group (35.6% vs. 25.0%, respectively; $P = .03$). In a subsequent study, 125 patients undergoing IVF, with recombinant FSH, on a Day 6 fixed antagonist protocol, were randomized to receive 10,000 IU of hCG when at least three follicles reached a diameter of ≥17 mm, 24 or 48 h later. In that study, patients were reviewed daily from Day 9 onwards until the criteria for hCG were met. Pregnancy rates were not significantly different between the three groups compared (17.9%, 27%, 25.8%, respectively). The difference in the results between the two studies might be due to the fact that in the second study, some patients may had already reached the criteria for hCG administration before Day 9 of stimulation. In an additional RCT, 120 patients were randomized to receive hCG either as soon as three or more follicles greater than 16 mm were present (early-hCG group) or one day after these criteria were met (late-hCG group). No significant difference in the probability of ongoing pregnancy was observed between the two groups (34.6% vs. 40.7%, respectively; $P = .55$). It should be noted that the above studies were performed before the concept of the freeze-all approach was introduced in clinical practice, especially in patients at high risk for OHSS [75]. Thus, although still clinically relevant, the importance of the criteria for triggering final oocyte maturation should be reevaluated in this new context.

GnRH antagonists and OHSS

Although the incidence of severe OHSS is significantly decreased in GnRH antagonist as compared to GnRH agonist cycles (RR: 0.63, 95% CI: 0.51–0.79) [9], OHSS can still occur. This is especially true in high-responder patients or those treated with excessive doses of gonadotropins and is invariably associated with administration of hCG for triggering final oocyte maturation [95].

One of the unique characteristics of GnRH antagonist as compared to GnRH agonist protocols is the ability to replace hCG with GnRH agonist, since the pituitary, following cessation of GnRH antagonist, remains sensitive and therefore responsive to a bolus dose of GnRH-agonist [96].

Replacement of hCG with GnRH agonist [97] eliminates the occurrence of severe OHSS, if it is accompanied with freezing of all embryos and no luteal support [75]. This is the so-called "freeze-all approach" [75,98]. In this strategy, all embryos are frozen for transfer in subsequent cycles. Besides in patients at high risk for OHSS, it is also performed in patients with high progesterone levels on the day of triggering final oocyte maturation [99,100], oocyte donors [101], as well as for fertility preservation, in cases of social freezing [102] or malignancy [103].

The modified natural cycle

The availability of GnRH antagonists allowed the evolution of the natural cycle IVF to the modified natural cycle [104], in which GnRH antagonists are administered concomitantly with FSH in the presence of a dominant follicle [105]. The modified natural IVF cycles is associated with decreased medication costs, represents the only choice in patients in whom FSH stimulation is not possible due to their high endogenous FSH levels, and is also a viable alternative in patients who wish to avoid supernumerary embryo production.

Use of GnRH antagonists in ovarian stimulation for intrauterine insemination

GnRH antagonists have been used to improve the outcome of stimulated intrauterine insemination (IUI) cycles. In a systematic review and metaanalysis including 15 RCTs (3253 IUI cycles, 2345 participants), ovarian stimulation without GnRH antagonists as compared with GnRH antagonist addition was associated with a higher probability of premature luteinization (OR 4.39, 95% CI: 2.73–7.05, $P < .01$), premature LH rise (OR 3.98, 95% CI: 2.53–6.26, $P = .02$), and spontaneous ovulation (OR 2.40, 95% CI: 1.12–5.15, $P < .05$) [106]. However, no difference in the probability of ongoing pregnancy/live birth was observed between stimulated IUI cycles with or without LH suppression by GnRH antagonists (OR 1.14, 95% CI: 0.82–1.57, $P = .44$).

Luteal phase support in antagonist cycles

Progesterone is recommended for luteal phase support in IVF. Progesterone can be administered in various forms including micronized vaginal progesterone (gel or oil capsules) [107], subcutaneous progesterone [108], and intramuscular progesterone [10,109]. Initiation of progesterone for luteal phase support should occur in the window between the evening of the day of oocyte retrieval and Day 3 postoocyte retrieval [10]. Progesterone for luteal phase support should be administered at least until the day of the pregnancy test [10].

Alternative to GnRH antagonists LH suppression regimens

The use of oral progestins has been suggested as an alternative protocol to that using GnRH antagonists, preventing effectively LH surge, with comparable oocyte yield [110,111]. However, further studies are necessary regarding the probability of pregnancy in frozen embryo transfer cycles following progestin suppression of LH surge before its wider implementation in clinical practice [112]. It should be noted that in contrast to GnRH antagonists, progestin suppression of LH surge can only be used when a fresh transfer is not performed [10].

In an attempt to optimize the existing stimulation protocols, several studies have explored various aspects of the use of GnRH antagonist in IVF. Such studies involved the optimal day GnRH antagonist administration should be initiated [24,56,57], the effect of the starting dose of exogenous FSH on pregnancy rates [113,114], the need to supplement the follicular phase with LH [116,118] as well as the need to increase the gonadotropin dose at GnRH antagonist initiation [119]. In addition, interest has been focused on the effect of the timing of human chorionic gonadotropin (HCG) administration on the probability of pregnancy [64], the replacement of HCG with GnRH agonist for triggering final oocyte maturation [97,115,117] as well as the possibility of direct effects of GnRH antagonists on extrapituitary tissues.

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CHAPTER

9

The long gonadotropin releasing hormone agonist IVF protocol

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Introduction

In vitro fertilization (IVF) is recognized as one of the major breakthroughs in reproductive medicine. The first IVF live birth was achieved in 1978 in a patient with tubal factor infertility who underwent retrieval of a single oocyte in the natural menstrual cycle [1]. Despite the initial success of natural IVF cycles, it soon became evident that retrieval of a single oocyte in the natural menstrual cycle limited the efficiency and success of IVF. Thus, investigators envisaged the incorporation of pharmacologic agents to stimulate the growth of multiple ovarian follicles, thereby increasing oocyte yield. Multiple oocytes could now be fertilized, and more than one embryo could be transferred at a time, thus increasing IVF pregnancy rates. However, the stimulation of multiple follicles with exogeneous pharmacologic agents was associated with supraphysiologic estradiol (E2) levels, often inducing a premature luteinizing hormone (LH) surge and/or ovulation, thereby compromising the timing and yield of oocyte retrievals. This problem served as an impetus for the development of protocols, specifically with gonadotropin-releasing hormone (GnRH) analogs aimed at inhibiting a premature LH surge during ovarian stimulation. This chapter begins by discussing the inception of early ovarian stimulation protocols for IVF and the shortcomings that necessitated the incorporation of gonadotropin-releasing hormone agonists (GnRH-a) in ovarian stimulation. In addition to reviewing the physiology of GnRH-a, we also appraise the existing medical literature pertaining to the indications and shortcomings of contemporary GnRH-a protocols. For the purpose of this chapter, we will limit our discussion to "long" GnRH-a protocols, but will also briefly review other GnRH-a protocols.

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A brief history of IVF attempts

Natural cycle IVF

The earliest description of oocyte retrievals in humans appeared as early as 1944 [2,3]. In their 1944 manuscript [2], Rock and Menkin described their clinical experience of retrieving a single oocyte from a 38-year-old para 4 woman and a 31-year-old para 6 woman undergoing laparotomy on cycle days 10 and 11, respectively. Their 1948 publication summarized the retrieval of 800 oocytes via laparotomy from women undergoing various gynecologic procedures [4]. These studies described the retrieval of oocytes and early stages of oocyte fertilization in humans.

Thereafter, Steptoe demonstrated his technique of laparoscopic oocyte retrieval during the natural menstrual cycle [4]. Subsequent scientific collaboration between Steptoe and Edwards resulted in the refinement of surgical and laboratory techniques for IVF [5,6], ultimately resulting in the birth of the first IVF baby in 1978 [1]. Early IVF cycles were limited to the natural menstrual cycle, as it was thought to be the best source of oocytes [7]. Despite the novelty and early success of natural cycle IVF, timing oocyte retrievals in relation to the endogenous LH surge posed a significant challenge [8].

The Norfolk group formulated a monitoring protocol to time oocyte retrievals in natural cycle IVF [7]. Patients deemed to be good IVF candidates were monitored for three menstrual cycles before proceeding with their IVF cycle. Patients were seen daily beginning cycle Day 10 and underwent cervical mucus scoring, vaginal cell maturation indices, basal body temperature measurement, and follicular development via transabdominal ultrasonography. They also underwent 24-hour urine and blood E2 and LH measurements. Blood samples were drawn every four hours in the period immediately before the LH surge to determine its precise timing, and these results were corroborated with the ultrasonographic presence of an early corpus luteum. Patients were monitored in a similar manner during their actual IVF cycle, and the information was correlated with laparoscopic findings [7]. With these data, the Norfolk group established two salient points—(1). the interval between ovulation and the LH surge was approximately 28–30 h after the beginning of the LH surge; (2). most follicles were 23–24 mm in diameter before ovulation, although some women ovulated at follicular sizes that were considerably smaller. The initial 41 oocyte retrievals conducted by the group yielded 21 metaphase II (M-II) oocytes [9]. Following insemination, 13 reached the 2-pronuclear (2-PN) stage and cleaved to the two- and four-cell stages. Embryo transfer (ET) was performed in 13 patients, but none conceived.

IVF with exogeneous pharmacologic agents

Given the yield of only single oocytes and frequency of cycle cancellations associated with natural cycle IVF, the retrieval of multiple oocytes during each treatment cycle became paramount to increase IVF success rates. Immediate clinical efforts, therefore, focused on generating multiple oocytes with ovarian stimulation followed by the transfer of more than one good quality embryo [10]. Thus, a cycle of IVF now incorporated ovarian stimulation, oocyte retrieval, fertilization of oocytes, and embryo transfer. Clomiphene citrate (CC) was perhaps the first pharmacologic agent used for ovarian stimulation in IVF [11]. CC is an oral antiestrogen consisting of a racemic mixture of two stereoisomers—enclomiphene and zuclomiphene. The two isomers demonstrate different patterns of agonistic and antagonistic activity, which results in increased follicle-stimulating hormone (FSH) secretion from the pituitary [11]. Some studies have estimated an approximately 50%–60% increase in baseline serum FSH levels with CC administration. Early investigations demonstrated that the combination of CC and timed oocyte retrieval with human chorionic gonadotropin (hCG) administration resulted in better retrieval timing, and therefore, higher yields of mature oocytes and embryos for transfer [12]. Studies from Australia demonstrated that IVF pregnancy rates had increased from 4% to 13% to 18% to 22% before and after using CC, respectively [13–15]. Using similar protocols, the Bourn Hall clinic in England reported increased implantation rates, that is, 16.5% without CC versus 30% with CC [16].

Due to the accrual of promising results from CC-based IVF cycles, investigators hypothesized that the administration of exogenous gonadotropins would further increase oocyte and embryo yield in IVF cycles, and subsequently pregnancy rates. It is important to note that while exogenous gonadotropin preparations existed even before the development of IVF, they were generally reserved for ovulation induction in hypogonadotropic women [17–19]. However, several refinements in gonadotropin preparations were required before its widespread use in IVF.

Initial studies involved the use of animal-derived gonadotropin extracts in patients with gonadotropin insufficiency. The first extract in 1930 was derived from swine pituitaries; however, later preparations from hog and sheep pituitaries, and the serum of pregnant mares, became available [17–19]. Animal-derived gonadotropin extracts did not assimilate into routine clinical practice due to its side effects, hypersensitivity reactions, as well as short-lived stimulation responses, which were thought to occur due to the production of antibodies to animal gonadotropins [11,17–29]. These limitations of animal-based gonadotropins propelled the discovery of a new source of gonadotropins. In 1958, Gemzell was successful in isolating gonadotropins from cadaveric human pituitaries [17,18]. Bettendorf demonstrated that human pituitary gonadotropins (hPG) could be utilized for ovarian stimulation as early as 1963 [17,18]. However, the production of hPG from human cadavers vastly limited its supply. Furthermore, cases of Creutzfeldt–Jakob disease in Europe were purportedly linked to the use of hPG, thereby resulting in its withdrawal from the commercial market [17,18]. Soon after, investigators demonstrated that FSH and LH could be isolated from the urine of postmenopausal women, that is, human menopausal gonadotropin (hMG) [21]. Clinical trials using hMG preparations for hypogonadotropic and anovulatory normogonadotropic women began in the 1960s [17–19]. Thus, based on the experience with ovulation induction, novel IVF protocols incorporating hMG were formulated for the treatment of normovulatory or tubal factor infertility patients. For example, the Norfolk group [21–23] initiated ovarian stimulation with 150 IU hMG daily beginning cycle day 2. E2 and LH levels were measured daily. Transabdominal ultrasonograms were added beginning cycle day 6, in addition to cervical mucus scoring and vaginal cell maturation indices. Blood samples were drawn at 8 a.m., 2 p.m., and 6 p.m. to measure LH levels on cycle day 8. In the absence of an endogenous LH surge, ovulation was triggered with a 10,000 IU bolus of hCG, and oocyte retrievals were scheduled 38 h later [7]. In one of their earliest studies utilizing hMG-based IVF protocols, the Norfolk group reported the successful retrieval of 48 fertilizable eggs in 26 out of 31 IVF cycles [22]. Cleavage of embryos occurred in 12 cycles, and two pregnancies occurred

[22]. Remarkably, one of these pregnancies resulted in the birth of Elizabeth Jordan Carr on December 28th, 1981, the first IVF baby born in the United States [7]. With further improvements in hMG stimulation protocols, the Norfolk team obtained fertilizable oocytes in 22 of 24 laparoscopic cycles [23]. Successful fertilization and embryo transfers occurred in 21 and 19 cycles, respectively. A total of five pregnancies were reported in this study [23]. The advantage of hMG-based IVF protocols was later reported by other centers [24]. With further advancements in molecular technology, hMG could now be passed through affinity columns containing highly specific monoclonal antibodies to FSH. The selectively bound FSH could then be extracted from the column providing highly purified FSH [17–19]. Soon, IVF protocols incorporating FSH began to emerge [25–31]. The Norfolk team were perhaps the first to introduce pure FSH protocols, also initiate early studies comparing different protocols such as concomitant combination of hFSH/hMG, sequential combination of the two gonadotropins, and pure FSH administration [31]. The group reported that FSH without additional LH could be used successfully for ovarian stimulation, and also allow oocyte maturation up to terminal maturation [28]. Their subsequent work also reported the advantages of combined FSH and hMG ovarian stimulation protocols [29,30].

Shortcomings of early IVF protocols

Despite the irrefutable advantages of gonadotropin-based IVF protocols over natural cycle IVF in increasing pregnancy rates, initial studies also reported a 20%–30% incidence of ovulation prior to oocyte retrieval in these early IVF protocols [11,32,33]. These findings were attributed to high LH levels, which occurred due to the positive feedback of high serum E2 during the mid-follicular phase of ovarian stimulation [11,32,33]. The exposure to high LH was associated with premature luteinization of follicles resulting in either cycle cancellation, poor oocyte quality, and severely compromised IVF outcomes [7,32,33]. These limitations necessitated the incorporation of pharmacologic agents that would suppress ovulation during ovarian stimulation without compromising IVF outcomes. GnRH analogs, in this context, emerged as promising treatment modalities.

Gonadotropin-releasing hormone agonists

Physiology, structure, and early studies

GnRH is a small decapeptide whose structure was elucidated in 1971 [34]. It is secreted by the hypothalamus into the pituitary portal circulation in a pulsatile fashion, thereby stimulating the anterior pituitary to synthesize and secrete FSH and LH [11]. Laboratory studies have revealed specific regions within the decapeptide that are responsible for receptor binding, activation of pituitary gonadotroph cells, and stability of GnRH itself (Fig. 9.1) [11]. GnRH-a were initially developed by replacing one or two D-amino acids [11]. Specifically, the potency of GnRH could be increased by replacing Gly (at position 6) and Gly-NH_2 (at position 10) with other D-amino acids and ethylamide, respectively [11,35]. For example, leuprolide is synthesized by replacing D-Leu at position 6 and ethylamide at position 10, while Buserelin is synthesized by replacing D-Ser at position 6 and ethylamide at position 10

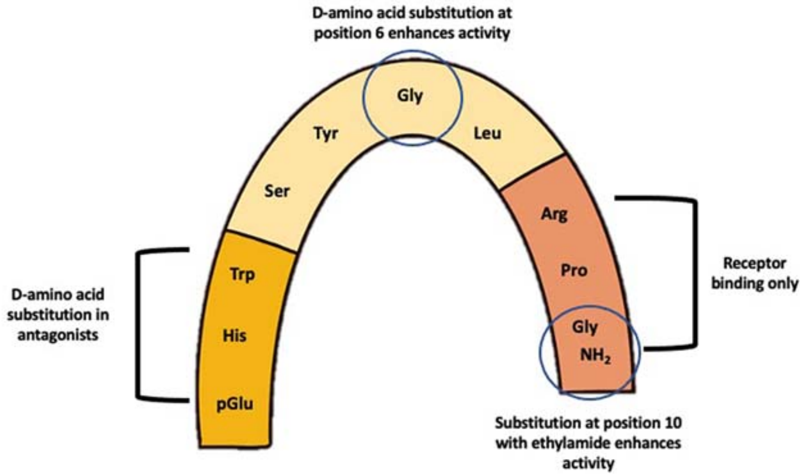

FIGURE 9.1 Decapeptide structure of GnRH revealing specific regions responsible for receptor binding and modification of potency.

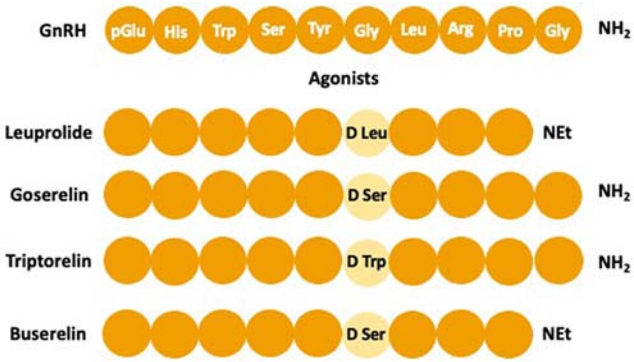

FIGURE 9.2 Structural similarities between GnRH, leuprolide, goserelin, triptorelin, and buserelin.

(Fig. 9.2). These amino acid substitutions make GnRH-a more resistant to enzymatic degradation and more hydrophobic [11].

Initial studies demonstrated that repeated administration of GnRH-a first increased the levels of FSH and LH, albeit transiently, followed by a decrease in FSH and LH and consequently a significant decline in sex steroid levels [11,36]. In contrast, mimicking the pulsatile secretion of GnRH restored physiologic levels of FSH and LH, and even ovulation, as evidenced by the treatment of women with hypogonadotropic hypogonadal anovulation [37,38]. These data indicated that the initial binding of GnRH-a activates GnRH receptors; however, persistent occupation of GnRH receptors by continuous GnRH-a administration results in desensitization and internalization of receptors, resulting in diminishing FSH and LH levels [39]. Thus, GnRH-a emerged as a treatment modality for different reproductive conditions where suppression of E2 or ovarian steroid production was necessary [40–43]. For example, five women with endometriosis were treated with a daily dose of a potent long-acting GnRH-a, D-Trp6-Pro9-Net-LHRH (GnRH-A), for 28 days [40]. Mean concentrations of E2, androstenedione, and testosterone on Day 28 in these women were in the postmenopausal range and similar to those measured in oophorectomized women [40]. Thus, investigators began to postulate whether GnRH-a could be integrated into IVF stimulation protocols, specifically to suppress premature LH surge and ovulation.

GnRH-a and early IVF outcomes

One of the IVF earliest studies utilizing GnRH-a included 12 women with a history of prior IVF cycle cancellation due to dominant follicle formation, suboptimal response, or premature LH surge [44]. None of these women had oocytes retrieved in previous cycles. The cohort was treated with long-acting GnRH-a for pituitary suppression followed by ovarian stimulation with hMG. Twelve IVF cycles were completed without cancellation. Approximately four oocytes were retrieved in each individual, and two pregnancies were reported. These findings were confirmed in a clinical trial of 44 consecutive women who did not have oocytes retrieved in prior IVF cycles with standard CC-hMG treatment [45]. Twenty-nine women received GnRH-a agonist Buserelin with hMG, while the remaining 15 received standard CC-hMG treatment, without GnRH-a. The median number of oocytes retrieved in the former group was 4 (range 0−19), while the latter was 0 (range 0−5). Although fertilization rates were similar in the two groups, 54% and 13% of patients underwent embryo transfer in the Buserelin with hMG and CC-hMG only treatment groups, respectively. Three pregnancies exclusively occurred in the Buserelin with hMG group. Following these studies, independent investigators began contributing to the emerging body of work demonstrating the benefits of GnRH-a in decreasing IVF cycle cancellation due to ovulation, increasing oocyte and embryo yield, as well as increasing IVF pregnancy rates [46−51]. A noteworthy metaanalysis of 10 clinical trials from 1992 reported that the use of GnRH-a during ovarian stimulation reduced cycle cancellations by 67% and increased clinical pregnancy rates by approximately 70% [50]. Furthermore, spontaneous abortion rates were similar with and without GnRH-a use [52].

Current GnRH-a protocols

One critical question about the use of GnRH-a during IVF was its timing in relation to exogenous gonadotropin administration. In the first strategy, daily injection of a GnRH-agonist is initiated in the mid-luteal phase of the preceding menstrual cycle [37,53,54]. Patients would present on cycle day 2 or 3 of the subsequent menstrual cycle for baseline hormonal measurements and ultrasonography. Exogeneous gonadotropins are only initiated when adequate pituitary suppression is confirmed by low E2 levels and a thin endometrial lining. At this point, the daily GnRH-a dose is halved until the day of trigger [53,54]. This mid-luteal, GnRH-a downregulation protocol is commonly referred to as a "long" GnRH-a protocol (Fig. 9.3). Contemporary long GnRH-a protocols utilize 1 mg leuprolide acetate in the mid-luteal phase, followed by 0.5 mg leuprolide acetate in the ovarian stimulation phase [54].

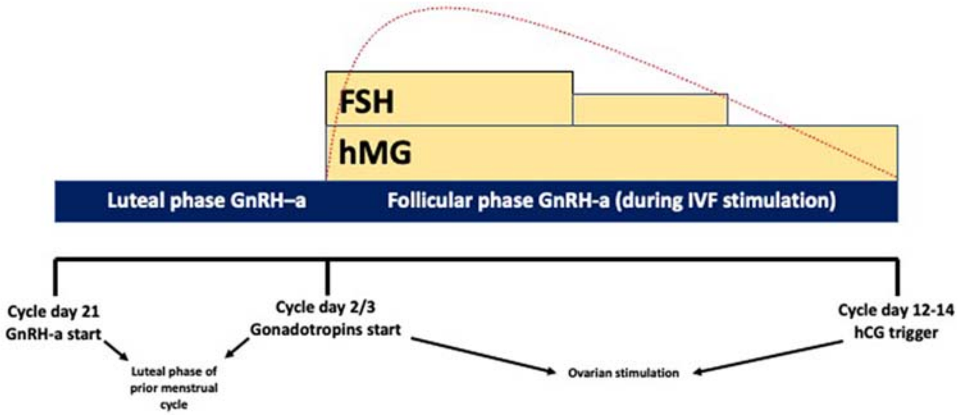

FIGURE 9.3 Simplified schematic of the long GnRH-agonist protocol.

The second strategy involves GnRH-a administration in the early follicular phase of ovarian stimulation, thereby availing of the initial stimulatory or "flare" effect of GnRH-a on endogenous FSH and LH secretion [55]. These protocols generally require shorter durations of GnRH-a administration and encompass coflare and microflare protocols [53]. The coflare protocol involves administration of 1 mg leuprolide acetate from cycles day 2–4 and then 0.5 mg leuprolide acetate beginning on cycle day 5 [54,55]. High-dose gonadotropins treatment (300–450 IU) were usually initiated on cycle day 2 or 3 [54,55]. Theoretical concerns of the coflare protocol included enhanced ovarian androgen production, rescue of the corpus luteum from the prior menstrual cycle, and premature luteinization [54]. Therefore, a lower dose of GnRH-a was proposed to circumvent these issues, that is, microdose of leuprolide (MDL), which is approximately 1/50th the concentration of commonly utilized leuprolide acetate [54]. In this protocol, oral contraceptive pills (OCPs) are administered for 14–21 days [54,56]. Twice daily MDL (40 μg) is started on the third day following the last day of OCPs, followed by high-dose gonadotropins (300–450 IU) on the third day of MDL [54,56]. Coflare and MDL protocols can be modified by the addition of estrogen patches in the preceding luteal phase to improve ovarian responsiveness in poor responders [54].

Initial comparisons between long and short GnRH-a protocols showed statistically lower cancellation rates, higher oocyte and embryo yield, and higher pregnancy rates with long GnRH-a protocols [50,57]. However, no difference in pregnancy rates was observed in long GnRH-a protocols when comparing GnRH-a types [58]. There was also a lower rate of ovarian cyst formation in long GnRH-a protocols [59,60], especially when OCP pretreatment was used in the early follicular phase prior to mid-luteal GnRH-a [60]. Data from the Norfolk group suggested that the long GnRH-a protocol was more beneficial in intermediate and high responder patients, defined as patients with a basal FSH to LH ratio of 1:1 [53]. In contrast, flare protocols exhibited better stimulation parameters in low- or poor-responder patients, with a basal LH to FSH ratio of >1.5 [53]. A 2015 Cochrane study of 37 randomized controlled trials (RCTs) compared different GnRH-a protocols, of which 20 RCTs compared long and short GnRH-a protocols [61]. The study found no conclusive evidence with regard to difference in live birth and ongoing pregnancy rates (OR 1.30, 95% CI 0.94−1.81) between long and short GnRH-a protocols. There was evidence of higher clinical pregnancy rates with long GnRH-a protocols (OR 1.50, 95% CI 1.18−1.92); however, these findings were considered moderate quality evidence. A different Cochrane study examined the differences between a single depot dose of GnRH-a versus daily GnRH-a doses in women undergoing IVF [62]. Twelve studies comprising 1366 women were included. There were no significant differences in live birth or ongoing pregnancy rates (OR 0.95, 95% CI 0.70−1.31) or ovarian hyperstimulation syndrome (OHSS) rates (OR 0.84, 95% CI 0.29−2.42) when comparing the two GnRH-a formulations. However, the study did postulate that depot GnRH-a could increase the overall cost of IVF treatment since this protocol required more gonadotropins and was associated with a longer duration of use.

Shortcomings of GnRH-a protocols

In addition to its desired effect of suppressing ovulation during ovarian stimulation, a further advantage of the long GnRH-a protocols was that the duration of suppression could be varied. Thereby, providing IVF clinics the ability to program or time IVF cycle stimulation starts [33]. However, with the widespread adoption of long GnRH-a protocols, many clinicians began to note recurrent side effects during the suppression period, including mood changes, hot flushes, severe headaches, and short-term memory loss [33,63]. These side effects were attributed to the acute drop in E2 levels. Other side effects included rapid bone loss and a slight increase in total cholesterol, though these effects occurred with >6 months of exposure to GnRH-a [64]. Additionally, daily GnRH-a injection in some women resulted in LH secretion immediately, which stimulated ovarian androgen production during IVF, potentially impacting follicular development and endometrial function [39]. The most concerning aspect of GnRH-a protocols was the high rate of OHSS. For example, a study of 516 women undergoing their first IVF cycle with a long GnRH-a protocol reported an 8.9% incidence of severe OHSS and 15.6% incidence of moderate OHSS [65]. These shortcomings of GnRH-agonists paved the way for the utilization of potent gonadotrophin-releasing hormone antagonists (GnRH-ant) in the early 1990s.

Pharmacologic studies revealed that deletion or substitution of histidine in position 2 of the GnRH decapeptide resulted in competitive antagonists of the GnRH receptors [66]. Therefore, unlike the initial stimulatory effects of GnRH-a, GnRH-ant produced immediate suppressive effects due to its competitive binding to the GnRH receptor [32,33]. However, the first-generation group of GnRH-antagonists such as Abarelix was associated with an increased incidence of local and severe systemic allergic and/or anaphylactic reactions due to histamine release [64,66], ultimately resulting in its withdrawal from the US market in 2005. These side effects in first-generation GnRH-ant, and to some extent in second-generation GnRH-ant, were attributed to the structural combination of a hydrophobic N-terminus and a basic/hydrophilic C-terminus [66]. Further modifications of the GnRH decapeptide at amino acid positions of 1, 2, 3, 6, and 10 resulted in third-generation GnRH-ant such as Cetrorelix and Ganirelix, which had higher metabolic stability and lower histamine release, thereby decreasing serious allergic reactions [64,66]. These GnRH-antagonists competitively inhibited the secretion of LH and FSH from the pituitary gland in a dose-dependent manner and could be easily reversed after treatment discontinuation

[64,66]. Thus, GnRH-ant could be initiated earlier in the follicular phase, rendering shorter and more patient-friendly ovarian stimulation cycles [32,39]. Notably, GnRH-ant protocols are associated with lower risks of OHSS due to the use of GnRH-a ovulatory triggers. This is illustrated in a 2016 Cochrane study of 73 RCTs with 12,212 participants [67], which reported a lower risk of OHSS with GnRH-ant protocols when compared to long GnRH-a protocols (OR 0.61, 95% C 0.51—0.72). Another metaanalysis confirmed these findings, reporting lower rates of OHSS with GnRH-ant protocols across normal and high-responder patients [68]. The most recent RCT, and the only reporting on cumulative live birth rates (CLBRs), demonstrated comparable CLBR between GnRH-a and GnRH-ant protocol (10.1093/humrep/dew358) and a higher OHSS rate for the long GnRH-a protocol (REF 69 in the current article).

Conclusions

The favorable side effect profile associated with GnRH-ant protocols has resulted in its accelerated adoption as the preferred ovarian stimulation protocol for the treatment of normal and high-responder patients. However, before attempting to abandon the long GnRH-a protocol, we must acknowledge the excellent pregnancy rates when compared with GnRH-ant protocols [68,69]. Prospective studies must aim to identify specific patient populations that can benefit from the long GnRH-a protocol, because, despite its limitations, it still remains one of the most commonly used ovarian stimulation protocols worldwide.

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CHAPTER

10

Double stimulation in the same ovarian cycle (DuoStim) and luteal phase start IVF protocols

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Introduction

The advances in vitro fertilization (IVF) such as blastocyst culture, aneuploidy testing, and vitrification have profoundly changed the approach to infertile couples, encouraging clinicians to maximize the exploitation of the ovarian reserve via tailored protocols, especially for poor prognosis patients [1]. At present, the measure of success in IVF for physicians is represented by cumulative live birth delivery rate (CLBdR) per started cycle [2]. Hard data support the evidence that the number of oocytes retrieved after controlled ovarian stimulation (COS) is an efficient strategy to increase the number of babies per intention to treat (ITT). Indeed, recent studies have demonstrated that, in this "vitrification era," CLBdR continuously increases with the number of oocytes retrieved after COS [3,4]. This is the reason why collecting the most proper number of oocytes for each category of patients is important, especially in poor prognosis patients, to increase the chance of success after an IVF treatment.

In this regard, maternal age is another important aspect to consider for the management of poor prognosis patients. Several dysfunctions related to aging have been associated with impaired fertility. Together with a progressive reduction of the ovarian reserve, it compromises also oocyte competence. Therefore, the target number of oocytes to collect should be

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defined according to maternal age so to counteract the decline in their quality. In other terms, the main goal to manage women of advance maternal age (AMA) and poor ovarian reserve is to tailor COS so to fully exploit the ovarian reserve.

The technical improvements in IVF (e.g., blastocyst culture, single embryo transfer [SET], cryopreservation of oocytes/embryos, preimplantation genetic testing for aneuploidies [PGT-A], freeze-all) allowed the implementation of nonconventional COS protocols not only for oncologic but also for poor prognosis patients undergoing IVF [5].

Theories of folliculogenesis

Follicular development within the ovary is a dynamic process: from a large pool of primordial follicles in the resting phase, some will develop as preantral and antral follicles until they undergo atresia, whereas just few of them will complete their development after some further steps, namely sequential recruitment, selection, and growth of follicles. According to the classic theory (single recruitment episode theory), a single cohort of antral follicles grows during the follicular phase of the ovarian cycle after luteal regression. However, this theory has been overtaken by the evidence of multiple waves arising during an ovarian cycle in many mammals [6]. Such evidence, at first reported in large animal models, has been confirmed also in humans leading to the definition of two further theories of follicle recruitment: the *continuous recruitment theory*, according to which the follicles start growing and regress continuously during the ovarian cycle; *and the waves theory*, according to which 2–3 cohorts of antral follicles are recruited per ovarian cycle. However, the mechanisms underlying follicular recruitment have not been fully elucidated yet. Several intraovarian regulators, follicle-stimulating hormone (FSH) and progesterone levels, and inflammatory markers (e.g., serum C reactive protein) were all proposed as modulators of the dynamics behind the origin of follicular waves [7].

From a clinical perspective, the growing knowledge of human ovarian follicular waves opened new options for COS to improve the efficiency and possibly the efficacy of IVF. Usually, COS with gonadotrophins is initiated the 2nd or 3rd day of the menstrual cycle in a gonadotropin-releasing hormone (GnRH) agonist or antagonist protocol. However, the evidence of multiple waves of follicular growth during one ovarian cycle paved the way to novel COS strategies designed to better exploit the ovarian reserve; in fact, ovarian stimulation can be started at any time of the menstrual cycle. In this regard, in oncologic patients who need to vitrify as many oocytes as possible in a very short frame of time, women can start ovarian stimulation at any time of the cycle, thus reducing the time to oocyte retrieval. Thanks to this flexibility, oocytes can be collected even within just two weeks, without further delaying anticancer therapies [8]. The successful application of the random start (RS) protocol or luteal phase stimulation (LPS) for poor responder patients ultimately led to the theorization of a novel ovarian stimulation protocol, known as double stimulation in a single ovarian cycle (DuoStim). This approach, by combining conventional follicular phase stimulation (FPS) with a second stimulation back-to-back in the same ovarian cycle, provides the unprecedented opportunity to increase the cohort of oocytes retrieved in less than one month [9].

Definition of poor prognosis patients

The management of poor responder patients is thorny because even though different strategies have been suggested, a consensus is missing. Poor responders represent 9%–24% of patients undergoing COS, meaning that up to one in four patients conceals a poor reproductive prognosis. The last decade has witnessed the attempts of the medical community to standardize the diagnosis of poor responders at first with the Bologna criteria [10,11] and then with the POSEIDON classification (Patient-Oriented Strategies Encompassing IndividualizeD Oocyte Number). This novel classification introduced a better stratification of "low prognosis patient" within four subgroups based on (1) quantitative and qualitative parameters such as age and the expected aneuploidy rate; (2) ovarian reserve biomarkers (AFC and/or AMH); and (3) response to a previous stimulation, if performed. In addition, the POSEIDON Group also stated that the main measure of COS success shall be "the number of oocytes needed for a specific patient to obtain at least one euploid embryo for transfer" [12]. From a clinical perspective, the adoption of age, oocyte yield, and ovarian reserve in the Poseidon classification allowed for the definition of two main categories, namely the "expected" (groups 3 and 4) and the "unexpected" poor responders (groups 1 and 2). The POSEIDON criteria might help clinicians exploring patient-oriented strategies to retrieve the number of oocytes required to obtain at least one euploid embryo in poor prognosis women undergoing IVF. As these patients are more prone to treatment failure and discontinuation, a clear definition of poor prognosis was essential to avoid heterogeneity and allow the use of personalized management to achieve the intended goal.

An unconventional ovarian stimulation protocol: luteal phase stimulation

The current knowledge of ovarian physiology has challenged the traditional concept of folliculogenesis, laying the foundation for novel COS protocols in ART. Indeed, unconventional strategies, such as LPS, have been developed to retrieve the highest number of oocytes in the shortest possible time. It could be especially important for some specific patients, such as poor responders, and women seeking fertility preservation before oncologic therapy. LPS starts 2–7 days following ovulation or oocyte retrieval. However, as the endometrium loses synchronicity, a fresh embryo transfer cannot be performed [13]. Although initiating COS in the luteal phase requires a longer stimulation and a higher total dose of gonadotropin, these differences are not clinically significant. Furthermore, COS initiated in the luteal phase does not compromise the quantity or quality of oocytes retrieved compared to outcomes of traditional stimulation in the follicular phase. Lastly, unconventional stimulation, such as LPS, offers similar outcomes as conventional protocols, but with increased flexibility and within a shorter timeframe [14]. However, more well-designed trials are required.

DuoStim for poor prognosis patients: the evidence up to date

Many COS strategies have been proposed for the management of poor responders. Unfortunately, none of them has yet been proven superior to the others. Indeed, COS cannot create follicles ex novo. Therefore, DuoStim has been proposed to maximize the exploitation of the ovarian reserve in a short timeframe. The first experience with this protocol dates to 2014, and it was known as the "Shanghai Protocol" [15]. The authors published a pilot study in which 38 women underwent mild ovarian stimulation, and after the first oocyte retrieval, human menopausal gonadotrophin and letrozole were administrated to stimulate a second wave of follicle development, and oocyte retrieval was carried out a second time when dominant follicles had matured. This study showed the feasibility of a double COS in the same ovarian cycle, and two years later Ubaldi et al. published a proof-of-concept study in 51 poor prognosis patients that confirmed that evidence in a setting combining PGT-A at the blastocyst stage, vitrification and SET [16]. The protocol for the first and the second stimulation was identical: recombinant-gonadotrophins (rec-FSH 300 IU/day plus rec-LH 75–150 IU/day) combined with GnrH antagonist. In both stimulations then, final oocyte maturation was triggered through GnrH agonist (a single subcutaneous bolus of buserelin at the dose of 0.5 mL or triptorelin 0.3 mL). Five days span the two stimulations to achieve complete luteolysis. The results showed that the oocytes collected after FPS and a paired second stimulation are similar after intracytoplasmic sperm injection (ICSI), blastocyst culture, trophectoderm biopsy, comprehensive chromosome testing, vitrification, and SET. Therefore, the study concluded that DuoStim might increase the number of patients obtaining at least one euploid blastocyst in a single ovarian cycle compared with a standard COS (from 42% to almost 70% per oocyte retrieval).

In 2018, a case-control study was conducted from the same group where the paired cohorts of oocytes collected from a first and a second COS in the same ovarian cycle were compared, showing a similar competence in terms of maturation, fertilization, blastulation, euploidy, and implantation rates. Yet, the second cohort was on average 1 oocyte larger [17]. In the same year, Vaiarelli et al. reported the reproducibility of DuoStim and its promising outcomes from a multicenter perspective, including 310 poor prognosis patients from four IVF centers [18].

The higher number of oocytes collected after the second stimulation is imputable to the high level of estradiol and progesterone after FPS that may (1) synchronize the cohort of antral follicles of the anovulatory waves; (2) stimulate the proliferation of FSH receptors in their granulosa cells, thereby resulting in an overall better response to COS; (3) change the ovarian microenvironment; (4) increase the angiogenic factors; (5) enhance the sensitivity of the granulosa cells to FSH; and (6) elicit a flare-up effect from GnRH agonist trigger that might induce a downregulation in the expression of AMH in the follicles from the anovulatory wave. However, all these speculations must be confirmed from future studies.

In 2020, a 3-year observational study including women fulfilling the Bologna criteria (namely at least two of the following characteristics: maternal age $\geq$40 years and/or $<$3 oocytes retrieved after previous COS and/or reduced ovarian reserve [i.e., antral follicle count, AFC$<$7 or antimullerian, AMH$<$1.1 ng/mL]) reported the clinical contribution of a second stimulation in the same ovarian cycle. The results showed that the CLBdR per ITT was 15% after DuoStim, while it did not exceed 8% among the 197 women who chose to undergo a conventional COS, as only 17 not pregnant patients returned for a second stimulation after the first attempt (drop-out rate, 81%) [19]. Indeed, from a strategic perspective, DuoStim is a powerful tool to prevent treatment discontinuation in poor prognosis patients who still have reasonable chance to conceive.

Evidence to support the safety of the second stimulation in the same ovarian cycle in terms of clinical, obstetric, and perinatal outcomes were then reported also from a multicenter perspective by Vaiarelli et al. in 2020 [20].

Over the time, several independent groups outlined the consistency and reproducibility of DuoStim safety and efficiency, thus confirming the evidence initially reported only from the Chinese and Italian authors [21].

Fig. 10.1A is a timeline of the most relevant papers published about DuoStim up to date.

Framework of a DuoStim protocol

Luteal estradiol priming (4 mg/day of estradiol valerate) is started in Day 21 of the previous menstrual cycle to promote the synchronization and coordination of follicular growth. After the transvaginal ultrasound and basal assessment of the ovaries, on Day 2 or Day 3 of the menstrual cycle, luteal estradiol priming is stopped, and FPS is started with fixed dose of rec-FSH 300 IU/day plus LH 75–150 IU/day for 4 days. Follicular growth is monitored on Day 5 and then every 2–3 days. GnRH antagonist is administered daily after the identification of a leading follicle with a diameter ≥13–14 mm during both FPS and second stimulation until the day of ovulation trigger. The final maturation of oocytes is triggered with a GnRH analog to reduce the time of luteolysis (Fig. 10.1B). Egg retrieval is performed 35 h after the trigger. ICSI, blastocyst culture, trophectoderm biopsy, and vitrification are performed as described in Refs. [22,23]. Five days after the first retrieval, namely, the time needed to complete luteolysis, the second stimulation is started with the same protocol and daily dose regardless of the number of antral follicles visible through ultrasound scan

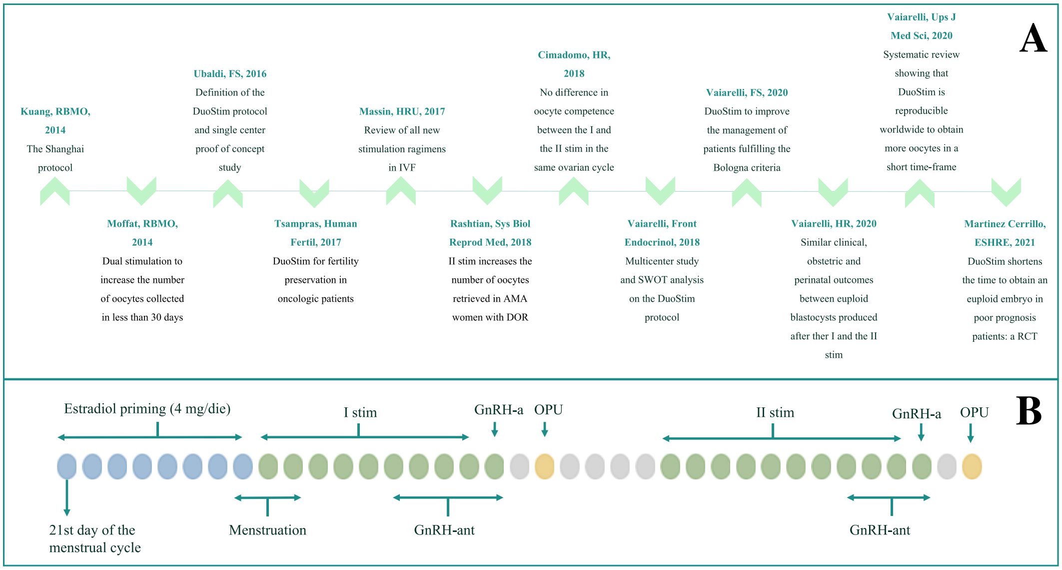

FIGURE 10.1 **(A) Timeline of the main papers published about DuoStim up to date; (B) schematic representation of the DuoStim protocol.** *AMA*, advanced maternal age; *DOR*, diminished ovarian reserve; *GnRH-a*, GnRH-agonist; *GnRH-ant*, GnRH antagonist; *OPU*, oocyte pick-up; *RCT*, randomized controlled trial; *stim*, stimulation.

in the anovulatory wave. A freeze-all approach is then adopted and the biopsy fragments from both stimulations are shipped together and analyzed in the same run at an external genetic lab. In presence of euploid blastocyst(s), vitrified-warmed SETs are performed in a modified-natural or artificial cycle.

Clinical indications for DuoStim

The main indication for DuoStim is AMA, especially when associated with reduced ovarian reserve (low AMH, low AFC, and/or suboptimal response to previous COS) [24]. In general, though, all patients requiring the maximization of COS to optimize their chance of achieving at least one pregnancy in a short timeframe might benefit from DuoStim. For instance, severe male factor, due to its impact on oocyte fertilization and blastulation rates [25], is also a potential indication to DuoStim. However, there is no unanimous consensus on the indications to DuoStim, and it can be conceived even as a strategy to adopt in progress (1) to prevent treatment drop-out in patients obtaining no blastocyst after FPS and with limited residual time to conceive with their own oocytes, or (2) to improve the prognosis of AMA patients obtaining 1–3 blastocysts after FPS, but with very high risk, they would be aneuploid. In fact, several groups across the literature suggested that, although low, the chance to conceive keeps increasing with an increasing cumulative number of oocytes retrieved, also in AMA.

At last, DuoStim can be applied whenever the time is limited, such as for fertility preservation. Ideally, 10–15 oocytes should be obtained in these patients, a number that shall be adjusted based on maternal age. In fact, while the RS protocol is useful to speed up fertility preservation, in many patients, the number of oocytes collected might be insufficient to allow a reasonably good chance of a future pregnancy. In these cases, DuoStim can be discussed with the oncologist and with the patient as a valuable option. Tsampras et al. for instance, published a pilot study in 10 oncologic patients undergoing fertility preservation via DuoStim [8], whose preliminary results were very promising in terms of efficacy and safety with no delay in the initiation of oncologic treatments. An outcome extremely important, since most of these patients have a single chance to preserve their fertility before starting their life-saving therapies.

DuoStim: is the second stimulation in the same ovarian cycle conducted in the luteal phase pf the cell cycle?

Some authors recently suggested that the term "luteal phase stimulation" is improperly adopted to define the second stimulation performed after the conventional FPS in the context of DuoStim [26,27]. Nevertheless, some points need future investigations on this topic. First, although the adoption of the GnRH analog reduces the length of the luteal phase, the hormonal levels in this period of the cycle are different from those detected during a conventional FPS. Moreover, the ovarian cycle is asynchronous with respect to the endometrial cycle. Luteolysis then is patient-specific and highly dependent on the hormonal levels, the number of oocytes retrieved, and the number of corpora lutea, making the date of menstruation unpredictable [28]. Second, in the absence of COS starting 5 days after the first oocyte collection, the follicles development in this phase of the ovarian cycle would never reach full maturity. In other terms, this second stimulation allows collecting those oocytes that would have grown and regressed in the luteal phase. For this reason, the term "LPS" has been conventionally adopted from Kuang's publication onwards [15]. Third, the data on pure LPS are very limited to date, and it is certainly worth investigating this protocol, especially in view of the high reliability of the vitrification technique [29]. A study comparing a pure FPS versus a pure LPS conducted in the same patient in consecutive ovarian cycles has the potential to reveal important data in near future. In conclusion, "LPS" terminology has been changed into "second stimulation in the same ovarian cycle," but an international consensus is desirable on the real nature of the anovulatory waves exploited in the DuoStim protocol.

Conclusions

Continuous stimulation throughout the ovarian cycle is an option to manage this unfavorable patient population undergoing IVF and in all clinical condition when the time is limited. Several studies demonstrated the feasibility of COS conducted in the luteal phase to obtain competent oocytes/embryos regardless of the menstrual cycle. Randomized controlled trials and cost-effectiveness analyses are still in the pipeline, though, to confirm all the promising evidence reported to date. In conclusion, although DuoStim has not been proven superior to two conventional FPS in terms of CLBdR per ITT, it clearly is a valuable tool to obtain a greater number of oocytes/blastocysts in a limited timeframe [30] and prevent treatment discontinuation [19].

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CHAPTER

11

The mild stimulation in vitro fertilization protocol

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Mild ovarian stimulation for IVF

Background

Mild ovarian stimulation for in vitro fertilization (IVF) is being practiced alongside conventional IVF (C-IVF) across the globe. Certain advantages of mild stimulation IVF (MS-IVF) have been highlighted in the literature [1–3], yet the debate surrounding the effectiveness of (MS-IVF) protocols continues. There has been a renewed interest in MS-IVF, because of recent evidence from large randomized controlled trials (RCTs) that showed MS-IVF to be as successful as C-IVF, but safer and less expensive [4]. In this chapter, we describe the basic concept behind MS-IVF, what the protocol entails, its benefits, and alleged limitations.

Concepts behind the mild ovarian stimulation protocol

Oocyte competency

"Livebirth rate (LBR) per oocyte" or "Oocyte utilization rates," defined as LBR per mature oocyte has been regarded as the yardstick of assessing the competency of the individual oocyte to produce a baby [5]. Several studies have demonstrated that a low oocyte yield is associated with a higher LBRs per oocyte, particularly in the realm of mild and natural cycle IVF [4,6,7]. A large (n= 23 354) retrospective study investigating the reproductive potential of mature oocytes found oocyte utilization rate to be highest with 1–5 mature oocytes and lowest when >10 oocytes

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were retrieved in women under 37 years of age [8]. Another study with MS-IVF protocol reported an inverse relationship between oocyte utilization rates and oocyte yield [9]. A study demonstrated oocyte yield to have a direct correlation with cytoplasmic dysmorphism of oocytes and an inverse relationship with the oocyte utilization rates [10]. All the evidence goes against the prevalent belief that obtaining as many oocytes as possible by intensifying the stimulation dose will improve success rates as the number of competent oocytes in a given treatment cycle appears to be fixed and is unlikely to be increased by increasing the oocyte number. In addition, it is often forgotten that women producing high numbers of oocytes following stimulation represent women with good ovarian reserve, who will present with good IVF success rates regardless how these women are stimulated [a schematic diagram of the effect of ovarian stimulation is shown in Fig. 11.1 [93]].

Studies that investigated the relationship between the stimulation dose and LBR reported an inverse correlation between the two: this holds true not only in IVF cycles with autologous oocytes [11,12], but also in oocyte donor's cycles [13] which are less likely to be affected by women's prognostic variables.

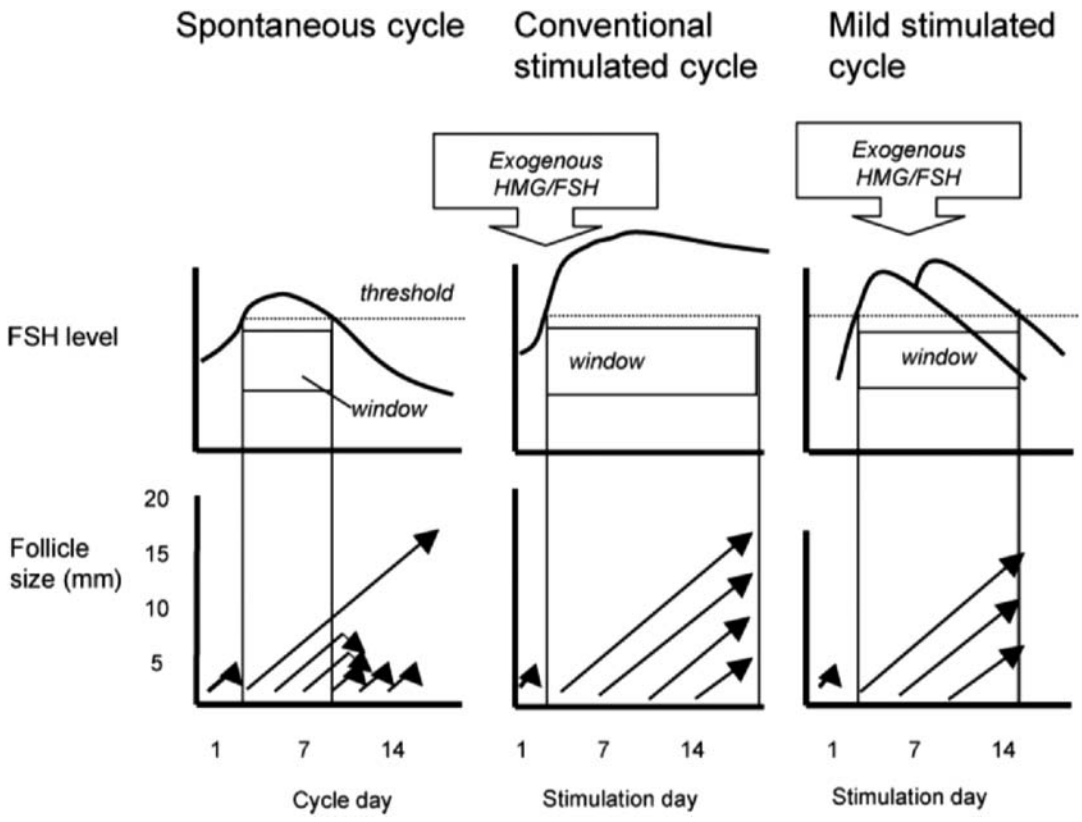

FIGURE 11.1 Principles of ovarian stimulation for IVF. Interference with the FSH threshold and window concept for secondary follicle recruitment during convention or mild ovarian stimulation. *Reproduced from Macklon et al.* [93]

Embryo quality and ploidy status

A study with polar body biopsy found increasing number of chromosomal aberrations with higher oocyte yield [14]. In contrast, another retrospective study showed a direct relationship between the number of euploid oocytes and the number of mature oocytes [15]. In terms of embryo euploidy, the only RCT to date found a higher proportion of euploid embryos with fewer oocytes generated from MS-IVF [16]. This study has been criticized for using older "fluorescent in situ hybridization" (FISH) technology which could analyze only 9 chromosomes. However, several more recent studies using advanced 24-chromosome "next generation sequencing" (NGS) or "Single Nucleotide Polymorphism" (SNP) reported the proportion of euploid embryos remains unchanged with increasing oocyte number or stimulation dose [17,18]. A study with day 3 embryo biopsy showed an increasing absolute number of euploid embryos as the number of oocytes increases [19]. This means that the absolute number of aneuploid embryos also rises with increasing oocyte yield. The number of nonbiopsiable embryos has also been reported to increase with increasing oocyte numbers [20].

With regard to stimulation dose, the aforementioned RCT by Baart et al. found higher proportion of euploid embryos with MS-IVF protocol. [21]. Other recent retrospective studies showed neither euploidy nor aneuploidy rate to get affected by the stimulation doses [17,20,22]. The study by Gianaroli et al. demonstrated an inverse relation between euploid oocyte and the stimulation dose per oocyte [15]. Thus, current evidence indicates oocyte or embryo aneuploidy rate is dictated by women's age and ovarian reserve [18,23], but not by the intensity of ovarian stimulation or oocyte yield.

Several RCTs reported higher proportion of high-quality embryos with MS-IVF compared with C-IVF [16,24,25]. One RCT found the number of good quality blastocyst did not rise with higher response by increasing the stimulation dose [26]. Metaanalysis RCT with low (150 IU/ day) and high stimulation ($\geq$ 150 IU/ day) failed to show a benefit with regard to the number of cryo-preserved embryos [27]. A metaanalyses and systematic review of RCTs on poor, normal, and high responder women found no difference in the number of high-grade embryos between MS-IVF (defined as $\leq$150 IU/ day of gonadotropin) and C-IVF, despite a higher oocyte yield with C-IVF [4] (Fig. 11.1). This may also explain why the cumulative LBR is also not depended on the intensity of ovarian stimulation.

Stimulation versus response

Ovarian stimulation is not synonymous of ovarian response. There is no direct causal relation between the two either. The concept of 'ovarian sensitivity index' which is a ratio between the response (oocyte number) and the stimulation dose, has been shown to have a better correlation with ovarian reserve than oocyte yield [28]. A higher ovarian response (oocyte yield) with a lower stimulation dose has been found to be positively related to the pregnancy outcomes [29]. Conversely, A low oocyte recovery despite high stimulation is associated with embryo aneuploidy [20] and poor IVF outcome.

Endometrial receptivity

Supra-physiological serum estradiol and progesterone has been shown to be detrimental to endometrial receptivity [30–32]. High stimulation dose itself has been reported to have direct adverse effect on the endometrium [33]. Frozen-thawed embryo transfer give better LBR in presence of high response [34]. This has been regarded as one of the explanations as to why LBRs do not rise further after a certain number of oocytes retrieved in fresh transfer cycles.

The protocol

Antagonist vs Agonist

Gonadotropin releasing hormone (GnRH) antagonists as an agent to suppress luteinizing hormone (LH) have largely replaced GnRH agonist 'long protocol.' It has been confirmed in multiple systematic reviews and metaanalyses that the requirement for gonadotropin as well as the risk of ovarian hyperstimulation syndrome (OHSS) is less with GnRH antagonist protocol as compared with downregulation with GnRH agonist [35,36]. This is unsurprising since gonadotropin stimulation is initiated in a state of suppressed endogenous gonadotropins using GnRH agonist cotreatment, whereas this is not the case using GnRH antagonist protocols. Between 'fixed' and 'flexible' start antagonist protocols, flexible favors MS-IVF, as this has been shown to be associated with reduced gonadotropin dose and duration [37,38].

What type of gonadotropin

Generally, no particular type of gonadotropin has been proven superior to the other(s), except in the clinical scenario of hypogonadotropic hypogonadism. There has been a debate whether older or poor responder women may benefit from adding LH in the stimulation protocol: some systematic reviews and metaanalysis are in favor of adding LH [39]. More recent RCTs could not demonstrate any clinical advantage of adding recombinant LH to FSH in older women [40,41].

Gonadotropin dose

It has been widely accepted that 150 IU or lower daily gonadotropin dose is classed as "mild" stimulation. Most of the studies that compared MS-IVF with C-IVF used this dose of gonadotropin alone or in combination with oral agents in the MS-IVF arm. In the context of treating women who are poor responders of IVF, the Practice Committee of American Society of Reproductive Medicine (ASRM) also recommended a daily gonadotropin dose of $\leq$150 IU to define MS-IVF [42].

Adjustment with women's age: Women's age has been recognized as single most reliable predictor of ovarian response, embryo euploidy rate as well as livebirth. However, there is no clear evidence that adjustment of the stimulation dose according to age of the women improves the pregnancy outcome. A RCT comparing 150 IU and 225 IU daily dose reported higher ovarian response with 225 IU dose in younger age-group (23−32 years), but no difference in the oocyte yield on older women between 36−41-year age-group [43]. Although more cycles were cancelled due to inadequate response in 150 IU/d group, the incidence of ovarian hyperstimulation syndrome (OHSS) was high in 225 IU groups in this study. Randomized trial recruiting women with previous poor response to IVF found no difference in the pregnancy rates across all age groups (<36 years, 36−39 years and >39 years), whether had IVF with highly stimulated cycles or with completely unstimulated cycles [44]. There is not enough data to demonstrate the effect adjusting the stimulation dose according to woman's age only in unselected population undergoing IVF; most of the studies took age as well as ovarian reserve in the nomogram of calculating the starting dose of gonadotropin.

Adjustment with ovarian reserve: Ovarian reserve has generally been taken into account when deciding the starting gonadotropin dose; inclusion of anti-Mullerian hormone (AMH) or antral follicle count (AFC) has been shown to improve the prediction of ovarian response [45]. However, ovarian reserve is not recognized as a good predictor of pregnancy outcome e.g., livebirth [46]. Therefore, dose increment to compensate for low ovarian reserve has not been found to improve LBR [47,48]. Although more cycles reported to have been cancelled with 150 IU compared to 225 IU or 450 IU daily dose, cumulative LBR per patient was not compromised [48]. Thus, the value of ovarian reserve testing is limited to identifying the high responders who may require a lower than 150 IU daily dose to prevent OHSS. Also, an individualized dosing algorithm based on AMH and body weight has shown to achieve a higher proportion of patients reaching the desired preset number of oocytes being retrieved [49].

Adjustment with body-mass index (BMI): The bioavailability of a drug is depended on the BMI. Therefore, women with high BMI may need a higher starting dose such as 225 IU/ day, while those with low BMI and high ovarian reserve require a lower than 150 IU daily dose. High BMI has been implicated for suboptimal response and resultant cycle cancellation and por pregnancy outcome [50].

Oral agents

All metaanalysis of RCTs comparing protocols with or without clomiphene citrate (CC)/ aromatase inhibitors found significantly lower requirement of gonadotropin stimulation by their inclusion [51–53]. This finding applies in unselected populations as well as in poor responder women [54,55]. Aromatase inhibitors, letrozole and CC, when combined with low-dose gonadotropins, have been shown to give similar LBRs, ongoing pregnancy rates, and clinical pregnancy rates while reducing the incidence of OHSS in normal and high-responder population [4,51,53].

Letrozole has been mostly used in women suspected for a high as well as poor response—it has an added advantage of further lowering the risk of OHSS owing to its ability to maintain the serum estradiol level at a low level, particularly in women with polycystic ovarian syndrome (PCOS). The RCTs with letrozole reported no incidence of OHSS when letrozole was added to gonadotropin [56,57], while letrozole has been shown to reduce the gonadotropin requirement in high-responders [56,58]. Letrozole is an off-license medication in many countries mainly as a result of publication of an abstract reporting higher congenital malformation of children born following maternal exposure to letrozole [59]. However, subsequent studies, including a recent systematic review and metaanalysis of 18 RCTs, 21 cohort-control and 7 uncontrolled studies confirmed no such association [60].

CC when administered continuously with or without gonadotropin until the day of ovulation trigger can potentially prevent premature LH-surge and ovulation. By avoiding the use of GnRH-antagonists in this protocol, the treatment cost can be substantially reduced [61].

Why mild?

Clinical effectiveness

MS-IVF is being practiced in different parts of the world, yet it has not gained momentum mainly due to the general belief that although it is offers a higher level of safety, it does so at the cost of compromised pregnancy outcomes. However, with few exceptions [62], every individual RCT found no difference in the LBRs, ongoing pregnancy rates or clinical pregnancy rates between MS-IVF and C-IVF whether in poor responder, normal or high responder women; a description of all RCTs can be found in our recently published systematic reviews [4,54] as well as in other reviews [55]. The benefit of MS-IVF has gained more recognition in treating both poor responder and high responders. For the former group of women, MS-IVF has drawn attention by virtue of it being better tolerated (patient-friendly) and less costly with equivalent pregnancy outcome; this has led the ASRM to recommend MS-IVF (defined as $\leq$150 IU of daily gonadotropin) for this category of women (Practice Committee of the American Society for Reproductive Medicine. Electronic address, 2018). In contrast, recent European Society of Human Reproduction and Embryology (ESHRE) guidelines recommend use of no more than 300 IU for poor responders [63].

For the high responders, a daily dose of 100 IU has been shown to result in similar LBR's to that with 150 IU/day while reducing the incidence of OHSS [64]. Systematic reviews and metaanalyses comparing "mild" versus "conventional" or "low" versus "higher" stimulation IVF have regularly been published. Except for an earlier one [65], all other metaanalyses found lower stimulation doses to be as effective as higher doses with regards to pregnancy outcomes (Table 11.1). The ESHRE guidelines are also in favor of using low gonadotropin dose but advise against use of letrozole due to aforementioned reason [63].

There has been a concern that MS-IVF, by producing fewer oocytes and embryos, may compromise the cumulative pregnancy outcomes. This led to ESHRE recommend 150 or higher gonadotropin dose for normal responders of IVF [63]. However, none of the RCTs that investigated cumulative outcomes has found C-IVF to be better than MS-IVF, and likewise so is the metaanalysis of the pooled data from these RCTs (Fig. 11.2). A recent study from our group found only 9 and 11 oocyte optimized the fresh-cycles LBRs (40.3%) and cumulative (fresh and subsequent frozen transfer) LBRs (42.9%) respectively, with a mild stimulation protocol in normal responder women [67]. When a freeze all embryos (FAE) and subsequent transfer cycles from the high responder women were included, the cumulative LBR reached 53.1% by the time 15–20 oocytes were retrieved [68]. The cumulative LBR reached an optimum level of 53.8% when 9 embryos were obtained with MS-IVF protocol in this study.

It is commonly stated that time to pregnancy is an important parameter for patients and is increased in mild stimulation IVF. While there are data to support this theory for natural/modified natural cycle IVF [69], there is no hard evidence to confirm or refute it with Mild IVF. Admittedly, women's age, the number of embryos transferred, and the efficiency of embryo selection are the factors that have profound impact on the success of the fresh (first) embryo transfer (and therefore time to pregnancy). In terms of the oocyte number, a low oocyte

TABLE 11.1 Systematic reviews comparing mild/low-dose and conventional/high-dose IVF protocols.

Systematic reviews	Interventions	LBR	CLBR	CPR	OHSS	CCR
Poor responders						
[94]	CC+ Gn ±. vs C-IVF	↔	–	↔	–	↔
ASRM (2018)	Gn ± CC/Let vs C- IVF	↔	–	↔	–	
[55]	Lower vs Higher dose IVF	↔	–	↔	–	↔/↑e
[54]	Gn ± CC/Let vs C- IVF	↔c	↔c	↔c	–	↔ a
[66]	Gn ± CC/Let vs C- IVF	↔ b	↔c	↔b	–	↑ b
Normal and poor responders						
[51]	CC/Let + Gn vs C- IVF	↔ b	–	↔ d	↓ a	?
[95]	CC ± Gn vs C-IVF	↔	–	↔	↔	↑ ↔f
[53]	CC/Let ± Gn vs C- IVF	↔ a	–	↔ a/b	↓ a	↑ a
Normal responders						
[27]	Gn only low vs high-dose IVF	–	–	↔	↔	↔/↑g
[52]	CC+ Gn vs C-IVF	↔ a		↔ b	↓ a	↑f b
[96]	CC+ Gn. vs C-IVF	↔ b		↔ b	↓ a	–
[65]	Gn only/CC+Gn vs C-IVF	↓h c			↓	↑
[54]	Gn ± CC/Let vs C- IVF	↔ c	↔ b	↔ c	↓ c	↑ a
High responders						
[54]	Gn only low vs high-dose IVF	↔ c	↔ b	↔ c	↓ c	↔ c

[a] *very low quality of evidence;*
[b] *low quality evidence;*
[c] *moderate quality of evidence;*
[d] *high quality of evidence;*
[e] *No difference with Gn only protocol, high with oral agent incorporated protocol;*
[f] *no difference if antagonist was used;*
[g] *high with 100 IU dose versus 200 IU dose, no difference between 150 IU versus 225 IU dose;*
[h] *based on 1 RCT.*
CC = clomiphene citrate; *C-IVF* = conventional IVF; *Gn* = gonadotropin; *Let* = letrozole; ↔ = similar; ↑ = high with MD-IVF; ↓= low with MD-IVF; – = not mentioned;

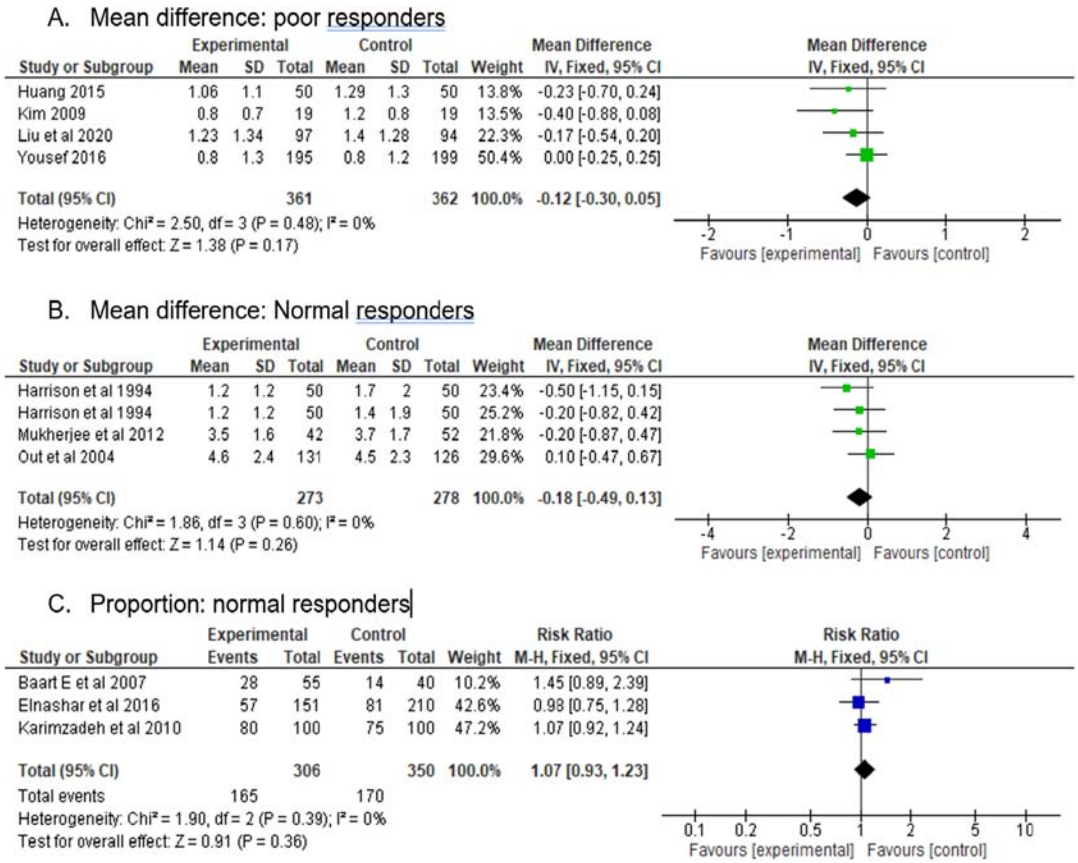

FIGURE 11.2 Mean or proportion of high-grade embryos- mild vs conventional IVF.

yield due to poor prognostic factor(s) would take longer time to achieve a pregnancy, while good prognosis patients, as mentioned earlier, may easily produce enough number of oocytes to achieve the same in the first attempt. In theory, milder stimulation may reduce the chance of freeze all embryos due to high-response and thereby, lower the risk of a long "time to pregnancy" interval. However, there is no data to prove this hypothesis.

Perinatal outcomes

A metaanalysis of RCTs showed natural and natural modified IVF was associated with a lower risk of preterm delivery and low birthweight babies [70]. A more physiological endocrine condition at the endometrial level has been implicated. Whether the same is applicable with mild ovarian stimulation is unproven. A large study from national database found a link between high number of oocytes and premature or low birthweight babies [71]. High serum estradiol level itself has been reported to result in low birthweight babies [72]. In contrast, an analysis of another national database found no such association with the oocyte yield [73].

Safety

In recent years, the risk of OHSS has largely been reduced by some very effective interventions including GnRH-agonist for ovulation trigger and FAE. However, cases of OHSS has regularly been reported despite agonist trigger, particularly when a fresh embryo transfer followed [74]. Individualized dosing has been found to decrease the incidence of OHSS; for example, in one RCT in high-responder women, a 100 IU/day dose of gonadotropin resulted in a lower OHSS rates when compared with standard 150 IU/ day dose, without compromising the LBRs [64]. Metaanalyses of pooled data from RCTs on the normal and high responder women has demonstrated a lower incidence of OHSS with MS-IVF compared with C-IVF [4]. Severe OHSS was not found in the retrospective analysis of 862 IVF cycles when a strategy of mild ovarian stimulation (≤150 IU/ day) was combined with a low threshold of FAE and subsequent frozen embryo transfer [4]. Aiming for no more than 15-18 oocytes has been suggested to be a balance between a reasonable cumulative LBRs and the risk of OHSS [75].

High oocyte yield (>15 oocytes) has been shown to be associated with increased risk of thromboembolism [73]. OHSS itself predisposes patients to the risk of venous or arterial thromboembolism during the IVF cycles as well as in consequent pregnancies. High ovarian response (>20 oocytes) subjects the women to complications during oocyte retrieval which include pain, bleeding and ovarian torsion [76,77]. Therefore, effort should be made to avoid such complications by employing milder stimulation.

Patient's experience

The importance patient's tolerance to IVF treatment is paramount. Stress of the treatment cycle, depression following a failed cycle and as a result, discontinuation of further cycle is reported to be less likely with MS-IVF [78,79]. OHSS itself renders patients to considerable physical stress and discomfort. Emphasizing the "patient-friendly" nature of MS-IVF, the ASRM recommended to consider it in treating the poor responder patients (Practice Committee of the American Society for Reproductive Medicine. Electronic address, 2018).

Cost

High treatment cost and disparity in the funding policy are a hindrance to uniform accessibility of IVF treatment worldwide [80,81]. Comparison of cost between MS-IVF and C-IVF have been done in several retrospective and prospective RCTs Table 11.2. The treatment cost between the two approaches is strikingly different in treating poor responders [54]. Even the cumulative cost of 5 cycles of mild IVF was estimated to be less than 4 cycles of C-IVF in a RCT [82].

Limitations

Cumulative pregnancy outcome

Based on recent large systematic reviews, some aspects of MS-IVF, once believed to be drawbacks, are not true. Most of the systematic reviews of RCTs have shown a lower oocyte yield associated with MS-IVF in normal responders [4,27,65], as well as in poor

TABLE 11.2 Studies comparing the treatment cost between mild and conventional IVF.

Studies	Stimulation dose (daily)	Cost mild IVF	Cost conventional IVF	Pregnancy outcome
Poor responders				
[97]	CC 150 mg only vs 450 IU gonadotropin	€ 81294 per livebirth	€ 113107 per livebirth	No difference
[48]	150 vs 300/ 450 IU gonadotropin	€ 5 289	€ 6 397	No difference
Normal responders				
[82]	150 IU gonadotropin day 5 vs day 5 start	€ 8 333	€ 10 745	No difference
[98][a]	150 (day 3) vs 150-300 IU gonadotropin	€ 136	€ 2 160	No difference
[99]	Let + 75 IU vs 150-225 IU gonadotropin	Mild stimulation: 34% cost-saving		No difference
[61]	CC 50 mg from Day 2, gonadotropin 150 IU from day 5	IVF with CC continued till trigger without GnRH antagonist/ agonist costs: $ 675		
High responders				
[64]	100 vs 150 IU gonadotropin	€ 4 622	€ 4 714	No difference

[a]*converted from Yuan to Euro.*
CC, clomiphene citrate; *Let*, letrozole.

responders [54,55]; however, this does not compromise the cumulative LBRs when MS-IVF is applied [4,66] (Fig. 11.3).

Goal of "completing family"

There is a recent trend of targeting high number of oocytes (under the perceived protective umbrella of GnRH-agonist trigger), which may enable a couple having more than one child to complete the family from one oocyte retrieval. This "one-and-done" approach sounds reasonable in the sense that it may avoid repeated IVF cycles and resultant complications and cost of the treatment. However, how many children is classed as "completing family" and how many oocytes are needed to achieve that goal is not well defined. Also, to what extent it is possible to achieve the target of >1 child from 1 IVF cycles is a big question. A study with conventional ovarian stimulation found approximately 1 in 5 women are capable of generating enough oocytes to fulfil the goal of 2 children [83]. Thus, this target can be achieved in only good prognosis patients with high ovarian reserve. Trying to achieve this goal for everyone would put the patients in both physical and financial distress with no benefit for the majority.

Crude data on the number of oocytes needed to achieve >1 child is lacking. The study by Vaughan et al. reported 53.9% and 57.9% women had ≥1 LBRs with the oocytes number of 15-25 and >25 oocytes, respectively [83]. The incidences of OHSS were 3.3% and 8.5% for the respective oocyte cohort in this study. Another study showed that about 25 oocytes are required to expect 2 children, and near 40 oocytes to except 3 children [74]. The same study reported the rate of severe OHSS to be 7% when >25 oocytes are retrieved following human

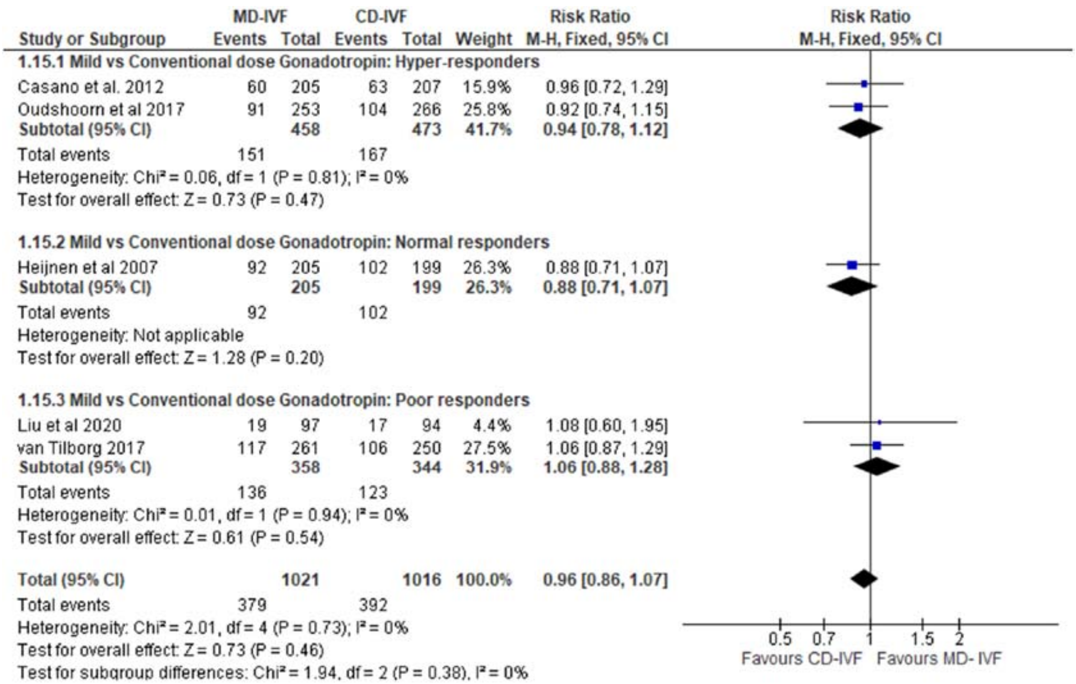

FIGURE 11.3 Cumulative livebirth rates—mild vs conventional IVF.

chorionic gonadotropin trigger, and around 1% even with agonist trigger [74]. Other studies that demonstrated an increasing cumulative LBRs with the number of oocytes also found a rising incidence of OHSS, particularly when more than 15–18 oocytes are retrieved [73,84–86]. It is apparent that the number of oocytes that may offer ≥1 children in a minority of women, also put all women who are being treated for this goal into a significant risk of OHSS.

Cycle cancellation

There has been a concern that MS-IVF may lead to a higher risk of cycle cancellation. Two basic principles critically influence this outcome: the criteria for cycle cancellation and the use of antagonists. Many earlier studies did not use antagonist but used GnRH agonist in the C-IVF arm [87–89]. As a result, more cycles were cancelled due to premature ovulation. Some systematic reviews found no difference in the incidence of cycle cancellation when compared with low- and high-dose gonadotropin regimens in poor responder population [54,55], while other reviews reported higher cancellation rates [66]. Among the normal responders, no difference in the cycle cancellation was noted when the studies that did not use of antagonist were excluded [52]. In addition, most of the studies set the development of fewer than 3 follicles as the criterion for cycle cancellation; while this could be acceptable in conventional protocols, it may not be applicable in a MS-IVF cycle that expects a lower oocyte yield.

Future prospects

Recently, the World Health Organization has recognized that fertility treatment including IVF services should be developed throughout the world to give people of all socio-economic background a uniform access to fertility treatment [90]. As a result, the quest for an effective, safe and affordable IVF treatment has now received some momentum. This awareness is reflected in different editorials [80,91]. Of late, a review of different IVF protocols for middle and low economic population calls for a consensus on cost-effective fertility care including IVF protocols for wider affordability and acceptability [92]. Mild stimulation IVF protocols have long been proposed as a cost-effective and safer "whole world" option [1,81]. From the randomized trials in recent years, it appears that mild stimulation IVF has the potential to combine all required elements for global acceptability, effectiveness, safety, and affordability. It has the potential to reduce the treatment burden as well as economic burden of the individual or state funds, with no compromise in the treatment outcomes compared to conventional IVF (Table 11.2).

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CHAPTER

12

The use of progestins to prevent the LH surge in IVF cycles

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Alternative protocols for ovarian stimulation

In vitro fertilization (IVF) has been increasingly employed over the last few decades. The so-called conventional ovarian stimulation (COS), which is the therapeutic core of IVF, has long been the gold standard of ovarian stimulation. COS consists of the administration of different drugs:

- Gonadotropins, starting from the second or the third day of the menstrual cycle (MC 2–3), with the aim of recruiting the growth of multiple follicles;
- Gonadotrophin-releasing hormone (GnRH) analogs, agonists or antagonists, to inhibit the untimely surge of luteinizing hormone (LH), which can be elicited by the excess of estrogens produced by the growing follicles in the ovaries;
- An ovulation trigger, which can be a bolus of human chorionic gonadotropin (hCG), a GnRH agonist (in GnRH antagonist protocols), or both.

Fig. 12.1 shows a scheme of a classic COS protocol, with a GnRH antagonist used in this specific case.

Despite their long-term worldwide use, GnRH analogs are burdened by a series of disadvantages. GnRH agonists act by binding to pituitary receptors in the hypophysis and inducing the release of large amounts of follicle-stimulating hormone (FSH) and LH (the so called "flare-up" effect), and an increase in the number of GnRH receptors (upregulation). After prolonged use, internalization of the GnRH agonist–receptor complex occurs, which results in a decrease in the number of GnRH receptors (downregulation). As a result, the pituitary becomes refractory to stimulation by GnRH, leading to a decrease in circulating gonadotropins and thus preventing a premature LH surge [1]. As a result of

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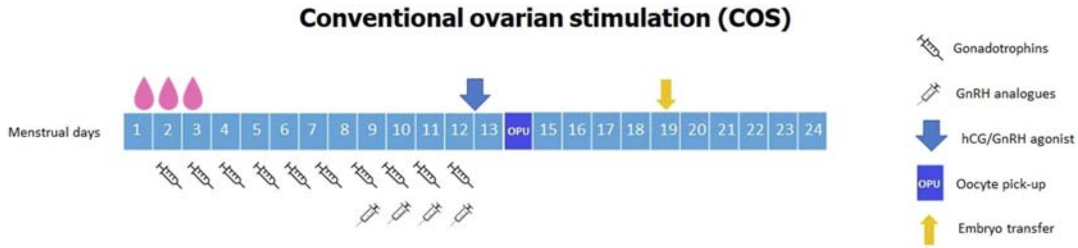

FIGURE 12.1 Conventional ovarian stimulation using a GnRH antagonist to inhibit ovulation.

the flare-up effect, they can induce the formation of ovarian cysts [2]. Since their use makes mandatory to induce ovulation with a bolus of hCG, they are associated by an increased risk of ovarian hyperstimulation syndrome (OHSS) from an hCG trigger [3]. Moreover, their administration leads to their accumulation and subsequent concentration in the peripheral circulation, with consequent possible extrapituitary effects [2].

GnRH antagonists act in a different manner, by promptly suppressing pituitary gonadotropins' secretion by GnRH–receptor competition [1,4]. Gonadotropin secretion is decreased within hours of antagonist administration, and no flare-up effect occurs. The discontinuation of GnRH antagonist treatment results in a rapid, predictable recovery of the pituitary–gonadal axis as the pituitary receptor system remains intact. Protocols with GnRH antagonists have fewer complications and are more convenient for patients rather than GnRH agonists because of the shorter treatment time and lower number of injections. However, the cycle cancellation rate seems higher in women who received GnRH antagonist protocols compared with GnRH agonist protocols [5]. A varied proportion (0.34%–38%) of patients using a GnRH antagonist has been demonstrated to experience a premature LH surge, especially seen in older women and women with diminished ovarian reserve [6,7]. Indeed, GnRH antagonists, at the doses currently used in IVF programs, have been demonstrated to be unable to block the stimulating effect of exogenous estrogens on the LH surge in women with unstimulated ovaries, suggesting that their clinical efficacy in IVF cycles is determined by the ovarian stimulation process [8,9].

Beside their specific side effects, GnRH analogs need daily subcutaneous administration with poor manageability, and they are burdened by conspicuous costs. These disadvantages have prompted interest in exploring convenient alternatives to prevent premature LH surges in ovarian stimulation.

After the publication of two studies on luteal phase ovarian stimulation (LPOS) [10,11] demonstrating consistent LH suppression with no spontaneous surge when the stimulation is performed in a climate of elevated serum endogenous progesterone, investigation has been carried out into whether exogenous progesterone could be applied in ovarian stimulation cycles with the aim of inhibiting ovulation. Progesterone's effect on pituitary suppression is well known, and it has been the basis of hormonal contraception over the last 60 years. However, its detrimental effect on endometrial receptivity makes its use during ovarian stimulation potentially harmful for a subsequent embryo transfer (ET), since it determines a shift in the window of endometrial receptivity. Vitrification techniques developed in the last 20 years have enabled the achievement of oocyte and embryo cryopreservation with an excellent

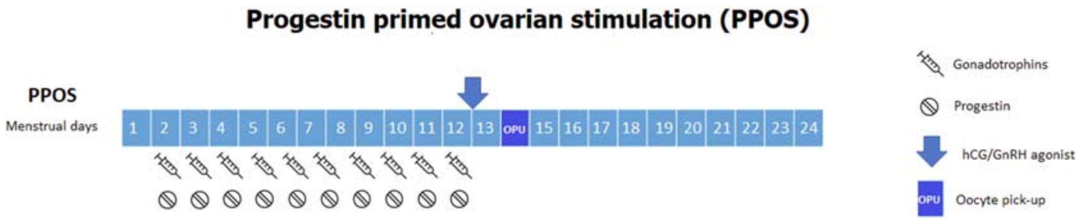

FIGURE 12.2 Progestin-primed ovarian stimulation.

survival rate of over 90%, and pregnancy chances equivalent to those with fresh transfers [12–14] or even sometimes better [15]. Improvements in such techniques for freezing embryos and oocytes for assisted reproductive techniques have enabled to move away from the standard sequence of ovarian stimulation–retrieval–transfer, with the possibility to "freeze" the entire cohort of embryos or oocytes, or a mixture, created from a stimulation cycle, and to perform the embryo transfer in a subsequent cycle (Fig. 12.2).

Mechanism of pituitary suppression by progestins

Despite progesterone's extended use in hormonal contraception and the large number of studies on its role in the control of gonadotropin secretion, the exact mechanisms through which progesterone regulate the LH surge, interacting with estradiol and other hormones, are not completely understood.

The interaction with estradiol is crucial for the ovulation dynamics. It is well known that while estradiol upregulates the expression of progesterone's receptors in the tissues, progesterone downregulates its own receptors. Progesterone action may be synergistic with, or antagonistic to, the actions of estradiol: this depends on hormone ratios and timing of exposure [16]. Various studies in fact report a facilitating role of progesterone on the preovulatory LH peak. Experiments have shown a rise in progesterone levels a few hours before the LH surge in the preovulatory period [17]; nonetheless, a peak in LH secretion is achieved only when progesterone is administered after an estrogen priming [18]. At the end of the follicular phase, increasing progesterone production by the granulosa cells is hypothesized to act as a signal to the hypothalamus of the follicle's readiness to rupture by stimulating gonadotrophin secretion [19]. On the other hand, higher levels of progesterone have long been shown to block the LH surge [20,21], and its inhibiting role on follicular growth has been the foundation of progestin-only contraceptives, which after a prolonged use suppress follicular growth and thus block ovulation. Further studies are needed to better define the interaction between progesterone and other molecules—hormones, inhibins, and cytokines on ovulation dynamics.

Progestins

Besides natural progesterone, produced and secreted normally in the human female by the corpus luteum, the placenta, and in small quantities by the adrenal cortex, there is a broad spectrum of steroids with progesterone-like actions, derived from different parent compounds, named progestins (Table 12.1).

TABLE 12.1 Classification of progestins.

Progestin	Example
Progesterone	Natural progesterone
Retroprogesterone	Dydrogesterone
Progesterone derivative	Medrogestone
17α-Hydroxyprogesterone derivatives (pregnanes)	Medroxyprogesterone acetate, megestrol acetate, chlormadinone acetate, cyproterone acetate
17α-Hydroxynorprogesterone derivatives (norpregnanes)	Gestonorone caproate, nomegestrol acetate
19-Norprogesterone derivatives (norpregnanes)	Demegestone, promegestone, nesterone, trimegestone
19-Nortestosterone derivatives (estranes)	Norethindrone, norethisterone acetate, lynestronol
19-Nortestostrone derivatives (gonases)	Norgestrel, levonorgestrel, desogestrel, dienogest
Spironolactone derivatives	Drospirenone

Close to the natural progesterone, there are retroprogesterone, followed by the pregnane-(17-hydroxyprogesterone, C-21) derivatives and the 19-norprogesterone-(19-norpregnane, C-20) derivatives. The 19-nortestosterone derivatives, subdivided in estranes (C-18) and gonanes (C-17), are a clinically important group and the basis for the success of hormonal contraception. Some spirolactone derivatives have also been developed for clinical use [22].

Any compound with progesterone-like activity has, of course, to be able to bind to the progesterone receptor. There are two forms of the progesterone receptor, derived from a single gene: PR-A and PR-B. The two forms have different specifications: while progesterone receptor A mediates inhibitory activities and also represents a transcriptional inhibitor of other steroid receptors, progesterone receptor B mediates progesterone's stimulatory activities [23].

All the progestins share the capacity of inducing characteristic changes in the estrogen-primed endometrium, the so-called progestogenic effect. The final progestogenic activity of any progestin also depends on the route and timing of its administration. This effect varies between different compounds, and it is often expressed by the difference in the dose required for endometrial transformation, called the transformation dose. Beside the progestogenic effect, another activity elicited by progestins is the capacity of ovulation inhibition. In the field of contraception, progestins are frequently selected for clinical use on the basis of the differences in the dose necessary for ovulation inhibition. While a dose of 10 mg/day of medroxyprogesterone acetate (MPA) is sufficient to inhibit ovulation, 300 mg/day of natural progesterone (micronized progesterone [MIP]) are necessary to obtain the same effect. This explains easily why over decades certain progestins have been favored over others: since a lower dose is required for the same effect, costs and potential side effects are, as a consequence, also lower (Table 12.2) [22].

TABLE 12.2 Progestogenic effectivity on the endometrium and ovulation inhibition dose of each progestin.

Progestin	Ovulation inhibition dose mg per day p.o.	Transformation dose mg per day p.o.
Progesterone	300	200–300
Dydrogesterone	>30	10–20
Medroxyprogesterone acetate	10	5–10
Cyproterone acetate	1	1
Norethisterone	0.5	/
Levonorgestrel	0.05	0.15
Desogestrel	0.06	0.15
Gestodene	0.03	/
Dienogest	1	/
Nomegestrol acetate	5	5

PPOS protocols: outcomes and schemes of treatment

Progestin-primed ovarian stimulation (PPOS) is shown to effectively inhibit spontaneous ovulation by blocking the premature LH surge in IVF cycles since the first papers published in literature [3,24–28]. There was no statistically significant difference in the number of oocytes retrieved, mature oocytes (MII), and top-quality embryos obtained when comparing patients treated with a PPOS protocol with patients treated with COS. Clinical pregnancy rates (CPRs) and live birth rates (LBRs), similarly, were not significantly different in the two groups of patients. OHSS risk has been demonstrated to be lower when progestins are used instead of GnRH analogs to inhibit ovulation. All the previous data have been confirmed by recent metaanalysis, conducted on large sets of patients [29,30].

Studies conducted in order to analyze neonatal outcomes from cycles with progestins administered during ovarian stimulation have shown that PPOS neither has any effect on neonatal outcomes in IVF nor does it elevate the rate of congenital malformations [31], reassuring on its safety.

The most described progestins in scientific literature on PPOS are MPA, MIP, and dydrogesterone (DYG). Their efficacy in preventing the LH surge and their effect on oocytes retrieved, embryos obtained, and pregnancy outcomes is similar [29]. A randomized controlled trial (RCT) by Begueria et al. showed a lower LBR when MPA was compared with a GnRH antagonist in an oocyte donation program, while a comparable number of MII oocytes was yielded in the two groups [32]; a recent RCT conducted on a larger number of patients, on the contrary, showed similar pregnancy outcomes in the recipients of oocytes from a PPOS protocol with MPA compared to those obtained from a COS with a GnRH antagonist [33]. MPA has been largely used in hormonal contraception for its ovulation

TABLE 12.3 Characteristics of included studies.

	Study group		Control group		
Study	Sample size	Progestin	Sample size	Intervention	Outcomes
Kuang et al., 2015 [24]	150	MPA 10 mg/die	150	Short protocol	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Zhu et al., 2015 [3]	187	MIP 200 mg/die	187	Short protocol	1, 2, 3, 4, 6, 7, 8, 9, 10
Wang et al., 2016 [11,25]	60	MPA 10 mg/die	60	Short protocol	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Zhu et al., 2016 [26]	123	MIP 200 mg/die	77	Short protocol	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Zhu et al., 2017 [27]	75	MIP 100 mg/die	75	MIP 200 mg/die	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Huang et al., 2019 [28]	63	MIP 100 mg/die	123	Short protocol	1, 2, 3, 4, 6, 7, 8, 9, 10
Beguería et al., 2019 [32]	86	MPA 10 mg/die	87	Short protocol	1, 4, 6, 7, 10
Giles et al., 2021 [33]	161	MPA 10 mg/die	156	Short protocol	1, 3, 4, 6, 7, 8, 9, 10
Yildiz et al., 2019 [34]	87	MPA 10 mg/die (fPPOS)	87	Short protocol	1, 2, 4, 6, 7, 8, 9, 10
La Marca et al., 2020 [43]	48	MPA 10 mg/die	144	Short protocol	6, 7, 8, 9, 10, 11

Outcomes. 1: Clinical pregnancy rate; 2: Premature LH surge; 3: Miscarriage rate; 4: Live birth or ongoing pregnancy rate per woman; 5: OHSS; 6: Gn duration; 7: Gn dose; 8: Oocytes retrieved; 9: MII oocytes; 10: Number of embryos; 11: Number of euploid blastocysts.

inhibition capacity (as can be seen in Table 12.2). After oral intake, it does not undergo any first pass effect; its bioavailability is nearly 100%. It has a strong glucocorticoid effect, while it does not have neither antiandrogenic nor antimineralocorticoid effect [22]. DYG appears to be a highly selective progestin which, due to its structure, binds almost exclusively to the progesterone receptor: due to its selectivity, effects not mediated by the progesterone receptor are minimal or absent. Its ovulation inhibition capacity is intermediate between that of MPA and that of MIP. Regarding the doses used in PPOS protocols, the lowest doses studied have shown to be sufficient in inhibiting the LH surge: 4 mg/day for MPA, 100 mg/day for MIP, and 20 mg/day for DYG (Table 12.3) [29].

Besides the classic PPOS protocol, consisting in the oral administration of a progestin from MC day 2–3, concomitant with the subcutaneous administration of a gonadotropin, a so-called "flexible" PPOS (fPPOS) has also been described: the gonadotropin is started on MC day 2–3, while the progestin is started on stimulation day 7 or when the leading follicle reached 14 mm, whichever comes first. When compared to a GnRH antagonist protocol, fPPOS has been shown to require the same stimulation days, a similar gonadotropin consumption and a similar oocytes and embryos outcomes (Fig. 12.3) [34].

Advantages and limits

The obvious advantages of the use of progestins to prevent the LH surge in IVF cycles are their low costs and their manageability. The comfort of use is given by the fact that in the PPOS cycles, the administration of progestins is performed by a daily oral assumption, while

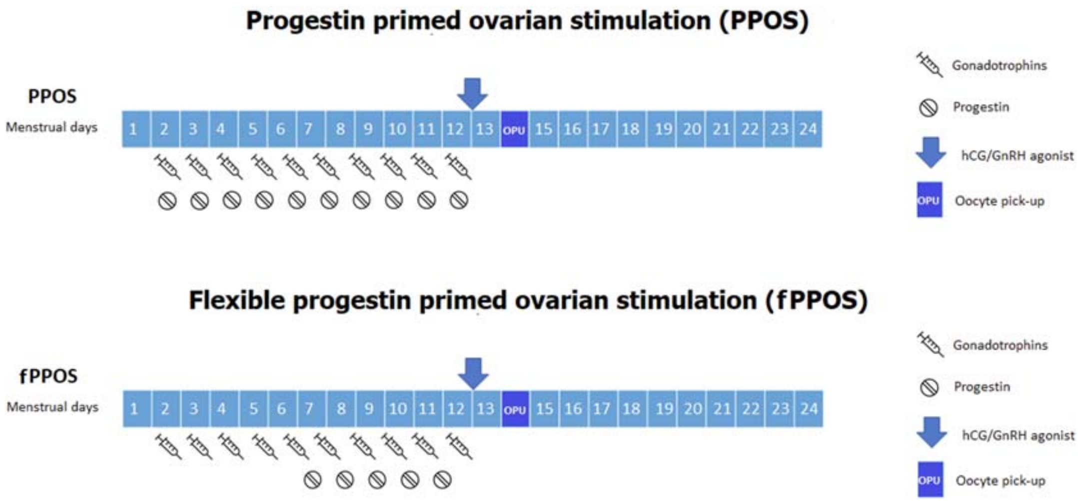

FIGURE 12.3 Progestin-primed ovarian stimulation versus flexible progestin-primed ovarian stimulation.

the analogs that are administered during COS have to be injected. The economic convenience in using progestins rather than GnRH analogs is easily understood by comparing the average cost of pituitary suppression therapy with both the drugs. The average cost of a single depot injection of a GnRH agonist or multiple injections of a GnRH antagonist (a single COS cycle often requires three to five injections) is about 40 times the cost of a package of an oral progestin, which contains a quantity of capsules sufficient to obtain pituitary suppression in a PPOS cycle.

The assumption of cost effectiveness of PPOS over COS cycles is based on the lower cost of progestins when compared with GnRH analogs, especially GnRH antagonists. However, PPOS requires elective cryopreservation of all embryos and deferring the ET to a subsequent menstrual cycle, which in turn brings in the direct costs of embryo thawing, monitoring for frozen ET, commuting to the clinic and indirect costs associated with loss of working hours for the patient. PPOS is therefore only more cost effective than the COS protocol in planned freeze all cycles [34]. As a consequence, PPOS is preferred in all the situations in which a freeze-all strategy is mandatory: oocyte donation cycles, predicted hyperresponder patients, patients candidated to preimplantation genetic testing (PGT), and other situations in which the fertility potential is desired to be saved for individual reasons (such as social freezing) or for medical reasons (i.e., fertility preservation in oncological patients).

In the past decades, some controversies have been raised over the effect of oocytes exposition to progesterone and progestins. Some studies have reported diminished rates of blastocysts formation from oocytes exposed to progesterone in vitro [35,36]. However, clinical studies regarding LPOS report reassuring observations in humans [37,38], suggesting safety of exposure of the developing follicles to high progesterone levels during the luteal phase. Likewise, IVF cycles with early follicular progesterone exposure due to premature luteinization have similar fertilization and obstetric outcomes as compared to cycles without premature luteinization [39,40], as long as ET is delayed to a subsequent cycle [41,42]. Since there is a direct relationship between the euploidy status of blastocysts and the probability of having a live birth, euploidy rate has been assumed as an intermediate marker for obstetric outcomes in a study comparing blastocyst euploidy rate between women treated with PPOS or with COS [43], with similar results obtained with the two protocols. Similarly, neonatal outcomes from cycles with progestins administered during ovarian stimulation have given reassuring results [44].

In conclusion, PPOS protocols for IVF are a safe alternative to GnRH analogs' administration for preventing LH surge without affecting pregnancy outcome or reducing the number of euploid blastocysts obtained. In the same manner, PPOS have not shown to be associated with an increase in adverse neonatal outcomes or congenital malformations. Nonetheless, more high-quality RCTs comparing PPOS with GnRH analogs reporting LBRs per women starting stimulation ideally in a cumulative manner are needed from different centers and countries; long-term outcomes of children born from PPOS cycles have to be collected, and different progestins, with different routes of administration, and PPOS protocols, for example, flexible PPOS, require further assessment.

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CHAPTER

13

Triggering final oocyte maturation. The role of human chorionic gonadotropin, gonadotropin-releasing hormone agonist, and dual trigger

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Introduction

In the course of the ovulatory cycle, ovulation is controlled by the hypothalamus and triggered by the midcycle follicle-stimulating hormone (FSH) and luteinizing hormone (LH) surges released from the pituitary gland. The mid-cycle LH surge causes a loss of gap junctions between the oocyte and cumulus cells, cumulus expansion, germinal vesicle breakdown, resumption of meiosis, and luteinization of the granulosa cells. Moreover, luteinization, that is, the consequent increase in progesterone synthesis, facilitates the positive feedback action of estradiol to induce the concomitant midcycle FSH peak [1]. This peak FSH assures an adequate complement of LH receptors on the granulosa layer and the synthesis of hyaluronic acid matrix that facilitates the expansion and dispersion of the cumulus cells, allowing the oocyte-cumulus cell mass to become free-floating in the antral fluid [1].

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Human chorionic gonadotropin, a surrogate to the naturally occurring LH surge

As part of a standard/conventional ovarian stimulation (OS) regimen, final follicular maturation is usually triggered by one bolus of human chorionic gonadotropin (hCG) (5000–10,000 units) that is administered as close as possible to the time of ovulation (i.e., 36 h before oocyte recovery) [2]. hCG, a surrogate to the naturally occurring LH surge, induces luteinization of the granulosa cells, final oocyte maturation, and resumption of meiosis.

Ovarian hyperstimulation syndrome

The use of hCG for final follicular maturation increases the risk of ovarian hyperstimulation syndrome (OHSS) due to its prolonged half time compared to LH, its distinct stimulation of intracellular signal cascades, and its effect on endothelial cells function [2,3]. Moreover, hCG causes higher degree of systemic inflammation and expression of vasoactive peptides compared to LH [4].

OHSS almost always presents either three–seven days after hCG administration in susceptible patients (early onset) or during early pregnancy, 12–17 days after hCG administration (late onset). Individualization of treatment according to the specific risk factor and the specific response in the current cycle with the option of freezing of all embryos, or replacement of only a single embryo, has the potential of reducing the risk and the severity of the syndrome in susceptible cases [5].

OS which combines gonadotropin-releasing hormone (GnRH) antagonist cotreatment and GnRH agonist (GnRHa) trigger has recently become a common tool aiming to *eliminate* severe early OHSS and to support the concept of an OHSS-free clinic [6,7]. However, due to the reported significantly reduced clinical pregnancy and increased first trimester pregnancy loss [8,9], efforts have been made to improve reproductive outcome. Three optional strategies were suggested aiming to improve outcome: freeze-all policy, fresh transfer and intensive luteal support, and fresh transfer and low-dose HCG supplementation [10].

hCG versus GnRHa trigger

While comparing the effect of hCG versus GnRHa trigger on the different follicular maturation variables following an in vitro fertilization (IVF) treatment cycle, studies have revealed that the number of oocytes retrieved, percentage of mature oocytes, and number of top-quality embryos were either comparable or in favor of the GnRHa trigger. Moreover, while studying the downstream effects of LH receptor activation by LH or hCG, it was demonstrated that LH has a greater impact on AKT (protein kinase B, also known as AKT, is the collective name of a set of three serine/threonine-specific protein kinases that play key roles in multiple cellular processes) and extracellular signal-regulated protein kinase (ERK1/2) phosphorylation, responsible for granulosa cells proliferation, differentiation, and survival, while hCG generates higher intracellular cAMP accumulation, which stimulates steroidogenesis (progesterone production) [11,12].

Following the aforementioned observations, GnRHa combined with hCG trigger, for final follicular maturation, has been implemented in clinical practice, and the different modes and timing of administration should be appropriately tailored to the various subgroup of IVF patients.

Triggering final follicular maturation in patients not at risk to develop severe OHSS

The aforementioned observations demonstrating a comparable or even better oocyte\embryos maturity and quality following GnRHa trigger, as compared to hCG trigger, and the different effects of LH and hCG on the downstream signaling of the LH receptor have led to a new strategy for final follicular maturation, the concomitant administration of both GnRHa and a standard bolus of hCG (5000–10,000 units) prior to oocyte retrieval, aiming to improve oocyte and embryo quality and the consequent IVF cycle outcome.

Dual trigger-standard hCG dose concomitant with GnRHa, 35–37h before oocyte retrieval

Standard hCG dose concomitant with GnRHa (dual trigger), 35–37h before oocyte retrieval. Lin et al. [13], in their retrospective cohort study, have compared IVF outcomes in normal responders patients undergoing controlled ovarian stimulation (COH) using the GnRH antagonist with either a standard dosage of hCG trigger (6500 IU of recombinant hCG) or the dual trigger (0.2 mg of triptorelin and 6500 IU of recombinant hCG) 35–36h prior to oocyte retrieval. The dual trigger group demonstrated statistically significantly higher number of oocytes retrieved, matured oocytes, and number of embryos cryopreserved, with the consequent significant increase in implantation, clinical pregnancy, and live-birth rates, as compared with the hCG-only trigger group.

In a subsequent prospective randomized controlled trial of normal responder patients, Decleer et al. [14] compared IVF outcome following either 5000IU of hCG trigger or a combination of GnRHa plus 5000IU of hCG concomitantly, 36h prior to oocyte retrieval. While no in-between groups differences were observed in the mean number of oocytes retrieved, mature oocytes, or pregnancy rates, the number of patients who received at least one embryo of excellent quality and the number of cryopreserved embryos were significantly higher following the dual trigger.

Griffin et al. [15] evaluated the effect of the dual trigger (GnRHa and hCG 5000 IU or 10,000 IU, 35–37h) prior to oocyte retrieval in patients with a previous history of >25% immature oocytes retrieved. Despite a significantly higher proportion of mature oocytes retrieved with the dual trigger, the observed IVF outcome remained poor, probably due to patients' underlying oocyte dysfunction.

In a systematic review and metaanalysis of four studies in 527 women undergoing IVF treatment, women receiving a dual trigger had a significantly higher pregnancy rate compared with those receiving hCG alone (RR 1.55 [95% CI 1.17, 2.06]) [16]. However, there was no significant difference in the number of oocytes retrieved or implantation rates between the groups. A recently published RCT, randomizing 155 normal responder patients to receive either hCG or a dual trigger for final oocyte maturation, demonstrated that using a dual trigger for final follicular maturation resulted in improved outcomes [17]. The use of a dual trigger significantly increased the number of oocytes retrieved (11.1 vs. 13.4, $P = .002$), MII oocytes (8.6 vs. 10.3, $P = .009$), total number of blastocysts (2.9 vs. 3.9, $P = .01$), and top-quality blastocysts transferred (44.7% vs. 64.9%; $P = .003$) compared with the hCG group. Moreover, the clinical pregnancy rate (24.3% vs. 46.1%, OR 2.65 [95% CI 1.43, 1.93], $P = .009$) and the live birth rate per transfer (22% vs. 36.2%, OR 1.98 [95% CI 1.05, 3.75], $P = .03$) were significantly higher in the dual trigger group compared with the hCG group [17].

Double trigger-GnRHa 40h and standard hCG added 34h prior to OPU, respectively, in patients with abnormal final follicular maturation

GnRHa 40h and standard hCG added 34h prior to oocyte pick-up (OPU) (double trigger), respectively. Recently, Beck-Fruchter et al. [18] have described a case of recurrent empty follicle syndrome, successfully treated by ovulation trigger with GnRHa 40h and hCG added 34h prior to oocyte retrieval. They assumed that by prolonging the time between ovulation triggering and OPU [19] and the GnRHa trigger with the consequent simultaneous induction of an FSH surge, the "double trigger" could overcome any existing impairments in granulosa cell function, oocyte meiotic maturation, or cumulus expansion, resulting in successful aspiration of mature oocytes, pregnancy, and delivery.

Prompted by the aforementioned observations, we offered the double trigger to two group of patients demonstrating abnormal final follicular maturation despite normal response to COH, those with low (<50%) number of oocytes retrieved per number of dominant follicles >14 mm in diameter on day of hCG administration [20], and those with low proportion of mature/metaphase-II (MII) oocytes (<66%) per number oocytes retrieved [21].

In the group of patients with low (<50%) number of oocytes retrieved per number of dominant follicles, following the double trigger, patients had significantly higher number of oocytes retrieved, number of 2 PN, number of embryos transferred, and significantly higher proportions of the number of oocytes retrieved to the number of follicles >10 mm and >14 mm in diameter on day of hCG administration, with a tendency toward a higher number of top-quality embryos, as compared to the hCG-only trigger cycles [20].

Moreover, in those with low proportion of MII oocytes (<66%) per number oocytes retrieved, following the double trigger, patients yielded significantly higher number of MII oocytes and proportion of MII oocytes per number of oocytes retrieved, with the consequent significantly increased number of top-quality embryos, as compared to the hCG-only trigger cycles [21].

Double or dual trigger in poor responder patients

According to the Bologna criteria, the minimal criteria needed to define poor ovarian response (POR) are the presence of at least two of the following three features: (i) advanced maternal age (≥40 years) or any other risk factor for POR; (ii) a previous POR (≤3 oocytes with a conventional stimulation protocol); and (iii) an abnormal ovarian reserve test [22]. One of the major unnoticed concerns in this group of poor responders is the observed high prevalence of premature luteinization\ovulation [23,24], which may be overcome by early triggering of final follicular maturation, while approaching a follicular size of 15–16 mm, and by shortening the duration between the trigger and OPU. However, since shortening the interval between hCG priming and oocyte retrieval may decrease the percentage of mature oocytes [19], an additional measure to improve the number of oocytes retrieved to the number of follicles >10 mm, and the proportion of mature oocytes should be implemented [20,21]. One of the suggested measures is the use of double trigger (GnRHa 40h and standard hCG added 34h prior to OPU, respectively) when the follicles reach the 15–16 mm diameter. An RCT evaluating the role of different modes and timings of final follicular maturation trigger, on IVF cycle outcome of 33 poor responder patients, has demonstrated that patients in the double trigger group had a significantly higher number of TQE (1.1 ± 0.9 vs. 0.3 ± 0.8 and 0.5þ0.7; $P < 0.02$) as compared to the hCG trigger and the

GnRH-ag trigger groups, respectively, with an acceptable nonsignificantly higher ongoing pregnancy rate [25]. Another option is to offer the dual trigger (hCG and GnRHa) administered 34h prior to OPU, whenever the follicles reach the diameter of 18 and above.

GnRHa and hCG in patients at risk to develop severe OHSS

One bolus of 1500 IU hCG 35h after the triggering bolus of GnRHa, that is, one hour after oocyte retrieval, was demonstrated to rescue the luteal phase, resulting in a reproductive outcome comparable with that of HCG triggering, and with no increased risk of OHSS [26]. However, when applied to patients at high risk to develop severe OHSS, 26% developed severe early OHSS requiring ascites drainage and hospitalization [27]. A figure that is comparable to the acceptable 20% prevalence of severe OHSS in ostensibly high-risk patients [28].

One bolus of 1500 IU hCG concomitant with GnRHa (dual trigger), 34–36h before oocyte retrieval, was suggested as a method which improves oocyte maturation, while providing more sustained support for the corpus luteum than can be realized by the GnRHa-induced LH surge alone [29,30]. While acceptable rates of fertilization, implantation, clinical pregnancy, ongoing pregnancy rates, and early pregnancy loss were achieved in high responders after dual trigger [29,30], the incidence of clinically significant OHSS was not eliminated, but rather reduced to 0.5% [30].

One bolus of 1500 IU hCG five days after the triggering bolus of GnRHa was suggested [5,31]. While the freeze-all policy was applied to all patients yielding more than 20 oocytes, those triggered with GnRHa, who achieved less than 20 oocytes, were instructed to start an intensive luteal support with estradiol and progesterone, the day following OPU, and were reevaluated three days after oocyte retrieval (on day of embryo transfer) for signs of *early* moderate OHSS (ultrasonographic signs of ascites as reflected by the appearance of fluid surrounding the uterus/ovaries, and/or Hct levels >40% for the degree of hemoconcentration). If no early signs of OHSS developed, one embryo was transferred, and the patients were instructed to inject 1500 IU of HCG. By deferring the hCG bolus by three days (five days following GnRHa trigger), the corpus luteum was rescued, with an observed extremely high midluteal progesterone levels [31], reasonable pregnancy rate, with no patient developing severe OHSS. However, while these preliminary results are promising, the small sample size mandates further large prospective randomized studies [31].

Conclusions

In this chapter, we analyzed and discussed the hitherto published studies relating to the different modes of GnRHa combined with hCG trigger—for final follicular maturation, aiming to elucidate how to tailor each mode to its appropriate subgroup of patients [32] (Fig. 13.1).

One bolus of 1500 IU hCG, concomitant, 35h or five days after the triggering bolus of GnRHa, was all demonstrated to rescue the luteal phase, resulting in improved reproductive outcome in patients at risk to develop severe OHSS, as compared to GnRHa trigger alone, with the questionable ability to *eliminate* severe OHSS.

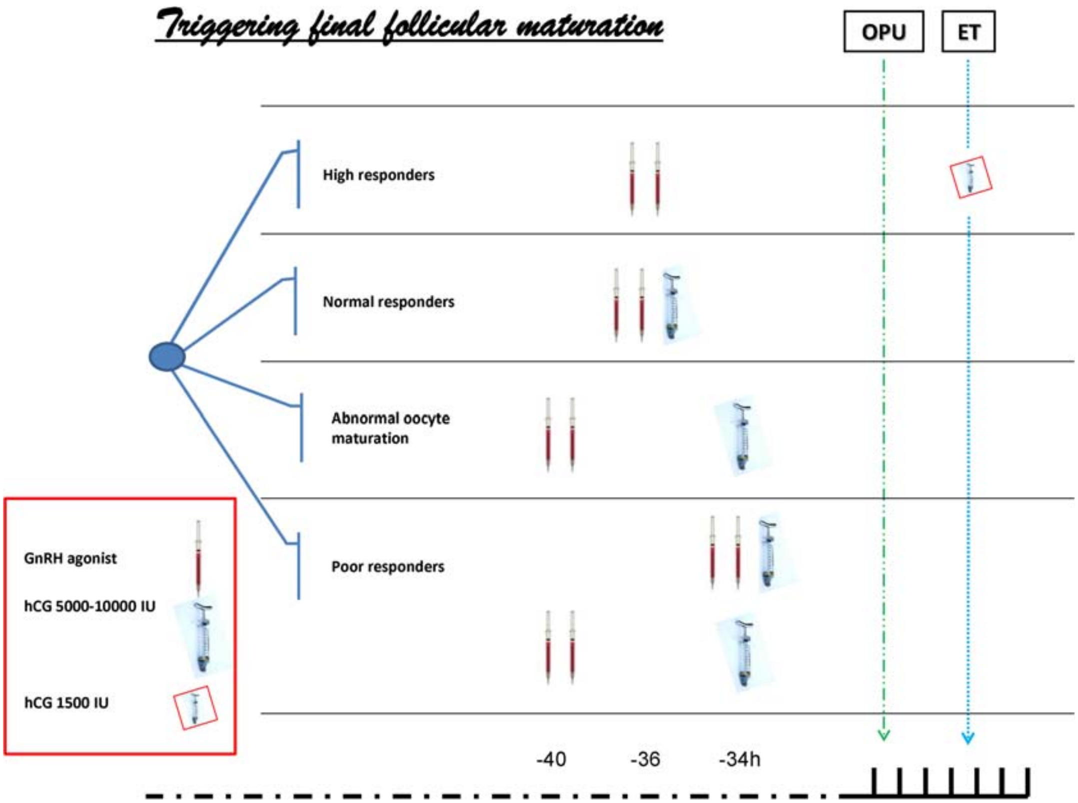

FIGURE 13.1 A suggested protocol for triggering final follicular maturation based on the magnitude of ovarian stimulation.

Moreover, following the observations demonstrating a comparable or even better oocyte\embryos quality following GnRHa trigger as compared to hCG trigger, and the different effects of LH and hCG on the downstream signaling of the LH receptor, GnRHa is now offered concomitant to the standard hCG trigger dose, to improve oocyte/embryo yield and quality. GnRHa and hCG may be offered concomitantly, 34–37h prior to oocyte retrieval (dual trigger) or 40 and 34h prior to oocyte retrieval, respectively (double trigger), in patients with abnormal final follicular maturation.

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CHAPTER

14

Pathogenesis and management in OHSS

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Introduction

Ovarian hyperstimulation syndrome (OHSS) is the most severe complication present in assisted reproduction, as a result of controlled ovarian stimulation (COS). Hence, it's an iatrogenic condition, affecting young healthy women undergoing assisted reproductive techniques (ARTs), in whom the objective is to obtain oocytes. There are less frequent causes, such as mutations in the follicle-stimulating hormone (FSH) receptor [1]. Even though the spontaneous OHSS is a rare event, compared to the iatrogenic form, both involve a unique pathologic entity called hyperreactio luteinalis, where multiple serous and hemorrhagic follicular cysts lined by luteinized cells are present [1].

Little is known about the pathophysiology of the OHSS, as its not fully understood, but we know that the natural history of the syndrome is mainly an effect of an acute serum shift from the intravascular space to the third space, consequence of increased capillary permeability [1]. The principal stimulus for OHSS is human chorionic gonadotropin (hCG), administered in ART for final follicular maturation and ovulation triggering. This leads to an increased expression of vascular endothelial growth factor (VEGF) in the ovary and the discharge of vasoactive-angiogenic substances, leading to increased vascular permeability and loss of fluid to the third space [2].

There are a vast number of signs and symptoms characterizing this syndrome. On one hand, there is intense abdominal pain, distension, and ascites, consequence of the leakage of fluid from ovarian follicles and the increased capillary permeability, leading to liquid accumulation in the third space. Abdominal pain can also be a result of enlarged ovaries, which in most complicated cases could lead to ovarian torsion, intraperitoneal hemorrhage, or rupture of cysts. Substantially, pulmonary function may be compromised causing dyspnea, as both

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enlarged ovaries and ascites restrict diaphragmatic movement. Dyspnea may also occur because of pleural effusion, pulmonary edema, atelectasis, pulmonary embolism, acute respiratory distress syndrome (ARDS), and pericardial effusion [1,2]. Although the most severe form of OHSS is rare, it can be a lethal condition in its most extreme form. A hypercoagulable state will increase the frequency of thromboembolic events, both arterial or venous, due to hemoconcentration and hypovolemia resulting from fluid shift to third space. Also, there is an increased risk of developing renal failure, due to a decreased organ perfusion. OHSS has also been related to maternal death [3], although this is exceptional.

There are two clinical forms of OHSS: the early-onset form, which occurs on the first eight days after hCG administration, and the late-onset form, related to pregnancy and therefore taking place nine or more days after hCG administration [4].

The determination of the true incidence of OHSS is challenging, as there is no accepted definition of the syndrome. Also, because OHSS depends upon the clinical setting (ovulation induction or COS), the gonadotrophins are employed, and the characteristics of patient are treated. Induced ovulation with clomiphene citrate or aromatase inhibitors could potentially precipitate a mild OHSS, but moderate and severe forms are extraordinary [4]. Contrariwise, in COS, moderate and severe cases may occur, even though they are rare: the incidence of moderate OHSS is 3%−6%, and in severe cases, 0.1%−2% [1,5,6]. Hospitalization has been described between 0.3% and 1.1% [7,8], despite an alarming increase in the incidence of severe OHSS cases in the 90s. Nonetheless, the introduction of risk markers and preventative measures has resulted in an important decrease of OHSS. The most meaningful preventive measure is freezing embryos (“freeze-all”) after COS, avoiding pregnancy, which has substantially decreased the incidence of severe OHSS.

Pathogenesis

In a physiological ovulatory cycle, the hypothalamic−pituitary−ovarian axis restricts follicular growth, and only a small number of early antral follicles will be preselected. Moreover, one only dominant follicle will be recruited, and consequently ovulate in response to the midcycle luteinizing hormone (LH) peak. However, in COS, this equilibrium is disrupted by exogenous gonadotropins administration, followed by ovulation induction with hCG. If COS is followed by pregnancy, the maintenance of serum levels of hCG will worsen and uphold symptoms [6].

Systemic peripheral arteriolar dilatation was proposed as the underlying mechanism of OHSS [9], but nowadays, it is accepted that OHSS is a local phenomenon restricted to the ovary. In fact, oophorectomy can be an effective treatment in OHSS, since when oophorectomy is carried out, normality is restored. Also, under experimental conditions, oophorectomized animals will not develop OHSS [10,11].

Furthermore, if ovulation trigger is not completed with hCG, OHSS does not occur [12−14], proving that hCG is what encourages all the pathophysiological events in OHSS. But hCG has no direct vasoactive properties, so the aim of past investigations was to detect the vasoactive molecule responsible for the syndrome. Initially, investigations suggested that high estradiol levels in COS were the root of OHSS [15,16]. However, it has also been described how women with low serum estradiol levels can likewise develop OHSS due to enzymatic mutations in the steroid pathway [17], so this theory has been discarded. Other investigations focused on substances present in follicular and ascitic fluid of women suffering from OHSS, such as: cytokines, growth factors, interleukins (ILs) (IL-2, IL-6, IL-8, IL-10, IL-18) [18], histamine, prolactin, prostaglandins, and renin–angiotensin [19]. Nowadays, it is accepted that the hCG mediator is VEGF [20], as its levels are proven to be higher in OHSS cases and increase even more after hCG administration. VEGF manages to enhance endothelial permeability x 50,000 times, when compared to histamine or other vasoactive substances [21].

The human VEGF gene is mapped in chromosome 6p12 and involves eight exons [22]. It comprehends the same structure in rodents and humans [22]. Also, investigations in both the rat's and human's ovaries have demonstrated a significant VEGF mRNA expression after the LH surge [23,24].

VEGF binds to receptors that belong to the tyrosine kinase receptor family: VEGFR-1 (Flt-1) and VEGFR-2 (Flk1/KDR) [25,26]. These receptors are present on the endothelial cells surface [25] The VEGFR-2 (Flk1/KDR) receptor is involved in vascular permeability regulation, angiogenesis, and vasculogenesis. Instead, VEGFR-1 opposes to VEGFR-2 by controlling the assembly and maintenance of tight junctions between endothelial cells in already existing vessels [27].

Even though evidence shows that VEGF is enhanced by hCG [23,24,30], some investigations revealed that both mRNA expression and protein levels of VEGF and VEGFR-2 surge previous to hCG administration, and consequently hCG increases even more these parameters [11,31]. Specifically, these effects were restricted to the ovaries and not to surrounding tissues.

In OHSS, VEGF is produced and overexpressed by granulosa-lutein cells [23,32], and consequently released into the follicular fluid, in response to the increased capillary permeability induced by hCG [23,33,34]. hCG induces the expression of VEGF in cultured granulosa-luteal cells: this VEGF expression is higher in women developing OHSS [35] compared to those who do not develop the syndrome. Moreover, investigations have proven that VEGF levels increase in both plasma and ascitic fluid, being a positive correlation with OHSS's severity. It has also been described that hCG administration in patients with risk factors for developing the syndrome rises VEGFR-2 expression in granulosa-lutein cells [36,37].

VEGFR-1 is also synthesized as a soluble receptor (sVEGFR-1), which appears to compete with VEGFR-1 and VEGFR-2 for VEGF binding, and heterodimerizes with VEGFR-2 to inhibit vascular permeability [28,29]. Soluble receptor sVEGFR-1 has been described as a modulator of VEGF bioactivity. Women who did not develop OHSS had significantly higher sVEGFR-1 plasma levels, while women with OHSS had lower sVEGFR-1 levels and significantly higher amounts of free and bound VEGF [38].

The endothelial cell-to-cell junctions are the down-stream targets of VEGF. With the increasing VEGF levels, there is a rise in cadherin expression, associated to a change in the conformational arrangement of endothelial cells. These changes have been described as the morphological confirmation responsible of the increased vascular permeability present in OHSS [39].

Recent investigations have proven that the dopaminergic system is strongly related to VEGF secretion. Studies in animals demonstrated that dopamine agonists could reverse increased capillary permeability in OHSS (consequence of gonadotropins) by VEGFR-2 phosphorylation and inactivation [40]. Moreover, investigations in humans, which compared different dopamine agonists, showed that these drugs can be used in women with OHSS to reduce capillary permeability [41,42], clarifying the potential relevance that dopamine has on OHSS development.

Here are different factors increasing the risk for developing the syndrome. These are as follows: having a previous episode of OHSS, polycystic ovary syndrome (PCOS), antral follicle count over 24, serum AMH over 3.4 ng/mL, young patients (<35 years old), low BMI, history of allergies, and Black race. Among patients who display OHSS, the group with highest risk of developing the syndrome are women with PCOS. During the last decades, experts have tried to explain the physiopathology and relationship between OHSS and PCOS, concluding that the dopaminergic system is responsible. Dopamine and its receptors have been explored in both healthy and PCOS patients. The dopamine receptor D2 (Dr2) is present in granulosa-lutein cells and in theca cells of antral and preovulatory follicles [43]. Dr2 concentrations have found to be significantly higher in normo-ovulatory women compared with PCOS patients. Furthermore, granulosa-lutein cells also secrete dopamine in much higher amounts in normal subjects than in PCOS [43]. Contrariwise, the secretion of 3,4-dihydroxyphenyl-acetic acid (DOPAC), the primary metabolite of dopamine, is significantly increased in PCOS, which indicates that dopamine metabolism is faster in PCOS patients compared to normo-ovulatory women. In summary, in PCOS women, the dopaminergic system is different than in normal-ovulatory women, as PCOS expose lower levels of dopamine and dopamine receptor D2 and a deregulated dopamine metabolism with increased DOPAC. Simultaneously, PCOS women display higher serum and follicular VEGF and VEGFR2 levels, compared to normo-ovulatory patients, which explains the increased vascularization [32,43,44]. All this results in an enhanced capillary permeability and increased fluid extravasation in PCOS woman, which leads to a higher expression of the syndrome (Fig. 14.1).

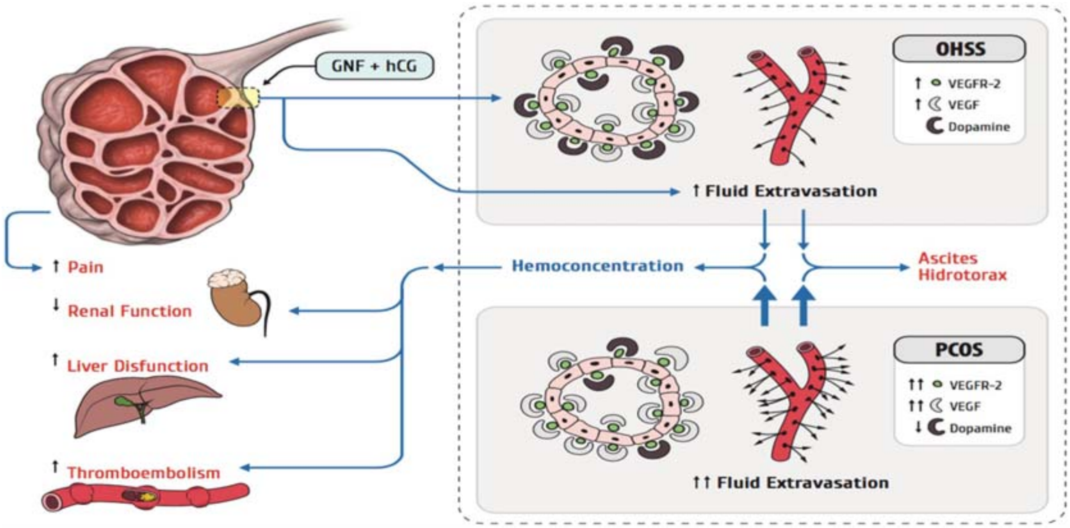

FIGURE 14.1 Pathophysiology of OHSS in women treated for controlled ovarian stimulation (COS) with gonadotropins and in women with polycystic ovarian syndrome OHSS. *Reproduced with permission from Pellicer N., Galliano D., Pellicer A. 'The ovary', Chapter 22—ovarian hiperstimulation syndrome. Edited by:Leung P.C.K and Adashi E.Y. Elsevier; 2018. vol. 11: 345-362.*

Administration of a dopamine agonist, as cabergoline, bromocriptine, or quinagolide, reduces, in a dose-dependent fashion, VEGF secretion by granulosa-lutein cells [45]. But the mechanism by which dopamine agonists reduce vascular permeability is even more complex than a simple blockage of VEFGR-2 phosphorylation. Instead, both in vitro and in vivo studies have revealed that dopamine agonist action over Dr2 in granulosa-lutein cells is post-transcriptional [45–47]. Recent genomic association studies have shown that dopamine agonists affect several pathways in granulosa-lutein cells, such as the VEGF signaling pathway, apoptosis, and ovarian steroidogenesis [48]. Dopamine agonists seem to reduce the incidence of moderate or severe OHSS in women at high risk of OHSS. But very importantly, if an embryo transfer is performed in the same cycle, the use of dopamine agonists to prevent OHSS does not affect pregnancy outcomes, such as live birth rate, clinical pregnancy rate, and miscarriage rate [40]. Nonetheless, dopamine agonists present some adverse events, such as gastrointestinal symptoms, and alterations of the central nervous system (for example, orthostatic hypotension).

Infrequently, OHSS may occur spontaneously, in the absence of ovarian stimulation. There are four major causes of spontaneous OHSS, and all generally appear in weeks 8th to 14th of spontaneous pregnancies [49]. The first cause is ovarian hypersensitivity to gonadotropins, consequence of mutations in the FSH receptor, which allow hCG binding [50–53]. A single amino acid substitution in the FSH receptor can mimic the pharmacological OHSS syndrome, by increasing the receptor's activity and sensitivity to hCG. In those cases, normally a family history is found, as the mutation is segregated with OHSS along the family [51,52].

Second, OHSS has been described in molar pregnancies and multiple pregnancies, consequence of abnormally high levels of endogenous chorionic gonadotropin, in a phenomenon known as hyperreactio luteinalis for the first trimester [49,54]. In third place, high levels of molecules structurally similar to gonadotropins, such thyroid-stimulating hormone (TSH) in hypothyroid states, may lead to OHSS [55,56]. The reason for this is a similarity between the beta subunits of the four glycoprotein hormones: thyrotropin, FSH, LH, and chorionic gonadotropin [50]. High levels of THS may lead to an occupation of FSH and LH receptors and stimulation of the ovaries in pregnant women [55,56].

Fourth and last, spontaneous OHSS can occur in women with pituitary adenomas, even when FSH levels are found to be normal. In these cases, the ovaries appear particularly sensitive to the FSH secreted by the hypophysis. This higher biologic activity of the FSH induces the formation of massive cysts in the ovaries, following high serum estradiol levels [57].

OHSS management

For the OHSS management, we propose a similar approach as suggested by the different societies, based on their clinical practice guidelines. These societies include the American Society of Reproductive Medicine (ASRM), the European Society of Human Reproduction and Embryology (ESHRE), the Royal College of Physicians of Ireland, and the Joint Society of Obstetricians and Gynecologists of Canada-Canadian Fertility and Andrology Society [7,58–62].

However, prevention strategies must be considered, as they can be the most important measure to avoid OHSS [42]. The first step should include identifying patients at risk of suffering OHSS: PCOS, history of OHSS, AMH above 3.4 ng/mL, and AFC over 24 follicles [60]. Second, clinicians must choose an adequate COS protocol for these patients, lowering the risk of OHSS, such as employing maximum doses of gonadotropins of 150 UI per day, GnRH antagonists for ovulation suppression or progestins (MPA, medroxyprogesterone acetate), and triggering with GnRH agonists. Adjuvant therapies during COS have been proposed: 100 mg aspirin/day (from the first day of COS until the next menses, in the case of a negative ART outcome, or the ultrasonographic detection of embryonic cardiac activity), and metformin. Finally, other strategies have been proposed, once OHSS is suspected and before embryo transfer: withholding embryo transfer ("freeze-all policy" or "cycle segmentation"), protected intercourse, aromatase inhibitors such as letrozole (5 mg/day for seven days), GnRH antagonist post pick-up(2−3 days after the pick-up), and dopamine agonists (cabergoline 0.5 mg/day from the day of hCG for eight days.) [60].

OHSS manifestations can be discriminated into mild form of OHSS, moderate, severe, and critical. However, the syndrome displays a sequence of signs and symptoms in increasing severity that defy any attempt at classification. In summary, established mild OHSS management should be conservative, and patients can be managed as outpatients. Patients should always be informed and directed about symptoms and/or signs of worsening. But women with severe or critical OHSS require hospitalization, and in worst case scenario, some cases require admission in intensive care units (ICUs). While early-onset OHSS is a self-limited disease, signs and symptoms may worsen and be prolonged if pregnancy occurs (late-onset OHSS), as the rising hCG during pregnancy exacerbates the course of the syndrome (Fig. 14.2).

Mild OHSS

As previously mentioned, most OHSS cases are mild or moderate and can be managed on an outpatient basis. Normally, these cases are self-limited and can be handled conservatively. The main objective is symptom relief [63].

In fact, mild OHSS is frequently seen in the daily practice, in women undergoing COS for assisted reproduction. In these cases, general recommendations are normally enough. Sometimes, supplementation with antiinflammatory drugs is required, where acetaminophen is recommended. Nonsteroidal antiinflammatory drugs (NSAIDs) are commonly contraindicated, as they can compromise renal function. General measures include avoidance of heavy physical activity and instructions to call for any signs or symptoms of worsening, such as oliguria, abdominal distention, dyspnea or shortness of breath, or weight gain (which is a sign of fluid accumulation) [63].

However, patients with mild OHSS should be always observed, as the syndrome can worsen and become moderate or even severe, especially if pregnancy follows. Therefore, these women must be monitored for either two weeks, or until menstrual bleeding occurs, recording symptoms, weight gain, and abdominal circumference.

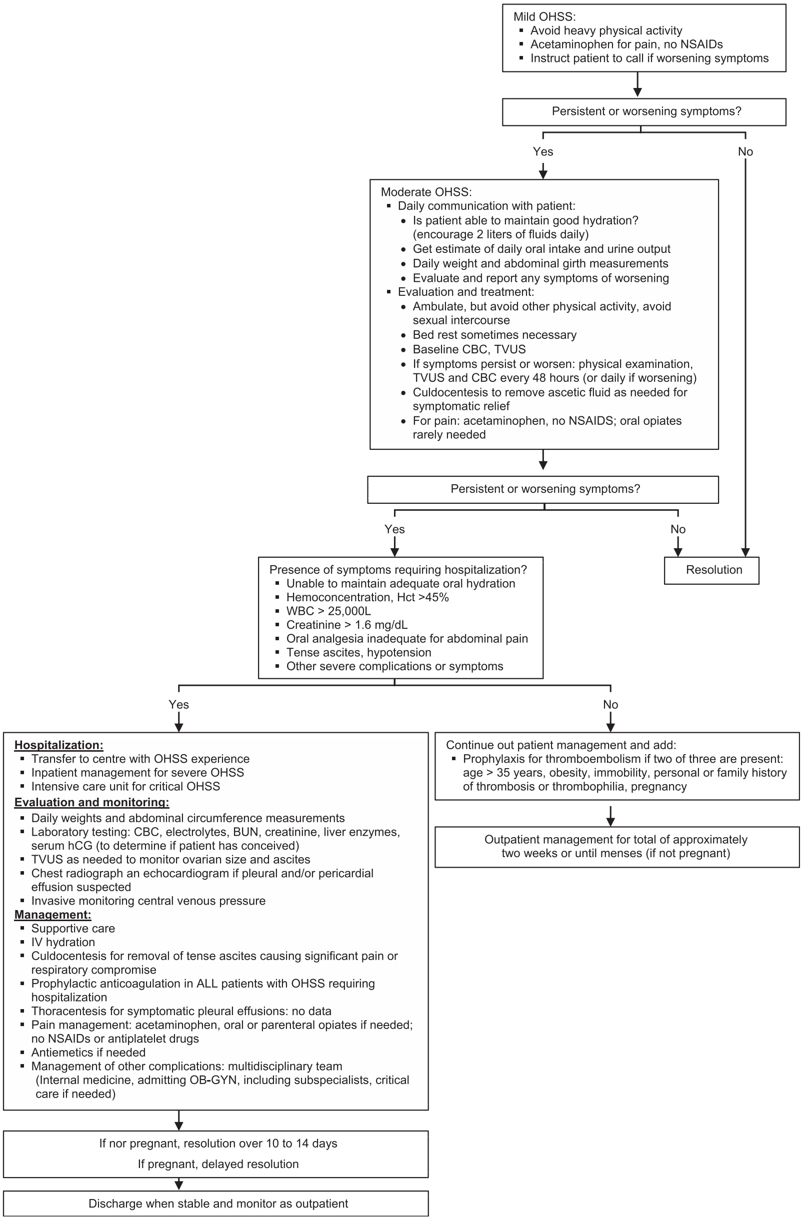

FIGURE 14.2 Management of OHSS. *Reproduced with permission from Busso CE, Soares SR, Pellicer A. Management of ovarian hyperstimulation syndrome. In: UpToDate, Post TW (Ed), UpToDate, Waltham, MA. (Accessed on June 19, 2017.) http://www.uptodate.com*.

Moderate OHSS

For women with moderate ovarian OHSS, hygiene and dietetic measures include oral fluid intake, from 1 to 2 L of water per day [63]. It's important that diuretics should always be contraindicated, as they can worsen the syndrome by decreasing intravascular volume. Ambulation should be recommended, but avoiding heavy physical activity, especially sexual intercourse (crucial, as pregnancy should be avoided). Instead, bed rest is sometimes necessary if symptoms are very intense, especially abdominal distension.

Women must be monitored at the clinic every 48−72 h. On each visit, the following will be performed: recording of vital signs, a physical examination (weight and abdominal circumference), performing an ultrasound to evaluate ascites and ovarian size, checking fluid intake and urinary output (fluid balance), and laboratory testing (complete blood count, electrolytes, creatinine, serum albumin, and liver enzymes). In view of progression or worsening of signs and symptoms, monitorization then will be then performed every 24 h [63]. Daily communication with patient is crucial, checking for an adequate hydration, and documenting any worsening of signs and symptoms.

It's essential that pregnant patients are followed very closely, as they are inclined to aggravate symptoms, due to the progressive rising levels of endogenous hCG. Because of this, pregnant patients experience a longer syndrome and take longer to recover than nonpregnant patients, who typically undergo resolution around 14 days after symptom onset or after menstruation.

Furthermore, three important measures should be considered. The first action is addressed to decrease the volume of liquid accumulated in the abdomen. Transabdominal paracentesis or transvaginal aspiration of ascitic fluid is sometimes necessary to relief symptoms. It is safe and successful and can even be performed on an outpatient basis [64]. Most centers prefer the transvaginal aspiration of the ascitic fluid from the cul-de-sac with transvaginal ultrasound guidance (also named as culdocentesis), as most in vitro fertilization (IVF) centers have more experience with this technique. Indications are tense ascites, orthopnea, rapid increase of abdominal fluid, or any other sign that may indicate worsening [63]. Removal of ascitic fluid provides abrupt amelioration of symptoms, and given the absence of other complications, women can continue to be monitored as outpatients. Infections should be avoided with prophylactic antibiotic coverage. However, the quantity of volume of fluid that needs to be removed is a matter of controversy. In general lines, aspiration of 500 mL of fluid can be enough to provide resolution of abdominal discomfort. Aspiration to up to 2000 mL of fluid can lead to intraabdominal pressure reduction, following an increase in renal artery resistance (probably consequence of a decompression effect) which will lead to a rise in urine output [65]. However, aspiration over 4000 mL of fluid at once is not recommended, as complications may occur, having a negative effect on intraabdominal pressure and renal flow. Blind paracentesis should never be performed because of the potential risk of bowel or vessel puncture and perforation. Statistically, it has been reported that more than 90% of patients will benefit from culdocentesis at some point of the syndrome to relief symptoms, and that an average of 3.4% procedures will be necessary until the syndrome is resolved [66,67].

Second, culdocentesis must always be followed by fluid reposition, with intravenous hydration, employing isotonic crystalloid solutions (for example, normal saline, Ringer's lactate). At least part of the fluid removed should be repositioned.

Finally, thromboembolism prophylaxis should always be considered, in both hospitalized patients and women managed as outpatients with two additional risk factors. Risk factors include age above 35 years, obesity, immobility, personal or family history of thrombosis, hereditary or acquired thrombophilias, and pregnancy.

We employ prophylactic low molecular weight heparin (LMWH), 20 mg subcutaneously every 12 h or 40 mg every 24 h (the most frequently employed), or heparin 5000 units subcutaneously every 12 h [68].

Severe and critical OHSS

Hospitalization is mandatory in women with severe OHSS. This encompasses severe abdominal pain, intractable vomiting, severe oliguria/anuria, tense ascites, dyspnea or tachypnea, hypotension, dizziness or syncope, severe electrolyte imbalance, or abnormal liver function tests. Other hospitalization criteria include hematocrit >45%, >25,000/L leukocytes, and creatinine >1.6 mg/dL [63].

Management of severe and critical OHSS is not easily accomplished; thus, patients should be hospitalized in units with extensive OHSS experience. Also, in the benefit of the patients or because of the severity of the syndrome, direct hospitalization in an ICU is strongly advised.

The objective of the treatment is supportive care, patient monitorization, and prevention and treatment of complications. The first goal of treatment will be to correct the intravascular blood volume and shift the direction of fluid from extravascular to the intravascular space. Typically, isotonic crystalloids are used, although some clinicians employ intravenous albumin due to its osmotic properties in critically ill, volume-depleted patients. However, evidence shows that intravenous albumin provides no additional benefit when compared with crystalloid solutions. The only real indication to use albumin is when there is concomitant oligoanuria. When there is oligoanuria, intravenous albumin (25% in 250 mL) should be administered, associating 20 mgrs. of furosemide, in order to increase urine output. Moreover, intravenous dopamine (2–3 μgr/kg/min) can be administered to force diuresis, and repeated every 8 h, but this medication requires ICU hospitalization and management [69,70].

Electrolyte balance must be controlled and corrected. Important complications should be prevented: ascites, hydrothorax, and thromboembolic events (thromboprophylaxis is mandatory in all hospitalized patients with OHSS). Also, for those in whom bed rest is recommended, an intermittent pneumatic compression device is typically suggested, to prevent thromboembolic events. Broad-spectrum empiric antibiotic therapy is advised when infection is suspected, using third or fourth-generation cephalosporin in combination with metronidazole [71].

Critical OHSS

Critical OHSS cases should be managed in an ICU, as significant complications may arise, such as massive hydrothorax, pericardial effusion, arterial thrombosis, pulmonary embolism, sepsis, acute renal failure, ARDS, and disseminated intravascular coagulation. These complications need to be managed by specialized staff [68]. In ICU, patients will be continually monitored with assessment of fluid balance (daily or less), records of weights and measurement of abdominal circumference, daily (or less) blood tests (including complete blood count, electrolytes, blood urea nitrogen [BUN], creatinine, and serum hCG measurements), invasive monitoring of central venous pressure, pelvic ultrasound to evaluate ovarian size and ascites, and chest radiograph and echocardiogram when pleural or pericardial effusion is suspected (as often as needed).

Nowadays, oophorectomy is an obsolete therapeutic option that is no longer practiced in life-threatening conditions. However, pregnancy interruption might be a therapeutic alternative in critical uncontrolled cases, after all conservative therapeutic approaches have been unsuccessful.

Resolution and prognosis

As previously mentioned, OHSS is a self-limited syndrome, and therefore increased ovarian vascular permeability regression will spontaneously take place in 10−14 days, or after menstruation is restored. Fluid in the third space will slowly reenter the intravascular space, hemoconcentration will overturn, and natural diuresis will be restored. As a result, hematocrit will normalize, and ultrasound will show no signs of ascites while general clinical symptoms alleviate. If pregnancy occurs, resolution will take longer [72], as hCG continue to rise. Some investigations have related those pregnancies occurring after OHSS to a higher rate of miscarriage, gestational diabetes, and pregnancy-associated hypertension [73,74]. Nonetheless, these complications have not been confirmed in other studies, which did not show increased fetal or maternal morbidity during the second and third trimesters among women who developed OHSS, compared with IVF control patients who had not established the syndrome [75].

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CHAPTER

15

Complications of in vitro fertilization (excluding ovarian hyperstimulation syndrome) and their management

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Introduction

The first successful in vitro fertilization (IVF) cycle was done in 1977, resulting in the birth of Louise Brown on July 25, 1978. The pregnancy was "established after laparoscopic recovery of an oocyte on Nov.10, 1977, in vitro fertilization and normal cleavage in culture media, and the reimplantation of the 8-cell embryo into the uterus 2 $^{1}/_{2}$ days later" [1]. Since then, over eight million babies have been born from assisted reproductive technology (ART) including IVF, and it is estimated that more than two million IVF treatment cycles per year are done globally [2]. The protocols for ovarian stimulation and the technique of oocyte retrieval have evolved significantly, continually increasing the safety of the IVF process for patients. However, just like in virtually every other area of medicine, complications still occur. This chapter serves as a review of all major complications of IVF and their management, with the exception of ovarian hyperstimulation syndrome (OHSS), which is covered in a separate chapter of this book.

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Summary of complications, morbidity, and mortality estimates

Over the past four decades, multiple studies have addressed the safety profile and associated complications of the IVF process, using the design of single-center, multicenter, or population cohort studies.

Soon after the first successful live birth from IVF, transvaginal ultrasound-guided (as opposed to laparoscopic) oocyte retrieval was introduced and rapidly became the preferred route of egg retrieval. One of the earliest studies regarding complications of this procedure reported no major complications during the first 181 attempts at follicle aspiration in a newly established IVF unit by gynecologists with no prior experience with ultrasound-guided follicle aspiration [3].

In the early 1990s, complications arising from a series of 2670 consecutive transvaginal ultrasound-directed follicle aspirations were monitored during a 4-year prospective study [4]. Vaginal hemorrhage occurred in 229 patients (8.6%) with an estimated blood loss of over 100 mL in 22 (0.8%). Postoperative pelvic infection occurred in 18 patients (0.6%). Ovarian hemorrhage with resulting hemoperitoneum formation was observed in two cases (0.07%) and necessitated emergency laparotomy in one of the two affected patients. In this cohort, a single case of "pelvic hematoma formation from a punctured iliac vessel" also occurred, and resolved without intervention. 9 of the 18 identified infections (50%) were severe with pelvic abscess formation.

A similar (retrospective) study from Egypt from the late 1990s reported all complications of medically assisted conception in 2924 patients undergoing a total of 3500 cycles of IVF or intracytoplasmic sperm injection (ICSI) [5]. The authors reported complications related to the treatment in 291 patients (8.3%). The majority of these consisted of moderate ($n = 206$) or severe ($n = 60$) OHSS. Thus, the rate of actual procedure-related complications was 25/3500 (0.7%). Pelvic infection occurred in 10 patients, deep vein thrombosis in 4, vaginal bleeding in 3, acute abdomen in 3, anesthetic complications in 2, and hemiparesis in 2 (Table 15.1). In this dataset, there was one fatality due to hepatorenal failure secondary to OHSS in a patient with preexisting hepatitis C.

A 1998 retrospective study from Belgium analyzing 1500 transvaginal ultrasonographically guided oocyte retrievals identified pelvic infections in 0.4% of patients, intraperitoneal bleeding in 0.2%, and adnexal torsion in 0.13% [6].

An Australian population study in the early 2000s studied the topic of mortality in patients undergoing IVF via identification of deaths in a cohort of 29,700 IVF patients by record-linkage with the National Death Index [7]. They included patients who actually received IVF treatment, as well as those who were referred, but not treated, with the mortality rate in the general population. The reported all-cause mortality rates in IVF patients (treated and untreated) were significantly lower than in the general female population of the same age. In treated women, 72 deaths were observed and 125 deaths were expected giving an age-standardized mortality ratio of 0.58 (95% confidence interval 0.48–0.69). They concluded that the data not only provided some reassurance about the safety of IVF treatment but also speculated that the findings may indicate that unhealthier women are deterred from pregnancy and infertility treatment. A similar study published in 2010 reported all deaths related to IVF in the Netherlands during the period from 1984 to 2008 [8]. In this registry study, the authors identified 6 deaths that were directly related to IVF (incidence 6 per 100,000), 17

TABLE 15.1 Complications of medically assisted conception in a large dataset from Egypt.

Complications of ART performed in 3500 treatment cycles	
Complication	No. (%) of complication
OHSS—moderate	206 (5.9)
OHSS—severe	60 (1.7)
Vaginal bleeding	3 (0.09)
Deep vein thrombosis	4 (0.1)
Hemiparesis	2 (0.06)
Pelvic infection	10 (0.03)
Acute abdomen	3 (0.09)
Anesthetic complications	2 (0.06)
Mortality	1 (0.03)
Total/overall complication rate (including OHSS)	291 (8.3)
Complication rate excluding OHSS	25 (0.7)

Adapted from Serour, G.I., et al., Complications of medically assisted conception in 3,500 cycles. Fertil Steril, 1998. 70(4): p. 638–642.

deaths directly related pregnancies resulting from IVF (42.5/100,000) and eight deaths neither related to IVF nor to the IVF-related pregnancy. The overall mortality in patients undergoing IVF procedures was lower than in the general population (reproducing the findings from the Australian study), whereas the overall mortality related to IVF pregnancies was higher than the maternal mortality in the general population. Of the six deaths directly attributed to the IVF treatment, three occurred in the setting of severe OHSS, two women died because of sepsis, and one death occurred due to a dose error of local anesthetic medication just before ovum retrieval.

A linkage study from Finland including 9175 women receiving IVF treatment found that the rate of bleeding complications necessitating hospital care per initiated treatment cycle was 0.09% [9]. The rate of infection requiring hospitalization per treatment cycle was 0.4%. A German study from 2006 reported vaginal bleeding in 2.8% of procedures, with no cases of clinically significant intraabdominal bleeding [10]. In this prospective study including over 1000 oocyte retrievals, one injury of a pelvic structure—a ureteral lesion—was observed, with no cases of pelvic infection in this dataset.

The incidence of serious and minor complications in the setting of anonymous or directed oocyte donation was studied in a 2008 retrospective study of 587 donors undergoing 973 cycles of controlled ovarian hyperstimulation (COH) and 886 oocyte retrievals [11]. The authors reported a rate of 6 in 886 (0.7%) for serious complications, which included OHSS, ovarian torsion, infection, and ruptured ovarian cysts. The rate of minor complications severe enough to prompt the donor to seek medical attention after retrieval was 8.5%. Regarding non-OHSS complications, two cases of ovarian torsion (0.2% of retrieval cycles) occurred, with successful preservation of the ovaries via laparoscopy. Two cases of infection (0.2%) were successfully managed with IVF antibiotics. One ruptured ovarian cyst (0.1%) resulted in a hospital admission for observation and resolution with expectant management. Minor non-OHSS complications such as urinary tract infection (UTI) or yeast infection occurred in 18 donor cycles (2.0%) and were managed with one or two outpatient visits.

Finally, one of the largest studies assessing the rate of clinical complications after transvaginal oocyte retrieval ($n = 7098$ IVF cycles) reported four cases of severe peritoneal bleeding requiring surgical treatment (0.06%, 95% confidence interval [CI] 0.00−0.12), and two cases of pelvic abscesses (0.03%, 95% CI 0.00−0.07), for a total of six cases of severe clinical complications (0.08%, 95% CI 0.03−0.18) [12].

While the above body of data on this topic allows us to conclude that the rate of non-OHSS complications from IVF treatment is low, it is important to be aware of specific aspects and managements of the various complications, which will be reviewed in the remainder of this chapter.

Complications of ovarian stimulation: specific aspects and management

Ovarian torsion

Ovarian torsion is a rare complication of IVF that may occur during ovarian stimulation or around and after the time of the egg retrieval procedure. It is defined as partial or complete rotation of the ovarian vascular pedicle causing obstruction to venous outflow and arterial inflow, and puts the ovary at risk of necrosis [13]. Ovarian torsion is usually associated with cystic enlargement of one or both ovaries; therefore, COH causing multifollicular ovarian enlargement puts patients undergoing IVF at risk of this condition [14−16].

The exact incidence of ovarian torsion during IVF is uncertain; one study reported nine laparoscopically reported cases of ovarian torsion among 10,583 IVF/ICSI cycles (0.08%), indicating that one clinically significant case of this condition occurs approximately every 1000 cycles [17]. Despite its rarity, providers need to be vigilant about this condition and educate patients to seek medical attention immediately with symptoms of ovarian torsion including sudden onset of lower abdominal pain or nausea and vomiting. Swift intervention with laparoscopic detorsion can prevent the potential loss of an ovary, which is especially disastrous and upsetting for fertility patients, many of whom already have decreased ovarian reserve.

While it is common practice to advise patients undergoing IVF not to exercise due to a theoretical increased risk of ovarian torsion, there is no definitive evidence supporting this recommendation beyond expert opinion and case reports. In one report, a patient who had undergone ovarian gonadotropin stimulation in the preceding cycle began experiencing acute pain while participating in a kick-boxing class and was subsequently diagnosed with ovarian torsion which was successfully treated surgically [18].

Given the fact that exercise has stress-relieving and other beneficial effects, it may be best medical practice for IVF providers to educate patients about this rare complication of ovarian stimulation and allow them to make the determination whether they wish to avoid strenuous activity during the stimulation cycle.

Venous thromboembolism

The risk of venous thromboembolism (VTE) is increased during the IVF process [19]. Risk factors include the hypercoagulable state mediated by COH and potentially associated OHSS, the egg retrieval procedure, and a more sedentary state during the stimulation process and after the oocyte pick up (OPU). It appears that the main risk increase is mediated via the occurrence of OHSS.

One recent systematic review on the topic of thromboembolism and IVF identified 21 relevant articles (nine cohort studies, six case−control studies, three case series, and three reviews of case series) [20]. The authors reported a doubling of the antepartum risk of VTE after IVF (odds ratio 2.18, 95% CI 1.63−2.92) compared with the background pregnant population, due to a 5- to 10-fold increased risk during the first trimester in IVF pregnancies, in turn related to a very high risk of VTE after OHSS. They estimated that in the setting of OHSS, there is up to a 100-fold increase of VTE, or an absolute risk of 1.7%. They did not identify a robust study on thromboprophylaxis in this context.

Current efforts to reduce the incidence of OHSS through preventative measures, with the goal of "OHSS-free clinics," should lead to a steady reduction in the incidence of VTE.

Segmentation of the IVF process, using a "freeze-all" strategy with cryopreservation of embryos and subsequent interval frozen embryo transfer (FET) [21], allows for the pregnancy to begin in a less hypercoagulable stage, with a reduction of first trimester VTE cases in the first trimester of IVF pregnancies. Additional measures to reduce risk include minimizing the egg retrieval procedure time (for example, through unnecessary extensive follicle flushing) and encouragement of early postoperative ambulation and resumption of regular activity.

Providers should be aware of the symptoms and signs of deep vein thrombosis and pulmonary embolism, and initiate a workup for these conditions when clinical suspicion arises, to allow for prompt treatment of these rare but serious events.

Psychological morbidity

While physicians tend to focus on somatic complications of medical procedures, it is important to be mindful of the psychological consequences of IVF treatment. The rare but reported occurrence of suicides during or after IVF highlights the need to take this potential complication seriously [7].

A Danish registry study reported that women who did not have a child after an initial fertility evaluation and potential treatment had a twofold (HR: 2.43; 95% CI: 1.38−3.71) greater risk of suicide than women who had at least one child after a fertility evaluation

[22]. A nationwide Swedish case–control study examined whether women undergoing IVF are at greater risk of postnatal suicide or postnatal depression (PND) requiring psychiatric care, compared with women who conceive spontaneously [23]. The authors found that whereas mothers who receive IVF treatment are not at increased risk of PND, the risk is increased among mothers with a history of mental illness. IVF can exacerbate mental illness in patients with preexisting disease, and the risk of hospitalization due to adjustment disorders is increased among infertile women [24].

Given the emotional rollercoaster couples go through during fertility treatments, providers should be sensitive to the potential occurrence or exacerbation of mental illness in their patients, and have a low threshold for referral to a mental health expert, ideally with access to in-house psychologist [25].

Complications of the egg retrieval procedure

Hemorrhage

Bleeding from the vaginal puncture sites is a common, but almost invariably harmless and minor complication of the egg retrieval procedure. Following the egg retrieval, the surgeon should inspect the puncture sites and, as an initial step, apply pressure with a sponge to achieve hemostasis. If this simple maneuver does not remedy the problem, a hemostatic agent such as silver nitrate may be applied. In rare cases, a stitch using absorbable suture material may be placed.

A more serious bleeding complication from the oocyte retrieval procedure consists of intra-abdominal hemorrhage. Fertility providers who have performed laparoscopic GIFT (gamete intrafallopian transfer) and ZIFT (zygote intrafallopian transfer) procedures at various intervals after the egg retrieval are aware that some degree of paraovarian bleeding is almost invariably noted. For the vast majority of procedures, this blood loss is self-limited, and the small amount of clot surrounding both ovaries gets absorbed in the days following the procedure.

In rare cases, ongoing bleeding from ovarian vessels may result in a life-threatening hemoperitoneum with the occurrence of potentially fatal hemorrhagic shock.

In a case series from Israel, 7 out of a total of 3241 patients undergoing transvaginal oocyte retrieval (0.2%) were diagnosed with ovarian hemorrhage requiring surgical intervention [26]. The authors noted that all patients were thin, with a body mass index (BMI) of 19–21 kg/m^2, and 4 had polycystic ovary syndrome (PCOS), leading them to conclude that these characteristics may be risk factors for this complication. The signs and symptoms in all the patients appeared soon after the completion of the egg retrieval procedure, in the recovery room. All patients were directly hospitalized. All women experienced symptoms of diffuse abdominal pain, nausea, vomiting, and severe weakness lasting beyond 1 h after the procedure, and signs of an acute abdomen were apparent in all the patients.

Injury to major vessels in the pelvis, such as the iliac artery and vein, is also possible and potentially life-threatening. Ultrasound guidance with close attention to the iliac vessels should avoid this rare but serious risk. Providers should have a low index of suspicion for hemorrhagic complications of egg retrieval and monitor patients closely immediately after the procedure for abdominal signs and hemodynamic compromise.

The management of intraabdominal bleeding follows the management principles of other conditions associated with hemoperitoneum formation, such as ruptured hemorrhagic cysts. In many cases, surgery can be avoided with close observation of the patient's hemodynamic status and serial abdominal examinations, as bleeding is self-limited in the majority of patients with this complication.

If expectant management fails to achieve hemodynamic stability, surgical exploration of the abdomen and pelvis with laparoscopy or laparotomy is indicated. Embolization of pelvic arteries such as the uterine arteries or the anterior division of the iliac artery has been described as an alternative to surgery in select cases [27].

Infection

The technique of follicular oocyte aspiration for IVF via "sonographically controlled vaginal culdocentesis" was first reported in 1983 [28]. By virtue of the fact that the invariably unsterile bacteria-colonized vaginal wall is punctured with the purpose of entering the peritoneal cavity, it could be assumed that infections including peritonitis should be quite common.

However, in a previously mentioned review of 2670 consecutive egg retrievals, there were only nine cases of "minor pelvic infection" defined as pyrexia and pelvic tenderness, but no evidence of abscess formation (0.3%), and nine severe cases of pelvic abscess formation [4]. In that British study, microbiological examination of the pus from the cases of pelvic infection suggested that the most common route of infection was direct inoculation of vaginal organisms into the peritoneal cavity by the collecting needle. However, due to the low incidence of this complication, the authors questioned the value of routine prophylactic antibiotics for egg retrievals. It is currently unclear whether flushing of the vagina with bacteriostatic agents such as povidone iodine prior to the procedure reduces the risk of serious infection, and whether the benefit of this may be offset by potential toxicity to the oocytes. There is no definitive evidence for the use of prophylactic antibiotics prior to egg retrieval procedures. However, it is advisable to use perioperative prophylactic antibiotics, including anaerobic coverage, in patients at high risk of pelvic infection, such as those with known endometriomas, a history of tubo-ovarian or appendiceal abscesses, and prior major abdominal surgery [29].

Damage to surrounding organs

All pelvic organs are theoretically at risk of damage during the egg retrieval procedure. While the myometrium occasionally has to be traversed to access an ovary with a retrouterine location, there is no real risk of persistent damage to the uterus from a thin aspiration needle.

The bladder may be injured from the egg retrieval procedure, and providers should ensure that the patient has a completely empty bladder during the case to minimize this complication. Despite this precaution, inadvertent puncture of the bladder wall during the procedure is likely common and underreported, as it remains inconsequential in the majority of cases. Experience from urogynecologic operations such as suburethral sling procedures demonstrates that the bladder is a "forgiving" organ: it is not uncommon for the trocar used to pierce the bladder wall; when this complication of sling procedures occurs, most providers nowadays do not alter the patient management, and choose to not leave a urinary catheter in place for longer than usual. Given that the aspiration needle is much smaller than the trocar used for suburethral sling procedures, providers can be reassured that permanent damage to the bladder is unlikely even in the case of inadvertent puncture.

Nevertheless, great care should be taken to avoid the bladder during the procedure; preoperative bladder emptying and ultrasound guidance are good practice in this context.

While ureteral injury from an egg retrieval has been reported, it is exceedingly rare [30].

Bowel injuries from the egg retrieval procedure are a more serious complication than bladder injuries, although it is likely that they also occur at a higher frequency than clinically recognized. Clinically recognizable and impactful bowel injuries occur at an extremely low frequency, demonstrated by the fact that several large datasets of cases did not report any patients with this complication [4,12].

Other rare complications, including ureterovaginal fistula and vertebral osteomyelitis, have been reported as isolated rare events [31,32].

Anesthetic complications

While it is theoretically and practically possible to perform egg retrievals with no or only local anesthesia, this practice is unpleasant and painful for patients, and associated with suboptimal egg yields. Conscious sedation is the standard of care and used by the vast majority of practices. In a study on the topic of anesthesia practices in the United States, conscious sedation was used in 95% of IVF centers [33].

Complications of conscious sedation are uncommon, especially given the fact that most IVF patients are young and healthy women. However, patients should undergo a thorough preoperative evaluation prior to the egg retrieval procedure and be managed by expert anesthetic personnel during the case. It is the duty of the fertility provider to identify potential high-risk patients and "flag" them for the anesthesia provider preoperatively. Due to the increased risks secondary to obesity, some practices choose to have a BMI cutoff for patients undergoing IVF.

Intermediate/long-term complications of IVF

Pregnancy-related risks

It is well established that IVF pregnancies are associated with higher rates of obstetric risks compared with spontaneous pregnancies, even when controlling for patient age and multiple gestations [34]. Recent research has attempted to identify whether certain variations in the process of IVF, such as segmentation of treatment with a "freeze-all" strategy and FET, or in the type of FET—"natural" versus "programmed"—can alter the obstetric risk profile of resulting pregnancies [35]. A detailed discussion of this complex topic is beyond the scope of this chapter. Providers should stay current on the literature in this context and counsel patients regarding this topic prior to initiating treatment.

Long-term health consequences of IVF

The topic of long-term health consequences in the mother and the resulting offspring has been studied extensively. Several earlier studies using large datasets including registry studies demonstrated an increased risk of adverse outcomes such as birth defects. One example is an Australian registry study published in 2002, with the finding that infants conceived with the use of ICSI or IVF have twice as high a risk of a major birth defect as naturally conceived infants [36]. While population studies have attempted to control for confounding variables, it is challenging to cleanly disentangle the effects of ART with regards to maternal and fetal (offspring) morbidity from the fundamental differences that exist between a cohort of couples undergoing assisted reproductive treatment for a variety of indications from the general fertile population. Studies including three groups of patients have attempted to shed light on this question: in addition to a group of patients conceiving with ART and a control group of patients with spontaneous conceptions, they include a group of patients who conceived spontaneously after being diagnosed with infertility [37]. Studies with this design tend to demonstrate that differences in outcomes such as fetal anomalies are due to characteristics of the infertile population, rather than the treatment using ART.

With regards to the risk of cancer in IVF patients and their offspring, the literature is constantly evolving. One review on the topic of cancer risk for patients using fertility medications concluded that the data are reassuring, although several methodological issues in the studies in this area limit definitive conclusions [38]. A US population-based cohort study regarding the association of IVF with childhood cancer found overall cancer rates (per 1,000,000 person-years) of 251.9 for IVF-conceived children and 192.7 for those conceived naturally (hazard ratio, 1.17; 95% CI, 1.00−1.36) [39]. The authors concluded that "an association of conception by IVF with childhood cancer is small and limited to rare tumors; however, continued follow-up for cancer occurrence among children conceived via IVF is warranted."

Analogous to the topic of increased obstetric risks in IVF pregnancies, fertility providers should stay up to date on the current literature regarding long-term effects of ART on the mother and the child and be prepared to answer questions patients may have prior to the initiation of treatment.

Conclusion

When managed appropriately, the process of IVF including the procedure of egg retrieval is very safe, with a low risk of complications. However, providers should be very aware of the risk profile of short- and long-term complications of ART, have a low index of suspicion for their occurrence, and educate their patients about risks prior to the initiation of treatment.

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CHAPTER

16

In vitro maturation of immature cumulus–oocyte complexes collected from antral follicles

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What is oocyte in vitro maturation?

Oocyte in vitro maturation (IVM) is a laboratory tool in reproductive medicine defined as the maturation in vitro of immature cumulus–oocyte complexes (COCs) collected from antral nondominant follicles. Although some clinicians in the field of reproductive medicine have proposed a revised definition based on follicular size including protocols that are intended to obtain oocytes that are matured in vivo and collected at the metaphase II stage [1], the original definition of IVM encompasses the meiotic immaturity of the oocyte at the start of the IVM process. Hence, compounds that unlock meiotic prophase arrest of the oocytes before the removal of COCs from the follicle are not administered to the patient in standard IVM protocols.

IVM of oocytes from the germinal vesicle (GV) stage to the metaphase II stage was first described by Ref. [2] and Edwards [3]. To facilitate spontaneous meiotic maturation in vitro, cumulus-enclosed oocytes have typically been cultured in a basic tissue culture medium or in media formulated for blastocyst culture, supplemented with a protein source (autologous patient serum or serum albumin), follicle-stimulating hormone (FSH), and often human chorionic gonadotropin (hCG), until they reach metaphase II stage, 24–48 h after aspiration from the antral follicles [4]. However, these IVM culture media have not been designed to support the metabolic needs of the cumulus and the oocyte during maturation, although the molecular signaling pathways leading to reinitiation of meiosis in spontaneous IVM differ substantially from those occurring during in vivo follicular oocyte maturation. Consequently, the developmental competence of IVM oocytes is generally lower compared with the competence of their in vivo counterparts. This translates into inferior clinical outcomes when compared to conventional ovarian stimulation (COS) [5–7]. Maturation rates of oocytes incubated in standard IVM systems registered for clinical use have not evolved

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since the early experiments of Edwards back in the sixties, and low maturation rates ~50% are still a major obstacle to the efficiency of IVM [8–10].

Clinical outcomes of standard IVM as an artificial reproductive technology

In view of the lower clinical efficiency of IVM when compared with COS, IVM has been proposed mainly to a subset of women who have high serum levels of AMH [11]. According to a retrospective analysis of outcomes in 320 women with polycystic ovary syndrome (PCOS) undergoing standard IVM, cumulative live birth rates (CLBRs) per started IVM cycle were highest in women with a PCOS phenotype A (i.e., PCO-like ovarian morphology on ultrasound scan, hyperandrogenism, and dysovulation) [12]. Overall, clinical success rates after standard IVM have improved with time, with live birth rates of up to 45% per started cycle in IVF clinics with longstanding experience with IVM, including clinics in Belgium and Vietnam [13]. The adoption of elective embryo vitrification to compensate for asynchronous endometrial receptivity following fresh embryo transfer [14] and blastocyst transfer [6] has contributed to improved clinical outcomes and renewed interest in IVM. Still, in comparison with COS, IVM protocols have not been capable of generating an equal number of blastocysts, although the implantation rate per blastocyst appears similar [6]. Sufficiently powered clinical trials investigating the clinical outcomes of standard IVM compared with COS have been lacking. The group of My Duc Hospital in Vietnam has reported the outcomes of a randomized controlled trial encompassing 546 women undergoing standard IVM following a biphasic culture system including a pre-IVM phase with meiotic inhibitors compared to conventional ovarian stimulation in a gonadotropin-releasing hormone (GnRH) antagonist protocol. According to their data, IVM was still inferior compared with COS after the first frozen embryo transfer (FET) after elective cleavage-stage embryo vitrification, with live birth rates of 35.2% and 43.2%, respectively (difference −8.1% [−16.6%, 0.5%]) [10]. CLBRs after IVM were also inferior compared with those after COS. Nevertheless, although an equivalent proportion of more than two-thirds of patients in each arm of that study had a diagnosis of PCOS, it is unknown whether a similar study design in a strict PCOS population would have yielded the same results. Although the results from Vietnam have not yet been replicated by other expert IVM groups, they have contributed to the renewed interest in IVM in clinical practice and may fuel the uptake of IVM as an alternative artificial reproductive technology (ART) in women with PCOS, in specific regions across the globe, depending on local reimbursement policies [15]. In view of the progress that has been made as far as clinical outcomes are concerned, the International PCOS Network has recommended that IVM could be offered to achieve pregnancy and live birth rates approaching those of standard hormone-driven ART without the risk of ovarian hyperstimulation syndrome (OHSS) for women with PCOS, when embryos generated using IVM are vitrified and thawed and transferred in a subsequent cycle, at least in units with sufficient expertise [16]. Finally, IVM is no longer considered an experimental technology, according to a committee opinion of the Practice Committees of the American Society for Reproductive Medicine, the Society of Reproductive Biologists and Technologists, and the Society for Assisted Reproductive Technology [17].

Selection of appropriate patients suitable for IVM treatment

IVM of oocytes has been advocated as a mild-approach alternative for conventional ovarian stimulation (COS) because of the ability to avoid the side effects and risks that are associated with COS in women with elevated functional ovarian reserve [18]. Ovarian response in these women can be excessive and may result in OHSS [19]. In recent years, techniques have been developed to reduce OHSS risk, and the adoption of GnRH agonist triggering in combination with a policy of freeze-all embryos can eliminate the severe type of OHSS [36]. Nevertheless, patient discomfort and hospital admissions because of OHSS remain a concern [20]. To some extent, the protocols developed to reduce OHSS risk may be underutilized, and physicians may still administer hCG in connection with or after a GnRH agonist to rescue the luteal phase and to allow patients to have a fresh embryo transfer. On the other hand, the observed relationship between high oocyte yield and favorable CLBRs per IVF cycle may be used as an argument by IVF practitioners to stimulate the ovaries of predicted high responders with higher doses than they used to do before the "freeze-all" era. Because of the observation that the iatrogenic complication of severe OHSS no longer exists if the use of hCG is avoided, clinicians are less concerned about hyperresponse after ovarian stimulation (OS). As a consequence, severe OHSS may have disappeared, but moderate OHSS persists in many European countries [21] and in the United States [20]. Although the majority of predicted high responders accept the inherent risk of OS, a considerable proportion of women would consider a less efficient fertility treatment if the burden of the treatment would be lower [22]. IVM could have an emerging role in this specific group of patients, although it is currently unknown to which extent women would accept a lower chance of pregnancy in return. Indeed, the original incentive of IVM was to offer a more "patient-friendly" approach with fewer hormonal side effects compared with OS. Subfertile women with PCOS who fail to respond to ovulation induction therapy, of who are not pregnant after ovulation induction, or those with an additional male factor problem, are probably the best candidates for IVM. Indeed, these women may expect to have sufficiently high numbers of immature oocytes to make up for the inherently lower efficiency of IVM compared with conventional ovarian stimulation for IVF [11,23,37]. Not only do the majority of women with PCOS exhibit excessive response to OS but also a subset of them have a narrow window of optimal ovarian response, especially those with hyperandrogenism and patients who are overweight or obese. This may require frequent monitoring of OS and may result in suboptimal outcome if hyporesponse is observed; we observed significantly lower live birth rates in hyperandrogenic women with PCOS compared to their normo-androgenic counterparts when conventional ovarian stimulation was undertaken with the aim to solicit normal ovarian response to avoid OHSS [24]. For these patients, IVM may be an attractive option because monitoring of follicular growth can be kept to a minimum and oocyte retrieval for IVM can be scheduled at the patient's convenience, especially in anovulatory women. Indeed, patients may struggle to combine fertility treatment with normal life, which could result in significant stress related to IVF treatment and could lead to treatment termination before a successful pregnancy is achieved [25]. In view of this, after patient counseling and discussion of pros and cons of conventional ovarian stimulation and IVM, a subset of patients with PCOS may embrace IVM as an alternative simplified, low-burden ART, although there is currently no evidence supporting this hypothesis.

The serum concentration of AMH, a biomarker that correlates well with the severity of the PCOS phenotype [26], is a strong predictor of the number of immature oocytes that can be retrieved by follicle aspiration. In view of this, AMH as a proxy of oocyte yield correlates with the probability of pregnancy after IVM [11], although a specific AMH cut-off at which level the efficiency of IVM may eventually surpass that of conventional OS has not been established. In a retrospective cohort study encompassing 320 women with PCOS who underwent IVM, Mackens et al. illustrated the importance of assigning a specific phenotype to women with PCOS, based on a combination of Rotterdam criteria. Indeed, after adjusting for potential confounders, the PCOS phenotype was significantly correlated with CLBR after IVM; patients with the classical PCOS phenotype A had the highest CLBR (40% per started cycle) [12]. Although no prospective studies have compared clinical outcome after IVM and conventional ovarian stimulation in women with PCOS type A, such a comparative study could be a valuable future endeavor.

Besides IVM, laparoscopic ovarian surgery (LOS) is another possible approach in women with severe PCOS who are resistant to ovulation induction and in whom targeted ovarian response using gonadotropins is unsuccessful. This procedure is often performed in settings where IVM is currently not available, with varying outcomes. Because the impact of LOS on ovarian function is difficult to titrate, it is important to have adequate documentation of serum AMH levels before embarking on such a procedure. Although the exact risk of ovarian damage or even premature ovarian failure after LOS is unknown, this risk may be nonnegligible [27]; therefore, only patients with excessively high levels of AMH may be considered eligible for such a procedure. Nevertheless, according to a recent metaanalysis, LOS may even decrease the live birth rate in women with anovulatory PCOS and CC resistance compared with medical ovulation induction alone [28].

There is emerging evidence that success rates after conventional OS and IVF [38], as well as those after IVM [6,29] may be improved when a freeze-all embryo approach is adopted instead of a fresh embryo transfer. However, a recent concern in the approach of women with PCOS who undergo FET is the observation that FET may be associated with an increased risk of early pregnancy loss [30] if FET is performed in hormonal replacement therapy (HRT) cycles. This is possibly due to inadequate progestin support [39]. Also, evidence has arisen that pregnancies following HRT-FET are at higher risk of hypertensive disorders of pregnancy (HDP) due to the absence of vasoactive molecules otherwise produced by the corpus luteum [31]. In view of this, it seems mandatory to develop efficient clinical protocols for FET that are associated with a lower risk of adverse obstetric events in general, and in women with PCOS who undergo IVM in specific.

Oncofertility preservation is another emerging indication for IVM. Cryopreservation of oocytes before the start of cancer treatment has been offered to postpubertal cancer patients in whom cancer treatment can be delayed with two to three weeks. When OS is not feasible, for example, because of time constraints (when an urgent start of cancer treatment is required), or in prepubertal girls, retrieval of immature oocytes and IVM of these oocytes is an alternative option. COCs can be harvested through transvaginal follicle aspiration, and/or they can be retrieved “ex vivo” from small antral follicles obtained from oophorectomy specimens, when ovarian tissue is surgically resected [34]. The latter procedure is referred to as IVM of ovarian tissue–derived oocytes (OTO-IVM), which may be offered in connection to ovarian tissue cryopreservation.

Finally, IVM may be the only valid reproductive treatment for women with ovarian resistance to FSH or LH who want to have genetically own offspring. Ovarian resistance to FSH or LH is a rare heterogenous condition characterized by impaired follicle growth due to a dysfunctional FSH receptor or LH receptor based on a genetic or immunological abnormality [35].

Standard IVM: clinical and laboratory protocols

As a typical clinical work-up for standard IVM, women with a normal cycle length (≤35 days) will receive injected recombinant FSH (rFSH) or highly purified human menopausal gonadotropin (hp-hMG) 150 IU/day starting on Day 2 or 3 of the spontaneous menstrual cycle. Women who do not have a normal cycle length (>35 days or amenorrhea) may take an oral contraceptive for 2–3 weeks, then receive rFSH or hp-hMG 150 IU/day injections for 2 or 3 days from 5 days after the last contraceptive pill onwards [12,13].

The clinical protocol of standard IVM as used in Brussels is depicted in Fig. 16.1. Typically, no hCG is administered before oocyte retrieval. Vaginal ultrasound-guided retrieval of cumulus–oocyte complexes from small antral follicles for IVM is scheduled 42 h after the last follitropin injection. In our IVF clinic in Brussels, all oocyte retrieval procedures are performed using a 17-gauge single lumen needle (Cook Medical, K-OPS-1230-VUB, Limerick, Ireland). Follicular aspirates are collected in Human Tubal Fluid (HTF) (IVF Basics VR HTF HEPES, Gynotec B.V. Malden, the Netherlands) supplemented with heparin (5000 IU/mL, Heparin Leo, Leo Pharma, Belgium; final heparin concentration 20 IU/mL) and filtered through a cell strainer (FalconVR, 70micron mesh size, BD Biosciences, CA, USA). After collection, COCs are washed and transferred to a four-well dish (Nunc; Thermo Fisher Scientific; MA, USA) containing IVM medium (IVM System, Medicult, Origio) supplemented with 75 mIU/mL HP-hMG (Menopur, Ferring, Saint-Prex, Switzerland), 100 mIU/mL hCG

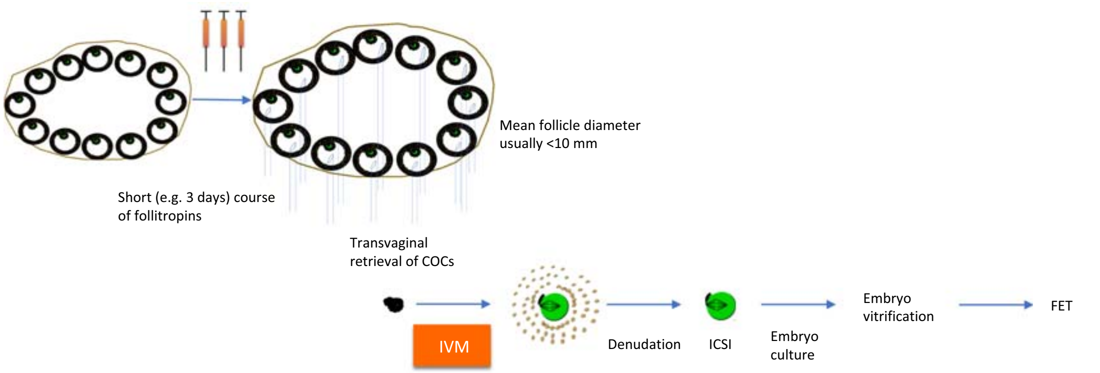

FIGURE 16.1 **Standard IVM.** Immature, germinal vesicle-stage cumulus–oocyte complexes (COCs) are matured in vitro (IVM) to the metaphase II stage in registered media supplemented with FSH, hCG, and proteins. Patients may or may not receive priming with follitropins (e.g., recombinant FSH or menotropins). Because of possible zona hardening of the oocytes, matured oocytes are typically inseminated with sperm using intracytoplasmic sperm injection (ICSI). Embryos are cultured to the cleavage or blastocyst stage, at which point they are usually vitrified before they are transferred (FET, frozen embryo transfer).

(Pregnyl, MSD, Oss, the Netherlands), and 10 mg/mL human serum albumin (Vitrolife, Göteborg, Sweden). As per standard IVM definition, all COCs are compact at the time of oocyte retrieval and contain cumulus-enclosed GV-stage oocytes, in contrast with hCG-primed IVM, where oocytes are collected at a mixture of meiotic stages [32]. COCs are cultured for 30 h in groups of 10 COC per well in 500 μL IVM medium with oil overlay (Ovoil, Vitrolife) at 37°C under 6% CO_2 and 20% O_2.

Following IVM, oocytes are mechanically and enzymatically denuded from their cumulus layers under a stereomicroscope, and oocyte maturation is assessed under the inverted microscope. Matured oocytes are inseminated using ICSI with partner sperm as described by Ref. [33]. Fertilization is assessed 16−18 h postinsemination by the presence of two pronuclei. Embryos generated after IVM are cultured in individual droplets of 25 μL of sequential media formulations (Quinn's Advantage TMFertilization, Cleavage and Blastocyst medium, SAGE or FertTM, CleavTM, BlastTMmedium, Origio) with oil overlay (Ovoil, Vitrolife) until Day 3 or until Day 5 (or Day 6) after ICSI, depending on the number of embryos available on Day 3; in cycles with at least four embryos on Day 3 that are classified as transferable/good-quality embryos according to the criteria described by Van Landuyt et al. [40], embryos are cultured until Day 5 or Day 6 after ICSI. Embryos are selected for vitrification according to the morphological criteria described in Belva et al. [41]. Vitrification is performed using closed CBS-VIT high-security straws (CryoBioSystem, L'Aigle, France) in combination with DMSO, ethylene glycol and sucrose as cryoprotectants (Irvine ScientificVR Freeze kit, Newtownmountkennedy, Ireland) according to the method previously described by Van Landuyt et al. [42]. IVM embryos are vitrified electively on Day 3 or 5/6 after ICSI. Embryo transfer is typically performed in an artificial endometrium priming cycle initiated when baseline hormone levels are reached after the IVM cycle.

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CHAPTER

17

Stimulated in vitro oocyte maturation

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Introduction

Notwithstanding the burgeoning success of in vitro fertilization (IVF) [1], it is important to recognize the seminal role of in vitro maturation (IVM) in the development of IVF as well as its current widespread application among the arsenal of assisted reproductive techniques (ARTs) for over 25 years [2,3]. While traditional IVF involves the recovery of in vivo-maturated oocytes, IVM refers to the recovery of immature oocytes from small antral follicles at the germinal vesicle (GV) or metaphase I (MI) stage and subsequent meiotic resumption under specifically controlled culture conditions. Cha et al. reported the first birth using IVM of immature oocytes collected at cesarean section within an oocyte donation program in 1991 [4], but it was only after Trounson and colleagues reported the first pregnancy using a woman's own immature oocytes collected by transvaginal ultrasound-guided follicle aspiration that IVM emerged as a viable alternative to IVF for patients with polycystic ovaries [5]. Currently, the exact definition of IVM remains a topic of ongoing debate depending on the utilization of gonadotropins, especially human chorionic gonadotropin (hCG) priming [6,7]. For the purposes of this chapter, IVM refers to any hormonally stimulated or unstimulated cycle for which the majority of oocytes are expected to be at the GV stage at the time of egg collection [8].

Oocyte physiology

Oocyte maturation is the physiological event that precedes, and is required for, successful fertilization and embryo development. Oocytes initially mature during the fetal period

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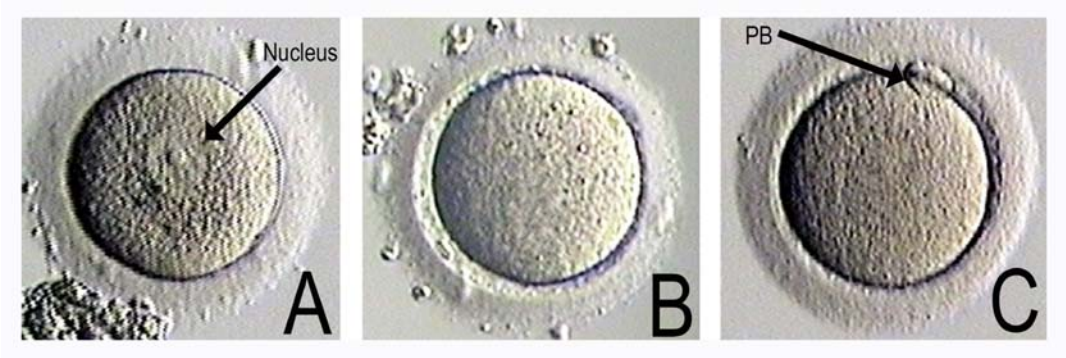

FIGURE 17.1 Different stages of oocyte development; (A) oocyte at germinal vesicle stage (GV), the nuclear membrane is intact and the nucleolus is visible; (B) metaphase 1 oocyte, the nuclear membrane dissolves and the nucleolus disappears; (C) metaphase II oocyte, the first polar body extrudes. Note: these oocytes are after the removal of cumulus cells.

and become arrested at the diplotene stage of prophase I (GV stage) until they are committed to ovulation or atresia. Resumption of meiosis and progression through maturation results in arrest at the metaphase II (MII) stage, with extrusion of the first polar body (Fig. 17.1). In vivo, the trigger for resumption of meiosis is the preovulatory surge of luteinizing hormone (LH). However, removal of the oocyte from the inhibitory influence of the follicle also allows spontaneous resumption of meiosis [9]. Importantly, the success of IVM relies on techniques that promote both nuclear and cytoplasmic maturation of the oocyte. Nuclear maturation consists of GV breakdown (induced by the LH surge in vivo) followed by resumption of meiosis and extrusion of the first polar body (MII). Cytoplasmic maturation is more difficult to assess microscopically as it involves the redistribution of various organelles, including cortical granules, and accumulation of factors that prepare the oocyte for fertilization and embryonic development [10].

Potential indications for IVM

IVM offers several key advantages over other assisted reproduction techniques, including a lower risk of adverse events and reduced financial and emotional burden owing to the short duration of monitoring and relatively low medication requirements. Historically, IVM has primarily been proposed as an alternative to IVF particularly for polycystic ovarian syndrome (PCOS) patients at risk for ovarian hyperstimulation syndrome (OHSS) given that they generally have the highest number of antral follicles for potential retrieval and simultaneously at highest risk of developing OHSS under traditional stimulation protocols. Other indications include patients with limited time for ovarian stimulation as well as those with contraindications to sustained elevation of estradiol (E2). However, studies have shown that IVM can be used in almost all areas where IVF and other ARTs are used [2,3,11].

IVM in the era of antagonist protocol with agonist trigger

Despite increasingly widespread use of gonadotropin-releasing hormone (GnRH) agonist trigger in GnRH antagonist cycles, OHSS remains an ever-present risk with gonadotropin stimulation [12]. Indeed, early severe OHSS can occur if there is agonist trigger with low-dose hCG (1500 IU) rescue, if there is agonist trigger with high-dose estrogen and progesterone supplementation for fresh ET, or even if there is agonist trigger and a freeze-all embryos policy. Only IVM can avoid OHSS completely.

IVM instead of natural cycle IVF

Natural cycle IVF involves no ovarian stimulation, and triggering is generally performed once the leading follicle reaches 18-mm diameter; however, it is associated with up to 30% risk of premature LH surge and ovulation. Alternatively, modified natural cycle IVF requires daily GnRH antagonist and follicle-stimulating hormone (FSH) injections once a follicle reaches 14-mm and continued until hCG triggering, which equates to at least three FSH and GnRH antagonist injections with a subsequent 15%–20% clinical pregnancy rate per cycle started given that only one MII oocyte is retrieved. In stimulated IVM, three injections of FSH are given on Days 4, 6, and 8, with subsequent hCG administration when the leading follicle is 12–14-mm, thereby forgoing the need of GnRH antagonist. Furthermore, several MII and GV oocytes can be obtained to generate multiple blastocysts, thereby increasing clinical pregnancy rates per cycle started of up to 45%–50% in women up to 37 years of age.

IVM instead of FSH/IUI

The first line of treatment in many fertility programs is FSH stimulation combined with intrauterine insemination (IUI). This treatment requires approximately 8 to 10 daily injections of FSH and has a pregnancy rate of 15%–20%, multiple pregnancy rate of 30%, and may cost upwards of $2000 in North America for medications and IUI. In comparison, IVM costs ~$4000 but requires fewer injections and generally yields higher pregnancy rates, lower odds of multiple pregnancy, and no risk of OHSS. In fact, Hatirnaz et al. have achieved a 35% live birth rate per IVM cycle with elective single ET [13], and unpublished data suggest similar results with letrozole-stimulated IVM cycles.

IVM and fertility preservation for cancer

When presented with an oncology patient wanting fertility preservation, it is important to first determine whether ovarian stimulation can be safely performed and how long the patient can wait before the start of chemotherapy [14]. If hormone stimulation is not contraindicated and chemotherapy can wait, ovarian stimulation followed by mature egg collection should be performed. However, IVM with or without ovarian tissue freezing is the only viable option if hormone stimulation is contraindicated or there is no time. Importantly, IVM allows multiple egg collections to be performed at any phase of the menstrual cycle including the luteal phase [15].

IVM and resistant ovary syndrome

A few small studies and case reports have investigated the utility of IVM among women with repeated ART failure owing to resistant ovary syndrome with promising results [16,17]. For instance, Galvao et al. reported a case series of nine women with repeated IVF failure who underwent 24 IVM cycles and achieved a live birth rate of 16.7% per started cycle and 33.3% per patient [16].

Protocols

Given significant advances in the field of reproductive laboratory medicine, it is important to emphasize that IVM protocols have significantly improved over the last few years which has translated to higher take-home baby rates and a more informed appreciation of the overall safety profile for IVM [18,19]. However, research is still ongoing regarding the optimal strategy for follicular priming, oocyte retrieval techniques, and IVM culture conditions.

Patient preparation

Generally, priming with hCG and/or FSH appears to improve implantation and pregnancy rates compared with no priming. Our randomized controlled trial showed that eggs collected at the GV stage successfully matured to MII after 24 h in women who received hCG priming compared to only 5% among women who were not given hCG [20]. By 48 h of maturation, rates were 84% versus 69% in hCG and control groups, respectively. Similarly, Sanchez et al. have achieved similar maturation rates with preincubation using C-type natriuretic peptide [18].

For ovulatory patients, a reasonable strategy would be to induce a withdrawal bleed, measure the antral follicle count, and repeat a second ultrasound scan when the largest follicle is approximately 12–14 mm and administer 10,000 IU hCG trigger. If the patient is anovulatory, FSH 150 IU can be given on Days 4, 6, and 8. In order to synchronize the development of the eggs and endometrium, E2 can be given from the time of egg collection and progesterone from the time of embryo fertilization. The dosage of estrogen depends on the endometrial thickness (ET): 10–12 mg of E2 is given for ET < 6 mm, whereas 8–10 mg of E2 is given in divided doses for ET between 6 and 8 mm.

Fertilization

Evidence is mixed with respect to fertilization strategy in IVM. We prefer intracytoplasmic sperm injection (ICSI) for fertilization, but recent studies have suggested that conventional IVF may suffice if semen analysis is normal [21]. Interestingly, Soderstrom-Anttila et al. also found that fertilization rates were 37.7% with conventional IVF compared to 69.3% with ICSI; however, implantation rates were 24.2% with conventional IVF versus 14.8% ICSI and subsequent clinical pregnancy rates were also in favor of IVM-IVF (23.8%) versus IVM-ICSI (17.1%). Hence, ICSI has been shown to promote higher rates of fertilization, although potentially at the expense of overall developmental competence compared to conventional IVF.

Aspiration technique and culture

Aspiration of small follicles in IVM differs from egg collection techniques used in traditional IVF both with respect to optimal aspiration pressure and needle design. In IVM cycles, smaller needles and lower aspiration pressure is selected than in standard IVF, with pressure ranging from 80–300mm Hg, and 16–20 gauge needles used. Higher pressures negatively impact embryo development. Double lumen needles and single-channel pseudo-lumen needles have been described to enable follicle flushing and has been shown in some studies to obtain superior oocyte retrieval results [22]. Dahan et al. [23] described a “rapid pass” technique, which involves aspiration under constant suction pressure (7.5 KPA) using a 19-gauge single lumen oocyte aspiration needle and collecting the visible follicles first. Although the stroma may appear devoid of follicles on ultrasound, suction is subsequently maintained while moving the needle through the long axis of the ovary, while repeatedly changing the vertical path of collection to have the needle tip pass through a significant proportion of the ovarian stroma. After 3–5 min of constant suction, the needle is removed from the ovary and flushed with heparinized saline to clear any possible blood clots. Using this technique, as many as 125 oocytes have been collected in a single cycle, although many were extremely immature and did not mature well by conventional IVM techniques [23]. Research advances by Telfer and colleagues in Edinburgh on in vitro growth of primordial follicles will hopefully improve this in the future [24].

Embryo transfer

Although fresh embryo transfer can be performed on Day 3 or 5, recent data suggest that frozen ET in an artificial cycle is better. Walls et al. [21] showed that live birth rate per cycle after fresh ET was lower with IVM than IVF, although the live birth rate per cycle with frozen embryos is comparable with IVM and IVF at over 30% per cycle.

Safety concerns

Although the long-term developmental outcomes of children conceived with IVM have only been studied in small numbers, current evidence suggests that fetal outcomes and incidence of congenital malformations are similar to pregnancies derived from IVF and spontaneous pregnancies in healthy women [25–32]. This reflects related cellular and molecular studies which found normal ultrastructural morphology by transmission electron microscopy (TEM) and no increase in imprinting errors rates at maternally or paternally methylated gene loci in IVM-derived oocytes [33,34].

Conclusion

IVM is a well-studied and safe procedure that has been practiced for several decades. It is a low intervention, mild approach to ART that offers improved safety and a simplified clinical approach compared with IVF. However, adoption has been mixed as incentives leading to enhanced uptake of IVM in the ART clinic vary widely around the world. As further innovations improve the recovery rate of immature oocytes and embryo yield from IVM, it has the potential to offer new approaches to infertility management, social fertility preservation, and toward a new clinical paradigm of minimal or zero-stimulation ART.

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CHAPTER

18

Fertility preservation

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Introduction

Fertility preservation aims at safeguarding the reproductive potential in men and women at risk of fertility loss. Both the quantity as well as the quality of gametes can be affected by the exposure to gonadotoxic treatments or impacted by genetic conditions or, especially in women, by biological aging. There are various indications for fertility preservation, ranging from malignant diseases, benign conditions, and gender-confirming procedures to age-related fertility decline [1,2]. Referral for and implementation of fertility preservation has increasingly been incorporated in health care guidelines for different patient groups [3,4].

Indications

Fertility preservation for malignant disease

Survival after cancer continues to improve with new treatments and supportive therapies. In Europe, nearly 80% of children and adolescents with cancer become "long-term survivors" [5,6]. Improving survival rates expands oncologic care to the management of long-term complications [5]. It is widely known that cancer treatments such as chemotherapy, radiotherapy, and/or surgery can induce premature ovarian insufficiency [7]. The gonadotoxicity of these treatments depends on multiple variables such as the ovarian reserve, patients' age at the time of treatment, administered therapies, and their doses and duration [8].

Chemotherapy

The ovaries are very sensitive to cytotoxic drugs, and damage can be caused by different mechanisms: direct loss of primordial follicles, accelerated activation of primordial follicles, follicular atresia, stromal tissue damage, and damage to the vasculature or inflammation [9]. As the oocyte is dependent of the follicular somatic cells, somatic cell death will ultimately cause oocyte apoptosis [10]. These cytotoxic drugs can either affect the resting

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primordial follicle pool or the growing follicle pool. The loss of the growing follicle pool results in less production of inhibitory substances by the growing follicles, leading to increased recruitment of primordial follicles and an accelerated depletion of the resting follicle reserve. This results clinically in a "two-hit" effect. The first hit is an immediate but temporary effect, occurring during cytotoxic treatment, and is the effect of losing the growing follicle population. This first hit is reversible, and if sufficient primordial follicles have remained in the dormant follicle pool, the population of growing follicles will be replenished upon cessation of gonadotoxic treatment and menses will resume. The second hit is the impact of losses in the primordial follicle pool, causing primary ovarian insufficiency (POI). A partial loss of primordial follicles will result in a future ovarian failure, whereas a complete reduction of the pool will result in a more acute effect where the patients experience POI shortly after the chemotherapy [10]. Some chemotherapy regimens are more toxic than others. The risk of POI is the highest with cyclophosphamide, being the alkylating agent known to cause the most damage to oocytes and granulosa cells, at a dose of $>7.5\ g/m^2$, busulfan >600 mg, or ifosfamide >60 g [8,11–13].

Radiotherapy

The impact of radiotherapy depends on the radiated field, total dose, dose per fraction, and age of the patient at the time of radiotherapy. Pelvic radiotherapy is known to cause irreversible ovarian damage in a dose–response relationship for decreased risk of pregnancy with increasing dose of ovarian/uterine radiation. The estimated dose of radiation at which half of the human primordial oocytes are lost (LD50) is <2 Gy [8,14]. Data from Childhood Cancer Survivor study showed that among cancer survivors, only those who received an ovarian/uterine radiation dose greater than 5 Gy were less likely to have ever been pregnant. Direct irradiation of the hypothalamus and/or pituitary may produce impaired secretion of follicle-stimulating hormone (FSH) and luteinizing hormone (LH), especially when the dose is > or = 30 Gy [12].

Surgery

Bilateral oophorectomy clearly results in premature menopause. Unilateral oophorectomy reduces the ovarian reserve, causing menopause to occur up to seven years earlier [15]. Therefore, fertility preservation should be considered, especially if the contralateral ovary is likely to undergo another gonadotoxic insult such as radiotherapy or chemotherapy. Also performing cystectomy on endometriomas may cause considerable reduction of the ovarian reserve, so fertility preservation or fertility-sparing surgery should be considered in these cases [16].

Fertility preservation for benign conditions

Some nonmalignant conditions carry the risk of premature ovarian insufficiency. Benign hematological diseases (thalassemia, sickle cell anemia, and aplastic anemia), and autoimmune disorders (rheumatoid arthritis, systemic lupus erythematosus, Behçet disease, and Wegener's granulomatosis) might require chemotherapy, radiotherapy, or bone marrow transplantation [1,8]. Ovarian pathology such as bilateral benign ovarian tumors (severe of recurrent) ovarian endometriosis, ovarian torsion, or a preventive oophorectomy for

BRCA1/2 also threaten gonadal function. Some pathologies carry a risk of premature ovarian insufficiency such as Turner syndrome, blepharophimosis, ptosis, and epicanthus inversus syndrome (BPES), or idiopathic POI [1,7,17].

Fertility preservation in transgender persons

Gender dysphoria refers to distress or discomfort caused by the incongruence between a person's sex assigned at birth and the gender a person identifies him or herself with [18]. Treatment options to facilitate the transition and to live in accordance with the identified gender comprise both hormone therapy (and/or) surgical interventions [18]. Unfortunately, these options might compromise fertility [18,19].

Pubertal suppression

Gonadotropin-releasing agonists (GnRH-a) preventing the development of secondary sexual characteristics inhibit germ cell maturation [20]. However, discontinuation results in normal pubertal progression [21,22]. Hagen et al. (2012) showed that GnRH-a administration in girls with precocious puberty resulted in a significant decline of anti-Müllerian hormone (AMH) after three months with recovery to pretreatment AMH serum levels six months after interruption of the treatment [20].

Cross-sex androgen therapy in trans men

Masculinizing hormonal therapy will in most cases lead to a reversible amenorrhea, but ovarian follicles are not depleted from the tissue [23,24]. Increased androgen levels may however adversely affect follicle growth, mostly affecting the more matured follicle stages as these follicles are surrounded by an androgen-sensitive theca cell layer [24–26]. Apparently, primordial follicles in cryopreserved and xenotransplanted ovarian cortical tissue do not lose their potential to resume growth and maturation [26].

A current debate is still ongoing whether or not masculinizing therapy induces polycystic ovary syndrome (PCOS) [25–27]. Descriptive histology by Pache et al. (1991) showed some evidence for altered morphology induced by testosterone treatment comprising enlarged ovaries, collagenized ovarian cortex, theca interna hyperplasia, and stromal hyperplasia [25,28]. The idea that androgens induce high AMH levels, which is a particular feature of PCOS, is however questionable [26]. Moreover, a (3D) transvaginal ultrasound study in trans men treated with long-term testosterone therapy showed no increased polycystic ovarian morphology [29].

Sex reassignment surgery

Sex reassignment surgery with hysterectomy with bilateral oophorectomy leads to irreversible sterility [30,31].

Fertility preservation for age-related fertility decline

Female fertility gradually decreases, inevitably compromising fertility over time. Ovarian aging causes a quantitative and qualitative oocyte decline by progressively losing the primordial follicle (quantitative) and increase chromosomal aneuploidy (qualitative) [32]. Fertility preservation can be offered to anticipate age-related fertility decline and finally gamete exhaustion [1].

Cryopreservation techniques

Gametes

While sperm cryopreservation has been an established technique for many decades, oocyte cryopreservation has only been optimized more recently. The major advancement has been the introduction of the oocyte vitrification technique replacing the slow freezing technique [33]. Direct contact with liquid nitrogen allows for an ultrarapid cooling of the oocyte, although closed vitrification devices have also proven to be efficient [34]. Initial clinical experience with regard to oocyte warming, however, has been obtained from oocyte banking in egg donation programs [35,36]. These studies ultimately resulted in the generalized acceptance of the oocyte vitrification technique and the removal of its experimental label [37].

Gonadal tissue

Ovarian tissue

Ovarian tissue (OT) cryopreservation is the surgical resection and cryopreservation of OT and is increasingly accepted as an established fertility preservation technique [7,38–42]. OT cryopreservation does not require controlled ovarian stimulation and can be performed immediately [43]. Hence, OT cryopreservation is the only option available for prepubertal young patients and for women who cannot delay the start of chemotherapy [38].

At this time, the only option to restore fertility using this tissue is by transplantation [39]. Transplantation of OT can be performed orthotopically or heterotopically. Orthotopic transplantation, transplantation in the pelvic cavity, provides the optimal environment compared to the physiological conditions [38]. The advantage of the orthotopic site is that the possibility of natural conception without the aid of assisted reproduction techniques [44]. Ovarian fragments can be transplanted upon the remaining ovary or in a peritoneal window [45,46]. However, it is an invasive procedure that may cause pelvic adhesions and the number of transplanted fragments is limited [44,47].

Heterotopic transplantation, transplantation in an extraovarian region, has been tested in humans in the uterus broad ligament [48], subperitoneally at the anterior abdominal wall [49], the rectus muscle [50], and the forearm [44,51]. Advantages of heterotopic transplantations include possible larger number of transplanted fragments, accessibility of the graft, and feasibility of transplantation in case of pelvic adhesion [44]. However, natural conception cannot be expected. In humans, the first birth after OT cryopreservation and grafting was reported by Donnez et al. in 2004 [52], and since then, more than 130 live births worldwide have been reported with the aid of transplantation of cryopreserved OT (not included unreported cases and ongoing pregnancies) [53]. Transplantation of cryopreserved OT also restores endocrine function in patients [38]. Restoration occurs within two to nine months postgrafting, and the mean duration of ovarian function is four to five years, depending on the follicular density in the graft [41,54]. Follicle burn out is one of the major issues to be addressed in OT transplantation. An important cause of the massive and accelerated loss of follicles after transplantation is the warm ischemia posttransplantation [55].

OT cryopreservation may not be suitable for all patients. In transgender men, transplantation of ovarian cortical strips and thus restoring female hormone activity is an unwanted side effect. Also, transplantation in cancer patients has a potential risk of reintroducing malignant cells [39,40,56–60]. This risk is disease dependent and, based on the available data, this risk is high in leukemia, moderate for gastrointestinal cancers, and rather low for breast cancer, sarcomas of the bone and connective tissue, gynecological cancers, and Hodgkin's and non-Hodgkin's lymphoma [59].

Testicular tissue

Testicular tissue cryopreservation means the freezing of the testicular tissue. For this technique, a surgical biopsy of testicular tissue from pre- or postpubertal transgender women is performed [2]. This option overcomes the need for masturbation and is possible in prepubertal boys [2]. It is a surgical procedure that can be combined with genital reconstructive surgery. Compared to cryopreservation of sperm, this is an experimental method. For future use, an in vitro maturation (IVM) procedure, which is currently not clinically possible, or transplantation is necessarily followed by assisted reproduction techniques.

As described for OT, also testicular tissue cryopreservation and transplantation may not be suitable for all patients due to the potential risk of reintroducing malignant cells in cancer patients or the undesired restoration of the male endocrine environment in transgender women.

Clinical approach

Female (Fig. 18.1)

Ovarian stimulation for embryo or mature oocyte cryopreservation is the most preferred method for fertility preservation in postpubertal women [61]. The GnRH antagonist protocol is considered the ovarian stimulation protocol of choice for fertility preservation [61]. Although the GnRH antagonist results in pregnancy rates comparable to the GnRH agonist protocol, it has important benefits that are particularly crucial for the fertility preservation population [62]; short duration of stimulation; and the prevention of the ovarian hyperstimulation syndrome (OHSS) by the association with the GnRH agonist trigger [63,64].

Ovarian stimulation in estrogen-sensitive cancers

It has been hypothesized that elevated serum estrogen levels during ovarian stimulation may induce breast tumor growth. This has resulted in the association of aromatase inhibitors (Letrozole) or selective estrogen receptor modulators (Tamoxifen), the GnRH antagonist protocol resulting in significantly lower estrogen levels [65,66].

Random start and duostim

Multiple major and minor waves in follicular development have been described allowing for the initiation of ovarian stimulation throughout the menstrual cycle [67,68]. In contrast to

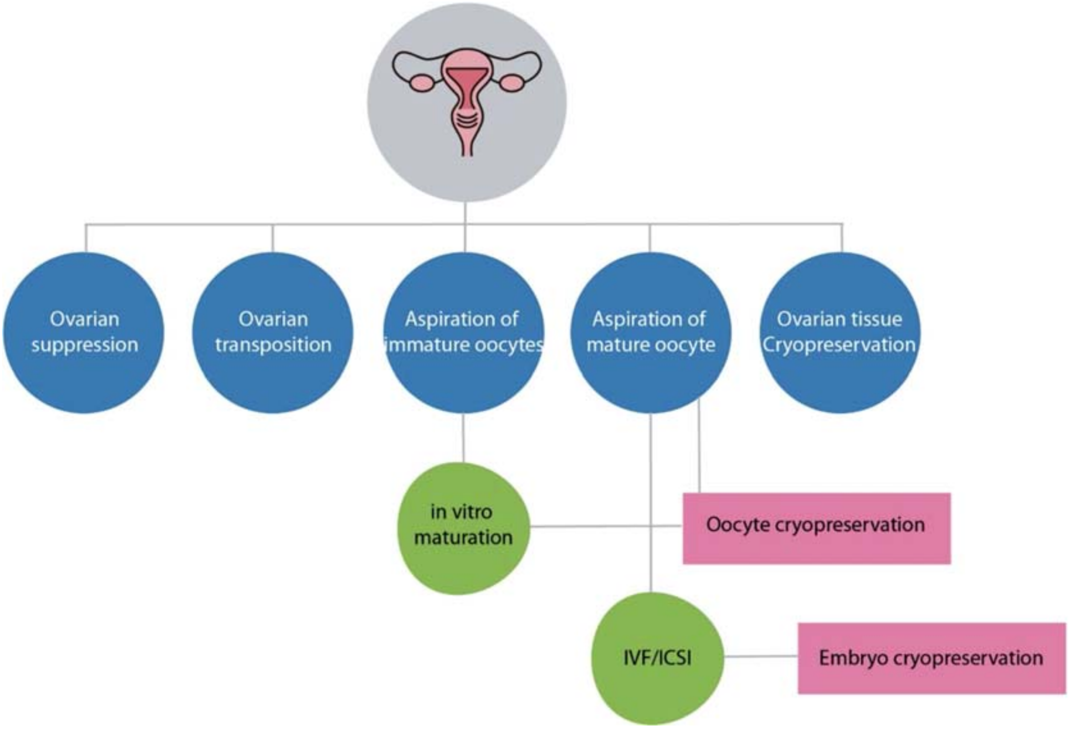

FIGURE 18.1 Options available for female fertility preservation.

earlier belief, the presence of a corpus luteum or luteal phase progesterone does not adversely affect the follicular development and oocyte yield in random-start patients [61]. Therefore, for patients not desirable to wait for the next menstrual period to start ovarian stimulation, random-start stimulation has been proposed [69].

Double ovarian stimulation or DuoStim allows for the optimization of the oocyte yield, especially in patients with an expected poor ovarian response and/or a limited amount of time before initiation of the gonadotoxic therapy [70]. Intriguing about this ovarian stimulation strategy is the reporting by several authors of an increased number of oocytes from the second stimulation as compared to the first [71,72].

GnRH agonist administration

The administration of GnRHa is used to downregulate the hypothalamic–pituitary–gonadal axis and thus to block ovarian function [73]. The use of fertiprotective agents during cancer treatment is increasingly being explored, particularly in breast cancer, showing a reduction in the prevalence of POI [74]. The effectiveness of GnRHa coadministration for other cancers is not clear [9,75]. However, ovarian downregulation yields other medical benefits such as reduced vaginal bleeding in patients with chemotherapy induced low platelets [7].

Transposition of the ovaries

Ovarian transposition or oophoropexy is the repositioning of the ovaries away from the radiation field before pelvic radiotherapy [7,76]. Ovarian transposition can be performed laparoscopically, and the ovaries are usually relocated and fixed to the anterolateral abdominal wall, 3–5 cm above the umbilicus and metallic clips are left at the inferior pole to mark the location [7]. Failure of this technique is due to scatter irradiation and blood supply damage [77]. The surgeon should thus pay attention dissecting both ovarian arteries and veins not to compromise the blood supply to the ovary while relocating the ovaries. Because of the risk of remigration of the ovaries, the intervention should be performed as close to the radiation treatment as possible [43]. Obviously, this technique does not protect against gonadotoxic effects of chemotherapy. Also, ovarian transposition should be restricted to carefully selected patients with a low risk of ovarian cancer involvement [76].

In vitro maturation of oocytes

IVM involves the in vitro maturation of immature cumulus–oocyte complexes (COCs) collected from antral follicles, with or without FSH priming, from the germinal vesicle (GV) stage to metaphase II (MII) [78–81].

IVM was initially introduced in clinical practice as a safer alternative for conventional ovarian stimulation to avoid OHSS and ovarian torsion [82,83]. IVM constitutes a promising alternative to in vitro fertilization of in vivo-matured oocytes conventional in predicted or expected high responders, such as PCOS patients [83–87]. Encouraging results have been reported in these patients, with maturation rates up to 84% [88], fertilization rates up to 80% [89], clinical pregnancy rates up to 50% per cycle [90], and live birth rates up to 33% [90] as reviewed by the Cochrane library in 2018 [83]. Subsequently, IVM indications have been broadened to include fertility preservation, with the advantage that controlled ovarian stimulation can be omitted in case of estrogen-sensitive tumors.

Ovarian tissue and ovarian tissue oocyte IVM

Although in its initial stages COCs were collected by transvaginal follicle aspiration, new procedures include in situ follicle aspiration from the ovary or the contralateral ovary during laparoscopy/laparotomy and ex vivo aspiration either from the excised OT or the spent media during cortex preparation for cortex cryopreservation. This latter protocol, called ovarian tissue oocyte (OTO)-IVM [91], is now increasingly being explored to maximize fertility preservation potential where the clinical application of OT transplantation may be contraindicated. Cancer patients in whom orthotopic transplantation of OT carries the risk of malignant cell contamination [56] and transgender men in whom transplantation is undesired as it restores the endogen productions of female hormones [24,92] are potentially benefiting of OTO-IVM.

Planned oocyte cryopreservation

Planned oocyte cryopreservation or AGE-banking (Anticipation of Gamete Exhaustion) differentiates itself from all other fertility preservation indications as it prevents the natural age-related decline of female fertility in the absence of medical condition [93]. In face of the increasing number of women confronted with age-related subfertility, an increasing number of women turn to oocyte cryopreservation to safeguard some of their reproductive potential [94]. The mean age at the time of first oocyte retrieval is around 37–38 years in most published reports [32,95–97]. Utilization rates of the cryopreserved oocytes tend to vary between different centers but may end up as high as 38.1% in studies with adequate follow-up time [98].

Male

Emergency sperm cryopreservation

Human sperm cryopreservation is a procedure to preserve sperm cells through freezing. For human sperm, the longest reported successful storage is 40 years [99]. This procedure is the simplest and most reliable method of male fertility preservation [2]. The sperm is obtained through masturbation. Electroejaculation or penile vibratory stimulation can be performed in case of difficulties to masturbate in order to produce a semen sample for preservation. However, electroejaculation requires anesthesia. Depending on the sperm quality, the freezing of the sperm cell or spermatozoa can be used for future intrauterine insemination or to perform IVF/ICSI.

Testicular sperm extraction and MESA

In cases of surgical sperm extraction, a percutaneous aspiration of sperm from the testis or the epididymis is performed [2]. The obtained spermatozoa can be used for future ICSI procedures.

Testicular tissue cryopreservation

Testicular tissue cryopreservation means the freezing of the testicular tissue. For this technique, a surgical biopsy of testicular tissue is performed. This option overcomes the need for masturbation and is possible in prepubertal boys. It is a surgical procedure that can be combined with genital reconstructive surgery. Compared to the other two options, this is an experimental method. For future use, an IVM procedure, which is currently not clinically possible, or transplantation is necessarily followed by assisted reproduction techniques. Transplantation can, however, restore the male endocrine environment, which clearly is an undesired effect for transgender women.

Conclusion

Fertility preservation is a rapidly developing and expanding discipline. This chapter aims at a deeper understanding of different indications, technical procedures, and clinical approaches to support a more comprehensive and specific decision-making for individuals in need for fertility (preservation) counseling or treatment options.

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CHAPTER

19

Preimplantation genetic testing

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The origins of PGT

Preimplantation genetic testing (PGT) is a specific area of genetic diagnostics investigating human embryo preimplantation period, and it is aimed at the identification of embryos not affected from monogenic conditions and/or chromosomal abnormalities among the ones produced during an in vitro fertilization (IVF) treatment.

The origins of PGT can be traced back to the period from 1952 to 1967, when Richard Gardner and Robert Edwards attempted to sex the blastocysts from animals such as cows and rabbits. In particular, the bovine embryo at the blastocyst stage can reach a size of several millimeters making it possible to remove some cells by hand with a small scalpel blade cutting a section of the outer trophectoderm (TE), not altering the embryonic disc. In 1968, their landmark study in the evolution of this technique was published. Adopting rabbit embryos and a novel method of immobilizing them though a holding pipette and a pair of fine iridectomy scissors normally used for eye surgery, they were able to biopsy small pieces of TE and aspirate them into a fine pipette [1]. Then, after a selective TE cell staining, they examined microscopically the presence of a dense Barr body on the periphery of the interphase nuclei indicating the heterochromatic inactive X chromosome in females. This approach was postulated also in humans for couples at risk of X-linked inherited diseases to select unaffected embryos. This brilliant intuition, however, was neglected for about two decades and finally credited only after the development of polymerase chain reaction (PCR). The main principle behind PGT requires the access to the embryonic DNA while avoiding any potential adverse effect on embryo viability or competence. However, back in the 1970s, it was hard to set-up a protocol for human embryo (200−300 μm) manipulation and downstream genetic analysis

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from a few cells [2]. It was in the early 1990s that the first successful biopsies of single cells for PGT were accomplished from human embryos and then sexed [3]. Through this pioneering clinical experience, in July 1990, some patients carrying X-linked inherited disorders gave birth to the first female twins after the application of a PGT approach. Following this initial success, this biopsy protocol was successfully replicated and implemented clinically in several clinics worldwide.

Different kinds of PGT and their indications

PGT was previously known as PGD for genetic conditions (preimplantation genetic diagnosis) and PGS for chromosomal imbalances (preimplantation genetic screening). Nevertheless, the ICMART and WHO in 2017 recognized that especially "screening" was a misleading term and introduced the terms PGT-M for monogenic conditions, PGT-SR for structural rearrangements, and PGT-A for aneuploidies [4]. In Fig. 19.1 are presented the indications for the different types of PGT.

Indications to PGT-M

To date, thousands of diseases caused by single-gene disorders have been identified, either autosomal dominant, recessive, or X-linked (e.g., cystic fibrosis, Tay-Sachs, fragile X, myotonic dystrophy) with an estimated overall prevalence of around 1% in the general population

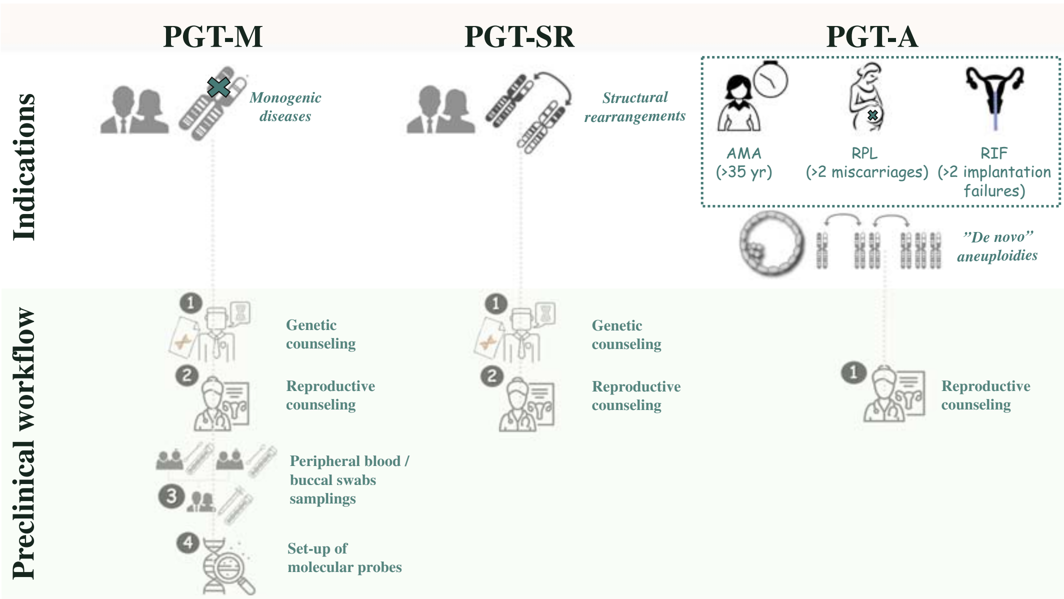

FIGURE 19.1 Summary of the different kinds of preimplantation genetic testing (PGT: M for monogenic conditions, SR for structural rearrangements, and A for chromosomal aneuploidies), their relative indications, and the specific preclinical workflows.

(Orphanet). It is estimated that each person is an asymptomatic carrier of approximately 2.8 recessive genetic mutations (WHO) that, when combined with the same recessive mutation in the partner, defines a couple at risk of having a baby affected from a genetic condition. A couple might become aware of this risk (whose prevalence is about 2.5% in humans) [5] through three different routes: (i) they have already conceived a baby affected from the disease, (ii) they have family history for the disease (of note, up to 90% of the babies born with a genetic condition have no family history), or (iii) they underwent preconceptional carrier screening (either suggested or mandatory). After genetic and reproductive counseling, once the disease and its causative mutation are known, any genetic condition might potentially be diagnosed on human embryos through PGT-M (unless the genomic region involved is not covered enough or it is too telomeric). Most of the workflows and protocols (for a comprehensive description, see Refs. [6,7]) require a preclinical set-up phase tailored on each couple (peripheral blood samplings are required), their parents, and, if any, their affected children (buccal swabs or peripheral blood are required). These documents and specimens must then be shipped to the genetic laboratory, whose professionals study the case and build specific molecular probes to detect the mutation and its flanking markers in the embryos produced during the IVF cycle. In some cases, such as the conditions caused by trinucleotide repeat expansion, deletion, and duplication, only linkage analyses are performed. To confirm their reliability, the probes are validated on the samples obtained from the partners and their relatives, whose genotype is known. If the results of this preclinical phase fulfill the parameters suggested by the international guidelines, the protocol can be applied on the embryonic biopsies [8,9]. An alternative approach for PGT-M is karyomapping, which will be described later in this chapter.

Indications of PGT-SR

Structural rearrangements (such as chromosomal translocations or inversion without loss of genetic material) can be present in the parental karyotype(s) in the absence of clinical manifestations, but being causative of reproductive issues. Chromosomal translocations figure among the most common structural rearrangements and involve the exchange of genetic material from one chromosome to another. They can be reciprocal, if they involve the breakage of two nonhomologous chromosomes with exchange of segments, or Robertsonian, if involving breakpoints close to the centromere of two acrocentric chromosomes. Inversion, instead, is a type of structural rearrangement that occurs when a chromosome segment changes its orientation within the same chromosome. The patterns of inheritance of these particular genetic conditions are complex and depend on the specific chromosomes involved and the size of the rearrangements. They cause a higher risk of producing gametes with unbalanced chromosome alterations that may generate embryos with unbalanced structural rearrangements (gain or loss of chromosomal segments) independently of parental age [10,11]. The prevalence of translocations in the general population is 0.2%, in infertile couples is 0.6%, varies from 2% to 3.2% in men suffering from severe infertility, and increases up to 9.2% in couples with an experience of multiple miscarriages. Specifically, couples carrying Robertsonian translocations produce on average only 30% of transferable blastocysts regardless of maternal age. This rate drops to 20% in the presence of reciprocal translocations. Therefore, their presence in the parental karyotype is an indication of couple infertility, higher risk for spontaneous abortion, and for the conception of fetuses affected from chromosomal abnormalities [12–14]. PGT-SR is designed to identify embryos from patient's carrier of chromosomal structural rearrangements. If the indication is Robertsonian translocations, any molecular analysis technique might be applied, as they are causative of whole chromosome aneuploidies in the embryos, while the diagnosis of imbalances resulting from reciprocal translocations requires high-resolution technologies, such as microarrays or next-generation sequencing (NGS) [15,16]. Regarding PGT-SR, the preliminary set-up is not required, and it is only required to verify that the specific genomic region involved in the translocation is detectable by the molecular platform applied. Likewise, the pathological copy number variations (CNVs) and the inversions in the parental karyotype must be known to ensure that the resolution of the technique is capable to detect them in the embryonic karyotype.

Indications of PGT-A

Advanced maternal age couples

Advanced maternal age (AMA) is the most common indication for PGT-A [17]. In women older than 35 yr, indeed, the increased impact of infertility, miscarriage, and chromosomal syndromes in the newborns is mainly imputable to meiotic chromosomal issues in the oocytes and the embryos resulting from fertilization [18–22]. AMA is negligibly associated with the fertilization rate, slightly associated with blastulation (especially beyond the age of 40 yr) [23], but highly associated with blastocyst chromosomal constitution. Several causes might concur to these consequences: reduced ovarian reserve, defective oogenesis, altered follicular metabolism, compromised competence, faulty physiological pathways, epigenetic deregulation, and lack of cell cycle checkpoints in the first phase of embryo preimplantation development. In the oocyte, the main impaired mechanisms are related with energy production and chromosome segregation during meiosis, starting from the premature separation of sister chromatids in meiosis I [24], throughout various mitochondrial dysfunctions [25], telomeres shortening [26,27], and meiotic spindle abnormalities [28].

Recurrent pregnancy loss

Another important indication for PGT-A is recurrent pregnancy loss (RPL). It has been estimated it affects 2%–5% of women trying to conceive. The multifactorial causes behind RPL are mainly related to uterine abnormalities, immunological, endocrine, metabolic, and genetic conditions, as well as their interaction with the environment. Nevertheless, despite all possible investigations, almost half of the RPL cases remain unexplained. ESHRE defined RPL as the occurrence of more than three consecutive pregnancy losses including the nonvisualized ones [29,30], while the ASRM defined it as two or more clinical pregnancy losses documented by ultrasonography or histopathologic evaluation, and not necessarily consecutive [31]. Regardless of how RPL is defined, the selection of euploid blastocysts before their transfer in utero reduces couples' risk for miscarriage even in case only few and/or poor-quality blastocysts are produced during an IVF cycle [32,33].

Repeated implantation failure

Repeated implantation failure (RIF) is defined as the failure to achieve pregnancy after several embryo transfers (at least three of a high quality in women younger than 40).

However, implantation failure is an event associated with so many anamnestic (e.g., maternal age) and embryonic (e.g., stage of transfer, embryo morphological quality, number of embryos transferred) features that a consensus on a universally valid definition of RIF is still missing [34–37]. Then, even if several possible etiologies have been hypothesized as causative of RIF (e.g., thyroid dysfunction, submucosal myomas, tobacco use, small polyps or small areas of intrauterine synechiae, progesterone resistance, shifted window of receptivity, decreased integrin expression, immunologic disturbances), many cases are simply idiopathic. Avoiding the transfer of aneuploid blastocysts is certainly beneficial, since aneuploidies are the most important cause of implantation failure in humans [38–40]; therefore, RIF patients are often addressed to a PGT-A cycle. Nonetheless, a lower live birth rate persists in RIF women also when euploid blastocysts are transferred [41]. More evidence is thus still required to reliably define RIF, its causes, and its treatment.

Severe male factor might not be an indication to PGT-A

Severe male factor (SMF) (i.e., oligoasthenoteratozoospermia or azoospermia) has been often suggested as an indication to PGT-A; yet, the data are controversial on whether aneuploidy testing shall be indicated or not also if the female partner is not AMA [42]. A study by Mazzilli and colleagues conducted through quantitative real-time PCR (qPCR)-based aneuploidy testing suggested that SMF is associated only with reduced fertilization and blastulation rates, but it does not involve increased risks for aneuploidies among the cohorts of biopsied blastocysts, even when the data were adjusted for maternal age [43]. Other studies from other groups are required on this topic to reach to a final consensus.

Biopsy techniques

Three main biopsy approaches for PGT exist: polar body (PB) biopsy from oocytes (metaphase II) and/or zygote, blastomere biopsy at the cleavage stage, and TE biopsy at the blastocyst stage. Although cleavage stage biopsy has been the mostly used approach for several years, the ESHRE recently released some updated guidelines stating that TE biopsy should be considered the current gold standard [44]. It is pivotal for a biopsy approach to be clinically valuable that it does not affect embryo competence and that it provides reliable and accurate information on the inner cell mass (i.e., the potential future individual). Ideally, then, the chosen approach shall be clinically sustainable by reducing the costs and the workload, so to be applicable to a larger population of patients and from a larger number of clinics (for a comprehensive review on the history of human embryo biopsy, see [45]).

PB biopsy

PB biopsy was conceived mainly as an alternative to embryo biopsy, especially in those countries where the latter is not allowed. The rationale for it is that especially aneuploidies are mainly due to meiotic errors of gametogenesis [46,47]. Two protocols exist: (i) a sequential approach that removed the first PB in day 0 and the second PB after fertilization from the zygote in day1, or (ii) a simultaneous approach conducted 6–9 h after ICSI in which both PBs are removed and distinguished according to their morphology [48].

Although PBs are waste products of oocyte meiosis, and therefore this biopsy approach is characterized by a limited invasiveness [49], it still suffers from several limitations. PB biopsy is time-consuming, and it is not cost-effective. It is a single cell-based analysis, therefore suffering from diagnostic inaccuracy and high amplification failure. Moreover, the genetic analysis is limited to the maternal contribution, without considering paternal and postzygotic mitotic errors [50,51]. At last, further diagnostic limitations exist related with chromosome segregation errors, higher rate of false positives, and trouble in detecting precocious separation of sister chromatids (PSSCs) [52—54]. Therefore, although interesting outcomes were reported from a multicenter randomized controlled trial entailing PB-based array CGH analysis [55], this approach is used clinically in less than 1% of European PGT cycles [17].

Blastomere biopsy

Blastomere biopsy is performed on Day 3 of preimplantation development when the embryo is constituted of about six to eight cells. Zona pellucida opening can be performed via a mechanical, chemical, or laser-assisted method, and blastomere(s) removal is facilitated through a Ca^{2+}-Mg^{2+} free medium that de-compacts the embryo [56]. The blastomere(s) can finally be removed through either aspiration or displacement [57]. Although blastomere biopsy is a fast procedure even compatible with fresh transfer at the blastocyst stage, it still suffers from several drawbacks: (i) the removal of blastomeres in this early stage of development significantly impacts embryo reproductive competence [58—60]; (ii) single cell analysis is associated with a higher prevalence of technical issues like allele drop-out (ADO), preferential DNA amplification, chimeric DNA molecules formation, and amplification failure [61]; (iii) over 90% of the zygotes reach to this stage, while several arrest their development before reaching the blastocyst stage, possibly because the activation of the embryonic genome in humans occurs at the four to eight cell transition thereby reactivating cell cycle checkpoints and the fidelity of chromosomal segregation and cell division, which are instead highly instable throughout first three days of preimplantation development [62]. This last point, in turn, involves two consequences, a putative higher prevalence of chromosomal mosaicism at the cleavage stage [63] and a larger number of embryos that must be biopsied with respect to the blastocyst stage. In fact, the latter normally benefits from the self-selection of developmentally incompetent ones [64] (including developmentally incompetent euploid embryos [65]), reducing the number of procedures to perform.

TE biopsy

TE biopsy is characterized by the highest stress tolerance. Moreover, a limited proportion of the whole embryonic biomass (only 5—10 cells) is removed, from a section of the blastocyst that gives origin to the embryonic annexes and not the embryo proper, and when embryonic genome activation has already occurred. Moreover, the absence of an impact imputable to this biopsy procedure has been elegantly reported in two nonselection studies [59,66], and the prevalence of full chromosome mosaicism (see the related paragraph) at this stage is as low as 4%—5% [63,67], which is the lowest rate throughout preimplantation development. In fact, the accuracy of PGT conducted on the TE to predict the constitution of the inner cell mass is higher than 99% [68–70]. Moreover, depending on the molecular technique adopted for chromosomal testing, the positive predictive value upon the implantation of euploid blastocyst can be as high as 60% [59,66] with a negative predictive value upon the nonimplantation of aneuploid blastocysts as high as 98%–100% [59,66].

Three protocols exist to conduct TE biopsy. The first described by McArthur and colleagues in 2005 [71] and entailing laser-assisted zona opening performed at the cleavage stage to induce the herniation of TE cells at the blastocyst stage. An easier and more operator-friendly approach that although presents some disadvantages, such as the need to remove the embryo from the incubator in Day 3 and the risk that the inner cell mass will herniate from the hole. A different TE biopsy protocol has been described by Capalbo et al., 2014 [72], which instead leaves the embryo undisturbed until the fully expanded blastocyst stage, when zona opening and TE biopsy are performed simultaneously [73]. The third method stands in between the former two, since it entails zona opening to induce artificial hatching, but conducted at the blastocyst stage [74]. Only a few studies have compared the different protocols to date, and although better performance has been reported when no assisted hatching is conducted [75,76], none can be yet considered safer than the others.

Despite the many advantages of the TE biopsy approach, critical prerequisites are mandatory before it might be implemented clinically. Specifically, extended culture to blastocyst and vitrification must be efficiently conducted, the laboratory must be well equipped, and the operators must be enough to face the workload required from a PGT setting, highly skilled and/or properly trained.

PGT-M: molecular analysis

Two methods for PGT-M are mostly adopted clinically: direct mutation analysis using PCR combined with molecular markers and karyomapping.

Direct mutation analysis combined with flanking markers

Single gene mutation analysis can be performed through the PCR of the sequence of interest. The product of amplification is then sequenced to directly define the mutation [77]. However, this technique is subject to diagnostic error, like ADO or contamination with exogenous cellular material [78]. ADO is the lack of amplification of one of the two alleles, and it can result in diagnostic errors with an incidence of 5%–10% in case of single cell analysis. To overcome these issues, it is currently used as a strategy that combines gene mutation analysis with the study of short tandem repeats (STRs) or single nucleotide polymorphisms (SNPs) linked to the gene of interest. The former are two to six bp sequences repeated in tandem, while the latter are variations of a single nucleotide in the sequence. They can be either intragenic or close to the mutated region of the gene of interest; for this reason, they segregate together with the mutation. These sequences often differ in the number of repetitions between one allele and another, as well as between individuals. Therefore, they are adopted as highly specific markers of different haplotypes for different individuals [79,80]. A diagnostic strategy combining direct mutation analysis with the linkage analysis of multiple polymorphic markers limits the risk for ADO, and diagnostic errors in general, making PGT-M more reliable [81].

Karyomapping

Karyomapping draws a chromosomal map of an embryo through the a priori characterization of thousands of SNPs in the paternal and maternal genome [82]. SNPs represent the most frequent genomic alterations in the population and are responsible for 90% of genetic variability in humans. Thus, based on their signature, an individual can be genotyped (or "fingerprinted"). Mutation analysis through karyomapping involves whole genome amplification (WGA) and SNP arrays for genotyping the embryo. Then, the SNPs of the embryo are compared with the map of the couple, and of a close affected relative. In other terms, mutation analysis by karyomapping does not require the development of patient- and mutation-specific probes. This technology allows also a simultaneous high-resolution visualization of the embryonic karyotype, highlighting four panels of SNPs for each chromosome. In this way, it is possible to highlight meiotic aneuploidies and their parental origin (based on the haplotype present in excess in case of trisomies/duplications, or in defect in case of monosomies/deletions). Conversely, trisomies/duplications of mitotic origin (i.e., deriving from missegregations in the divisions after fertilization) are not detected because they are characterized by an identical duplicated DNA sequence [82,83]. The main limitations of this technique are its cost, and the fact that the mutation itself is not analyzed directly, but indirectly through the segregation of the haplotype at risk.

Comprehensive chromosome testing techniques

For indications such as parental structural rearrangements and/or de novo embryonic aneuploidies, initially cytogenetic technologies, like fluorescent in situ hybridization (FISH), were adopted. However, FISH was limited to the analysis of only nine chromosomes and was easily replaced with comprehensive chromosome testing (CCT) platforms to conduct full karyotype analyses in the embryos [84]. In the paragraphs below, we will discuss the main techniques used throughout the years for PGT-A/PGT-SR.

Microarrays (array-CGH and SNP-array)

Microarray technology is based on the capacity of nucleic acids to specifically associate with complementary sequences, according to the base pairing rules. It is characterized by a solid support to whom a considerable number of DNA probes are anchored. The number of spots of the matrix (called "feature") on each microarray, the size of the probes, and the genomic distance between the probes determine the complexity and the resolution of a microarray. The probe sequences contained in each feature will hybridize with the complementary target sequences of the examined sample, previously labeled and therefore detectable and quantifiable. Microarray technology requires WGA before its application to guarantee the analysis of all chromosomes [85]. Mainly two microarray platforms have been used for PGT, that is, array-CGH (comparative genomic hybridization) and SNP-array.

Array-CGH allows the detection of chromosomal alterations, such as deletions, amplifications, or translocations, at the whole genome level; the probes of the arrays contain clones corresponding to specific loci of each chromosome, covering almost the whole genome.

The chromosomally normal genome (reference DNA) and the genome under investigation (sample DNA) are then labeled with two different fluorochromes and compete for the hybridization to the features on the microarray. The microarrays are then read with a laser scanner, and a dedicated software estimates the ratio between the two signals: an increase or decrease of one signal with respect to the other at each feature indicates the presence of amplification or deletion of that portion of DNA. For several years, array-CGH has been successfully adopted to detect aneuploidies in human embryos [86], with robust and consistent results especially when applied to TE biopsies.

SNP-arrays contain specific probes for SNPs uniformly distributed throughout the genome. In this technology, the sample DNA is labeled and hybridized on the array, and the genetic alterations are identified by evaluating the intensity of the emitted fluorescent signal. These arrays analyze a considerable number of SNPs simultaneously, allowing the full characterization of chromosomal imbalances like deletions, duplications, and translocations at high resolution [87,88]. Treff and colleagues pioneered this approach reporting an accuracy as high as 99% even on a single cell validation analysis [89] and its positive and negative predictive values in a prospective randomized nonselection study [90].

Quantitative real-time PCR

qPCR represents a rapid method for the simultaneous amplification and quantification of a specific DNA sequence in real time. By evaluating the number of amplification cycles and the amount of fluorescence produced from the amplified product during the exponential phase of the reaction, it is possible to calculate the amount of starting DNA. In 2012, Treff and colleagues developed a highly specific method for assessing embryonic full-chromosome aneuploidies based on this technology [91], which was then compared also with microarrays on multiple biopsies conducted from the same blastocysts with very high concordance rates [92]. Of note, qPCR does not allow the analysis of the whole chromosomal sequence but only of four sequences nonvariable in the population. Although subject to a lower resolution and an application limited to full-chromosome aneuploidies (it cannot be used in case of reciprocal translocations in the parental karyotype) [91,93,94], this targeted amplification protocol ensures a lower risk of artifacts than WGA-based approaches, such as array-CGH [92]. This method has been also successfully adopted by Scott and colleagues in an RCT in 2013 [95]. qPCR is the technique with the shortest turn-around time from biopsy to diagnosis (four hours) and is suitable also for PGT-A and PGT-M conducted on a single biopsy by simply adding the specific primers for the mutation and the flanking markers in the preamplification phase of the procedure.

Next-generation sequencing

For many years, the classic Sanger method represented the gold standard for sequencing-based diagnostics, until it was replaced with NGS, namely a high-throughput approach to sequence many DNA fragments in parallel, quickly and at a much lower cost [96]. Nowadays, two NGS protocols exist: WGA-NGS, where the whole genome from the embryonic biopsy is amplified and fragmented, and each sequence is aligned to a reference genome and quantified to assess an excess/defect [97]; targeted-NGS, which does not involve WGA, but adopts PCR to preamplify only specific nonvariable loci located on each chromosome and analyze just them through the NGS technology [98].

Clinical evidence supporting PGT-A

When PGT was conducted through 9-chromosome FISH on a single blastomere, several RCTs proved it less efficient than the standard care, especially when applied to AMA patients [99]. These disappointing outcomes were imputed to the technical issues related with the old-fashioned approach entailing a biopsy strategy that impacted embryo competence and an inefficient diagnostic workflow subject to several technical and biological drawbacks [8,100—104]. To date, TE biopsy combined with CCT for the analysis of full-chromosome meiotic aneuploidies, instead represents a safe and reliable workflow for PGT, as showed by its high positive and negative predictive values [59,66]. The clinical utility of CCT-based PGT-A to reduce the risk for miscarriage and increase the chance of live birth per embryo transfer with respect to the standard care has been confirmed also in two metaanalyses published by Dahdouh et al. and Chen et al. in 2015 [32,105]. Moreover, the more confident adoption of a single embryo transfer policy when euploid blastocysts are replaced is useful even to minimize the risk for multiple pregnancies and their gestational and perinatal consequences [106,107]. The main persisting limiting factor of blastocyst stage biopsy strategy is that only experienced IVF laboratories can safely perform this approach. Specifically, it requires an extended embryo culture, an efficient cryopreservation program, and skilled embryologists to conduct TE biopsies, which are all critical factors that may compromise the efficacy of the IVF-PGT treatment if not conducted properly (Fig. 19.2 is a summary of the prerequisites to implement PGT-A in an IVF clinic, the gold standard workflow, and the expected benefits deriving from its application).

Chromosomal mosaicism, intermediate copy numbers, and segmental aneuploidies

Chromosomal mosaicism is defined as the presence of karyotypically different cell lines in the same embryo [108]. This event is due to postzygotic chromosomal missegregation occurring independently of maternal age. Clearly, the earlier these errors occur, the greater the level of "mosaicism" in the resulting blastocyst (provided that the embryo develops to this stage) (Fig. 19.3A). Chromosomal mosaicism represents a troublesome biological issue for the clinical practice of a PGT-A cycle, since in case of a euploid—aneuploid mosaic blastocyst, the diagnosis will be affected from the sampling bias (i.e., different results might be obtained depending on the section of the blastocyst from which the TE biopsy is retrieved) [63]. Nevertheless, several studies reported very high concordance between the TE and the inner cell mass, when blastocysts were donated to research, disaggregated and analyzed in all their sections, with full-chromosome euploid—aneuploid mosaicism affecting about 5% of the embryos [63,67,68] (Fig. 19.3B). A rate compatible with the prevalence of mosaicism as

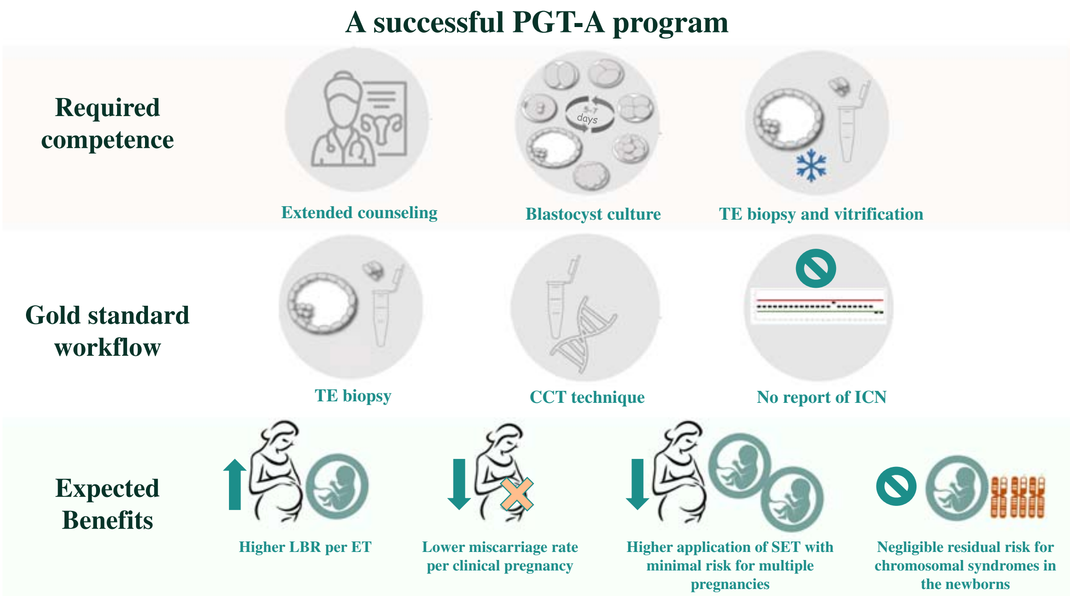

FIGURE 19.2 Summary of (i) the requirements for the clinical implementation of a successful PGT-A program, (ii) the gold standard workflow, and (iii) the expected benefits. *TE, trophectoderm; CCT, comprehensive chromosome testing; ICN, intermediate copy number; LBR, live birth rate; ET, embryo transfer; SET, single ET.*

reported by Huang et al. at the end of the first trimester in either spontaneous or IVF-derived gestations is 1.32% and 1.22%, respectively [109]. Therefore, chromosomal mosaicism involves a tolerable risk for diagnostic errors and should not involve a major diagnostic impact on PGT-A. In this regard, the main issue with mosaicism might not be mosaicism per se, but attempting to diagnose it clinically based on a single TE biopsy and intermediate copy numbers (ICNs) (i.e., a chromosome copy number in between the ranges established to call a disomy and a trisomy or a disomy and a monosomy) [110]. ICNs are in fact imputable to several technical and biological causes (e.g., poor biopsy quality, over- or under-amplification, cells in the S-phase of the cell cycle, DNA contamination, etcetera), other than the presence of both euploid and aneuploid cells in a TE biopsy [110] (Fig. 19.3C). Consequently, defining an embryo "mosaic" based on an ICN exposes to the risk of false-positive calls (a prevalence of mosaicism even higher than 20% has been claimed at the blastocyst stage based on this approach [67]) (Fig. 19.3B), involving fully euploid blastocysts potentially being discarded [111,112]. A recent nonselection study by Capalbo et al. has elegantly reported that human blastocysts that were defined "mosaic" (ICN in a range between 20% and 50%) blindly to the IVF clinics showed the same clinical outcomes as "euploid" control blastocysts, thus supporting that no clinical utility derives from reporting clinically ICN [113].

Differently from full-chromosome aneuploidies, partial (or segmental) aneuploidies, that is, gain or loss of small fragments of a chromosome, might occur more commonly during embryo postzygotic preimplantation development and involve mosaicism at the blastocyst stage. In this regard, a recent study by Girardi et al. reported a 5% prevalence of

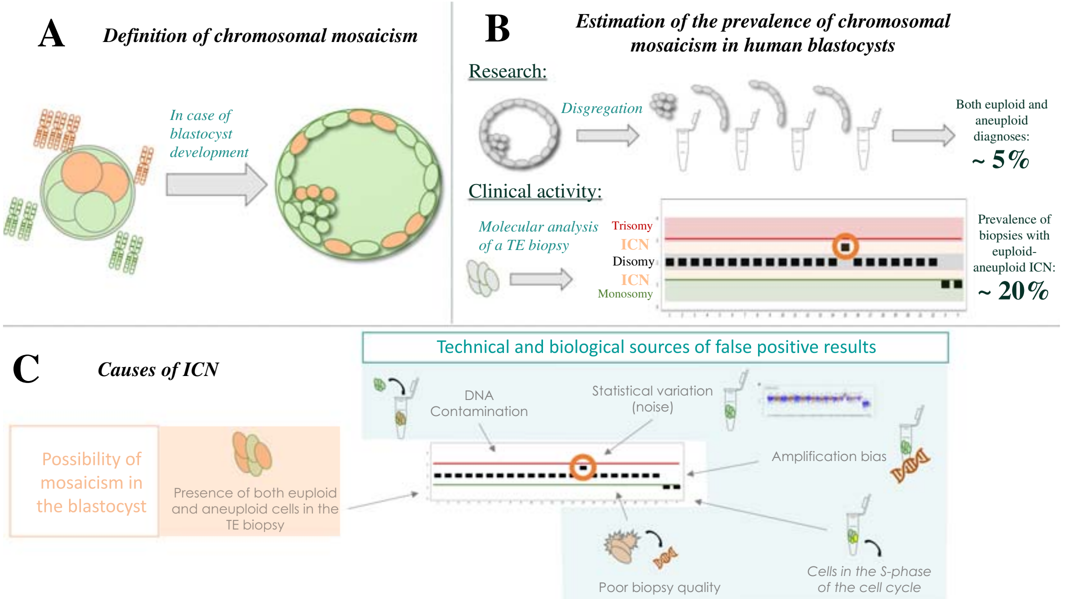

FIGURE 19.3 (A) Definition of chromosomal mosaicism. Orange cells, aneuploid; green cells, euploid. (B) The estimated prevalence of chromosomal mosaicism in human blastocysts based on research and clinical workflows is largely different. In the scheme of a plot obtained from a trophectoderm (TE) biopsy, each black bar represents a chromosome copy number, the black area identifies the range for disomy, the green area represents the range for monosomy, and the red area represents the range for trisomy. The orange area, instead, represents the area of intermediate copy numbers (ICNs) (an example in the orange circle), which are clinically used to call "mosaic" embryos based on a TE biopsy. (C) Summary of the features causative of ICNs in a plot after a TE biopsy, namely either the presence of both euploid and aneuploid cells in the specimen (suggestive of mosaicism in the blastocyst) or a plethora of different biological and technical sources of false-positive results.

subchromosomal aneuploidies at the blastocyst stage, with about 2% of human embryos affected only from these chromosomal imbalances [69]. The disaggregation of several blastocysts donated for research in the same study highlighted that over 60% of the segmental aneuploidies are imputable to either mitotic errors or false-positive calls. More data are therefore required on this topic to guide the clinical management of patients whose embryos are diagnosed with a segmental aneuploidy in the future.

Alternative sources of DNA

Blastocentesis

In 2013, Palini and collaborators reported the presence of embryonic DNA in the blastocele and evaluated its potential clinical use [114]. Blastocentesis (as this approach was called [115]) is the aspiration of approximately 1 μL of the blastocele fluid from expanded blastocysts through an ICSI needle, thereby avoiding the aspiration of cellular material that could affect the results. Up to date, interesting results have been published, but the approach was not consistent and reproducible across all the groups that studied it, with amplification rates variable between 56% and 82% and concordance rates with TE biopsy variable between 37% and 97% [116]. Although this might be imputed to a limited expertise of the operators [117], the method has not been implemented clinically for PGT-A purposes. Nevertheless, it is a promising source of other -omic data that in the future may complement aneuploidy testing to boost embryo selection efficiency [118,119].

Morula biopsy

Morula biopsy was first described in 2014 by Zakharova et al. [120]. The embryos that reach the morula stage in Day 4 after fertilization were subject to a mechanical drilling of the zona pellucida, decompaction through Ca^{2+}-Mg^{2+}-free medium, and the removal of three to seven cells. However, this is a stage critical along embryo preimplantation development [121], and the data are very limited to support the safety of this approach. Therefore, it shall be considered just experimental.

Noninvasive PGT on spent blastocyst media

Even if safe and effective, TE biopsy requires embryo manipulation, experienced and skilled operators, a higher workload for the laboratory, and costly equipment. Therefore, the future perspective of conducting PGT on embryonic DNA retrieved from spent blastocyst media is extremely intriguing [122]. Also in this case, several authors investigated this possibility, reporting highly variable results in terms of amplification rates, risk for contamination, and concordance between the media and the embryo [116]. The issues are mainly imputable to the different protocols adopted for both sample collection and DNA analysis across all these studies. To date, the most promising data have been published in a pilot study and a multicenter study conducted by Rubio et al. [123,124], where the authors outlined a reliable protocol at first, and then reported its consistency across eight IVF centers with a global concordance with the TE biopsy as high as 80% across all clinics involved. Yet, the protocol entails some slight modifications of the standard approach for blastocyst culture (media changeover is required in Day 4 into a smaller 10 μL drop where the embryo must be cultured for at least further 40 h), full karyotype concordance is still low, and the risk for amplification failure and maternal contamination is still high. For these reasons, other scrupulous preclinical and nonselection studies, as well as RCTs, must be performed before noninvasive PGT can be implemented clinically and be considered a potential game-changer in IVF.

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CHAPTER

20

Nutritional supplements and other adjuvants in fertility care

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Introduction

The live birth rate (LBR) for in vitro fertilization (IVF) in the United States in 2019 was 30.9%, which also indicates that 69.1% of patients did not deliver a baby [1]. On account of the emotional and financial toll associated with IVF, both providers and patients are incentivized to add measures to optimize even the first cycle. However, the majority of adjuncts or "add-ons" are not backed by robust evidence and may add significant cost. In addition, proponents argue that forgoing the discussion of options with patients may be seen as paternalistic and may withhold promising treatments [2].

In this chapter, we will review the evidence behind some of the most common IVF adjuvant therapies, both over-the-counter and prescribed, which may improve outcomes in patients undergoing IVF. Table 20.1 summarizes these recommendations.

Growth hormone

Growth hormone (GH) plays a role in reproductive physiology, including steroidogenesis, folliculogenesis, and embryogenesis, primarily by its ability to increase circulating insulin like growth factor-1 (IGF-1) [3,4]. It has been used off-label as an adjuvant therapy during the controlled ovarian stimulation (COH) of IVF, particularly for patients classified as "poor responders" or low oocyte yield.

There is certainly no shortage of observational studies suggesting that GH may improve outcomes in poor responders undergoing IVF [5–8]. In contrast, there have been several randomized controlled trials (RCTs) and metaanalyses evaluating the effectiveness of GH in COH, and yet the results are inconsistent. In 2010, Duffy et al. published a Cochrane review

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TABLE 20.1 Summary of supplements considered for adjuvant therapy in IVF patients.

	Dosage	Timing	Recommendations
Growth hormone	4–24 IU	Initiation cycle day 21 prior to stimulation—stimulation day 8	May consider in poor responders
DHEA	25 mg TID, 75 mg/qday	Pretreatment 2–3 months	May consider in poor responders
Testosterone	2.5–12.5 mg/qday	Pretreatment 2–4 weeks	May consider in poor responders
Coenzyme Q10	200–600 mg qday	Pretreatment 2–3 months	May consider in poor responders
Myo-inositol	1–4 mg/day 40:1 myo-inositol to D-chiro-inositol	Pretreatment 2–3 months	May consider in women with PCOS
Aspirin	75–100 mg/qday	Beginning with stimulation versus prior to embryo transfer	Do not recommend
Corticosteroids	Methylprednisolone 4–16 mg/day Prednisolone 5–60 mg/day Dexamethasone 0.5–1 mg/day	Initiation at start of COH to four days prior to transfer	Do not recommend

and metaanalysis, including 10 RCTs with 440 couples, evaluating pregnancy outcomes with the addition of GH to IVF cycles [9]. Two studies compared the routine use of GH (n = 80), while the remaining eight evaluated GH in patients described as poor responders (n = 360). They found that the routine use of GH did not improve LBR, pregnancy rate, number of oocytes retrieved, or number of embryo transfers (odds ratio [OR] 1.32, 95% confidence interval [CI] 0.40–4.43). In women defined as poor responders by study criteria, they found higher LBR (OR 5.39, 95% CI 1.89–15.35) and pregnancy rate (OR 3.28, 95% CI 1.74–6.20). Cozzolino et al. updated these findings in their 2018 systematic review and metaanalysis [10]. A total of 12 RCTs including 1139 patients classified as poor responders were included. They found a significantly higher clinical pregnancy rate (CPR) per cycle (OR 1.34, 95% CI 1.02–1.77; $P = 0.03$), but no difference in CPR per transfer (OR 1.34, 95% CI 0.92–1.96; $P = 0.13$) or LBR (OR 1.34, 95% CI 0.88–2.55; $P = 0.17$). When only studies that used the Bologna criteria to define poor responders were included, the higher CPR per cycle was no longer noted (OR 1.13, 95% CI 0.50–2.54; $P = 0.77$) [11]. In 2020, another metaanalysis including all of the studies from Cozzolino, plus an additional three, suggests that LBR (relative risk [RR] 1.74, 95% CI 1.19–2.54; $P = .004$) and CPR (RR 1.65, 95% CI 1.31–2.08; $P < .0001$) are improved with GH treatment in poor responders [12]. While attempts at combining the available RCTs are certainly worthy, this is not a straight forward task as there are significant incongruities among the data sets, namely, the differences in definition of a poor responder, the variation in dose and timing of GH administration (although Duffy and Cozzolino attempt to account for this in their analyses), and differences in stimulation protocol. Further studies, specifically RCTs, powered to detect a difference in live birth with GH treatment are still required before definitive recommendations can be made.

For providers considering this treatment, it is important to note that there is no standard recommended dose or timing of GH administration. The studies discussed initiated injectable GH anywhere from cycle day 21 prior to COH to stimulation day 8 of IVF. Dosages varied depending on frequency of administration and were between 4 and 24 IU. As with many IVF add-ons, GH treatment can also add a significant cost to an already costly treatment, particularly in patients without insurance coverage. Lastly, it is important to discuss with the patient the off-label use of this medication during IVF cycles. GH may be considered in poor responders; however, no strong evidence of improved outcomes is available.

Dehydroepiandrosterone

Dehydroepiandrosterone or DHEA is an endogenous androgen produced mainly not only by the zona glomerulosa of the adrenal cortex but also by ovarian theca cells and peripheral conversion [13]. Possible benefits of DHEA supplementation, thought to have "antiaging properties" [14], have been reported since the early 2000s [15−17]. Early studies showed that DHEA is a prohormone in the ovarian steroidogenesis pathway and that it may increase IGF-1, similar to GH supplementation. In turn, supplemental administration is thought to potentially improve follicular development [17,18].

In 2010, Wiser et al. published one of the first RCTs studying the addition of DHEA in IVF cycles [19]. They found that live birth was significantly greater in the group supplemented with DHEA; however, LBR was not an established primary outcome measure, and there were only seven pregnancies in the study. Several subsequent RCTs found no difference in ovarian response markers (such as antimullerian hormone [AMH], antral follicle count [AFC], or follicle-stimulating hormone [FSH]) [20−22] or number of oocytes retrieved [23−25] with DHEA presupplementation, while others did note an improvement in both measures [26,27]. Of the two larger RCTs to evaluate pregnancy outcomes, one noted a significantly higher CPR (32.8% vs. 15.7%, $P = .029$), ongoing pregnancy rate (OPR) (28.5% vs. 12.8%, $P = .036$), number of retrieved oocytes (6.9 ± 3 vs. 5.8 ± 3.1, $P = .03$), and fertilization rate (62.3% $\pm$ 27.4 vs. 52.2% $\pm$ 29.8, $P = .039$) in those supplemented with DHEA compared to those without [27]. Another study (n = 208) found greater number of oocytes retrieved and improved fertilization rate in the study group, but higher CPR in the control group [24]. One study comparing pregnancy outcomes in patients exclusively between 36 and 40 years old (n = 109) found a significantly higher LBR with the addition of DHEA [28].

A 2019 systematic review and metaanalysis evaluated a varied patient population, including those with diminished ovarian reserve, poor responders, and normal responders [29]. Eight studies, including 820 patients, reported CPR, while five studies, including 379 patients, reported LBR with no heterogeneity identified in either group. Metaanalysis showed a significant increase in CPR (RR = 1.27; 95% CI 1.01 − 1.61; $P = .04$) and LBR (RR, 1.76; 95% CI 1.17 − 2.63; $P = .006$).

While there may a potential benefit to DHEA supplementation, further studies powered to detect a difference in live birth are still needed. For those considering adjuvant DHEA supplementation, the dose is either 25 mg by mouth three times a day versus 75 mg orally once a day up to three months prior to IVF. Side effects reported include hair loss, hirsutism, and changes in voice; however, these are negligible at the suggested dose [30].

Testosterone

Androgen receptor gene expression has been identified in primary stage follicles, gradually increasing until present in all preantral follicles. Testosterone is an androgen that may enhance follicular responsiveness to FSH by upregulating FSH receptors on granulosa cells [31,32]. Considering this proposed mechanism of action, testosterone supplementation has been proposed as an adjuvant therapy prior to IVF in patients expected to be poor responders.

Several smaller RCTs have demonstrated an improvement in CPR and/or LBR with the administration of testosterone prior to stimulation in poor responders; however, they vary by timing, route, and dosage of administration and are not powered to detect differences in live birth [33–37]. A Cochrane review published in 2015 assessing the effectiveness of pre- or cotreatment with testosterone found that pretreatment with testosterone was associated with higher LBR (OR 2.60, 95% CI 1.30, 5.20; four RCTs, N = 345). Upon sensitivity analysis, removing the studies at high risk of performance bias, no difference in live birth was noted in the remaining study [38]. A subsequent systematic review and metaanalysis including seven RCTs found a higher LBR (RR 2.29, 95% CI 1.31–4.01, $P = .004$), CPR (RR 2.32, 95% CI 1.47–3.64, $P = .0003$), total number of oocytes retrieved (mean difference = 1.28, 95% CI 0.83–1.73, $P < .00{,}001$), metaphase II oocytes (mean difference = 0.96, 95% CI 0.28–1.65, $P = .006$), and total embryos (mean difference = 1.17, 95% CI 0.67–1.67, $P < .00001$) in the study group receiving testosterone compared to those without [39]. Subsequent sensitivity analysis did not alter these findings.

In the current studies, administered dosages vary and include transdermal testosterone gel 10–25 mg/day and transdermal patch 2.5 mg/day. Timing also varies from several days prior to COH to several weeks making it difficult to provide definitive recommendations. It is important to note the potential for side effects from androgen treatment in reproductive-age women. This may lead both to symptomatic hyperandrogenemia, but also diminished oocyte response [40,41]. A large ($n > 400$) RCT is currently underway in Europe (T-transport trial) that will hopefully provide much-needed edification to this treatment option [41].

Coenzyme Q10

Coenzyme Q10, or CoQ10, is a naturally occurring antioxidant and component of the mitochondrial electron transport chain and oxidative phosphorylation. Mitochondrial dysfunction and oxidative stress along with decreased levels of CoQ10 are thought to be tied to oocyte aging as well as poor ovarian response [42,43]. This relationship between antioxidant levels and poor oocyte quality, due to aging or otherwise, is well studied [44,45]. Extrapolating from these findings, supplementation with CoQ10 prior to COH has been proposed as a mechanism for improving outcomes in poor responders.

Few high-quality studies are available on this topic. A 2014 RCT designed to compare aneuploidy rates in patients receiving CoQ10 versus placebo showed an improvement in LBR in the study group; however, this study was prematurely terminated due to polar body biopsy and only included 24 transfers [46]. Another RCT, including 101 women with clomiphene-resistant polycystic ovary syndrome (PCOS), compared pregnancy outcomes in women receiving CoQ10 with clomiphene versus clomiphene alone [47]. They found a significant improvement in ovulation and CPR in the study group. A 2020 Cochrane review attempted to compare the effects of a variety of antioxidants, but the only studies included for CoQ10 are the two listed above [48]. They reported no improvement in LBR (OR 0.82, 95% CI 0.19−3.54), but improvement in CPR (OR 0.82, 95% CI 0.19−10.26). An RCT published in 2018 (n = 169) compared the number of high-quality embryos from IVF between patients who received 200 mg of CoQ10 orally three times daily for 60 days versus a placebo [49]. The CoQ10 study group had a greater number of high-quality embryos available for transfer ($P = .03$), and fewer patients who did not make it to embryo transfer (8.33% vs. 22.89%, $P = .04$). Unfortunately, there was not a difference in LBR per fresh transfer (31.82% vs. 21.88%, $P = .33$) or cumulative LBR (28.95% vs. 15.54%, $P = .08$); however, it was likely underpowered due to a small sample size.

CoQ10 is available over the counter and is an overall low-cost adjuvant therapy. Safety data suggest that the studied doses (200−600 mg per day) are below the observed safety level [50]. It may have the potential to improve outcomes for poor responder patients, but further research is needed before it can be routinely recommended.

Vitamin D

Vitamin D is a steroid hormone that is central in promoting bone mineralization and maintaining calcium and phosphorus homeostasis; however, its role in reproduction has also been demonstrated [51−53]. Vitamin D may affect reproductive physiology in several proposed ways including primary follicle recruitment through the regulation of AMH, folliculogenesis, and embryogenesis [54,55]. Furthermore, it has been suggested that vitamin D may increase ovarian steroid hormone production and deficient levels may impair implantation and proper placentation [56−58]. Serum vitamin D levels are measured by checking calcidiol (25OH-D) levels and are classified as sufficient (25OH-D $\geq$ 30 ng/mL), insufficient (25OH-D 20−29 ng/mL), and deficient (25OH-D $<$ 20 ng/mL) [59]. Approximately 21%−27% of women with infertility are classified as vitamin D deficient, and yet despite it's known association with fertility, the evidence of benefit of vitamin D supplementation in ART is inconsistent [60,61].

Ozkan et al. demonstrated a correlation between follicular fluid and serum 25OH-D levels in women undergoing IVF and found that patients who achieved a clinical pregnancy had a significantly higher follicular fluid 25OH-D level [60]. A retrospective cohort study including 188 infertile women undergoing their first IVF cycle found that vitamin D deficiency is associated with lower pregnancy rates, specifically in white women. However, ovarian stimulation parameters including dose of medications required, peak estradiol levels, number of oocytes retrieved, and number of mature oocytes remain unchanged [61]. This prompted the suggestion that vitamin D status may be more heavily involved in implantation. However, a prospective cohort study including 280 frozen embryo transfers found comparable CPR between vitamin D replete, insufficient, and deficient women [62].

Several systematic reviews also sought to elucidate the relationship between vitamin D and IVF outcomes. Chu et al. published a systematic review and metaanalysis in 2018 including 11 cohort studies and 2700 patients [63]. They found that LBR and CPR were more likely in women replete in vitamin D compared to those with insufficient or deficient levels; however, later commentary found errors in the data collection and conclusions [64]. Cozzolino et al. published a systematic review and metaanalysis in 2020 including 14 studies with 4382 patients [65]. While their primary analysis showed higher CPR and higher OPR/LBR in vitamin D replete women, subsequent sensitivity analysis called these results into question. The exclusion of one study (Chu et al.) led to nonsignificant differences in CPR and OPR/LBR in women who were replete versus insufficient or deficient, and exclusion of another (Ciepiela et al.) showed improvement in CPR but not OPR/LBR in women who were replete or insufficient versus deficient [66,67].

Vitamin D serves many functions in reproductive-age women aside from reproduction, and supplementation is recommended for those who are deficient; however, there is no clear evidence that supplementation will improve IVF outcomes.

Inositol

Inositol is a polyalcohol that is found in nine different stereoisomer forms. Two of these, myo-inositol and D-chiro-inositol, have been shown to mediate the postreceptor effects of insulin and act as an insulin sensitizer. Due to the increased insulin resistance seen in women with PCOS, this population may benefit from inositol supplementation. Inositol also has some antioxidant properties that may decrease oxidative damage in cells [68]. Inositols are found in beans, fruits, and nuts and may be endogenously produced. They are also available as an over-the-counter dietary supplement.

One of the first studies investigating inositol and PCOS found improvement in insulin sensitivity and ovulation when 1200 mg D-chiro-inositol was taken daily for 6–8 weeks [69]. Several subsequent studies showed the potential benefit of myo-inositol for PCOS patients undergoing natural and stimulated cycles [70–73]. In 2011, Unfer et al. compared treatment with myo-inositol versus D-chiro-inositol in PCOS patients undergoing IVF and found that, despite no difference in total number of oocytes retrieved, the number of mature oocytes was much higher with myo-inositol [74]. It was then found that normal women had a myo-inositol/D-chiro-inositol ratio of 40:1 and supplementation with that ratio was superior to D-chiro-inositol alone. In fact, when given in higher doses, D-chiro-inositol is actually detrimental to reproductive outcomes [75,76].

There have been several RCTs and subsequent metaanalyses comparing outcomes with myo-inositol treatment in women with PCOS undergoing IVF treatment. A metaanalysis including seven RCTs with 935 women published in 2017 found that pretreatment with myo-inositol increases CPR (OR 1.45, 95% CI 1.08–1.95, $P = 0.01$), decreases miscarriage (OR 0.26, 95% CI 0.12–0.59, $P = 0.001$), and increases the proportion of Grade 1 embryos (OR 1.73, 95% CI 1.10–2.74, $P = .02$). They did not find a difference in number of oocytes retrieved (mean difference, 0.11; 95% CI, 0.79 to 0.58; $P = 0.75$) [77]. A subsequent 2018 Cochrane review including 13 RCTs with 1472 patients with PCOS compared pretreatment with inositol to placebo, no treatment, another antioxidant, a fertility agent, or another type of inositol [68]. They found no difference in LBR (OR 2.42, 95% CI 0.75, 7.83) or CPR (OR 1.27, 95% CI 0.87, 1.85) with myo-inositol versus placebo or no treatment. A significant increase in clinical pregnancy in those receiving myo-inositol versus D-chiro-inositol was noted; however, analysis included only the previously mentioned study by Unfer et al. They concluded that there was insufficient evidence to conclude myo-inositol supplementation is beneficial in women undergoing IVF with PCOS.

Myo-inositol has typically been administered in 1–4 g/day doses; however, there is no established preferred dose. Clinical trials testing myo-inositol supplementation for a variety of diagnoses from 1 to 12 months at doses ranging from 4 to 30 g/day found only that the risk of mild gastrointestinal symptoms seemed to occur at doses > 12 g/day [78]. Although the data do not show definitive benefit in terms of clinical pregnancy or live birth in women undergoing IVF, myo-inositol may improve some outcomes in women with PCOS. Along with a potential fertility benefit, myo-inositol may also help decrease the risk of long-term complications associated with PCOS, including type 2 diabetes and cardiovascular disease, due to its insulin sensitizing properties [74]. It should be noted that although inositols are relatively inexpensive and easily accessible, patients should be counseled on the appropriate formulation to purchase (40:1 myo-inositol to D-chiro-inositol) as an increased D-chiro-inositol proportion may actually be of detriment.

Aspirin

Acetylsalicylic acid or aspirin is a cyclooxygenase inhibitor in platelets, thus inhibiting prostaglandin synthesis. In low doses, it has a vasodilatory effect by inhibiting the synthesis of thromboxane A2. Because of its mechanism of action, low dose aspirin has been well studied as an IVF adjuvant to potentially improve ovarian blood flow and/or uterine vascularity and receptiveness [79]. The most common population studied for this application is couples with recurrent implantation failure. The hypothesis is that enhancing uterine blood flow and decreasing thrombosis may improve pregnancy outcomes; however, it has been studied in general infertility populations as well.

In one of the larger RCTs to date (n = 1380 cycles), Waldenstrom et al. compared live birth per transfer in an unselected IVF population undergoing fresh embryo transfer. The experimental group received 75 mg of daily aspirin from day of transfer to the day of pregnancy test. After adjusting for a difference in the number of embryos transferred per group, they found that the aspirin group had significantly higher LBR (OR 1.2, 95% CI 1.0–1.6) [80]. A 2009 RCT (n = 169) comparing 100 mg aspirin daily to placebo, started with oral contraceptive pills prior to stimulation, in women with nontubal factor infertility found no difference in implantation rate or CPR when treatment was initiated compared to placebo [81]. Two similar RCTs (n = 193 and n = 487) also noted no difference in CPR [82] or LBR [83] between aspirin and placebo groups. A Cochrane review including 13 RCTs and 2653 patients confirmed these findings—treatment with low dose aspirin does not improve outcomes for an unselected infertile population undergoing IVF [84]. A large, double-blinded RCT, known as the EAGeR (the Effects of Aspirin in Gestation and Reproduction) trial evaluated if aspirin benefits women who previously experienced one to two pregnancy losses [85]. There was no difference in LBR (58% vs. 53%, $P = .0984$) or pregnancy loss (13% vs. 12%, $P = .7812$) between the preconception aspirin group and placebo.

At present, there is no compelling data to suggest improved IVF outcomes with adjuvant aspirin therapy and it is not currently recommended [86].

Corticosteroids

Corticosteroids are steroid hormones produced by the adrenal cortex and include glucocorticoids and mineralocorticoids. Glucocorticoids, such as prednisolone or dexamethasone, have been studied as fertility adjuvants as early as the 1980s, particularly for women with PCOS undergoing COH [87–89]. Their mechanism of improving ovarian response is by suppressing elevated androgen levels, which is known to be detrimental to normal folliculogenesis, and by increasing the production of growth factors such as IGF-1 [90].

Later studies focused on glucocorticoid's potential to decrease intrauterine inflammatory markers, particularly natural killer (NK) cells, which had been shown to increase the risk of implantation failure and miscarriage [91–93]. It was suggested that the administration of steroids around the time of implantation may improve ongoing pregnancy [94].

First, we will discuss any potential benefit during ovarian stimulation. A 2001 RCT ($n = 290$) found that administering 1 mg dexamethasone significantly decreased the IVF cancellation rate for poor ovarian response; however, no differences in implantation rate or OPR were seen. Of note, the population study included PCOS and normal responders, not specifically patients at a high risk of cancellation for poor response [95]. A 2017 Cochrane review with only two studies ($n = 212$), including the previously mentioned 2001 RCT, found insufficient evidence that glucocorticoid treatment during ovarian stimulation improves CPR (OR 1.69, 95% CI 0.98–2.90) or LBR (OR 1.08, 95% CI 0.45–2.58) [90].

Several studies have evaluated the role of steroids for improved implantation. In 2003, Ubaldi et al. published an RCT ($n = 360$ cycles) comparing prednisolone 10 mg/day taken for four weeks starting the first day of stimulation to no treatment. They found no difference in the number of metaphase II oocytes retrieved, pregnancy rate, or implantation rate between groups [96]. A subsequent Cochrane review including 14 studies and 1879 participants showed no evidence that glucocorticoids improved pregnancy rates (OR 1.16, 95% CI 0.94–1.44) [94]. With only three studies evaluating LBR, the metaanalysis found no differences between groups (OR 1.50, 95% CI 1.05–2.13).

The typical dose administered depends on the type of steroid used: methylprednisolone 4–16 mg/day, prednisolone 5–60 mg/day, or dexamethasone 0.5–1 mg/day [94]. Possible side effects of systemic steroids include hypertension, hypernatremia, hypokalemia, peptic ulcers, and osteoporosis are typically dose and duration dependent. A short course administered during an IVF cycle is unlikely to cause significant adverse events. The 2012 Cochrane review described no significant increase in adverse events with steroid use, but note that they were inconsistently and poorly reported. At this time, routine use of glucocorticoids to improve ovarian stimulation or implantation is not recommended, a position echoed by the ASRM practice committee [86].

Conclusion

A review of the discussed IVF adjuvants reveals a common theme: larger RCTs are necessary for the majority of these treatments before they can be routinely recommended. Due to the costly and emotionally taxing nature of IVF, many patients are interested in any adjuvant therapies available and may learn of them from a literature search, internet forum, or from friends and family who have undergone IVF. It is important to have an honest and open discussion of why an adjuvant may be helpful or harmful. In some cases, the data remain unclear, and if a treatment is not cost prohibitive, it may be worth considering. Other times, as is the case with aspirin and corticosteroids, the data are sufficient to suggest that adjuvant therapy is not recommended. Lastly, it is important to note that even luteal support with hCG (human chorionic gonadotropin) or progesterone was once considered an adjuvant therapy. Perhaps in time, some of these treatments will become standard of care for certain populations, but as is the case with many topics in our field, more research is required.

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CHAPTER

21

Maximizing fertility outcomes in poor ovarian response patients

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Introduction

The term poor responder along with its associated nomenclature that has been used in the literature such as "impending premature ovarian insufficiency" and "poor ovarian responder" can be stigmatizing for patients. The European Society for Reproduction and Embryology (ESHRE) consensus and International Committee for Monitoring Assisted Reproductive Technologies (ICMART) definition use the term poor ovarian response (POR) which has therefore been adopted in this chapter [2,3]. Most definitions of POR are conceptualized as a low number of oocytes (≤3) retrieved following a cycle of ovarian stimulation (OS) or a low number of follicles (≤3) present on the day of oocyte maturation trigger.

The number of patients with POR seeking assisted conception treatment is increasing, due to women delaying childbearing, with between 9% and 24% of all patients undergoing OS being classified as POR [4]. Various etiologies and biological mechanisms behind POR have been hypothesized including chromosomal, genetic, metabolic, enzymatic, iatrogenic, toxic, autoimmune, and infection, but is beyond the scope of this chapter to explore in detail [5].

Until a decade ago, there was no universally agreed consensus on the criteria for diagnosis of POR. There are over 40 different definitions of the term POR in the existing literature [6]. The first commonly proposed definition of POR is described by the Bologna classification from the ESHRE consensus on the definition of poor response to OS in in vitro fertilization (IVF) [2]. The classification advises that two out of three following criteria must be met to define POR:

1) Advanced maternal age or any other risk factor for POR such as gonadotoxic chemotherapy or radiotherapy or ovarian surgery such as for endometrioma excision.

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2) Previous POR (retrieval of three or fewer oocytes from a conventional cycle of OS as part of IVF or three or less follicles on the day of oocyte maturation trigger).
3) An abnormal ovarian reserve test—an antral follicle count (AFC) (of less than seven follicles or anti-Müllerian hormone [AMH] less than 3.5–8 pmol/L).

The Bologna classification is based on a patient who has already undergone a cycle of OS. However, when OS has not happened, the classification advocates diagnosing the patient as an expected POR patient.

Criticisms of the Bologna criteria include a lack of specifically defining risk factors, failure to acknowledge the role of oocyte quality, and most importantly the significant degree of heterogeneity between patients who have been diagnosed using the Bologna criteria [7]. This level of heterogeneity can also make it challenging to conclude which therapeutic interventions are effective in POR patients.

A more recent classification system for POR is the POSEIDON (Patient-Oriented Strategies Encompassing Individualized Oocyte Number) [8–10]. This classification system uses four subgroups:

1) Quantitative and qualitative parameters such as age and expected aneuploidy rate
2) Ovarian reserve markers (AFC and AMH)
3) Ovarian response
4) Number of oocytes needed to obtain one euploid embryo for transfer

Strengths of this classification system are that it is a more holistic approach incorporating qualitative and quantitative data and provides a stronger stratification system for POR patients. Furthermore, there is evidence to support that when the POSEIDON classification is used, cumulative live birth rates following IVF vary between the different POSEIDON groups, supporting its role in stratifying POR patients [11].

Treatment

Number of oocytes retrieved is perhaps one of the most important prognostic factor for IVF success with numerous studies showing a positive association between oocyte number and live birth rate as well as cumulative live birth rate [12,13]. The main aim of treatment for POR patients is to improve oocyte number with an increase of a single oocyte potentially increasing the live birth rate by up to 5%. The mechanisms to maximize and reach optimal response of OS are discussed in this section.

Pituitary suppression regimes

There has been much debate in the literature regarding the ideal pituitary suppression regime to maximize oocyte numbers in POR patients. The two commonest pituitary suppression regimes used in assisted reproductive technology (ART) are the gonadotrophin-releasing hormone (GnRH) agonist and GnRH antagonist protocols.

Advantages of the GnRH agonist protocol include reducing the risk of premature luteinizing hormone (LH) surge which can occur more frequently in POR patients, as well as the allowance of cycle programming for individual clinics. However, on the other hand, the antagonist protocol has the benefit of a lower duration of gonadotrophin stimulation and an overall lower duration of treatment as well as being financially less expensive and also being more patient acceptable.

A 2017 metaanalysis of three randomized controlled trials (RCTs) suggests that there is no difference between GnRH agonist protocols versus antagonist protocol RR 0.89, 95% CI 0.56−1.41 [14].The ESHRE 2021 guideline on OS also advises that GnRH agonist and antagonists can equally be recommended in the management of poor responders [15].

Another protocol used for pituitary suppression is the flare protocol. This has been used in ART for over 20 years targeting predominantly the POR population and was first described in 1997 by Karande et al. [16]. The flare protocol aims to use GnRh agonist pituitary suppression for a short period in the early follicular phase of the cycle, with the aim of releasing endogenous pituitary follicle-stimulating hormone (FSH) after stopping the agonist. However, despite the duration that flare protocols have been in use, robust evidence to support its use in POR patients is lacking.

Type and dose of gonadotrophin

There has been much debate in the literature on both the types of gonadotrophins such as the formulation of the FSH or the supplementation with recombinant LH, as well as the dose of gonadotrophin on whether a high or low dose supports the management of POR patients.

There is some suggestion that recombinant FSH should be used over urinary gonadotrophins due to a higher number of oocytes retrieved possibly due to increased bioavailability with recombinant FSH [17,18]. However, the increased number of oocytes has unfortunately not translated into a higher live birth rate in the studies investigating this [19].

Long-acting FSH, such as Corifollitropin, which has a longer half-life showed initial promising data by demonstrating increased oocyte numbers when used in OS [20]. In one prospective randomized study of 409 patients, the results showed improved pregnancy outcomes when corifollitriptan was used in either GnRh agonist or antagonist protocols compared to the no corrifollitriptan group [21]. However, a subsequent multicenter RCT failed to show any difference in pregnancy outcomes within the POR population when corifollitriptan was used over recombinant FSH, though the number of cryopreservable embryos when corifollitriptan was used was greater [22].

Higher doses of FSH have been suggested to increase the oocyte number and hence improve treatment outcomes for POR patients. However, exogenous gonadotrophin has been shown to only support recruitment of an existing cohort of follicles and not result in de novo synthesis of new follicles. This is supported by the 2018 Cochrane review which identified that an FSH dose of more than 300 iu, the live birth rate does not improve [23]. Furthermore, it should also be noted that in animal studies, high-dose stimulation has been associated with aberrant chromosomal development and subsequent embryogenesis, though this finding in humans remains unclear [24,25].

However, the data regarding dosage above 150 iu −300 iu remain unclear with RCTs giving conflicting answers as to whether a raised FSH dose alone increases either the live birth rate or cumulative live birth rate [26]. The ESRE 2021 OS guideline advises not using FSH doses more than 300 iu and remains in consensus that the evidence for dosing between 150and 300 iu remains unclear [15].

Given the lack of data, decisions on gonadotrophin type and dosing remain predominantly determined based on clinician preference, drug availability, and cost.

Recombinant luteinizing hormone

The use of recombinant LH has been postulated to overcome the effect of ovarian resistance to OS by minimizing the effect of genetic polymorphisms that exhibit resistance to exogenous FSH. One retrospective study of 13 patients evaluating follicular fluid steroid composition, comparing FSH monotherapy versus FSH supplemented with recombinant LH, showed modified steroid production that was more comparable with physiological steroidogenesis and led to an overall improved ovarian response [27].

Overall, however, while the use of recombinant LH has biological plausibility, there is minimal strong evidence supporting its use in the management of POR patients [28].

Timing of GnRH antagonist

There has been much debate on when to start the GnRH antagonist in short antagonist protocol cycles. One of the challenges in POR patients is that there can be a greater variation in rate of growth of follicles and subsequent response leading to follicular growth asynchrony and therefore a difficulty in determining optimal trigger time.

A metaanalysis of four RCTs showed that using a "delayed start" antagonist protocol, whereby pretreatment with estradiol for seven days occurs in the late luteal phase, followed by seven days of daily GnRH antagonist, resulted in an overall lower duration of gonadotrophin use and a higher clinical pregnancy rate compared to conventional gonadotrophin protocols (RR 2.90, [95% CI 1.52, 5.51], $p = 0.001$). The proposed underlying mechanism behind this protocol is that the pretreatment and early GnRH antagonist limits follicular prerecruitment and subsequent asynchronous growth that is seen in POR patients [29].

Oocyte maturation trigger

Oocyte maturation is a critical step in IVF cycles. A low yield of immature oocytes reduces the chances of developing embryos resulting in a reduced live birth rate. There has been much focus on how to improve the oocyte maturity rate in POR.

One proposed intervention is to combine two of the commonest oocyte maturation triggers—an agonist trigger with a human chorionic gonadotrophin (HCG) trigger, conceptualized as the dual trigger. The concept of dual trigger has been in existence since 2008; however, its use in aiming to improve the oocyte maturity rate in POR patients was first explored in 2015 by Orvieto et al. [30,31].

Several retrospective studies since then have advocated better cycle outcomes including number of oocytes, oocyte maturity rate, as well as pregnancy outcomes including higher live birth rate when a dual trigger is used in POR patients compared with a single agent trigger alone [32–34]. In one prospective randomized study comparing single agent trigger versus dual trigger of 160 women, a higher number of retrieved mature oocytes and a higher clinical pregnancy rate was seen when dual trigger was used [35]. However, in another RCT of 140 patients, there was no significant difference in oocyte number or maturation rates when the dual trigger was used over HCG alone [36]. Well-designed prospective RCTs reporting on live birth as an outcome are needed to interpret the role of different oocyte maturation triggers in the management of POR patients.

Natural stimulation or modified natural stimulation protocols

The aim of the modified natural cycle or natural cycle protocol is to generate at least one good quality oocyte that develops into a high-quality embryo and subsequently implants into a more receptive endometrium [37]. There are no good-quality, RCTs available to support the use of modified natural cycle or natural cycle IVF in POR patients. One retrospective cohort study of 161 modified natural cycles suggested a much higher live birth rate with modified natural cycle protocols compared to conventional high-dose stimulation antagonist cycles [38]. However, other studies have suggested that modified natural cycle protocols do not offer a realistic chance of a prospective live birth in POR patients [39]. The ESHRE 2021 OS guideline does not recommend modified natural cycle protocols over conventional OS [15].

Mild stimulation protocol

Mild stimulation is defined by ICMART as "a protocol in which the ovaries are stimulated with gonadotrophins, and/or other pharmacological compounds, with an intention of limiting the number of oocytes following stimulation for IVF" [40]. Mild stimulation protocols can consist of using FSH dose $\leq$150 iu alone or in combination with oral compounds such as clomiphene citrate or aromatase inhibitors such as letrozole or oral compounds alone. A recent systematic review and metaanalysis found no difference in live birth rate or pregnancy outcomes between conventional OS and mild stimulation. The review suggests that mild OS should be considered for POR, and this is also in agreement with the American Society of Reproductive Medicine guidelines [41,42].

The ESHRE OS guideline also reinforces this advising that clomiphene citrate alone or in combination with gonadotrophins is equally recommended as gonadotrophins alone in the management of POR [15]. However, it suggests that the addition of letrozole in a similar manner to clomiphene citrate should not be recommended for the management of POR patients due to the very limited evidence base to support this [15].

An additional advantage of using oral compounds is that it reduces the financial burden associated with gonadotrophins. However, the superiority of which mild IVF stimulation protocol is most efficacious in POR patients remains yet to be determined.

Double stimulation protocol

Double stimulation or dual stimulation involves two rounds of OS within the same menstrual cycle and has been proposed as an innovative approach to increase both the number of oocytes as well as subsequent euploid embryos in POR patients [43]. However, a fresh embryo transfer cannot occur following a double stimulation cycle.

The concept of double stimulation protocol is to be opportunistic of using the multiple follicular waves present within a menstrual cycle. Patient dropout rate and time delay between cycles are theorized to be significantly reduced given that two OS regimes occur within the same menstrual cycle. This is a crucial potential advantage where time is critical on the outcomes of POR patients.

The first proof-of-concept study for dual stimulation was an observational study of 43 POR patients which demonstrated a significantly higher yield of transferable blastocysts when this protocol was used for stimulation [44]. Subsequent to that, an observational case series compared 297 patients undergoing dual stimulation versus conventional OS. The most important finding from this study was the patient dropout rate was extremely high in the conventional OS group (81%) [45].

Finally, a multicenter experience of using the double stimulation protocol demonstrated an increase in the number of oocytes and embryos per menstrual cycle with no difference identified in the competence of either the oocytes or embryos compared to conventional OS regimes [46].

Criticisms of the double stimulation protocol include a high cancellation rate of cycles during the luteal phase of stimulation, cost effectiveness, challenges in cycle monitoring, and a need to cryopreserve all embryos generated [46]. Well-designed prospective RCTs are needed to assess the efficacy of double stimulation protocol compared to conventional OS protocols that take place in two separate menstrual cycles.

Adjuvant treatments

Many adjuvant treatments have been proposed for POR patients to improve IVF treatment outcomes with varying levels of evidence to support them. The most frequently encountered treatment adjuvants include growth hormone, androgen supplements such as dehydroepiandrosterone (DHEA), coenzyme Q1O, and these are summarized here.

Growth hormone

Growth hormone supplementation is thought to act through three chief mechanisms—promoting preantral follicle growth and differentiation, stimulation of preantral follicles to gonadotrophin-dependent stage, and finally an improved oocyte maturation rate [47].

Initial metaanalyses including a 2003 Cochrane review suggested improved treatment outcomes including live birth when growth hormone was used as an adjuvant treatment [48,49]. However, the most recent metaanalysis in 2020 did not show an improvement in live birth rate for POR patients following IVF, as defined by the Bologna criteria [50]. Furthermore, the heterogeneity in terms of growth hormone dosing, duration, and the lack of data on long-term safety restricts the use of this as an adjuvant treatment.

Androgens

Many trials have evaluated the use of androgen supplementation and androgen-modulating agents in POR patients undergoing IVF. The underlying biological mechanism proposed is that androgen receptor mRNA and androgen levels in follicular fluid correlate with FSH receptor mRNA in granulosa cells. Therefore, exogenous androgens stimulating androgen receptors will increase FSH receptors in granulosa cells and hence enhance follicular sensitivity to exogenous FSH as well as follicular recruitment [51].

Pretreatment with transdermal testosterone or DHEA supplementation, given with the aim of increasing intraovarian androgen levels, has shown conflicting evidence at improving treatment outcomes including live birth rate. Data from a 2020 network metaanalysis showed that out of all the adjuvants, DHEA had the highest clinical pregnancy rate and transdermal testosterone produced the greatest number of embryos [50]. However, there is insufficient evidence on the dosage, duration, route of administration, and long-term safety for androgen supplementation.

Coenzyme Q10

Coenzyme Q10 is thought to support treatment in POR patients by reducing the oxidative stress insult against oocytes and embryos. However, only one prospective RCT was identified in the literature which showed an improvement in OS outcomes and subsequent embryology parameters [52].

Challenges in determining evidence-based interventions

As discussed in this chapter, the greatest difficulty facing POR patients is the paucity of robust evidence supporting therapeutic interventions. Four principal reasons are thought to be behind this: the first challenge is the multitude in variation of definitions for POR. It is envisaged that using a classification system such as POSEIDON will reduce the heterogeneity of different study populations. Second, the sample size of many studies has been small resulting in an increased risk of type II errors affecting the results. Using large dataset analysis may be one avenue for being able to reach more definitive conclusions. Third, the etiology and underlying biological mechanisms behind POR, especially within the younger age demographic, remain uncertain. Finally, there is wide heterogeneity in outcome reporting from studies, it is recommended that studies report outcomes as per the recommended minimum core outcome set reporting for infertility research.

Conclusion

The main challenge in the management of POR is the paucity of the evidence base to support therapeutic intervention. Agonist or antagonist protocols appear to have equal efficacy with high-dose gonadotrophin stimulation unlikely to confer any additional benefit. For adjuvant therapies, there are insufficient data on dosage, duration, efficacy, adverse effects, and long-term safety. Crucially what remains at the core of the treatment of POR patients is that not all POR patients will respond the same way and individualization and stratification of patients are essential. Finally, it should be noted that ultimately egg donation is the most successful treatment for these group of patients with a live birth rate of >50% compared to

1%–10% with autologous eggs, and patients need to be sensitively counseled where appropriate regarding this option as well.

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CHAPTER

22

Endometrial receptivity

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Introduction

The human womb exhibits exceptional plasticity and undergoes cyclic changes at the morphological and functional levels that are driven by the hypothalamus–pituitary–ovarian axis. Each sex hormone acts on uterine endometrial wall cells in different ways and at different times, resulting in distinct endometrial stages throughout the menstrual cycle. The first description of this cyclic behavior came during ancient civilization, when Soranus (98–117 AD), a Greek scholar, noted in his book, *Soranus' Gynecology*, the fertile period commencing a few days after menstrual bleeding [1]. Little had been discovered since the time of Hippocrates (460–377 BCE), who brought a rational perspective with high relevance to many infertility treatments in his books: *Diseases of Women*, *Nature of Women*, and *The Generating Seed and the Nature of Child* [2]. Later, the understanding of fertility changed with the illustrations of the female pelvis by Leonardo da Vinci (Carporis Fabrica, 1543). The 1930s brought the discovery of the hormones progesterone and estradiol; 30 years later, human chorionic gonadotropin (HCG) was discovered, and many diagnostic methods began to emerge (e.g., Huhner's postcoital test and Ascheim Zondek test), marking a cardinal step in assisted reproduction [3,4].

Today, we recognize that the main goal of cyclical changes during the menstrual cycle is to be prepared for embryo arrival and implantation, leading to pregnancy. The mechanisms underlying these cyclical changes are timed precisely, except during pregnancy or when certain pathologies disrupt the implicated organs in reproductive-age women. However, the menstrual cycle phases are not identical across all women; indeed, some women have longer or shorter cycles or respond more quickly or slowly to certain treatments. Therefore, the menstrual cycle depends on the correct synchronization of every hormone and the corresponding tissue responses. Further, each delimited and dynamic menstrual phase has important roles to play in fertility.

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This chapter is dedicated to one of these menstrual phases, the receptive stage, the only moment of the cycle in which the embryo is able to implant. This period is also known as window of implantation (WOI). We will discuss the molecular and anatomical changes that occur in endometrial cells during the WOI and the personalized affordable therapy to improve reproductive outcomes in individuals struggling with infertility.

Relevance and concept of endometrial receptivity

Implantation during in vitro fertilization

Current global live birth (LB) rates per started in vitro fertilization (IVF) cycle are only 25%–30% [5], and implantation is considered the major rate-limiting step for achieving a successful pregnancy [6]. Successful implantation requires appropriately timed synchronization between a competent blastocyst and a receptive endometrium. However, because embryo development and endometrial receptivity are regulated independently, there is no guarantee that these two events will occur at the same time during any given treatment cycle.

Traditionally, embryo development is the primary factor guiding the timing of a transfer in an IVF cycle. This practice assumes that the WOI is constant in all women, starting on Day 19 or 20 of the cycle and remaining open for 4–5 days [7]. In the case of natural conception, embryonic development and the WOI are controlled by the follicle, where, following ovulation, progesterone begins to rise. If both, the resulting embryo and endometrium develop normally, then they will be synchronous and implantation may occur. However, in an IVF cycle, this natural coordination may be lost, even when controlling for appropriate embryo development on the day of transfer.

The WOI and endometrial receptivity

The existence of the WOI was first proposed in 1956 [8]. This concept defines the period during the mid-secretory phase when endometrial receptivity ensues, allowing for embryo adhesion and subsequent pregnancy [9]. This transient status exists only for a short period of time. Early studies suggested that the window may last up to 10 days after ovulation [10]; however, these studies poorly defined the day of ovulation, resulting in a systemic bias in their conclusions [11,90]. Later studies redefined the length of the WOI based on what was optimal versus what was possible, demonstrating superior clinical rates during a two-day window [12]. More recently, a study based on transcriptomic assessment of the WOI defined a slightly shorter window, lasting on average 29–36 h, and, in some women, less than 24 h [13]. Regardless, the WOI ends following initiation of stromal cell decidualization, subsequently permitting trophoblast invasion and placental formation. It is not completely known whether there is individual or intercycle variation in the duration of the WOI within each cycle, although molecular studies found that transcriptomic profiles pertaining to endometrial receptivity remain constant for at least 40 months when replicating cycle protocols and conditions [14].

Endometrial receptivity is the product of a well-orchestrated and integrated interaction involving paracrine signaling with ovarian hormones, growth factors, lipid mediators, transcription factors, and cytokines [15]. Progesterone is considered the single most important hormone pertaining to receptivity and the WOI. Once progesterone levels reach a critical threshold, a meticulously timed and orderly secretory transformation begins. Even very low levels of serum progesterone (as low as 2.5 ng/mL) can trigger differential expression of key genes associated with secretory transformation and control of the WOI [16,17], indicating that the onset of progesterone exposure is more important in initiating receptivity than serum levels reached during the mid-secretory phase.

Although the WOI was once assumed to be consistent for all women, earlier studies did report worse outcomes in cycles of delayed implantation, occurring near the completion of the window [11]. Displacement of the WOI could cause implantation failure, especially in IVF cycles when supraphysiological levels of steroid hormones that occur during controlled ovarian hyperstimulation deteriorate uterine receptivity [18–20], and even more so in cases where premature increases in progesterone levels lead to early secretory transformation [21] and slow-to-blastulate embryos increase the risk of desynchrony. Certain groups of women, specifically those experiencing repeated implantation failure (RIF), may be more likely to suffer WOI displacement. Fortunately, when accounting for a displaced WOI by modifying embryo transfer timing to coincide with time of endometrial receptivity, women with RIF show significantly improved implantation rates (IRs) [22,23].

Mechanisms of endometrial receptivity

Ovarian steroid hormones

The endometrium is receptive to an implanting embryo during the mid-secretory phase, and receptivity is chiefly regulated by estradiol and progesterone. The actions of these hormones are mediated by the presence of progesterone receptors (PR-A and PR-B) and estrogen receptors (ERα and ERβ) expressed within the epithelium and stromal compartments of the endometrium. During implantation, signaling of these receptors is executed by juxtacrine, paracrine, and autocrine factors that are coordinated by various growth factors, cytokines, lipid mediators, homeobox transcription factors, and morphogens [15].

Estrogen is the first hormone to exert its effects on the endometrium. During the first half of the menstrual cycle (referred to as the proliferative phase), increasing estrogen levels promote endometrial cell proliferation. Ovulation of the ovarian follicle occurs mid-cycle, leading to the release of progesterone from the resultant corpus luteum, which subsequently triggers endometrial epithelial differentiation and maturation. Initiation of the WOI by progesterone has been demonstrated clinically [24], epidemiologically [11], and morphologically [25].

Plasma membrane transformation

Implantation requires intercellular interaction between the apical membranes of the endometrial epithelium and the trophoblastic ectoderm. In the endometrium, it is imperative for the luminal epithelium to undergo a plasma membrane transformation from a nonadhesive surface to an adhesive one. Endometrial remodeling triggering this transformation requires meticulous coordination between multiple systems and encompasses the integration of numerous molecules, including prostaglandins, selectins, integrins, mucins, cadherins, cytokines, and immune response molecules (see below for details). Together, these molecules, along with others, define a complex network that when balanced, promotes endometrial receptivity and successful implantation.

Prostaglandins

Prostaglandins are common molecules found within tissues throughout the human body that actively participate in numerous biological processes, including in the endometrium where they induce epithelial cell proliferation [26,27]. Although the presence of prostaglandins persists throughout the entire menstrual cycle, prostaglandins have a direct influence on endometrial receptivity. A significant increase in the concentration of specific prostaglandins (PGE_2 and PGF_{2a}) has been detected in endometrial fluid (EF) of fertile women during the WOI, however not in infertile women. Additionally, 24 hours before embryo transfer in IVF cycles, higher concentrations of these prostaglandins occurred in women who became pregnant versus in those who did not [28]. These findings along with others, such as a correlation between defective prostaglandin synthesis and RIF [29], highlight the effect of these specific molecules during the WOI.

Selectins

Selectins belong to the cell adhesion molecule (CAM) family and are glycoproteins that mediate leukocyte–endothelium interactions. On the endometrium, L-selectin interacts with oligosaccharide ligands to mediate tethering and rolling of leukocytes on the endothelium [30,31]. Additionally, L-selectin is highly expressed on the surface of hatched blastocysts [31]. The dual presence of this molecule suggests the importance of L-selectin and its ligands in early implantation to facilitate the adhesion of a floating blastocyst to the endometrium. Conversely, abnormal expression of selectins and their ligands is associated with infertility, such as in RIF where an absence of the L-selectin ligand MECA-79 during the mid-luteal phase is linked to no chance or a low chance of pregnancy [32]. Abnormal expression is also implicated in cases of endometriosis [33] and ectopic pregnancies [34].

Integrins

Integrins are transmembrane CAMs found in the endometrium within the luminal and glandular epithelium [35–38]. Integrins are regulated by steroids and cytokines and play a critical role in embryo–endometrial interactions at the time of implantation. The integrin $\alpha_v\beta_3$ is expressed during the mid-secretory phase (along with other integrins) and is used as a purported marker for the WOI. The expression of $\alpha_v\beta_3$ is rate-limited by the production of the β_3 subunit, which is regulated by HOXA10, a transcription factor whose expression in the uterus is imperative for fertility and decidualization [39]. An increase in β_3 subunit mRNA occurs after Day 19 of the cycle, although not before [40]. In the case of endometriosis, the β_3 subunit is absent [37]; in unexplained infertility, there is aberrant β_3 expression [41]. These observations serve as the basis for the diagnostics applied in the E-tegrity test that analyzes endometrial samples for endometriosis and luteal phase defects.

Mucins

Mucins are transmembrane CAM family glycoproteins that line the apical surface of the endometrial epithelium. There are three specific mucins found within the endometrium (MUC1, MUC4, and MUC16) and are the major constituents of mucus, wherein their primary functions include lubrication and protection. MUC1 possesses antiadhesion properties, where the formation of a physical mucus barrier created by a high density of MUC1 may prevent blastocyst attachment by masking the adhesion molecules present on the endometrial epithelium [42]. Intuitively, the downregulation of MUC1 during implantation should be expected, as what is observed in a mouse model; however, the opposite has been found to occur in humans. Fertile women experience a progesterone-mediated increase in MUC1 levels during the WOI [43,44], whereas women with RIF typically do not [45]. Potential reasons for this contradiction include recognition of MUC1-associated glycans by the embryo, potentially resulting in a blastocyst-induced localized clearance of MUC1 during adhesion to facilitate implantation at that specific site [44]. Based on this proposed theory, MUC1 could essentially act as a gatekeeper, preventing the implantation of low-quality embryos that possess poor reproductive potential.

Cadherins

Cadherins are CAM glycoproteins responsible for calcium-dependent cell-to-cell adhesion and consist of three subclasses: E, P, and N. In mice, E-cadherin appears to play a role in proper trophoblast invasion, as a significant enrichment in E-cadherin expression is observed at the apical membranes of uterine epithelial cells before implantation [46]. Additionally, the presence of a mutation in the E-cadherin gene results in defective embryo implantation [47]. In humans, E-cadherin mRNA expression significantly increases after ovulation in response to progesterone [48], and E-cadherin is the dominant molecule found in adherens junctions, which play a critical role in the mediation of cell-to-cell adhesion [49]. However, immunohistochemical analyses have failed to detect cyclic variations in E-cadherin protein levels [50–53].

Cytokines

Cytokines are multifunctional glycoproteins that act as signaling molecules via their interaction with specific cell surface receptors. Cytokines are found throughout the endometrium, including in the luminal and glandular epithelium and the stroma, enabling communication between endometrial cells and between the endometrium and blastocyst. Several different cytokines regulate uterine receptivity and implantation (including decidualization), as marked by increasing cytokine levels during the mid to late secretory phases. Appropriate cytokine equilibrium must be maintained for these processes to occur successfully. In mouse models, interleukin-12 (IL-12), IL-15, and IL-18 are essential for proper activation of uterine natural killer (uNK) cells, the loss of which is associated with unexplained infertility, recurrent pregnancy loss (RPL), endometriosis [54], and lossed control of angiogenesis. When out of balance, cytokines can exert deleterious effects [55,56]. In humans, mutations in specific cytokines, abnormal expression patterns, and deficiencies of certain cytokines and their receptors are associated with unexplained infertility, RPL, and implantation failure [57–60].

Immune factors

Multiple immune regulators play essential roles in endometrial receptivity and implantation. The expression of immune regulators is directly controlled by estrogen and progesterone. During the secretory phase, there is an influx of immune cells into the endometrium that converts local immunity from adaptive to innate [61], allowing for the shift from protecting the host from infection to permitting the invasion of a blastocyst [62]. During the mid to late secretory phase, the majority of immune cells present within the endometrium belong to the innate immune system and include uNK cells (making up 65%–70% of the immune cells present at this time), macrophages, and dendritic cells; adaptive immune T cells are also detected, albeit to a lesser degree [63–67]. uNK cells provide numerous growth factors and cytokines that are integral to inducing local secretion of angiogenic factors by endometrial cells and building a healthy placenta [65,68]. Not surprisingly, abnormal levels of uNK cells are associated with unexplained infertility, RPL, and endometriosis [54]. Although uNK cells appear to be crucial for successful implantation and pregnancy, the poor prognostic ability of uNK cell counts has prevented the development of any clinically relevant diagnostic test involving this particular immune factor.

Transcriptomics in bulk tissue

Histological and molecular changes that occur during acquisition of the receptive phenotype are due to underlying changes in gene expression. The emergence of transcriptomics analysis in the 2000s coupled to other "-omics" sciences (secretomics, proteomics, or lipidomics) provided a new strategy to evaluate endometrial receptivity [69]. To date, several groups have described gene expression profiles across the menstrual cycle, and many studies have focused on identifying the transcriptomic signature during receptivity [70]. These findings provided a starting point to develop a more objective molecular tool to identify the WOI, resulting in the development and clinical application of several endometrial receptivity tests based on transcriptomics in bulk tissue. These tests use endometrial mRNA extracted from an endometrial biopsy (EB) to determine if the endometrium is receptive or not in a specific moment of the cycle. The first of these tests, the endometrial receptivity analysis (ERA), provided a novel, personalized approach to embryo transfer based on endometrial status [71].

To develop the ERA, researchers compared the transcriptomic signature between the receptive and nonreceptive phenotypes in bulk tissue from EBs. Those genes whose expression levels had an absolute fold change of >3 and a false discovery rate of <0.05 across the menstrual cycle were used to train a computational predictor. Currently, ERA uses next-generation sequencing (NGS) to assess the expression of 248 genes related to successful embryonic implantation and classifies the obtained transcriptomic profile as receptive or nonreceptive (proliferative, prereceptive, or postreceptive). A nonreceptive result indicates that the WOI is displaced, meaning that the endometrium requires more or less time with progesterone exposure to reach receptivity. So, gene expression indicates at what specific moment of the cycle the WOI is open, thereby personalizing embryo transfer timing, which classically was considered standard for all women [22].

To test the accuracy of this analysis, EBs collected throughout the menstrual cycle were analyzed in a blinded way by both, ERA and two independent pathologists (using Noyes criteria based on histology). Analysis of results included application of the quadratic weighted Kappa index, calculating concordance with the real endometrial stage. The concordance reached by ERA was 0.922 (0.815–1.000), while the pathologists reached 0.618 (0.446–0.791) and 0.685 (0.545–0.824) [13]. Reproducibility was also evaluated by analyzing paired biopsies from the same patient with the same progesterone exposure timing but collected 29–40 months apart. This analysis showed no variation between cycles [13].

The implementation of endometrial transcriptomics into the clinical practice has allowed the synchronization between the ready-to-implant blastocyst and the receptive endometrium. The clinical utility of this approach was first proven in patients with RIF to test if the lack of synchronization could be responsible, at least in part, for failed implantation. ERA was performed on 85 RIF patients who had experienced three or more previous failed cycles (4.8 ± 2.0) with at least four morphologically high-grade embryos transferred in total. From this group of patients, 25.9% had a displaced WOI (84% prereceptive and 16% postreceptive), while only 12% had a displaced WOI in the control group, which consisted of 25 patients with ≤ 1 previous failed transfer attempt. Those patients with a displaced WOI had a personalized embryo transfer (pET) according to the moment at which their endometrium was receptive based on the ERA results. After pET, the pregnancy rate (PR) and IR of RIF patients were 50.0% and 38.5%, respectively, similar to those obtained in control patients [22].

Later, it was published a follow-up case report describing a patient with seven failed embryo transfers [23]. ERA revealed that this patient was prereceptive at progesterone (P) + 5 (the standard timing of the cycle when previous embryo transfers were performed), but was receptive 2 days later, at P + 7. After the first pET following seven days of progesterone exposure, a successful implantation of two transferred blastocysts was achieved. This case report was complemented by a pilot study of 17 patients with one to six previous failed cycles. Clinical outcome between the last transfer before ERA (performed on nonreceptive endometrium) and the subsequent pET was compared after ERA identified the personalized WOI for each patient. Results of this comparison showed that while just 19% of the patients achieved clinical pregnancy when transferring into a nonreceptive endometrium, 60% of the same patients had a clinical pregnancy when the embryo transfer occurred during their personalized WOI.

Several clinics that apply endometrial transcriptomics to their RIF patients have published similar results, including singular case reports and retrospective cohort studies (Table 22.1). Additionally, ERA has resulted useful in specific cases where the patient has uterus didelphys [72] or endometrial atrophy [73].

Classical case reports help to understand the clinical management of patients in which personalization of the embryo transfer resulted in a successful pregnancy. One patient achieved pregnancy with pET guided by ERA after 11 previous failed attempts [74]. Clinicians observed similar results in three Indian patients who had several failed transfers using good quality blastocysts [75]. Two patients had an initial ERA at a different IVF center, but pET was not properly followed, resulting in unsuccessful transfers. However, once pET was properly performed in a new clinic, both patients achieved a successful pregnancy [75].

A retrospective study analyzing patients in India with and without RIF showed that 27.5% of RIF patients had a displaced WOI, and patients with just one previous failure showed a 15% displacement rate. Clinical outcome in RIF patients after pET demonstrated a 42.2% ongoing PR and a 33% IR [76]. A subgroup of patients with atrophic endometrium (<6 mm) was also studied, wherein after pET, the PR was 66.7%, close to patients with normal endometrial thickness [76]. In 50 Japanese RIF patients, a WOI displacement rate of 24% was observed, and these patients reached a 50% PR once transfers occurred during the confirmed receptive endometrium [77]. Additionally, 36.5% of RIF patients in the Czech Republic had a displaced WOI and achieved a PR of 69.2% following pET [78].

Other reported data compared the clinical outcomes of patients who were receptive earlier or later than the standard WOI with those that did not show a WOI displacement. In 210 RIF

TABLE 22.1 Published studies about ERA clinical use.

Study type	Year	Title	Authors	Journal
RCT	2020	A five-year multicentre randomized controlled trial comparing personalized, frozen, and fresh blastocyst transfer in IVF.	Simon C. et al.	*Reproductive BioMedicine Online*,. 2020; 41(3):402–415.
Prospective	2013	The endometrial receptivity array for diagnosis and personalized embryo transfer as a treatment for patients with repeated implantation failure.	Ruiz-Alonso M. et al.	*Fertil Steril*. 2013; 100(3):818–24.
	2021	Routine endometrial receptivity array in first embryo transfer cycles does not improve live birth rate.	Riestenberg C. et al.	*Fertil Steril*. 2021; 115(4):1001–1006.
Retrospective	2014	What a difference two days make: "personalized" embryo transfer (pET) paradigm: a case report and pilot study.	Ruiz-Alonso M. et al.	*Hum Reprod*. 2014; 29(6):1244–7.
	2015	Endometrial receptivity array: clinical application.	Mahajan N.	*J Hum Reprod Sci*. 2015; 8(3):121–9.
	2017	Efficacy of the endometrial receptivity array for repeated implantation failure in Japan: a retrospective, two-centers study.	Hashimoto T. et al.	*Reprod Med Biol*. 2017; 16(3):290–296.
	2017	Window of implantation transcriptomic stratification reveals different endometrial subsignatures associated with live birth and biochemical pregnancy.	Diaz-Gimeno P. et al.	*Fertil Steril*. 2017; 108(4):703–710.e3.
Retrospective	2018	The role of the endometrial receptivity array (ERA) in patients who have failed euploid embryo transfers.	Tan J. et al.	*J Assist Reprod Genet*. 2018; 35(4):683–92.
	2018	Does the endometrial receptivity array really provide personalized embryo transfer?	Bassil R. et al.	*J Assist Reprod Genet*. 2018; 35(7):1301–1305.
	2018	Window of implantation is significantly displaced in patients with adenomyosis with previous implantation failure as determined by	Mahajan N. et al.	*Journal of human reproductive sciences*. 2018; 11(4):353.

TABLE 22.1 Published studies about ERA clinical use.—cont'd

Study type	Year	Title	Authors	Journal
		endometrial receptivity assay.		
	2019	Personalized embryo transfer helps in improving in vitro fertilization/ICSI outcomes in patients with recurrent implantation failure.	Patel JA. et al.	*J Hum Reprod Sci*. 2019; 12(1):59–66.
	2019	Endometrial receptivity Analysis—a tool to increase an implantation rate in assisted reproduction.	Hromadova L. et al.	*Ceska Gynekol*. 2019; 84(3):177–183.
	2019	What is the clinical impact of the endometrial receptivity array in PGT-A and oocyte donation cycles?	Neves AR. et al.	*J Assist Reprod Genet*. 2019; 36:1901.
Retrospective	2020	Evaluation of the endometrial receptivity assay and the preimplantation genetic test for aneuploidy in overcoming recurrent implantation failure.	Cozzolino M. et al.	*J Assist Reprod Genet*. 2020; 37(12):2989–2997.
	2020	Comparing endometrial receptivity array to histologic dating of the endometrium in women with a history of implantation failure.	Cohen AM. et al.	*Syst Biol Reprod Med*. 2020; 66(6):347–354.
	2021	Clinical utility of the endometrial receptivity analysis in women with prior failed transfers.	Eisman LE. et al.	*J Assist Reprod Genet*. 2021; 38(3):645–650.
	2021	Evaluation of pregnancy outcomes of vitrified-warmed blastocyst transfer before and after endometrial receptivity Analysis in identical patients with recurrent implantation failure.	Kasahara Y. et al.	*Fertility and Reproduction*. 2020; 3(2): 35–41.
	2021	The use of propensity score matching to assess the benefit of the endometrial receptivity analysis in frozen embryo transfers.	Bergin K. et al.	*Fertil Steril*. 2021; 116(2):396–403.

(*Continued*)

TABLE 22.1 Published studies about ERA clinical use.—cont'd

Study type	Year	Title	Authors	Journal
Case report	2014	Live birth after embryo transfer in an unresponsive thin endometrium.	Cruz F. and Bellver J.	*Gynecol Endocrinol*. 2014; 30(7):481–4.
	2018	Different endometrial receptivity in each hemiuterus of a woman with uterus didelphys and previous failed embryo transfers.	Carranza F. et al.	*J Hum Reprod Sci*. 2018; 11(3):297–299.
	2019	Why results of endometrial receptivity assay testing should not be discounted in recurrent implantation failure?	Simrandeep K. et al.	*The Onco Fertility Journal*. 2019; 2(1): 46–49.
Case report	2019	The reproductive outcomes for the infertile patients with recurrent implantation failures may be improved by endometrial receptivity array test.	Ota T. et al.	*Journal of Medical Cases*. 2019; 10(5):138–140.
Letter to editor	2018	Intrapatient variability in the endometrial receptivity assay (ERA) test—Letter to editor.	Cho K. et al.	*J Assist Reprod Genet*. 2018; 35(5):929–930.
	2018	Variations in the endometrial receptivity assay (ERA) may actually represent test error—Letter to editor.	Dahan MH. et al.	*J Assist Reprod Genet*. 2018; 35(10):1923–1924.
	2018	Intercycle consistency versus test compliance in endometrial receptivity analysis test.	Stankewicz T. et al.	*J Assist Reprod Genet*. 2018.

patients who underwent pET, there were no statistical differences in clinical outcome between patients who were receptive at P + 5 and patients who were receptive at a different time [79]. These results illustrate the importance of personalizing embryo transfer to when the endometrium is receptive instead of standardizing it to the same cycle day.

Although the primary focus has been on the endometrium, the contributions of the embryo toward successful implantation and pregnancy cannot be neglected. Following transfer of only euploid embryos, patients with a corrected WOI had higher IRs and ongoing PRs than those without displacement (although not statistically significant) [80]. A similar outcome was obtained in a subgroup of RIF patients that had failures with euploid embryos. After correcting the WOI in cases where a displacement was detected, the IR was 66.7% and the ongoing PR was 58.3%, while in patients without a displacement, the IR was 44.4% and the ongoing PR was 33.3%. These differences highlight that in the first group, when the embryo factor is corrected, the lack of synchronization was responsible, at least in part, for their previous failures.

Other studies evaluated pET by retrospectively comparing patients with an ERA indication treated by pET with patients without ERA indication, reporting no statistical differences in clinical outcome [81–83]. Because patients with an ERA indication are considered more complex cases (without known reason for their RIF condition), it is expected that pET equalizes their outcomes to those patients without ERA indication.

The utility of pET was also evaluated in patients without previous implantation failures in a prospective, open-label, multicenter, randomized clinical trial (RCT) with 16 IVF centers participating in Europe, South America, and Asia recruiting 458 patients ($\leq$37 years). The study had three arms comparing fresh embryo transfer (ET) under a stimulation protocol, frozen embryo transfer (FET) at P + 5 in hormone replacement therapy (HRT) cycles, and pET guided by ERA with frozen embryos in HRT cycles (pET). The intention-to-treat analysis only showed statistical differences when comparing cumulative pregnancy rate (CPR) between pET (93.6%) and FET (79.7%) ($P = .0005$) or ET (80.7%) ($P = .0013$) groups. However, the per protocol analysis showed that PR at the first attempt was statistically higher in pET (72.5% pET and 54.3% FET; $P = .01$) as well as the IR (57.3% pET, 43.2% FET, 38.6% ET; $P < .05$). Although LB rate at the first attempt was not significantly different (pET 56.2% vs. FET 42.4%, $P = .09$; pET 56.2% vs. ET 45.7%, $P = .17$), the cumulative LB rate (after 12 months) was statistically higher in the pET group (71.2%) than in the FET (55.4%; $P = .04$) and ET (48.9%; $P = .003$) arms [84].

On the other hand, endometrial transcriptomics identified different variables that could affect the acquisition of the receptive phenotype. A prospective cohort study using ERA showed that the WOI could be affected in patients with a body mass index (BMI) of $>30\ kg/m^2$, revealing a higher incidence of displaced WOI within this population group [85]. While normal-weight individual and those with overweight showed a displaced WOI rate of 9.1% and 7.7%, respectively, the displacement increased considerably in those who were obese (22.5%) or morbidly obese (37%) (although not significant). Also, patients with obesity showed significant alterations at the transcript level in their receptive profile [85]. A later prospective study [86] detailed similar results for 170 infertile women. These patients were grouped into nonobese ($18.5-29.9\ kg/m^2$; $n = 73$) and obese ($>30\ kg/m^2$; $n = 97$) categories, finding that the nonreceptive rate was significantly higher in the obese group (9.7% vs. 25.3%; $P = .02$).

Transcriptomics at the single-cell level

Transcriptomic changes that occur across the menstrual cycle have been studied using single-cell RNA sequencing (scRNA-seq) [87], obtaining findings consistent with those in bulk tissue. For this study at single-cell level, mRNA obtained from cells isolated using microfluidic circuits and nanodroplets was sequenced individually. The spatial distribution of RNA or translated proteins can also be mapped within the endometrial tissue (https://data.humancellatlas.org). Using microfluidics with Fluidigm technology and nanodroplets with 10× Genomics, an analysis of 73,180 single cells from endometrial tissue samples of

29 healthy ovum donors identified six major endometrial cell types (epithelial cells, including a previously uncharacterized epithelial ciliated cell type, stromal fibroblasts, endothelial cells, macrophages, and lymphocytes) based on canonical markers and highly differentially expressed genes [87]. These findings also showed that transcriptional activation occurs in the epithelium when the WOI opens, which coincides with that observed in bulk tissue [71]. Characterization of gene expression across different endometrial stages at the single-cell level provides relevant information about the biological behavior in this tissue and how bulk tissue can represent a true evaluation of endometrial factors.

Future directions

As technology and medicine are in constant development, and ERA straddles both fields, changes in some respect can occur, such as the type of samples analyzed, the ability to obtain the samples, the time in which the samples are collected, processing techniques, or physician expertise. These changes may all positively impact testing accuracy.

Samples: Currently, ERA is performed on endometrial tissue obtained from an EB, wherein the sample is taken by scratching the inner walls of the womb, potentially resulting in aches, bleeding, and pain during the procedure. This traditional biopsy is considered an invasive procedure, even if it does not require a surgery room, anesthesia, prophylactic antibiotic treatment, or other procedures required for major invasive surgery. Additionally, it is not possible to perform an embryo transfer in the same menstrual cycle in which the biopsy is obtained to allow for healing.

To solve the above limitations, a noninvasive technique is required. The closest sample to the endometrial tissue is the EF, which is produced by endometrial cells and possibly reflects the transformational status of endometrial cells, providing the same information as a traditional biopsy while avoiding patient pain and discomfort, while also allowing for an embryo transfer within the same cycle.

An analysis to validate the use of EF as a suitable diagnostic for endometrial staging separately assessed 56 paired EF and EB samples. In 13 EF samples (23.2%), RNA quantity and/or quality was poor, while the remaining 43 EF samples (76.8%) generated good-quality libraries for sequencing. These 43 EF results were concordant with their paired EB samples analyzed by ERA: 33 were receptive (76.7%), 7 were prereceptive (16.3%), and 3 were postreceptive (7.0%). EF testing resulted in 100% sensitivity and specificity with no false-positives or false-negatives versus ERA [88].

EF is also a validated diagnostic tool to study the endometrial microbiome, another important factor that may impact embryo implantation. Bacterial DNA is detectable in 61.9% of EF samples, a similar percentage to that observed in EB samples (64.2%). Microbiota diagnosis yielded by both type of samples was also found to be similar [89].

Clinical use of EF samples is challenging. Many physicians experience difficulties recovering enough sample for analysis. The technique is simple but requires some expertise and is not a standardized procedure; therefore, some doctors are not accustomed to performing it, limiting the wider use of EF as a diagnostic tool.

Timing: The timing of sample collection is important because ERA identifies the optimal moment for embryo transfer. This is not an issue if the schedule is well timed from the beginning (when P supplementation will be introduced in an HRT cycle or when the luteal peak is well diagnosed). Coordination of the 120 h of progesterone exposure should be based on the moment at which the biopsy is collected (according to the doctor's schedule); 120 h before that moment, the patient must begin taking progesterone. This should also be done during the embryo transfer cycle, starting with identifying the embryo transfer time (according to the doctor's schedule) and counting back the suggested number of hours indicated by the ERA report to determine the time of progesterone initiation.

Technique: Improvements in techniques can occur as the science continues to develop and new laboratory methods are introduced. Because we can use ERA and assess endometrial cell status in real time, it may be possible to find another approximation to study the response of endometrial cells to progesterone when acquiring the receptivity profile. Initial diagnosis of endometrial receptivity leveraged RNA arrays, but has since been replaced by NGS technology. Accordingly, NGS may be replaced by another technique in time, despite that NGS is easy, fast, effective, and relatively inexpensive.

Physician expertise: Although there are contradictory results among different clinicians using ERA in their clinics [81–83], it is essential to highlight the knowledge and follow-up of this diagnostic tool. Some guidelines must be followed to obtain the best results and accuracy of the test. First, the biopsy must be performed at the right moment in each patient's cycle. Not all patients are the same, and it is strongly advisable to obtain the sample at their real P + 5 or LH + 7 day. To ensure this, each cycle must be properly controlled, in example, in an HRT cycle endogenous progesterone must be <1 ng/ml within the 24 h prior the first progesterone intake. After confirming that the patient does not have an early progesterone escape, the biopsy can be scheduled counting 120 h in advance from the first progesterone dose in an HRT cycle. The sample then reflects the real P + 5 of that specific patient. The same procedure must be followed to transfer the embryo in the right moment, as indicated by the ERA results.

Data from some published papers do not reflect this important issue. It is imperative to consider both the serum progesterone level before starting progesterone supplementation (which should always be less than 1.0 ng/mL within the 24 h prior to the first intake of exogenous progesterone to ensure no previous endogenous escape has occurred) and the exact number of hours that have passed between the beginning and the end of supplementation in both ERA and embryo transfer cycles. When these variables are not properly controlled for and reflected in studies, there is no true evidence that ERA was correctly performed, so these results cannot be completely accepted.

Conclusions

Several changes take place at the molecular level to allow for embryo implantation in the endometrium. Thus, the acquisition of the receptive phenotype cannot be evaluated based on one marker alone but rather using a holistic methodology. The use of transcriptomics in endometrial bulk tissue has allowed for the identification of the optimal time at which to perform an embryo transfer. Several published studies highlight the utility of evaluating the endometrial factor in patients with RIF, in which synchronizing a receptive endometrium with a blastocyst has normalized their clinical outcome. However, evolution does not stop, and this evaluation of the endometrium should be improved to a noninvasive technique. Transcriptomic profiling of EF samples provides high sensitivity and specificity compared with EB samples, offering a possible alternative to diagnose endometrial receptivity. However, there remains a need for technical improvement in the collection of EF. On the other hand, to continuously improve the usability of endometrial receptivity evaluations in the clinic, we must keep learning from published studies, being sure that these studies are properly performed following an exhaustive control of multiple variables, especially the main activator of the receptive phenotype acquisition: the progesterone.

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CHAPTER

23

Impact of the microbiome in fertility

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Since 1892, thanks to the German gynecologist Albert Sigmund Gustav Döderlein, we know that the vagina has a complex bacterial ecosystem abundant in *Lactobacillus* spp.

It was not until the 1960s, however, that the axiom that the female genital tract beyond the cervix was sterile began to be questioned. It was not until the late 1980s that publications began reporting positive cultures of endometrial samples.

Some authors suggested that the cervical mucus plug might not be an impermeable barrier for bacteria and that there was even, thanks to uterine peristaltic movements, a negative pressure effect to facilitate the ascent of sperm through the cervical canal that could also be a route of ascent for bacteria [1,2].

The advent of prokaryotic 16s RNA gene amplification and sequencing techniques has opened a Pandora's box by suggesting the presence of bacterial populations in areas such as the uterus or the fallopian tubes, much smaller in number than the populations in the vagina, but complex in variety.

Furthermore, we have moved on from the concept of microbiota, defined as the set of microorganisms in a given anatomical location, to that of the microbiome, which is a much broader notion. This concept refers to the microbiota, its genetic information, as well as the microenvironment that surrounds it and with which it is intimately related [3]. There are some authors such as Lynn Margullis that go even further, speaking of the holobiome as the sum of the genome of a eukaryotic individual and its symbiotic microbiome [4].

In recent years, there have been many reports about how the inhabitant microorganisms of different niches of the human body establish mutualistic relationships with us. It is not only that they do not harm us, but we also need them for our normal physiology due to the thousands of years of pacific coexistence that has become interdependence.

We will try to define the concept of a healthy microbiome and its influence on the results of assisted reproduction treatments (ARTs), as well as on the course of gestation.

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Microbiome

In 2007, the Human Microbiome Project (HMP) was presented with the aim, among others, of defining the criteria for a healthy microbiome. It considers that the human being has been evolving in symbiosis with its microbiome for thousands of years and, therefore, the HMP can be considered as an extension of the Human Genome Project [5].

Dysbiosis, a concept that refers to the disturbance of the balance of a given microbiome, has been associated with numerous diseases such as pelvic inflammatory disease, multiple sclerosis, diabetes, allergies, asthma, and cancer [6,7].

A healthy microbiome is generally defined as one that does not generate symptoms and does not give rise to overt disease, contributing to the good health of the person of which it is a part [8].

Lower genital tract microbiome

We know that the normal vaginal microbiota has a very abundant bacterial population with a concentration between 10^8 and 10^9 colony-forming units per gram of vaginal discharge. The HMP estimates that it represents approximately 9% of the total microbiota of women [5].

Compared to the gastrointestinal microbiome, the vaginal microbiome has a low variety of bacteria, that is, low alpha diversity. However, it has a somewhat higher beta diversity, which reflects interindividual differences. The most representative and predominant genus is *Lactobacillus* spp., with its species *L. crispatus, L. iners, L. jensenii, and L. gasseri*.

In healthy women of reproductive age, *Lactobacillus* spp. represents up to 95% of the total bacteria. The other most commonly represented genera are *Prevotella, Bifidobacterium, Gardnerella, Atopobium, Megasphaera, Sneathia, Anaerococcus*, and *Corynebacterium* [9–11].

The dominance of *Lactobacillus* spp. favors an acidic environment due to their metabolization of glycogen with the consequent production of lactic acid and H_2O_2 that hinder the overgrowth of anaerobic and nontolerant bacteria at this pH. In addition, they have the capacity to secrete bacteriocins that are substances with bactericidal activity. Thanks to these different mechanisms, *Lactobacillus* protects against bacterial vaginosis and even sexually transmitted infections [12–15].

It is therefore essential that *Lactobacillus* spp. be the predominant genus to ensure a healthy vaginal microbiome.

Prior to menarche, the vaginal microbiome consists primarily of skin and gut bacteria such as *Escherichia coli*, *Enterococcus* spp., *Diphtheroids*, *Lactobacillus* spp., *Staphylococcus epidermidis*, and *Streptococcus viridans*, with relatively few *Lactobacillus*. After puberty, elevated estrogenic levels make *Lactobacillus* spp. the dominant genus. This is maintained throughout a woman's childbearing years. When menopause arrives, the drop in estrogen production also causes the proportion of *Lactobacillus* to decrease again.

In the field of microbiology, the way to identify the different types of bacteria has traditionally been based on microscopic visualization of the samples, as well as their culture on agar plates or other media.

In the case of vaginal exudate samples, the fresh sample is visualized to detect the presence of easily identifiable germs such as *Trichomonas*, *Candida*, or key cells (vaginal epithelial cells lined with bacteria), and a part of the sample is Gram stained and observed under the microscope (Fig. 23.1).

Depending on the presence or absence of clue cells and the morphology and type of bacterial staining (gram positive, negative, or variable), the relevant bacterial genera are identified.

The sample is cultured according to the type of microorganism suspected. However, one of the problems of this traditional method is that many microorganisms in the vaginal sample are not culturable.

When there is an imbalance in the microbiota, that is, dysbiosis, this can lead to bacterial vaginosis.

For the diagnosis of bacterial vaginosis, we can use the clinical microbiological criteria of Amsel (homogeneous, grayish vaginal discharge, pH >4.5, positive amine test, and microscopic visualization of clue cells). We can also apply the purely microbiological criteria of Nugent, which are based on microscopic visualization of a sample of vaginal exudate with Gram stain. This allows us to calculate the proportion of *Lactobacillus* (gram-positive elongated bacilli), *Gardnerella*-type bacteria (small Gram-variable bacilli), and mobiluncus-type bacteria (Gram-variable curved bacilli) [16].

Prokaryotic cells share the 16 S ribosomal gene complex. It contains highly conserved regions common to all bacteria, and its amplification allows us to detect whether a particular

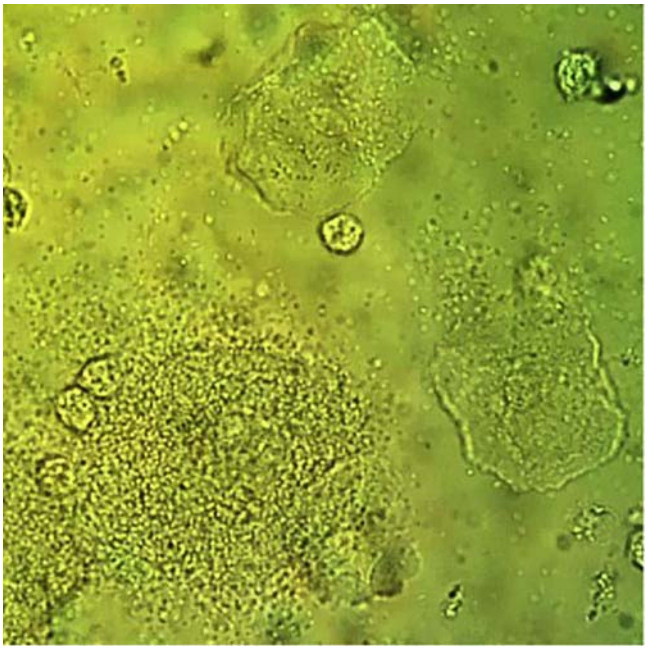

FIGURE 23.1 Fresh mounting with clue cells.

sample is sterile or not. In addition, this same gene contains nine hypervariable regions (V1 to V7) that allow us to identify bacteria down to the species level [17,18].

One of the great advantages of next-generation sequencing (NGS) is that we have the possibility to identify nonculturable bacteria.

This technique allowed Ravel's team to define four types of microbiota in the lower genital tract:

Type I	Prevalence of 26.2% of the population studied. Dominance of *Lactobacillus crispatus*.
Type II	Prevalence 6.3% of the studied population. Dominance of *L. gasseri*.
Type III	Prevalence 34.1% of the population studied. Dominance of *L. inners*.
Type IV	Prevalence of 33.4%. There is not so much dominance of *Lactobacillus* (<90%). It presents a higher percentage than the other groups of anaerobes (*Gardnerella*, *Corynebacterium*, *Prevotella*, *Atopobium*, *Megasphaera*, etc.). The vaginal pH of this group is slightly less acidic, 5.3 ± 0.6 [10].

The biotype depends, among other things, on intrinsic factors such as ethnicity or genetics.

However, the vaginal microbiome is not stable but evolves with transitions between the different biotypes due to factors such as menstrual cycles, fluctuating estrogen levels, contraceptive or antibiotic use, type of hygiene, and sexual activity.

Microbiome of the upper genital tract

The classical idea that beyond the cervix the upper female genital tract is sterile began to be questioned more than three decades ago, when positive endometrial cultures obtained by hysterectomy were reported [19,20].

The advent of new generation 16S prokaryotic RNA gene sequencing techniques has helped confirm the existence of a bacterial population in the female upper genital tract, both at the endometrial level and in the fallopian tubes and even in the ovaries. However, the bacterial load in the upper genital tract is 10,000 times lower than that of the lower vaginal tract [21].

This is thought to be due, in part, to the cervical barrier hindering the ascent of bacteria, as well as to immune reactions and a microenvironment specific to this anatomical location. Even so, the composition of the microbiota of the upper vaginal tract mirrors that of the lower vaginal tract. *Lactobacillus* is also the dominant genus here, but it shares the presence with other anaerobic bacteria such as *Gardnerella, Streptococcus, Staphylococcus, Bifidobacterium, Prevotella,* and *Atopobium.* Inmaculada Moreno's group compared the vaginal and endometrial microbiota of fertile and healthy women, confirming similar profiles in both anatomical locations in 80% of cases [22].

There is still controversy about the role that all these microscopic inhabitants play in the upper genital tract environment and how their presence modulates the host immune system.

Recent microbiome research has evolved from simply analyzing the composition of microbiota to study its genome expression profiles in order to also shed light on their functionality and behavior. The transcriptomic and metaproteomic characterization of the human microbiota are promising tools to clear up this point [23,24].

Microbiome and infertility

The incidence of infertility has been increasing in recent years, reaching 15% of couples of reproductive age. Despite advances in the field of reproduction, there are cases of failure in ART that continue to be a mystery. It is in this context that the study of the microbiome has become increasingly important.

The relationship between genital tract microbiome dysbiosis and infertility has been proposed by several authors such as Haahr T. and Moreno I. et al. [25,26].

Embryo implantation is one of the fundamental factors that determines their success in ART, since the 1990s; it has been shown that the uterine microbiota could play a role in infertility when lower gestation rates were detected in women who presented positive cultures with enterobacteria, staphylococci, or streptococci in the embryo transfer cannula compared to those in whom *Lactobacillus* was recovered [27].

Franasiak's group performed an interesting study in which they analyzed the distal 5 mm of the catheter after embryo transfer, using NGS for 16SRNA gene sequencing. They demonstrated that the uterine microbiome can be analyzed without altering the clinical practice of embryo transfer. However, they were unable to find differences between the group that achieved gestation and the group that did not, the dominant bacterium in both cases being *Lactobacillus* [28].

At the same time, Moreno's group determined that the composition of the endometrial microbiota influenced the success of ART. Admittedly, this was a small study involving only 35 women, but they observed that *Lactobacillus* dominance below 90% (non-*Lactobacillus* dominance) in the uterine microbiota was associated with lower rates of implantation, gestation, and live newborns (Fig. 23.2) [26].

Subsequently, Hashimoto's group analyzed the microbiota of 99 patients, taking an endometrial biopsy at the time of the transfer test. The patients were under 40 years of age and underwent transfer of a thawed blastocyst. They diagnosed dysbiosis in the microbiome of 31 of these patients, but observed no difference in the rate of gestation per transfer or miscarriage between the two groups [29]

Microbiome and gynecological diseases

On the other hand, it is logical to think that the composition of the genital microbiome may also have an impact on infectious diseases related to infertility such as bacterial vaginosis, pelvic inflammatory disease, and chronic endometritis.

FIGURE 23.2 *Lactobacillus* spp. *Credit:Dr. Horst Neve, Max Rubner-Institut.* *http://vaam.de/infoportal-mikrobiologie/mikrobe-des-jahres/kontakt-presse/pressebilder.html*

Bacterial vaginosis

This is a very common lower genital infection. As we have already described, it is associated with the disruption of the optimal vaginal microbiome due to a decrease of lactobacillus and increase of strict and facultative anaerobes, including *Gardnerella vaginalis*, *Megasphaera* spp., and *Atopobium vaginae*.

Pelvic inflammatory disease

In the case of pelvic inflammatory disease, which can cause tubal factor infertility, the most frequent pathogens are, on the one hand, those involved in bacterial vaginosis and, on the other, those related to sexually transmitted infections such as *Chlamydia trachomatis*, *Neisseria gonorrhoeae,* and *Mycoplasma genitalium.*

Chronic endometritis

This is another infectious disease that could have a negative impact on the embryo implantation rate. It consists of a persistent inflammation of the endometrial mucosa, caused by bacteria such as *Enterobacteriaceae, Enterococcus, Streptococcus, Chlamydia, Gardnerella, N. gonorrhoeae, Mycoplasma,* and *Ureaplasma* spp. [30].

Some authors have suggested that 56.8% of women with infertility have chronic endometritis. The prevalence increases to 67.5% in patients with implantation failure (IF) and to 67.6% in women with recurrent miscarriage (RA); the fundamental problem lies in the fact that it often goes unnoticed since patients are asymptomatic, making it an underdiagnosed and undertreated disorder.

Diagnosis is complicated and has traditionally been carried out by hysteroscopy, as well as anatomopathological and microbiological study of endometrial biopsies. Unfortunately, the concordance between these three methods is poor.

In recent years, the use of massive sequencing techniques on endometrial biopsy samples has made it possible to detect both culturable and nonculturable microorganisms. This method has a 77% concordance with the three classical methods [31].

Once this disorder has been diagnosed, antibiotic treatment can be prescribed, ideally directed at the microorganism or microorganisms detected, thus, allowing for the normal microbiome to be restored [32].

However, an additional problem is that there is a lack of standardization of antibiotic treatments. There is no consensus as to the type of adequate treatment or its duration.

In some guidelines, such as those of the CDC, treatment is equated to that of PID (ceftriaxone 250 mg intramuscularly in a single dose plus doxycycline 100 mg orally twice a day for 14 days plus metronidazole 500 mg orally twice a day for 14 days) [33]. In other articles, doxycycline 200 mg/day for 14 days as a first-line therapy and ciprofloxacin and metronidazole 500 mg/n 14 days are proposed as a second-line treatment [34]. Still, on many occasions, the use of broad-spectrum antibiotherapy is resorted to. On the one hand, this can lead to the appearance of resistance, and, on the other hand, it can contribute to the sweeping away of the healthy microbiota, thus, contributing to dysbiosis.

A recent study by Espinós' group (SWOT: Strengths, Weaknesses, Opportunities, Treats) evaluated whether the assessment of the presence of chronic endometritis should be included in the basic study of the infertile couple. They concluded that chronic endometritis could affect implantation by different mechanisms, including the expression of proteins that affect endometrial receptivity. Even so, in the absence of standardized techniques for both diagnosis and treatment, there is not enough evidence to recommend it systematically. Nevertheless, it would be advisable to detect or rule out the presence of chronic endometritis in cases of recurrent implantation failure or repeated miscarriages [35].

Endometriosis

More striking is the fact that women with endometriosis, a gynecological disease closely linked to infertility, show differences in the genital microbiome compared with healthy women. Wei's group concluded that the microbiota of cervical intakes could be an indicator of endometriosis risk. On the other hand, Perrotta's group recently concluded that the study of the vaginal microbiota could be a predictor of the stage of endometriosis when present [36,37]

Microbiome, obstetrics, and the offspring

Furthermore, the existence of a relationship between adverse obstetric events and the uterine microbiome has been proposed. It seems that bacteria at the vaginal and cervical level, as well as those in the seminal fluid, could enter the uterine cavity and the fallopian tubes during conception. This bacterial transmission could influence fertility and increase the risk of miscarriage and preterm delivery [38,39].

It is interesting to note that a woman's microbiome may not only play an important role in relation to her own health and fertility but also influence her offspring. The Dominguez Bello team points out that, during evolution, the human microbiome has been adapting to the environment, being transmitted maternally. Indeed, the vaginal microbiome of the mother plays a fundamental role in the initial colonization of the newborn, and it has been seen that this can have consequences for the immune system and the neurodevelopment of the neonate [40].

Seminal microbiome

From the point of view of the fertility of a couple, we cannot forget the male factor, also with regard to the microbiome.

Mandar's group studied the vaginal and seminal microbiomes of 23 couples to evaluate how coitus affects the vaginal microbiome, observing that the prevalence of *G. vaginalis* was higher in women whose partners had leukospermia. They also found a high concordance between the vaginal and seminal microbiome after coitus, constituting the concept of the seminovaginal microbiome. Thus, the seminal microbiome can influence the vaginal microbiome, the reproductive health of the couple, and the microbiome of their offspring [41,42].

Treatment possibilities

Several studies raise the question of whether there is any treatment that can influence the genital microbiome and, consequently, the success rate after ART. The use of antibiotics is undoubtedly beneficial when used for targeted treatment of disorders such as chronic endometritis, bacterial vaginosis, or other active infectious disease.

It is hypothesized that oral antibiotic prophylaxis may reduce the rate of colonization of the upper genital tract. However, we know that antibiotic therapy, especially broad-spectrum antibiotics, can affect the healthy microbiota, in particular, *Lactobacillus*, causing dysbiosis and the opposite effect to the one we desire. Furthermore, antibiotic prophylaxis has not been shown to increase the gestation rate in ART [43]. For this reason, the prophylactic use of antibiotics prior to embryo transfer is not recommended.

Another interesting therapeutic strategy may be the use of probiotics. At present, there is no clear evidence that they have a direct impact on pregnancy rates in ART. However, probiotic supplementation has been shown to restore the flora of women with vaginal dysbiosis, increasing the percentage of *Lactobacillus*. Additionally, probiotics do not have some of the undesirable side effects of antibiotics, such as the appearance of resistance or alteration of the balance of the microbiome.

A review of the use of probiotics in infertility has recently been published. They noted that while there was growing evidence that the use of probiotics could modify the microbiome of the female genital tract, only two articles, by Kyono and Gilboa, respectively, analyzed their influence on pregnancy rate [44].

Kyono found a nonsignificantly higher clinical pregnancy rate in NLDM (Non-*Lactobacillus* Dominant Microbiota) women treated with vaginal probiotics versus no probiotics (58.8%, 10/17 versus 48.0%, 12/28, $P = .47$) [45]. Gilboa found no significant differences with intravaginal probiotics in clinical pregnancy rate (36.0% versus 34.3%, $P = .85$) [46].

Unfortunately, in the probiotics studies, there is no standardization in terms of type of probiotics, route of administration (oral or vaginal), duration of treatment, and use with antibiotics, so results are difficult to evaluate. For all that, there is insufficient evidence to recommend their use in ART to increase the pregnancy rate. However, it seems reasonable to consider treatment with probiotics in the pre- and periconceptional period in patients who have a tendency for bacterial vaginosis or a history of implantation failure, repeated miscarriages, or premature delivery [22,47].

Even so, more studies are still needed to determine whether there is an effective treatment that will allow us to improve the results of ART. In this context, Humaidan's team is conducting an interesting prospective multicenter, double-blind, placebo-compared study aimed at assessing the efficacy of treatments with clindamycin and probiotics in patients with altered microbiome who are going to undergo in vitro fertilization treatment.

Conclusion

Nowadays, easier access to massive sequencing techniques has resulted in great progress in the study of the female genital tract microbiome, discarding, once and for all, the belief that the uterine cavity is sterile.

Lactobacillus is the predominant genus in both the lower and upper genital tract. This predominance has been shown to be crucial in preventing dysbiosis and ensuring a healthy microbiome, which reduces the risk of pathologies such as bacterial vaginosis, pelvic inflammatory disease, chronic endometritis, and endometriosis, all intimately linked to fertility.

There is growing evidence that the female genital tract microbiome could influence a woman's fertility. Eubiotic microbiome may also have a positive effect reducing the risk of some adverse obstetric events, and even at birth, the colonization of the neonate could influence a child's immunological development.

However, to this date, there is no clear evidence that the use of either probiotic or antibiotic treatment improves the results of ARTs. Moreover, due to the lack of standardization of diagnostic methods and treatments, it is not recommended to perform a systematic analysis of the microbiome as part of the basic study of the infertile couple.

Nonetheless, there is growing evidence that in complicated cases with repeated ART failures, dysbiotic genital tract microbiome ought to be ruled out. This is especially recommended in cases of recurrent implantation failures and recurrent miscarriages.

Despite great advances in recent years, the complexity of the female genital tract microbiome and its evolution throughout a woman's life, including gestation, as well as the influence exerted on it by the seminal microbiome, have not been sufficiently described yet. More research is needed to assess how we can influence the genital microbiome and eventually improve the results of ART, as well as decrease the risk of adverse obstetric events [48].

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CHAPTER

24

How to manage recurrent implantation failure, what do we know?

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Definition and incidence

Perhaps one of the most frustrating conditions in in vitro fertilization (IVF), both for clinician and patient, is recurrent implantation failure (RIF). Further complicating the assessment and management of RIF is the fact that there is no consensus on the definition for this condition, and as such comparing studies and writing guidelines is difficult [1].

In an international survey of clinicians and embryologists, most adopted three unsuccessful embryo *transfers* as RIF; however, there was less of a consensus regarding the importance of the number of *embryos* transferred, nor was there a consensus regarding the role of the patient's age [1]. Most respondents to the survey gave equal value to a good-quality embryo irrelevant of whether it was a blastocyst or a cleavage embryo. A commonly used definition is the failure to achieve a clinical pregnancy (gestational sac on ultrasound) after transfer of at least four good-quality embryos, in a minimum of three fresh or frozen cycles, in a woman less than 40 years of age [2]. This definition is likely to change with the increased use of preimplantation genetic testing (PGT) [3].

The true incidence of RIF remains unknown and depends on the definition. It is believed that around 10%–15% of patients undergoing IVF suffer from RIF [4–6]. However, a study assessing the rate of RIF in a population of women aged 18 to 45 with no known congenital uterine anomalies, with an endometrium thickness greater than 7 mm and after transfer of euploid embryos, found that less than 5% would fail to achieve clinical pregnancy after three successive transfers [7].

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There are several pathologies known to be associated with RIF. Fibroids, particularly submucous and possibly intramural, are associated with a reduced implantation potential [8–11]. Hydrosalpinges, possibly due to the toxicity of the fluid inside the tube or to the washing of the embryo from the uterus into the vagina, reduce implantation rates [12–15]. Endometrial polyps have been associated with lower pregnancy rates following intrauterine insemination and are suspected to reduce implantation in IVF [16]. Polyps, hydrosalpinx, and submucous fibroids, if present, in RIF should be removed. It is unclear if intramural myomectomy improves pregnancy outcomes in patients with RIF. In some cases where a large fibroid uterus with a distorted cavity is present in RIF, it is the opinion of the authors that multiple myomectomy is a legitimate option for care. Uterine adenomyosis has been associated with a reduction in clinical pregnancy rates (CPRs) in some studies [17]; yet, evidence for treatment which improve outcomes remains elusive. A thin endometrium, especially when related to uterine adhesions, may also reduce implantation rates [18–23]. Treatment of a thin endometrium to improve implantation rates remains obscure although resection of synechia, if present, should be attempted.

Assessment of the uterine cavity

Intrauterine pathologies such as polyps, adhesions, submucous myomas, and uterine septa could all potentially affect implantation.

The prevalence of abnormal intrauterine findings in cases of RIF with a normal baseline ultrasound has been reported as up to 40% [24,25]. In a prospective randomized controlled trial (RCT), women with a history of repeated implantation failure were randomized to either hysteroscopy or no hysteroscopy in the month prior to starting a treatment cycle of IVF. 85 of the 323 women referred to hysteroscopy were found to have an abnormality of which 34 were considered severe [26]. Recognized pathologies were corrected. Importantly, no difference was found in the live birth rate (LBR) between the two groups, suggesting that the routine use of hysteroscopy in cases of repeated implantation failure and a normal uterine ultrasound offers little benefit.

Nevertheless, many clinicians would evaluate the cavity. 82% of clinicians use the "gold standard" hysteroscopy to investigate anatomical malformations in RIF (1). However, other modalities could be used. In a study comparing the diagnostic accuracy of sonohysterography, hysterosalpingography, and two-dimensional ultrasound, sonohysterography was found to be comparable to hysteroscopy in the diagnosis of endometrial polyps and be of higher sensitivity than hysterosalpingography or regular two-dimensional ultrasound. Hysteroscopy remained superior for the diagnosis of intrauterine adhesions [27]. Three dimensional sonohysterography was found to have a higher degree of concordance with hysteroscopy as compared to two-dimensional sonohysterography and required less time than hysteroscopy [28,29].

Assessment of parental karyotype

The notion that a balanced translocation in the gametes of the male or female partner could be a cause for RIF stems from the more defined role of balanced translocations in cases of recurrent pregnancy loss (RPL) [30,31]. However, when it comes to RIF, the data are limited and based mainly on observational studies. In two studies, one of 101 couples and the other of 615 patients (317 women and 298 men) with RIF, an abnormal karyotype was present in

3% of couples and 2.5% of women with RIF compared to a significantly lower rate in the general population (0.78%) [32,33]. A similar trend, although not statistically significant, was found in men (1.68% in men with RIF compared to 0.91% in the general population). In a study of RIF with at least 10 embryos transferred [34], translocations were found in 7/219 (3.2%) of couples with RIF in which both partners were tested and in 7/514 (1.4%) individuals with RIF, this was significantly higher when compared to historical cohorts of infertility patients (0.3%). Finally, Raziel et al., in a study including 65 couples with a history of six or more IVF trials or a history of 15 or more embryos transferred without achieving a clinical pregnancy, found chromosomal abnormalities in 10 of the 65 couples (15.4%) [35]. This evidence suggests that with higher order RIF, there may be a place for parental karyotyping.

Thrombophilia investigation in cases of RIF

It has been hypothesized that microvascular occlusion of the decidua could interfere with the process of implantation and thus lead to RIF [36]. With that said, the relationship between RIF and congenital or acquired thrombophilia is controversial, and studies reported conflicting results [37–42]. A prospective cohort study on 510 women requiring IVF compared to a cohort of 490 nulliparous women who conceived naturally found that the presence of prothrombin or Factor V Leiden mutation did not influence IVF outcome [43]. A metanalysis of 33 studies including a total of 1763 patients demonstrated a threefold increase in assisted reproductive technology (ART) failure among Factor V Leiden carriers (both heterozygous and homozygous) [44]. However, most of the studies in this analysis were case control and of poor methodological quality. When restricting the analysis to three cohort studies, comparable pregnancy rates were found irrelevant of Factor V Leiden mutation status. The same metaanalysis found no significant association between the other inherited thrombophilias and ART failures. When addressing the relationship between *acquired* thrombophilias and RIF, the metaanalysis (29 studies, 5270 patient) found a threefold higher risk for ART failure in the presence of antiphospholipid antibodies [44]. However, when limiting the analysis to studies whose control included only women who achieve live birth after ART (and not parous fertile women), no significant difference was found with the presence of antiphospholipid antibodies. Another retrospective study, published after the metaanalysis, compared 594 women undergoing IVF and who had a thrombophilia workup to 637 fertile women and 17,337 women, members of the same health care organization, with no history of venous thromboembolism. This study found no association between the common thrombophilias (factor V Leiden, prothrombin gene mutation, activated protein C resistance, or phospholipid antibodies) and failure to achieve pregnancy in IVF patients [45].

To summarize, the relationship between thrombophilia and RIF is still controversial. The British Fertility Society does not recommend testing for inherited thrombophilia but acknowledges that further research is needed to establish whether antiphospholipid antibody testing is indicated [46].

Assessment of sperm factors in RIF

Sperm aneuploidy testing has been studied in RIF. The rationale in testing sperm aneuploidy is that men with a normal karyotype may still harbor anomalies in the meiotic germline process, resulting in the production of aneuploid sperm. Several small studies have suggested that sperm aneuploidy may play a role in RIF [47,48]. While the role of sperm aneuploidy in RIF is possible, the available data are too scarce to recommend this test as part of the routine RIF investigation The American guidelines acknowledge that some would consider sperm aneuploidy testing in cases of failed ART cycles [49].

While there are some studies supporting an association between RPL and elevated sperm DNA fragmentation (SDF) [50], there is paucity of data regarding an association between high SDF and RIF. There is also no clear consensus regarding the preferred test for diagnosing elevated SDF, the upper limit of normal SDF, and importantly the recommended treatment in such cases [49,51]. Coughlan et al., in a study on 35 male partners in a couple suffering from RIF, 16 with RPL and 7 controls (recent fathers), found higher SDF in the RIF group when using the sperm chromatin dispersion test but not when using the terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay [52]. In a retrospective study on a total of 386 couples who underwent 1216 embryo transfers, high SDF was not associated with RIF [53]. Current guidelines do not support the use of SDF in routine clinical practice. The ASRM/AUA guidelines state that consideration should be given to testing SDF in couples with failed ART cycles, yet the guideline reference to data regarding the utility of SDF in RPL only [49].

Chronic endometritis

Chronic endometritis (CE) is a condition where a chronic inflammation, likely of infectious origin, is present in the endometrial tissue. The lack of universally accepted diagnostic criteria for CE makes studying the relationship between CE and infertility and RIF difficult [54]. The prevalence of CE in RIF depends on the definitions of both and varies across different retrospective studies with rates between 14% and 57.6% [55–60]. In a prospective study, only 7.7% of women with RIF were diagnosed with CE, a rate not different from the prevalence in the control group of fertile women [61]. In a metaanalysis addressing the effects of CE therapy on the outcome in women with RIF, antibiotic therapy resulted in higher CPRs (OR 4.02 CI 1.35–11.94); however, the metaanalysis was limited by the small number of patients and the poor methodological quality of the studies [62]. While some societies recommend evaluating and treating CE in women with RIF, other guidelines do not support this approach [63].

Assessment of the endometrial receptivity

Throughout the menstrual cycle, and in response to hormonal changes, the endometrium undergoes structural and functional changes preparing itself for the implantation. The period in the menstrual cycle in which the endometrium is receptive is termed the window of implantation (WOI). It is possible that a nonreceptive endometrium, due to an out-of-phase endometrium or asynchrony in endometrial development relative to the embryo development at transfer, could be a cause for RIF. Published histologic criteria specific to different phases of the endometrial cycle [64] were not found to be precise for diagnosing the WOI [65]. Sonographic markers such as endometrial thickness, pattern, blood flow or contractions, and hysteroscopic findings had poor prognostic accuracy for clinical pregnancy [66]. Recently, transcriptomics have been used to diagnose the WOI [67]. Using gene expression microarray, on a biopsy during a mock cycle, analyzing the genes whose expression changes significantly during the WOI could diagnose a subset of patients with RIF related to an out-of-phase endometrium. The potential for treatment by indicating the optimal time for embryo transfer exists. Results from initial studies demonstrated promise with a displaced WOI found in 27.5% of women after three failed embryo transfers [68,69]. These studies also reported an apparent benefit for correcting the time of transfer to the displaced WOI [70]. In contrast, a retrospective analysis on 2598 patients with RIF did not show any benefit for endometrial microarray analysis [71]. While the debate in the medical literature continues, the test is not recommended for RIF by the different societies.

Immunological testing

Implantation and pregnancy present a challenge to the maternal immune system, having to tolerate a semiallogenic fetus or, in the case of donated oocytes, a completely allogenic fetus. Both proinflammatory and antiinflammatory properties have a role in these events [72,73].

The T lymphocyte helper cells (T_H) are subcategorized into two main groups: the proinflammatory T_H1 cells and the antiinflammatory T_H2 cells. Pregnancy is a time of T_H2 dominance, and in early pregnancy, this dominance is induced by progesterone and by the embryo itself [74]. Indeed, small nonrandomized studies found an association between an altered T_H1/T_H2 ratio and proinflammatory/antiinflammatory cytokines and RIF [75–77]. However, whether these findings are the cause of or secondary effects of a failed gestation is still unclear. Change in the T_H1/T_H2 is also a normal pregnancy-related event. There is no T_H2 dominance prepregnancy, not even peri-implantation, and thus no validated preconceptual screening test is available.

The role of natural killer (NK) cells in RIF has been studied. There are two subgroups of NK cells that are of interest: the peripheral NK cells and the uterine NK cells. The peripheral NK cells are part of the immunes system innate response, and they are involved in infection and neoplasia defense. Several studies have found an association between a higher level of peripheral NK cells and RIF [78–80]. However, these studies were small and of varied methodologies, making a recommendation for a clinical diagnostic test difficult. Further, in one of the studies, using a specific percentage of NK cells as a cut-off resulted in a 97% specificity but only 11% sensitivity, limiting usefulness in RIF population [78].

Uterine NK cells show limited cytotoxic activity but are the dominant immune cell in the endometrium, and it is believed that they play a role in implantation [73]. Most studies on uterine NK cells were done on patients with RPL, and data are conflicting [81]. In a prospective study on 32 women with "idiopathic" RIF, 17 were found to have a higher number of uterine NK cells as compared to only 1 case out of the 17 control patients [82]. However, measurement of uterine NK cells is complicated since levels of peripheral NK cells and uterine NK cells are not correlated [83]. Furthermore, the levels of uterine NK cells fluctuate in response to reproductive hormones. Therefore, a single measurement may not represent the peri-implantation window.

A metaanalysis evaluating the association between NK cells (uterine and peripheral) and infertility found no difference in NK cells (as percentage) between the fertile and infertile population [84]. The same metaanalysis found no association between NK cells and IVF LBR. Of note, only two studies, both evaluating peripheral NK cells, were included in this subanalysis.

Recent studies focus on the function of rather than the number of NK cell. There is an interaction between the killer immunoglobulin like receptor (KIR) that is expressed on the uterine NK cells and human leukocyte antigen−C (HLA-C) present on the trophoblast cell. This interaction may affect early placentation and, as a consequence, is associated with pregnancy disorders such as RPL, fetal growth restriction, preeclampsia, and hypothetically RIF [85]. The KIR gene family is highly polymorphic but is generally divided into two haplotypes—KIR A which included mainly inhibitory KIRs and KIR B which includes mainly activating KIRs. The lack of activation of an immune response early in pregnancy may contribute to reproductive failure, whereas activation contributes to early placentation [86,87]. Different maternal KIR subtypes and different embryo HLA-C subtypes (which are at least semiallogeneic to the mother) likely contribute to implantation and the placentation process and could be targets for future therapy or at least embryo selection in RIF [88,89].

To date, while a correlation between an immunological mechanism and RIF is plausible, the diagnostic procedures and treatments are still investigative and not recommended as part of the clinical workup of RIF [90].

Treatment options in RIF

Lifestyle factors

While a high body mass index (BMI) is related to pregnancy complications, its association with a decreased likelihood of embryo implantation is more controversial. Obesity necessitates higher gonadotropin doses and lengthier stimulation cycles. It may be a cause for a more technically demanding embryo transfers and thus plausibly influence the chances of pregnancy. With that in mind, studies results regarding the association between different BMI categories and implantation success have been contradictory with some studies showing lower implantation rates, particularly with BMIs above 40 kg/m^2 and other studies resulting in comparable implantation rates [91−97]. Male obesity might also affect implantation; however, this too is controversial. A metaanalysis on the effects of paternal obesity on assisted reproductive outcomes found a nonsignificant decrease in clinical pregnancy for obese men [98]. No study addressed the effect of maternal or paternal obesity in the RIF population and the effect that interventions might have on implantation in RIF.

While associations between lifestyle factors and fertility have been studied [99], no studies specifically addressing the RIF population have been done. In an international survey, more than two thirds of clinicians take lifestyle factors into account, mainly focusing on illicit drugs, smoking, and patient's weight [1]. It seems reasonable to advise patients to maintain an optimal weight, and to cease smoking, but it is no less important to remember that patients with RIF may be dealing with heightened anxiety and depression and adding feelings of self-blame is counterproductive, especially when an association between these factors and RIF has not been proven.

Endometrial scratch

Endometrial scratch refers to an intentional endometrial injury, usually performed with a pipelle catheter. While this procedure has been commonly used in cases of RIF [100], the evidence behind it is controversial. Interest in endometrial injury as treatment started after an unexpected observation of higher pregnancy rates was found in a group of women undergoing multiple endometrial biopsies as part of a trial investigating the endometrial expression of a gap junction protein. The authors then conducted a prospective trial in good responders who failed to conceive in one or more previous embryo transfer cycles and found more than a twofold higher LBR in the endometrial scratch group [101]. The authors proposed several theories for their finding, including decidualization provoked by the injury or perhaps the inflammatory reaction involved in wound healing with the recruitment of cytokines which play a role in embryo implantation. Subsequent trials and metaanalyses reported conflicting results [102,103]. A metaanalysis evaluating subjects with two or more failed embryo transfers and after excluding trials with a risk of unintentional injury in the control group showed a benefit in terms of clinical pregnancies but not in LBR [104]. Interestingly yet another metaanalysis published around the same time of the previous metaanalyses found the procedure to be beneficial in women with two or more failed transfers, but no effect was found in women undergoing frozen-thawed embryo transfer [105]. Following the publication of these metaanalyses, a large RCT was published demonstrating no benefit for endometrial scratch in a subgroup analysis of patients with more than two previous failed embryo transfers [106]. Another, smaller RCT found in a subgroup analysis of patients with at least three previous failed transfers a benefit in those randomized to the endometrial scratch [107]. To conclude, whether endometrial scratch is beneficial in cases of RIF is still a matter of debate. While opponents state that high-quality data point to lack of efficacy, proponents will argue that the efficacy depends on the RIF patient population being evaluated.

Immunotherapy

While a review of all immunomodulatory approaches is beyond the scope of this chapter, a summary of the more common ones is provided.

Corticosteroids

Few studies focused on the role of corticosteroids specifically in patients with RIF. Geva et al. studied the effect of prednisone and aspirin on patients with 3 or more failed transfer cycles and the presence of nonorgan-specific autoantibodies and found a higher pregnancy rate when compared to a control group from their preliminary study [108]. There was no direct control group. In a prospective quasirandomized trial, Fawzi found the combination of prednisolone and low molecular weight heparin (LMWH) to improve implantation rates and pregnancy rates in patients with unexplained implantation failure (failed one or two intracytoplasmic sperm injection attempts) [109], while a retrospective study by Siristatidis et al. found that the addition of prednisolone and LMWH resulted in no benefit in biochemical rate or CPR in patients with RIF [110]. In a retrospective matched case control study, Motteram et al. found no difference in LBR with the addition of prednisolone, doxycycline, and aspirin in patients with 3 or more failed IVF cycles [111]. In summary, there are currently insufficient data to recommend corticosteroids to improve implantation in patients with RIF.

Intravenous immunoglobulin

Intravenous immunoglobulin contains polyclonal immunoglobulins derived from donor plasma and has potential immunomodulatory effects. The largest RCT to study the effect of IVIG in cases of RIF found no improvement in LBR with the use of IVIG [112]. A metaanalysis on the effect of IVIG in patients with RIF and immunological abnormalities (assessed differently in the different studies) found improved pregnancy rates and LBR with IVIG administration. It is worth noting that this metaanalysis did not include any RCT [113] and that there was a high level of heterogeneity among the studies.

Intralipid

Intralipid is an intravenous fat emulsion with immunomodulatory properties. An RCT studying the effect of intralipid on pregnancy outcomes in women with implantation failure (defined as one or more previously failed transfers) found a benefit for intralipid therapy with an adjusted odds ratio of 3.1 for clinical pregnancies in the treatment group [114]. However, in a subgroup analysis, the beneficial effect was only found in patients with one previous IVF failure. Another RCT including patients with a history of RIF found no difference in LBR between the treatment and placebo groups [114]. A metaanalysis on the matter found a benefit, in terms of LBR, for intralipid infusion; however, quality of evidence was deemed low and when the analysis excluded two publications that were from conference abstracts, the benefit was no longer statistically significant [115]. Currently intralipid is not recommended for patients with RIF.

Peripheral blood mononuclear cells

In 2006, Yoshioka described higher implantation rates, CPRs, and LBR in women with a history of RIF after uterine infusion of autologous peripheral blood mononuclear cells (PBMCs) cultured with human chorionic gonadotropin (HCG) for 48 h [116]. Since then, several RCTs utilizing different methods of PBMC preparation came to similar conclusions [117,118]. In a metaanalysis, PBMC infusion resulted in increased CPRs in patients with RIF [119]. Three more metaanalyses found a statistically significant benefit for PBMC in terms of implantation rate, CPRs, and LBR in patients with RIF [120–122]. However, the studies differed in the methods of PBMC preparation, in patient selection, and in the stage of embryo transferred. Most included studies were not randomized and were unblinded. The control arm in the studies included did not have a “sham procedure” and as such ruling out that local injury caused during the PBMC infusion rather than the PBMC themselves was the cause of benefit was not possible. The number of published studies included in the metaanalyses was small, and thus a publication bias is possible. In a double-blind RCT, subsequently published, with a placebo (phosphate buffer saline) infusion prior to endometrial transfer in the control group, significantly higher pregnancy and LBR in the PBMC treated group were noted [123]. While there seems to be potential in PBMC treatment, further studies, better defining patient selection, and standardization of the use of PBMC are needed.

Other immunomodulatory approaches in IVF in general and in RIF patients in particular include subcutaneous or intrauterine granulocyte colony-stimulating factor administration, intrauterine HCG administration, use of TNF-alpha-blocking antibodies, platelet-rich plasma, seminal plasma infusion, and immunosuppressants such as Tacrolimus. Studies on these treatments report conflicting results, and there is no evidence in favor of such treatments for RIF.

Aspirin

Potential mechanisms for aspirin’s role are the vasodilatory effects it has on the uterus and its antiinflammatory activity. Evidence from studies on the use of aspirin in IVF is inconsistent, and comparing trials is complicated because they differ with regards to the time in the cycle in which treatment with aspirin was started, the duration and dose of treatment, and the population being studied [124]. A recent metaanalysis on the use of aspirin in IVF (not limited to RIF patients) found no significant difference in implantation rates or CPRs with or without aspirin, but a small benefit when a sensitivity analysis was done [125]. When limited to RCTs only, no benefit was found in LBRs. There is a paucity of data regarding the role of aspirin in the RIF population. A retrospective study on patients with RIF demonstrated reduced uterine arterial blood flow resistance with the use of aspirin; however, analysis on treatment outcome was not done [126]. Currently, there is no evidence for a benefit of aspirin in RIF patients.

Low molecular weight heparin

The use of heparin and LMWH in patients with RIF is controversial. LMWH's potential benefit is beyond its anticoagulative effect and is likely related to its modulating effect on the implantation process and on trophoblast survival [127,128]. An RCT including 83 patients with RIF and thrombophilia (inherited or acquired) demonstrated a significantly higher LBR with the administration of LMWH when compared to placebo [129]. An open-label RCT including 150 patients with RIF and without known thrombophilia failed to demonstrate a significant difference in CPR or LBR with the use of LMWH [129]. A metaanalysis on the role of LMWH in RIF showed a trend toward an increase in implantation rate and a significant increase in LBR with its use however cautioned that the overall number of participants was low [130]. The beneficial effect was not significant when only studies with unexplained RIF were analyzed. Lastly, a recent large, multicenter, retrospective cohort study found no benefit for the use of LMWH in women with two or more unsuccessful IVF cycles [131].

Preimplantation genetic testing

The hypothesis behind the use of preimplantation genetic diagnosis (PGT) in patients with RIF is that an increased rate of aneuploidy might be responsible for RIF in a subgroup of these patients. However, most studies on the use of PGT in the IVF population focused on patients of advanced maternal age and excluded patients with RIF [132]. In a prospective, non-randomized trial using array comparative genome hybridization on trophectoderm biopsy, PGT did not improve LBR per patient but did improve LBR per transfer [133]. In a prospective randomized trial evaluating the effect next-generation sequencing PGT has in IVF outcome, PGT did not improve overall pregnancy outcome when analyzed per embryo transfer nor per intention to treat [134]. This trial excluded patients with RIF. A retrospective multicenter study evaluated the clinical usefulness of PGT (using NGS) and/or endometrial receptivity array in patients with moderate or severe RIF (failure to achieve pregnancy after 3 or 5 transfers, respectively) [135]. While the endometrial receptivity test did not significantly improve outcome in either group, PGTA did improve implantation rate and ongoing pregnancy rate per transfer in the group of patients with moderate RIF but not in the group of patients with severe RIF. Of note, the study was underpowered for detecting differences smaller than 20% in the severe RIF group. In summary, while PGT may have a role in improving the implantation rate *per transfer* in a subgroup of patients with RIF, the improvement is not in the cumulative outcomes. With that being said, reducing the number of failed transfers in patients with a history of RIF may have important benefits, considering the emotional consequences of failure. The difficulty of defining which subgroup of patients with RIF would benefit from this remains.

Other potential treatment options

A Cochrane review on the role of assisted hatching in assisted conception did a subgroup analysis of patients with repeated IVF attempts. No difference was found in LBR [136]. The quality of evidence in the review ranged from very low to low.

There is conflicting evidence regarding the use of intracytoplasmic morphologically selected sperm injection (IMSI), using higher magnification for sperm selection, in patients with RIF [137–139] as for other methods of sperm selection including magnetic activated cell sorting, hyaluronic acid-bound sperm selection, and use of surgical sperm retrieval.

Different methods for ovarian stimulation and endometrial preparation are continuously being studied. One such method that shows promise is dual suppression with the use of GnRH agonist plus aromatase inhibitor prior to frozen embryo transfer in patients with RIF [140,141]. It is worth mentioning that these studies were retrospective, and further research is warranted.

Recently, there is renewed interest on the dose and mode of progesterone administration in assisted reproduction. Prospective cohort studies on artificial endometrial preparation found progesterone levels on the day of embryo transfer to be associated with ongoing pregnancy rates [142,143]. Two RCTs found that intramuscular progesterone, alone or in combination with vaginal progesterone, for luteal support was superior to vaginal only in nonovulatory frozen embryo transfer cycles [144,145]. Further studies are needed to define the role of progesterone in patients with RIF.

Prognosis

The prognosis of patients with RIF depends on many factors such as age, the diagnostic criteria used to define RIF, and more. In a study on 118 women under the age of 39 diagnosed with RIF after having 3 failed fresh embryo transfers or 10 (fresh and frozen) failed embryo transfers of good-quality embryos, and after excluding patients with uterine anomalies, hydrosalpinx, or autoimmune disease, the reported cumulative incidence of live birth during a 5.5 year follow-up was 49% with a calculated median time to pregnancy of nine months after the diagnosis of RIF [146].

Conclusion

The impact of RIF on patients' psychosocial state is significant with studies showing higher levels of depression and anxiety [147]. For the clinician, this diagnosis presents a challenge amplified by the lack of universal diagnostic criteria and the heterogeneity in the studies on the matter. This combination of both patient and clinician frustration often leads to unproven diagnostic procedures and interventions with the entailed costs and possibly iatrogenic harm. Future studies would hopefully focus on better defining the different subgroups of patients with RIF. Perhaps this would allow better tailoring of the treatment to our patients.

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CHAPTER

25

How to set up an andrology laboratory for a fertility center?

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Introduction

The andrology lab is the dedicated space to perform semen analysis, sperm preparation for intrauterine insemination (IUI), and in vitro fertilization/intracytoplasmic sperm injection (IVF/ICSI), as well as sperm cryopreservation. Depending on the size of the assisted reproductive technology (ART) clinic, this should be a separate laboratory, preferably in close contact with the IVF lab. Usually, an andrology lab is not a single entity, but an active collaboration between the physicians, a genetic, endocrinology, and IVF lab. Though constructing an andrology lab may be less challenging compared with an IVF lab, the design also needs to respect the specific regulations, which are usually different between countries or even states. This chapter serves as a general guide in the construction of an andrology lab, including the specific requirements for diagnostic and therapeutic activities, without detailed regulations per country.

The design

Constructing a new andrology lab is not a one-man's job but involves a team with all concerned parties. At least one dedicated person of each concerned party (andrologist, embryologist, engineers, architect, management, and medical doctors) should be involved. Together they need to critically review all requirements and wishes, considering the expected workload within the provided space. Whether an existing facility is rebuilt or whether a new site is being constructed, the space limit, new regulations, safety as well as easy expansion in case of future growth should be considered. The andrology lab is ideally split between a diagnostic and therapeutic lab, allowing the possibility to perform the therapeutic procedures

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in a sterile environment, and a separate cryopreservation unit. Planning the lab is more challenging for the therapeutic lab, which is—like the IVF lab—constructed as a cleanroom; hence, it is more difficult to make changes once the sealed closed box is built. Layout, number of patients and future growth, type of treatments (current and future), and type of patients (including or excluding infectious patients) should all be discussed before planning the andrology lab. A *Guide to the Quality and Safety of Tissues and Cells for Human Application* [1] is available from the European Directorate for the Quality of Medicines and should be used as a backbone to define standards, guidelines, equipment, staffing, qualification, and validation. A detailed plan of all equipment should be available before the construction starts, to plan the position of a sufficient number of power outlets.

Equipment

The equipment needed in an andrology lab depends on the specific procedures that are performed in each lab. The most common standard procedures in the therapeutic lab are semen preparation for IVF/ICSI/IUI, and for the diagnostic andrology lab, this is the basic semen analysis. Complementary to the basic semen analysis, multiple other tests are available, for which we refer to the sixth edition of the the World Health Organization (WHO). The lab should be arranged in such a way that one sample is analyzed per predefined working area.

In the therapeutic lab, samples should be treated in a sterile way; all equipment is placed on stainless steel tables. Once liquefied on the heated stage, samples are handled in laminar flows that should be equipped with automated patient recognition software (for instance, RI witness). Each workstation has its own centrifuge and microscope with Makler, Neubauer, or similar counting chambers. A fridge and incubator (37°C) are available to store all required chemicals or solutions. In case IVF samples are processed, fertilization tubes should be gassed or an incubator with 5% O_2, 6% CO_2, and 89% LN_2 should be present, to deliver tubes with optimal pH. All other disposables are based on the specific selection of techniques per lab (gradient, swim-up, MACS, microfluidics, etc.). In case infectious samples are handled, biosafety cabinet II should be used.

The diagnostic andrology lab is less stringent in air quality criteria and laminar flows. Usually, the following equipment is standard and should be present: a balance for the sperm pots, refrigerator and freezer, incubator (37°C) as well as a heated stage (37°C), phase contrast microscope which includes $\times 40$ (or $\times 63$) positive phase objectives, $\times 100$ oil-immersion objective and heated stage, centrifuges, microscope slides and covers, different types of pipettes, Makler or the improved Neubauer or any alternative, computer-assisted semen analysis (CASA) machine, and specific equipment to cryopreserve sperm (based on the specific procedure applied per the lab). This equipment is of course based on the different types of tests that are performed in the lab. All equipment are placed on impermeable work surfaces, like stainless steel.

Quality assurance

Quality assurance in the andrology lab is a set of tools that is used to define whether all procedures deliver reliable and robust outcomes. The main aim is to predefine deviations that should be classified as discrepancies to detect systemic errors and reduce technical/operational variability. Only regular examination and critical review of outcomes of different aspects of all laboratory techniques/handlings will allow a lab that the established standards are being delivered to each patient. An international standard (ISO 15189) for medical laboratories exists, to identify the procedures of laboratory quality management. Besides the implementation of accurate standard operating procedures (SOPs), the personnel should also be adequately qualified and monitored for each procedure. Internal quality assurance (focusing on imprecision/variability) and external quality assurance (focusing on correctness) are required by law in some countries, while standardization is requested in other countries. Commercially available internal quality control samples exist and are ideally, performed on a monthly basis, while external quality can be performed quarterly/biannually.

All procedures should be described in SOPs, updated regularly, and strictly followed by all personnel. These SOPs also describe acceptable degrees of error. The magnitude of these is described by a confidence interval with upper and lower limits, often presented as Levey Jennings plots, X_{bar} or S charts, and will express the variation that is allowed for each procedure. The quality assurance program should also describe preventive and corrective actions. The continuous monitoring should detect and correct problems, but the corrective and preventive actions will also prevent the further occurrence of these problems. All this information has to be described in a comprehensive quality manual. Further detailed information on quality and risk management in the lab is described elsewhere [2].

Accreditation

Laboratory accreditation is a procedure by which an authoritative body gives formal recognition of technical competence for specific tests, based on third-party assessment and following international standards. It aims at evaluating the laboratory's quality, performance, reliability, and efficiency. Accreditation also enhances the credibility of a laboratory's results and services by demonstrating that it meets the quality and competency criteria required for accurate, reliable, and safe testing [3].

Accreditation bodies are specialized authoritative organizations that assess laboratory compliance to quality standards that reflect the laboratory's performance. Laboratory accreditation does not guarantee optimal outcomes, but it validates the policies and procedures implemented in the laboratory that would identify any deviation from standards and provides opportunities for continuous improvement [4,5].

Accreditation process:

1. Contracting with an accreditation body: The accreditation body is responsible for assessing the quality system and technical aspects. After choosing the accreditation body, an application is submitted, and following approving the quotation, an agreement is signed between the two parties.
2. Preparation for accreditation: The first step is the implementation of internal and external quality systems (IQS/EQS). Normally, the accreditation body will share a checklist of all required documents and protocols. That will include quality manual that

determines how the laboratory complies with the standards, standard operating protocols for all procedures carried out by the laboratory, job description, and responsibilities of all laboratory staff.

3. Audit: The audit will be carried out by designated inspectors assigned by the accreditation body. All documentations and protocols will be assessed to determine the laboratory's compliance with the established standards.
4. Assessment and corrective action: The assessment team will share a report indicating all nonconformances and deviations from standards and will give the laboratory team a specific time period to respond and address all the requirements.
5. Accreditation: Once the executive committee reviews all the material and confirms meeting all the requirement, an accreditation certificate will be issued. Periodic audits will be required to ensure compliance with the standards are required to sustain the accreditation.

Diagnostic andrology laboratories abide to the medical laboratory accreditation process. There are several accreditation bodies that are authorized to assess and certify diagnostic laboratories. Following are some of the international accreditation bodies:

- College of American Pathologist (CAP)
- International Standardization Organization (ISO)
- Joint Commission International (JCI)
- Clinical Pathology Accreditation (CPA)

Clinical Laboratory Improvement Act (CLIA)

- United Kingdom Accreditation Services (UKAS)
- International Laboratory Accreditation Cooperation (ILAC)

In addition to the diagnostic andrology laboratory accreditation bodies, therapeutic andrology laboratory accreditation bodies are also considered, such as follows:

- Society for reproductive technology (SART)
- European Society of Human Reproduction and Embryology ART centre certification for Good Practice (ESHRE)
- College of American Pathologist Reproductive Accreditation Program

Air quality

IVF is a high complexity, multistep procedure that involves the manipulation of gametes and embryos, which are relatively sensitive and can be impacted by the environmental conditions. Several studies have demonstrated that laboratory air quality and ambient air pollution can affect embryogenesis and implantation [6]. For that reason, scientific societies and governing bodies have mandated a specific air quality grade in the embryology and andrology laboratory.

Air quality is classified according to the number and size of particles within a specific sample of air. ISO 14644-1 classifies air quality into nine classes (1–9), where Class 1 is the cleanest. US Federal Standard 209E classifies air quality, based on the maximum acceptable number of $\geq$0.5 microns per cubic foot of air, into: Class 1, 100, 1000, 10,000, 100,000. Class 1 represents the best air quality. Good manufacturing procedures (GMPs) classify air quality into four grades (A–D). Grade A is equivalent to ISO 5 and Class 100 (US Federal Standard), and Grade D is equivalent to ISO 8 and Class 1,000,000.

According to the European Union Tissues and Cells Directive, Grade A environment is recommended for tissue and cell processing with a background of at least GMP Grade D. However, Grade D environment was accepted if Grade A was not possible. On the other hand, the association of biomedical andrologists mandates Grade D background air quality and a minimum of Grade C within the immediate processing area.

Laboratory air quality can be controlled through the application of positive pressure in the most crucial rooms compared to the adjacent ones with a pressure differential of 10–15 Pa. In order to maintain the flow of air particles away from the highly pressured rooms, air pressure also creates turbulent air, which circulates still air in spots under equipment and workstations.

The implementation of high-efficiency particulate air (HEPA) filters in the air handling unit (AHU) allows the removal of 99.97% of particles greater than or equal to 0.3 microns present in the external air. Air changes per hour should also be controlled to not less than 15 changes per hour and humidity levels of 35%–55%.

Guidelines suggested by regulatory and legislative authorities for andrology and embryology laboratory's air quality are based on the sensitivity of biological material and the complexity of procedures performed. It is advised that diagnostic laboratories follow the guidelines of clinical andrology laboratories (IVF laboratory) to provide an optimal environment for testing. Therapeutic andrology laboratory is considered part of the IVF laboratory where oocytes and embryos are cultured. For that reason, the air quality requirement in a clinical andrology laboratory is more stringent than that of a diagnostic andrology laboratory. In addition to controlling the pressure, particles and air changes inside the laboratory, volatile organic compounds (VOCs), which are embryotoxic, need to be reduced.

AHU is equipped, in addition to the HEPA filters, with chemical filter containing activated carbon and/or potassium permanganate. The pores between carbon particles contain delocalized electrons that trap the chemical contaminants. Potassium permanganate detoxifies compounds that pass through the carbon pores such as alcohols and ketones [7].

Ultraviolet photocatalytic oxidation can be combined with the active carbon and potassium permanganate filters for reductions of VOCs. Photocatalytic oxidation process utilizes the ultraviolet light to break down the organic compounds to harmless water molecules, carbon dioxide, and detritus.

Hygienic maintenance before use

Aseptic conditions are mandatory for andrology laboratory operations. This requires the elimination of all microbial contaminants and VOC in the laboratory. The VOC levels of a newly designed laboratory are high due to the use of building materials such as adhesives, glue, paints, and sealants. In order to reduce the levels of VOCs, a burn-in procedure is recommended. The burn-in principle is to increase the room temperature with 10–20°C for 10–28 days, while increasing the ventilation rate, to allow for the removal of the organic compounds. If this temperature cannot be achieved using the available AHU, an auxiliary electrical heating can be used.

Walls, ceiling, and floor need to be cleaned several times using specific detergents that are nonembryotoxic. 3% hydrogen peroxide solution and quaternary ammonium-based solutions are recommended for cleaning and are low in VOC levels. Cleaning is advised prior and after the burn-in procedure. Following laboratory cleaning, microbial sampling for aerobic bacteria and fungi should be done in new facilities using an Andersen sampler followed by microbiological culturing and identification. Burn-in procedures are also applied on incubators, if to be used in andrology. All equipment should undergo hygienic maintenance as per manufacturer's recommendation prior to use. Further information can be found in Refs. [8–12].

Quality control check for automated and manual sperm counting methods

This can be done using beads solutions that are made of small latex spheres (4 μm in diameter) suspended in an isopycnic medium. The physical properties of this medium allow the beads to float and promote even distribution of the beads throughout the solution. The solution is used in the same manner normally used for standard sperm counting. Performing this quality check prior to using the counting chamber for therapeutic or diagnostic cases will allow identification of any issues with the results generated from the chamber.

Cryopreservation

Since the introduction of cryopreservation of human spermatozoa in the 1970s, there has been extensive progress in cryopreservation methods that optimized spermatozoa survival, leading to considering this technique an essential service in any ART program. Cryopreservation allows preservation of male fertility, especially in cases of cytotoxic male treatments, such as chemotherapy and radiotherapy, or in nonmalignant diseases such as diabetes or autoimmune disorders that may cause testicular failure or ejaculatory dysfunction. It is also beneficial for donor ejaculatory storage for future use. Sperm cells obtained by testicular extraction or percutaneous epididymal aspiration are normally cryopreserved for future insemination [13–15]. Different techniques are available for cryopreservation, which can be found in more detail in the sixth edition of the WHO.

Slow freezing

In this method, sperm cells are cooled in two or three steps progressively over a period of 2–4 h. The optimal initial cooling rate of the specimen from room temperature to 5°C is 0.5–1°C/min. The sample is then frozen from 5°C to −80°C at a rate of 1–10°C/min. The specimen is then plunged into liquid nitrogen at −196°C. Alternatively, the control of the cooling rate can be done automatically using a semiprogramable machine, which has been found to be simple to use and does not require continuous operator intervention [16].

Rapid freezing

This technique requires direct contact between the straws and the nitrogen vapors for 8−10 min and immersion in liquid nitrogen at −196°C. The sample is initially mixed dropwise with equal volume of cryoprotectant; the mixture is loaded into the straws and left to incubate at 4°C for 10 min. The straws are then placed at a distance of 15−20 cm above the level of liquid nitrogen (−80°C) for 15 min; after this stage, the straws are immersed in liquid nitrogen [17].

Vitrification

The sperm suspension is plunged directly into liquid nitrogen, and the sperm cells are cooled in an ultrarapid manner. During the procedure, water is cooled to a glassy state through extreme elevation of viscosity without intracellular ice crystallization making this procedure less labor-intensive, quicker, and presumably safer compared to traditional slow-freezing protocols [18].

Cryostorage standards and requirements

Cryopreservation facilities should be safely located outside, but close to the laboratory and, for safety reasons, with visible access to the interior [19]. Access to cryostorage tanks and LN_2 supply tanks should be restricted only to authorized personnel and trained laboratory staff. Protocols should be developing for visual inspections of tanks to assess tank integrity. Water condensation, frost or ice, and cracks on the bank body or neck are signs of vacuum failure and compromised tank integrity. The level of LN_2 must be measured three times per week, as per requirements and minimum standards for cryostorage management published by the College of American Pathologists (CAP). Reserves of LN_2 should be readily available to fill cryostorage tanks as needed, either physically or by autofill systems. All storage tanks must be electronically monitored with probes that will sense a rise in temperature and/or a decrease in LN_2 levels. In case of any variation in temperature or LN_2 levels, the alarm system will notify the laboratory personnel with an alarm in the laboratory and a text message or phone call, specifying the affected tank and the current values. Constant monitoring of oxygen levels in the cryoroom where LN_2 tanks are stored or LN_2 is decanted is mandatory. Off-gassing of LN_2 from tanks can create a hypoxic environment that can cause asphyxiation. The decrease in oxygen levels can go unnoticed; hence, oxygen monitoring system is mandatory in cryostorage rooms to protect laboratory staff.

Cryopreservation of infectious samples

The most important infectious microorganisms that may be found in semen are HIV and hepatitis B and C viruses (HBV and HCV). In case no serology is known, laboratory personnel should treat all biological samples as potentially infectious and use appropriate caution in handling them. Cryopreservation of semen samples of infected patients might be required, in order to reduce the cross-contamination. Tests for viruses and sexually transmitted infections should be performed on all males at the time of banking as follows: (i) HIV, types 1 and 2 (using nucleic acid testing [NAT], which also tests for group O antibodies), (ii) HBV test for hepatitis B surface antigen (HBsAg), total antibody to hepatitis B core antigen (anti-HBc) (IgG and IgM), and a NAT assay for HBV or combination that includes HBV (HCV using anti-HCV and a NAT assay for HCV), (iii) treponema pallidum (i.e., syphilis), (iv) chlamydia trachomatis (using test design for detecting in asymptomatic, low-prevalence populations), (v) *Neisseria gonorrhea* (using test design for detecting in asymptomatic, low-prevalence populations), (vi) human T-lymphotropic virus (HTLV) types I and II using anti-HTLVI/II, and (vii) cytomegalovirus (CMV) using anti-CMV (total and IgG and IgM).

Men who are seropositive for hepatitis B or C or HIV should check the viral load prior to cryopreservation of semen sample. Since viruses can survive and be transmitted via liquid nitrogen, viral positive samples should be stored in separate tanks according to the type of viral infection. The use of high security cryopreservation straws with thermal sealing reduces the risk of cross-contamination in storage; vapor phase cryopreservation is preferably used over liquid nitrogen. It is recommended to process the samples using density gradient centrifugation (for instance with ProInsert kit) followed by swim-up procedure, to discard nonsperm cells present in the seminal plasma. Personnel working with potentially infectious samples should be vaccinated against hepatitis B or other viral disease, for which the vaccine is available. Sperm preparation should be performed in a Class II laminar flow cabinet to protect both the operator and the contaminated specimen.

Sperm functioning tests

The primary assessment test for male fertility is the routine semen analysis. Semen analysis provides essential information about sperm cells concentration, motility, morphology, and viability. In addition to routine semen analysis, more tests were developed to assess the sperm function competency including the ability to bind to zona pellucida, penetration of cervical mucus, and DNA integrity that provide a deeper and more comprehensive assessment of male's fertility. In the following section, we will discuss some of the common sperm function tests. More detailed information and more sperm tests can be found in the sixth edition of the WHO.

Hypoosmotic swelling test

This test is used to check the viability of immotile sperm. It is a relatively simple test that measures the ability of the sperm to transport fluid. Viable sperm membrane transports water in. The sperm is incubated in a hypoosmotic solution (150 mOsm/L or less). Viable sperm cells will maintain an osmotic gradient and absorb water resulting in swelling of the plasma membrane causing curling of the tail. Dead sperm will not react, and the tail will not curl.

Cervical mucus/sperm interaction assay

This is a simple test that provides important information about female hormonal status and sperm function. An aspirate of cervical mucus is retrieved within 30 min of intercourse at midcycle. The mucus is then examined microscopically for sperm count and motility, and mucus features such as viscosity and ferning.

Acrosome reaction assay

In order for the spermatozoa to be able to fertilize an oocyte, it has to acquire specific characteristics referred to as sperm capacitation. The last step of capacitation is acrosome reaction, during which the sperm plasma membrane and the outer acrosomal membrane fuse. Acrosome reaction is an exocytotic process fundamental for successful penetration of the oocyte. In vitro, acrosome reaction can be induced by sperm exposure to progesterone or calcium ionophore, which is referred to as acrosome reaction to ionophore challenge. The acrosome reaction is measured using fluorescent dyes. If there is a bar of fluorescent in the acrosomal part or it looks blank (no staining), then acrosome reaction occurred; however, if there is a cap of stain, then the acrosome is intact, and the acrosome reaction has not occurred.

Sperm zona-binding assay

Spermatozoa capability to bind to the zona pellucida is an obligatory step in natural conception. Tests were developed to assess sperm cells competency to bind to the zona. However, these tests are dependent on the availability of donor oocytes or unfertilized oocytes. The principle is to compare the sperm cells against donor fertile sperm cells in their ability to bind to oocyte's zone. The hemizona assay protocol requires splitting the zona in half, incubating each half in a tube for four hours, one tube using donor sperm and the second using the patient's sperm. Then expressing the results as a ratio of patient to donor sperm bound. Due to the technical difficulty of cutting the zona in half, a new method was developed that uses the whole zona incubated with donor and patient sperm, but each sample is stained with different color fluorescent dye [20,21].

Sperm DNA fragmentation

Sperm DNA fragmentation is damage in the sperm's DNA seen as breaks or separations in a single or double stands of DNA. During spermatogenesis, around 50% of germ cells are destined to be phagocytosed by Sertoli cells; these sperm cells exhibit an apoptotic marker such as Fas, p53, and phosphatidylserine. However, the inefficiency of apoptosis process can result in the presence of these defective sperm cells in the ejaculate.

DNA fragmentation in sperm cells can be induced by different mechanisms, such as apoptosis during spermatogenesis, DNA strand breaks during sperm chromatin remodeling, posttesticular damage caused by oxygen radicals, radiotherapy and chemotherapy, and damage induced by environmental toxicants such as air pollution.

DNA damage in the sperm can be induced by six mechanisms: (1) apoptosis during spermatogenesis; (2) DNA strand breaks during sperm chromatin remodeling; (3) posttesticular DNA fragmentation caused by oxygen radicals; (4) DNA fragmentation induced by endogenous caspases; (5) radiotherapy and chemotherapy; and (6) DNA damage induced by environmental toxicants, such as air pollution.

Several tests are available for sperm DNA fragmentation; following are some of the common assays.

Comet assay

Also known as single-cell gel electrophoresis, in this assay, spermatozoa embedded in the agarose gel are lyzed with detergent and the fragmented DNA moves out of the head toward the tail, while intact DNA remains in the head.

Spermatozoa are dispersed individually and suspended in low-melting agarose at 37°C. This mixture is placed on a microscopic slide and covered with a glass coverslip. These slides are placed at 4°C to undergo solidification process followed by lysis of spermatozoa with buffer containing Triton X-100 detergent and proteinase K. Electrophoresis of the microslides in neutral buffer for 20 min at 25 V separates out the fragmented DNA from intact DNA toward the anode pole [22]. This assay can detect multiple types of DNA fragmentation only in fresh semen samples and requires only 5000 spermatozoa.

TUNEL assay

The TUNEL assay is based on the identification of DNA breaks by the addition of template-independent DNA polymerase called terminal deoxynucleotidyl transferase (TdT) to the 30 hydroxyl (OH) breaks-ends of ssDNA and dsDNA. TdT is an enzyme that catalyzes attachment of deoxynucleotides, tagged with a fluorochrome or another marker, to 3′-hydroxyl termini of DNA strand breaks. Fluorescein isothiocyanate is conjugated with 20-deoxyuridine 50-triphosphates, and the fluorescent signal measured by the flow cytometer is directly proportional to the DNA fragmentation in the analyzed spermatozoa. The counterstain propidium iodide, a red-fluorescent dye, is specifically used for nucleic acid staining.

Acridine orange assay

The concept of this assay is that exposing the sperm to acid denaturation will allow the intact DNA to bind to acridine orange (AO) as will be displayed as green and the damaged DNA will be in red. The analysis can be done using microscopy or flow cytometer.

In microscopic examination, the air-dried semen sample smears are fixed in Carnoy's fixative for two hours, then stained with AO for 5 min. In the case of flow cytometry analysis, 1×106 spermatozoa are fixed in 70% ethanol for 30 min and permeabilized using 0.1% Triton X-100 for 30 s. Then, spermatozoa stained with AO excited at 488-nm wavelength and the green fluorescence from the dsDNA (double-stranded) and red fluorescence from ssDNA (single-stranded) are measured [23].

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CHAPTER

26

How to set up an in vitro fertilization laboratory

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As discussed in the previous chapter, an andrology laboratory is an important and essential entity within any assisted reproduction technology (ART) clinic. Maybe with the exception of very small in vitro fertilization (IVF) clinics (less than 300 cycles per year), it is advised to build a dedicated laboratory for semen analysis, preparation for insemination or IVF/intracytoplasmic sperm injection (ICSI), and cryopreservation. In this chapter, I will share my experiences in designing ART clinics and embryology laboratories of different capacity in different parts of the world. In this context, an ART clinic is seen as a full-time clinic with integrated clinic and laboratories offering all modern ART techniques.

There are many ways to design a good laboratory, and different concepts can work although success mainly comes with experience of staff that will carry out the procedures and not the infrastructure as such although a prerequisite requirement to obtain optimal results is a clean, ergonomic, and functional laboratory infrastructure and environment.

Designing and installing a new ART laboratory is not something one does every day, and the challenges one faces are many. For most embryologists, this project will be new and far from the daily routine, so it seems like a seemingly impossible task. At the start of the project, expectations (by clinicians and management) are often set very high and maybe not completely realistic due to underestimation or misunderstanding of the problems facing a modern IVF laboratory. The purpose of this chapter is to help those faced with this task with practical tips and advice and arguments in making necessary choices and decisions.

IVF is a multidisciplinary activity, and the design and setup must not only take into account the requirements and wishes of the clinicians, embryologists, and nursing staff, but it is off course also crucial that regulations and the legal framework and licensing criteria are respected, all within the available but always limited budget. Think before you start definitely applies here; therefore, it is wise to put together a planning team with all concerned parties and stakeholders already in the very early phase of the project.

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The planning team

This planning team should include the hospital management (responsible for strategic choices and budget), medical doctors (responsible for medical planning), embryologists, nursing, engineers (including a design-build engineer specialized in heating, ventilation and air conditioning [HVAC] design), and architects. It is surprising how far apart the wishes and opinions of the different parties involved can be and this can lead to heated discussions, and how little understanding is shown for other opinions when each party wants to defend his/her priorities. Here a consultant can play an important role in looking at positions in a broader context and acting as a referee. The aim of the planning team is to critically consider all wishes and requirements, to weigh all the pros and cons and make realistic choices.

First of all, a need assessment should be done (what do we want to achieve, what are the goals), then review whether it is possible to realize these in existing facilities. Often existing facilities will be rather limited in space and maybe not really adapted to current requirements (increase air quality, construction, safety), so it is wise to assess if a new facility maybe a better option to accommodate the needs. The output of this planning discussions should be a set of possible scenarios with a list of alternatives to choose from and with pros and cons for each alternative and with detailed budget calculations for each alterative.

When considering and preparing the future planning, one should try to anticipate future changes or amendments in legislation and should take into account the future increase in the number of treatments that one wants to do (with the associated potential increase in the number of staff) as well as possible changes in new or more complex methods and equipment. Many embryology laboratories get too small over time; therefore, for the responsible embryologist in this planning team, it will be a struggle to fight for extra laboratory space.

Regulatory requirements and professional guidelines

In many countries, some sort of license or approval is required to run an IVF laboratory. It is beyond the scope of this chapter to go in detail on current standards, guidelines, or national legislations, but an excellent overview is given by Sjöblom [1]. So, no matter what the final scenario is, it should comply with regulations and/or legislation. An excellent reference document is the *Guide to the Quality and Safety of Tissues and Cells for Human Application* (fourth edition published in September 2019) [2]. This document from the European Directorate for the Quality of Medicines consolidates existing guidance and minimum standards and contains good practice guidelines on the quality management system, personnel and organization, premises, equipment and materials, and qualification and validation and is a basis for inspectors. For premises, a GMP grade D air quality is specified as the minimum background environment for tissue handling. This implicated that IVF laboratory facilities should be constructed according to cleanroom regulations to control airborne contaminants. The current consensus on IVF air quality [3], however, is even more stringent, and background particulate level of ISO Class 7/GMP Grade B "in operation"/Grade C "at rest" is accepted as the target. To obtain this, many aspects of cleanroom design should be used in its construction.

The ISO 14644-4 standard [4] specifies requirements for the design and construction of cleanroom and clean air devices, as well as requirements for start-up and qualification.

The ART laboratory, however, has different requirements than cleanrooms for integrated circuit board or pharmaceutical manufacturing. Supplementary measures should be taken to respect optimal conditions for temperature, humidity, and volatile organic compound (VOC) levels as specified in the Cairo Consensus Guidelines on IVF culture conditions [5].

Cleanroom design and space requirements

When considering the future location, one should realize that the HVAC installation for a cleanroom is a complex and voluminous machinery and a suitable location must be available nearby. Often, this installation will be placed outside (Fig. 26.1).

Another important factor is the vibration sensitivity of the building. Vibrations during micromanipulation will have a negative impact on operations. The higher the building, the greater the risk of vibrations occurring. Ground floor or underground laboratories are least sensitive, so these are preferred locations. If this is not possible and the lab is built in a vibration-sensitive zone, the use of antivibration tables is recommended.

The layout and planning of an IVF laboratory according to cleanroom principles is one of the hard decisions to be taken by the planning team in the very beginning. A cleanroom is an airtight closed box and once build, it is very difficult, if not impossible to make changes.

In addition to the correctly designed air handling unit (high efficiency particulate air [HEPA] filters, positive pressure, air changes per hours, air velocity and air speed, and VOC filtration according to the Cairo consensus), the layout and required space need to be carefully considered. In addition to the anticipated capacity and the type of treatment which will be offered (sperm/oocyte banking, genetic diagnosis, treatment of viral-positive patients, oncofertility program, etc.), variables to be taken into account to calculate the required space are the number

FIGURE 26.1 HVAC installation.

of staff, number and type of workstations, and type and number of incubators. For the treatment of viral-positive patients, the biological fluids and cells should be kept separate from other patients, and the construction of a separate small laboratory with negative pressure might be indicated. It is wise to decide how the access between the embryology laboratory and the operating room (OR) will be organized. According to cleanroom principles, access of staff is through an airlock with interlocking doors and circulation of people should be limited. This principle limits the possibility to carry tubes with follicular fluid directly to the laboratory during egg retrieval. A pass-through (with two interlocking doors to maintain positive pressure inside the laboratory) is a better option. The same problem exists at embryo replacement. It needs to be decided whether the embryologist will bring the loaded catheter to the transfer room via a double door air-lock or if it will be handed through the pass-through.

The recommended staffing levels for a clinical embryology laboratory are one full-time equivalent "bench" or "hands-on" embryologist per 120 stimulation cycles per year [6], so enough workstations should be available for this staffing level. Furthermore, IVF is an activity with strong fluctuations in numbers over time, so a certain percentage of overcapacity should be considered to cope with peaks in the number of cycles and to avoid bottlenecks during the laboratory phase. Dreaming is allowed, but one should also be realistic in his wishes; building is expensive! Taking all costs into account (construction, electricity, HVAC, and medical gases), building a laboratory space per square meter is about the same as the cost of an OR. That is why it's so important to discuss all this in advance in the planning team. To avoid future bottlenecks during operation, it might be wise to construct multifunctional facilities, that is, a room dedicated to embryo replacements can be designed without much extra cost in such a way that it is also suitable for performing egg retrieval if necessary.

The embryologist and technical specialists on cleanroom design must ensure that a balance is struck between the technical specifications and the specific problems of an embryology laboratory (strong air flows, cooling, and risk of contamination). Correct placement of HEPA filter outlets in the cleanroom so that gametes or embryos are not exposed to strong air currents is crucial for future functioning.

Once the cleanroom is built and the HVAC is working, compliance with the designated classification has to be demonstrated in the formal process of qualification (or validation). The process of qualification demonstrates that the environment and process meet its previously defined requirements. The HVAC system is a complex machinery and needs strict maintenance, filter replacement, and monitoring. Monitoring includes particle testing and microbiological monitoring (active and passive sampling of viable particles) and monitoring with contact agar dishes for the efficiency of daily cleaning and disinfection. These qualification and monitoring are discussed in detail by Guns and colleagues [7].

The total space requirement to house a modern and an up-to-standard ART clinic is around 1000 m^2 per 1000 cycles. Approximately 30% of this total space is required for operation rooms and laboratory cleanrooms (nonpublished own data).

Cryostorage facility

Do not forget the cryostorage facility. Cryostorage involves multiple risks [8], and the tendency is that the amount of embryos or gametes stored only increases over time. This has to be taken into account when calculating the space and technical requirements of the cryospace, storage tanks, cryostorage tanks, filling, monitoring, and alarming system. The storage room should be a dedicated, locked room ideally not too far from the laboratory cleanroom and should have floors in liquid nitrogen-resistant stainless steel or tiles, and it should be easily accessed for liquid nitrogen supply. As an alternative, a centralized liquid nitrogen distribution via a vacuum-isolated pressured line (VIP) can be considered. Installation of such a VIP is expensive, and the feasibility depends from the distance of the crystorage room from a liquid nitrogen tank, and this is an important factor to be taken into account when deciding on the location of the crystorage room. It is an investment, but in the long run, a VIP line might be more cost efficient and certainly logistically a better and certainly a safer option than labor-intensive manual nitrogen supply with transport vessels. Nitrogen supply via VIP allows the use of modern large capacity storage tanks with automatic filling. Next to the less risk for staff (no handling of liquid nitrogen), these large tanks not only have better temperature and level control but also safer in use than the classical 40 or 60 L dewars and also require relatively less floor space than a large number smaller dewars. For safety reasons, the cryostorage room should be equipped with oxygen deficiency monitoring, low oxygen alarm, and forced ventilation to evacuate nitrogen fumes.

Workstations

Sufficient workstations and incubators must be available to absorb fluctuations in workload.

One of the most important choices in terms of space requirements and layout is the type and number of workstations. An important aspect of the lab process is the avoidance of gamete mix-ups, and for this purpose, a separation in time and space should be respected at all times when working with gametes/embryos from different patients. Semen preparation in particular is a risky activity because semen samples are often collected and prepared in a concentrated manner in the morning. Based on own experience in lab design and taking into account the duration of acts (i.e., 30 min for semen density gradient centrifugation), Table 26.1 can give an indication of the appropriate number of work stations needed.

A sperm preparation workstation should include a class II grade A laminar air flow (LAF) and dedicated microscope, centrifuge, set of pipettes, and preferentially identification hard/

TABLE 26.1 An estimate of the number of work stations needed as a function of the work load.

Number of cycles	Semen	Oocyte collection	ICSI biopsy	Dish preparation	Vitrification Thawing
$\leq$1000	2	2	2	1	2
$\leq$2500	3	2	3	1	3
$<$5000	6	2	4	1	4
$>$5000	8	3	5	1	5

software. Class II laminar flow hoods provide an aseptic environment in addition to the protection of the operator from potential infectious agents.

An ICSI workstation includes a stereomicroscope and heated stage, an inverted microscope, micromanipulators and injectors, and digital imaging. The ICSI set can be installed on an antivibration table. In most countries, regulation does not require the use of a grade A laminar flow cabinet during ICSI; however, doing ICSI in an LAF is possible, thus reducing contamination risk. Once the number and exact type of workstations are known, their exact position should be drawn on the plans in order to determine the necessary outlet points for medical gasses, electricity plugs, and network connections. Egg collection workstations are best placed as close as possible to the puncture room(s), so that tubes with follicle fluid (via a transfer hatch) can be handed over directly for egg identification (Fig. 26.2).

Direct communication between OR and laboratory is difficult in this setup; therefore, a hands-free interphone system or other means of direct communication such as headsets should be available.

The other workstations should be positioned with respect of a logical workflow, so that no unnecessary distances have to be traveled from the workstations to and from incubators. Future positioning and correct placement of the workstations is also important when designing the HVAC. Care should be taken that the HEPA filters are not placed immediately above a workstation to avoid strong air currents. Also, LAF hoods and benches should respect distances from walls and enough circulation space should be left for routine

FIGURE 26.2 Two door passage to enable transfer of the oocytes from the operating room to the laboratory while minimizing contamination.

operations. A free space of 100 cm is recommended in front of a LAF, 30 cm distance should be kept between the side of a LAF, and the wall and a distance of 150 cm should be left between the front of a LAF and a door or a bench.

Equipment, furniture, and storage

The planning team should prepare a list of equipment and the intended location in order to plan positions of furniture and necessary power outlets.

The choice and number of incubators are important factors in determining the need for space and laboratory layout. Classic "big box" incubators have an enormous footprint in relation to the number of patients cultured and are now more and more replaced by more compact desktop incubators and/or time-lapse incubators. The advantage of desktop incubators is that they can be stacked or installed in cupboards, so that multiple incubators can be placed on a rather small surface. Recent (time-lapse) incubators have a larger capacity so requiring less space. In order to determine the correct placement and the position and number of required fittings (medical gasses, power plugs, and network access points), it is therefore advisable to make this decision regarding the future incubator technology in an early phase.

All these laboratory equipment and monitoring require a huge number of power plugs and network access points. Critical equipment (incubators, refrigerators, etc.) should be connected to uninterrupted power supply with a power generator and sufficient battery capacity. A safe estimation for the number of power plugs is what you count as needed and multiply this by 3. Power plugs are usually placed about 1m20 (47 inches) from ground level, but in a clean room, it is advised that power cords can be kept away from the floor. It is therefore better and more practical to place power plug and access points for medical gases close to the ceiling. This position has an extra advantage of visibility (Fig. 26.3).

FIGURE 26.3 Note the raised power plugs and access to medical gases.

Be aware that some equipment such as LAF workstations are quite large, so to allow easy entrance to the laboratory, it is best to have outer doors of 1m20.

Storage and supply of disposables

Art laboratories consume huge amounts of disposables, so it is crucial to have enough storage space. Storing these disposables in the laboratory is not really compatible with proper cleanroom practice and should be avoided so ample external space is needed. Also plastics can release monomers and other VOCs over a long period, so a separate storage room with negative pressure allows outgassing without contamination laboratory air. Daily supply of disposables to the laboratory is best organized with mobile shelves and boxes which can be introduced through a hatch with airlock. These mobile shelves can be best kept in stainless steel furniture; stainless steel does not release VOCs (as many other furniture made of compressed wood do) and is easy to clean and decontaminate as is required in cleanrooms.

Medical gasses

In smaller laboratories, supply of medical gasses (CO_2, N_2, and premixed gas) with gas cylinders is mostly used. These cylinders should be placed in a dedicated, ventilated space, accessible from outside. Supply to the laboratory is then through suitable tubing or cupper or stainless steel lining. Low oxygen incubators run on N_2 and CO_2, so there should be one connector for each incubator. For future renovation, expansion, or reorganization, it is wise to have a number of spare connection points.

As an alternative for gas cylinders, external bulk tanks with liquid CO_2 and N_2 can be considered.

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CHAPTER

27

Introduction to the in vitro fertilization laboratory: management of sperm, oocytes, and embryos

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A holistic perspective of the IVF laboratory

It is the main role of the in vitro fertilization (IVF) laboratory to maintain the inherent viability of the oocyte and sperm and the resultant embryo (one cannot make a viable embryo from compromised gametes). In vitro, gametes and embryos are susceptible to stress inherent when working outside of the human body. These stresses include changes in temperature and pH, exposure to atmospheric oxygen, media lacking amino acids, and physical stress during manipulation [1]. Fig. 27.1 documents some of the sources of stress which should be minimized, or if possible avoided, to facilitate the effective functioning of the laboratory. Furthermore, many active variables within the IVF laboratory impact the overall outcome of an IVF cycle, and to make matters more complicated, many of these variables interact with each other, with the presence of one stress factor rendering the embryo more susceptible to a second stressor [1,2]. In Fig. 27.2, the relationship between several key variables is highlighted. We have learned over the years that it is essential to take a holistic perspective to laboratory and patient treatment if we are to optimize processes and outcomes. In other words, there is no one panacea to optimizing outcomes, rather it is essential to ensure that each and every aspect of the laboratory and treatment cycle is considered [3].

Equipment

To adequately maintain optimal temperature and pH, it is essential to ensure that not only are sufficient numbers of incubators available, all capable of regulating both carbon dioxide

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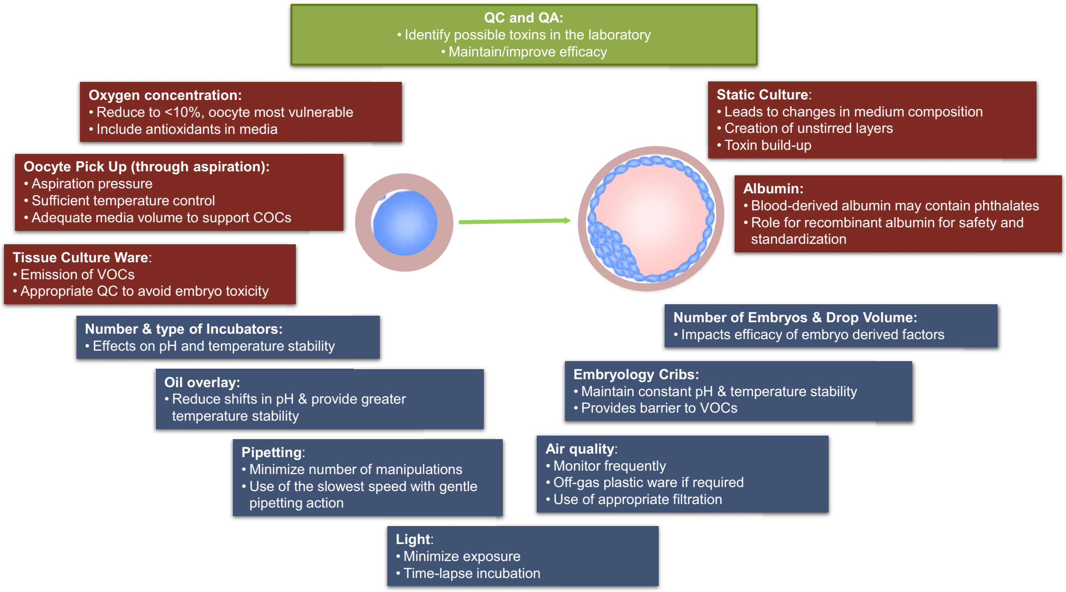

FIGURE 27.1 **An overview of key chemical and physical factors that affect mammalian embryo development in vitro; red represents chemical factors and blue represents physical factors.** QC, quality control; QA, quality assurance; VOCs, volatile organic compounds.

and oxygen concentrations, but also that during gamete and embryo manipulation outside of the incubator, precautions are taken to maintain temperature and pH. The latter can be accomplished through the use of gassed embryology workstations such as those originally modified from pediatric isolettes. If such equipment is not available, then the use of certified heated stages combined with an appropriately buffered medium utilizing either MOPS or HEPES can facilitate the maintenance of medium pH. It is essential that whenever possible, the oxygen concentration is lowered from atmospheric to 5% and that equilibration of medium and subsequent embryo cultures are performed at 5% oxygen for all developmental stages [4,5]. Several types of incubators have been used in human IVF—from the original "big box" tissue culture chamber incubators to IVF-specific designed benchtop incubators, and more recently the introduction of time-lapse incubation systems. Whereas the larger chamber style incubators are useful for media equilibration, they are not considered optimal for embryo culture, due to the associated changes in temperature and pH as a result of frequent incubator door openings. Benchtop incubators or time-lapse incubation systems have the advantage of continual gassing and direct heat transfer to the culture dish. Hence, such systems are now the equipment of choice in a 21st century IVF laboratory. Importantly, as well as reducing handling stress, time-lapse incubator systems provide detailed morphokinetic data throughout embryo development without the need to remove embryos from their incubation environment [6].

Quality management

Temperature, osmolality, and pH are the three fundamental external conditions that influence the gametes and embryos homeostasis (the self-regulating process by which biological

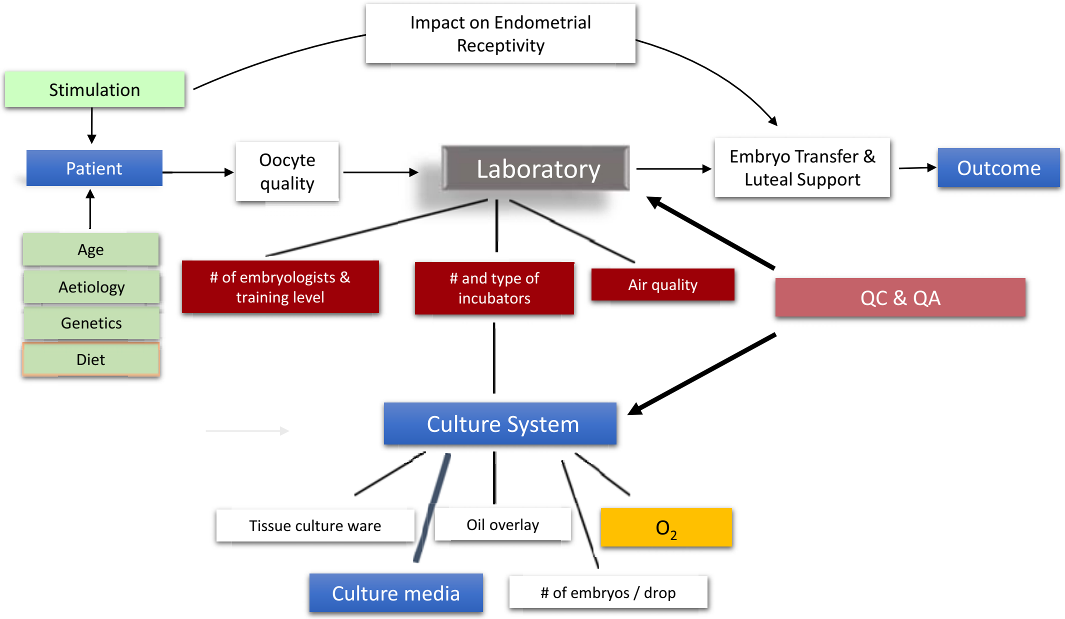

FIGURE 27.2 **A holistic analysis of human IVF.** This figure serves to illustrate the complex and interdependent nature of human IVF treatment. For example, the stimulation regimen used not only impacts oocyte quality (hence embryo physiology and viability [76]) but also affects subsequent endometrial receptivity [54,77,78]. Furthermore, the health and dietary status of the patient can have a profound effect on the subsequent developmental capacity of the oocyte and embryo [79]. The dietary status of patients attending IVF is typically not considered as a compounding variable, but growing data would indicate otherwise. In this schematic, the laboratory has been broken down into its core components, only one of which is the culture system. The culture system has in turn been broken down into its components, only one of which is the culture media. Therefore, it would appear rather simplistic to assume that by changing only one part of the culture system (i.e., culture media), that one is going to mimic the results of a given laboratory or clinic. A major determinant of the success of a laboratory and culture system is the level of quality control and quality assurance in place. For example, one should never assume that anything coming into the laboratory that has not been pretested with a relevant bioassay (e.g., mouse embryo assay) is safe merely because a previous lot has performed satisfactorily. Not all of the contact supplies and tissue culture ware used in IVF comes suitably tested. Therefore, it is essential to assume that everything entering the IVF laboratory without a suitable pretest is embryotoxic until proven otherwise. In our program, the 1-cell mouse embryo assay (MEA) is employed to prescreen every lot of tissue culture ware that enters the program, that is, plastics that are approved for tissue culture. Around 25% of all such material fails the 1-cell MEA (in a simple medium lacking protein after the first 24 h) [50]. Therefore, if one does not perform QC to this level, one in four of all contact supplies used clinically could compromise embryo development. In reality, many programs cannot allocate the resources required for this level of QC, and when embryo quality is compromised in the laboratory, it is the media that are held responsible, when in fact the tissue culture ware is more often the culprit. *Modified from Gardner D.K., Lane M., Towards a single embryo transfer. Reprod Biomed Online 2003;6:470-481. with permission from Reproductive Healthcare Ltd.*

systems maintain stability while adjusting to changing external conditions) and are maintained through rigorous quality control in the IVF laboratory. All incubators, embryology workstations, and heated stages need to be temperature calibrated with the temperature probe at the site where the oocyte or embryos reside within the medium in the dish used. This ensures the temperature is correct to maintain the optimal environment for the oocyte or embryo. As many of the dishes used for IVF and culture have lips at the bottom which creates an air pocket, if the temperature is calibrated simply to the stage, then the actual temperature where the oocyte or embryo resides will be 1–2° cooler. This is particularly important given that poor quality management has been associated with an increase in the rates of aneuploidy [7].

Mammalian embryos can develop over a wide range of osmolalities, but human embryo culture media typically range from 265 to 295 mOsmol [8]. Of practical significance is that culture dish preparation can impact media osmolality [9]. It is therefore imperative that when placing droplets of media in a dish, they should be immediately covered with oil before going on to prepare a second dish and so on. The pH of a culture medium is related to the concentrations of bicarbonate in the medium and the CO_2 percentage in the laboratory incubator, which in turn depends on atmospheric pressure. A CO_2 of 6% will provide a working pH of ~7.35–7.40 (depending on the composition of the medium being used). Hence, it is important to measure the pH of all media used to ensure their performance. In recent years, the replacement of thermocouple sensors with infrared sensing technology for CO_2 has removed the importance of humidity in the incubation chamber to accurately determine CO_2 concentration. Concerns have been raised about increases in media osmolarity if dishes are incubated in "dry" chambers, if culture dish preparation is performed correctly, and then five-days of dry incubation translates to an increase of ~5mOsmol provided an oil overlay is used [10].

Gamete retrieval and preparation—oocytes

Oocyte collection protocols vary widely between clinics; however, they are based around a standard transvaginal oocyte aspiration procedure, in which the oocytes are collected between 34 and 38h after the trigger of either hCG or GnRH agonist. The collection procedure for oocytes is important to maximize oocyte yield, while maintaining oocyte integrity and minimizing stress on the oocytes (Fig. 27.3). The first aspect of this is to apply the appropriate aspiration pressure, <120 mmHg, as too high a pressure can induce irreversible trauma to the oocyte. The oocyte collection medium must contain amino acids to minimize stress on the cumulus–oocyte complex (COC) and the competence of oocytes. If flushing of the follicles is to be performed, then media used to flush the follicles must be buffered and also contain amino acids. It is paramount to oocyte integrity that temperature is maintained from the moment the COC is removed from the follicle; hence, it is highly recommended to utilize an embryology workstation when searching follicular fluid and to house the collected COCs before being returned to an incubator.

A vital function of the cumulus cells is to control the nutrient balance of the developing oocyte [11]. Glucose is the principal substrate of cumulus cells which is metabolized to lactate and pyruvate, the predominant energy source of the oocyte [12]. Hence, once removed from the follicle, the COCs should be incubated in media at sufficient volume to support the cumulus cells as well as the oocyte; it is recommended that 0.5 mL of media per COC be used, depending on the glucose level in the medium. The provision of adequate preequilibrated media for the collection procedure is facilitated by providing the IVF laboratory with an accurate indication of the expected number of COCs, derived from the follicle number seen on the ultrasound prior to the collection procedure. The next step, after returning the COCs to the laboratory properly, is to allow them to rest for a few hours. This time is extended if standard/conventional insemination is the method of fertilization. Again, the

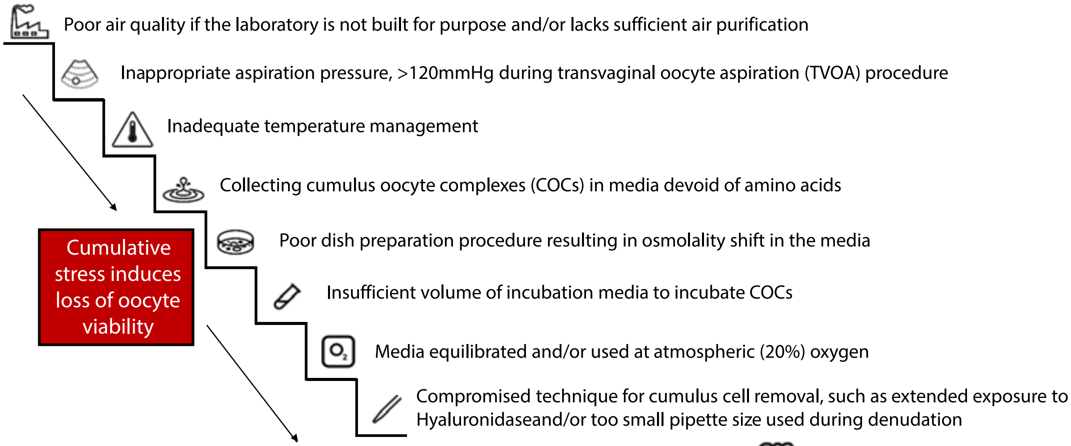

FIGURE 27.3 Cumulative stressors during oocyte retrieval and preparation. Contemporary ART laboratories need to be purposely designed to reduce the risk of introducing external contaminants into the culture area. Control of airborne contaminants is achieved using HEPA filtration, a dedicated air handling unit (AHU) or a dedicated supply duct, chemical filtration (either in the AHU or freestanding), and positive pressure [80]. The same design features need to also be applied to the oocyte retrieval room. Inappropriate aspiration pressure, >120 mmHg, during transvaginal oocyte aspiration procedure. Too high a pressure can induce irreversible trauma to the oocyte. Paramount to oocyte integrity that temperature is maintained from the moment it is removed from the follicle, hence it is highly recommended to utilize a gassed embryology workstation when searching follicular fluid and to house the collected COCs before being returned to an incubator. Simple flush medium is problematic as it is not suited to support metabolic requirements of cumulus oocytes complexes, and the utilization of a specially formulated, amino acid-enriched flushing media is essential. Poor preparation technique: for example, after placing the droplets in the dish not immediately covering them with oil before commencing second dish will increase osmolality way above the level that oocytes and embryos can tolerate for development and survival. Glucose is the principal substrate of cumulus cells which is metabolized to lactate and pyruvate, the predominant energy source of the oocyte [12]. COC uptake of glucose has an upper of ~50 nmol/h. This value, combined with glucose level in the medium, the volume of media, and length of culture can be used to determine the maximum number of COCs which can be placed in a culture vessel. This is why it is essential to provide the IVF laboratory with a precise indication of the expected number of COCs, derived from the follicle number seen on the ultrasound prior to the collection procedure. Even a transient exposure to atmospheric oxygen can have a negative impact on oocyte competence. Denuding—enzyme concentration and exposure time must be kept to a minimum, as both factors have been linked with pathogenetic activation of the oocyte. The pipette diameter should never be smaller than the average diameter of the human oocyte. The diameter of human oocytes, excluding the zona pellucida (ZP), usually is 110–120 μm, while the thickness of the ZP is 15–20 μm [81].

metabolic requirements of the cumulus cells and the length of culture need to be considered to ensure appropriate amounts of culture media are provided to support to coculture of the COCs and sperm as is the case with standard/conventional insemination.

Gamete retrieval and preparation—sperm

Human spermatozoa while suffused in seminal plasma are unable to fertilize oocytes. Once removed from seminal plasma, sperm undergoes maturational changes during which they acquire the ability to fertilize oocytes, capacitation. In addition, prolonged exposure of sperm to seminal plasma inhibits the ability of spermatozoa to undergo the acrosome reaction in vitro resulting in reduced fertilization capacity. The inhibitory effects of seminal plasma and the deleterious effects of other factors in the ejaculate on sperm function mean it is important that spermatozoa are removed from the seminal plasma as quickly as possible after ejaculation and liquefaction occurs. Table 27.1 outlines the four types of techniques currently employed to remove spermatozoa from seminal plasma. As with all techniques involving IVF media, it is important to choose a culture medium that is buffered appropriately for the atmosphere in which the technique takes place. If the incubator is atmospheric, then the medium should be buffered with HEPES or MOPS (note: atmospheric incubators are undesirable as the oxygen concentration will be nonpsychological).

Standard/conventional insemination

Standard/conventional, usually, insemination is performed 4−5 h postoocyte retrieval (40−41 h after ovulation trigger). As highlighted, the laboratory staff need to consider the length of culture time for COCs to provide an optimal environment for standard/conventional insemination. Insemination is performed by adding prepared sperm to wells or droplets under oil containing COCs, with the aim of a final concentration between 100,000 and 250,000/mL. Higher concentrations may be used if previous IVF produced low fertilization. Conversely, lower concentrations can be employed if previous cycles yielded a high rate of triploidly. Fertilization is conventionally checked $\sim 17 + 1$ h postinsemination (hpi) [13]; however, a more recent analysis using time-lapse evaluation has indicated a slightly earlier time of $16.6 + 0.5$ hpi might be more reliable [14].

Intracytoplasmic sperm injection

When patients present with certain male-factor conditions, fertilization rates using conventional insemination can be much lower. These conditions include lower concentration (oligozoospermia), lower motility (asthenozoospermia), or lower rate of normal morphology (teratozoospermia), or any combination of these factors. For such patients, intracytoplasmic sperm injection (ICSI) has become the standard of care [15]. Further, when there is high DNA fragmentation in sperm, ICSI is typically utilized.

TABLE 27.1 Procedures for sperm preparation.

Name	Technique	Advantage	Disadvantage
Sperm washing	Dilution of semen with culture media (typically a 1:5 ratio), separation of spermatozoa, from seminal plasma by centrifugation to form a pellet. The centrifugation is usually performed at 300–500 g.	Simplest and least expensive	Nonviable and immotile spermatozoa together with any contaminant in the original semen (such as leukocytes, epithelial cells, or noncellular debris) are present in the washed sample.
Swim-up from washed pellet	This procedure sperm washing as outlined above, then the pellet of spermatozoa can be either left intact or resuspended in a very small volume of media to create a noncompacted pellet. Taking extreme care to ensure no mixing occurs, overlay the sperm pellet (or noncompacted pellet) with culture medium and then incubated for 45–60 min in physiological conditions (37°C in an appropriately buffered medium) to allow the spermatozoa to swim-up from the pellet. Placing the test tube on a 45°C angle will increase the surface area ratio between the culture medium and sperm pellet and increase the yield of motile spermatozoa.	Recovery of a relatively high percentage of motile sperm in the absence of other cells and debris. This technique has also been demonstrated to yield suspensions of spermatozoa with increase swimming velocity and more normal sperm morphology [82].	The efficiency of the technique is related to the initial sperm motility in the ejaculate. Good handling skills are required to achieve consistent high-quality overlaying with no mixing—if mixing occurs, the final supernatant will be contaminated with immotile sperm, debris, and nongerm cells. Increased ROS levels as a result of centrifugation on unselected populations of spermatozoa [83].
Direct swim-up from semen	Aliquots of liquefied semen are layered *underneath* culture medium in a series of test tubes. Again, as with the swim-up from washed pellet, placing the test tubes on 45°C angle increases surface area ratio between the semen layer and culture media. The number of test tubes used is dependent on the initial ejaculate volume, sperm concentration, and sperm motility. The test tubes are incubated for 45–60 min in physiological	The greatest advantage is the elimination of centrifugation. Recovery of a relatively high percentage of motile sperm in the absence of other cells and debris.	Low percentage recovery of total progressively motile spermatozoa.

(*Continued*)

TABLE 27.1 Procedures for sperm preparation.—cont'd

Name	Technique	Advantage	Disadvantage
	conditions (37°C in an appropriately buffered medium) to allow the spermatozoa to swim-up from the liquefied semen.		
Density gradient	Silica beads are probably the most common density gradient material used in clinical IVF laboratories. Density gradients are most often made with two later; a lower (denser, usually 90−95) and upper (less dense, usually 40−45) layer. The effect of the two layers is that due to the mass (density) of the spermatozoa, they pass through the upper layer first. Further, motile spermatozoa swim in the direction of centrifugal force and therefore penetrate the gradient boundaries faster than immotile spermatozoa, and as a result, the pellet at the bottom of the centrifuged density gradient contains a high percentage of motile spermatozoa. The pellet is then washed with culture medium to remove any residual density gradient.	Elimination of leukocytes and reported reduction of ROS. The technique has also been reported to yield improved DNA quality sperm population [84,85].	Cost and additional centrifugation steps. Increased ROS levels as a result of centrifugation on unselected populations of spermatozoa [83].

Following retrieval, oocytes are typically cultured for several hours prior to injection. Around two hours postretrieval, oocytes are denuded from their surrounding cumulus and coronal cells, and wherever possible, oocytes should be rested for one hour after this procedure. This visualization of the oocyte allows both assessments of the maturity and handling of the oocytes during micromanipulation. Initially, the COCs are placed in media containing hyaluronidase which aids digestion of the hyaluronic acid (HA), the key matrix component which keeps the cumulus cells together. The enzyme concentration and exposure time have to be kept to a minimum, as both of these factors have been linked with pathogenetic activation of the oocyte [16]. COCs are then transferred into fresh culture media (buffered appropriately for the atmosphere in which the technique is taking place), and the remaining coronal radiata cells are removed by repeated pipetting of the cumulus-oocyte through a fine bore pipette (only slightly larger than the diameter of the oocyte). Oocytes should be washed serially through additional drops of media and placed in culture media under oil to maintain as close to physiological conditions as possible. The denuded oocytes are then examined under the inverted microscope (×400 magnification) to assess their maturation stage. Three unique maturation stages are found with the ratio of the oocyte in these three stages, a direct reflection of the accomplishment of the stimulation protocol is prescribed. The majority of oocytes (75%–90%) should be in metaphase-II (MII) as revealed by the presence of the first polar body [17], and the remaining oocytes will be either metaphase-I (MI) having undergone breakdown of the germinal vesicle (GV) but yet to have extruded a polar body or in prophase identifiable by the presence of a GV.

ICSI is usually performed 3–4 h after oocyte retrieval (39–40 h after ovulation trigger). The ICSI procedure is carried out on the heated stage (or chamber) of an inverted microscope at 400× magnification. The microscope is typically equipped with two coarse positioning manipulators and with two three-dimensional micromanipulators (or similar tools). ICSI should be performed only on those oocytes that have reached MII and are therefore haploid and able to be fertilized normally. MI oocytes may be cultured in vitro and frequently achieve MII after a few hours; however, these oocytes may not exhibit synchrony between nuclear and cytoplasmic maturation. GV oocytes will require overnight culture to reach the MII stage; however, the use of in vitro matured GV oocytes yields poor results with only a few reported pregnancies and live births [18].

The prepared sperm is used at a manageable concentration (∼0.1 million/mL) with or without PVP; the viscosity of PVP decreases the motility of sperm and thereby aids with their capture and manipulation. HA is a physiological alternative offering the same properties as PVP. HA is the major component of the matrix in between the cumulus cells of the mature oocyte and is involved in the mechanism of sperm election via specific receptors on the sperm head. It has been demonstrated that spermatozoa with HA-binding characteristics exhibit good nuclear morphology and decreased DNA fragmentation which can lead to improvements in both embryo quality and development. A single spermatozoon is selected from the sperm droplet of the ICSI dish and is rendered immobile by breaking the tail from pushing it with the injection pipette against the bottom of the dish before being aspirated into the tip of the injection pipette. Appropriate immobilization of sperm by squeezing the tail was initially emphasized by Palermo et al. [15]. The oocyte is held with the polar body at 6 o'clock (or 12 o'clock), and the micropipette is pushed through the zona pellucida and the oolemma into the ooplasm at 3 o'clock (or 9 o'clock if left-handed) ensuring that the plasma membrane is broken, and a single spermatozoon is injected into the ooplasm in the smallest amount of medium possible. Afterward, injected oocytes are washed in culture medium and transferred into a culture dish before being placed into the incubator.

Embryo culture

In the early days of human IVF, there were no media designed specifically for the human embryo, but rather either tissue culture media (such as Hams F10), or simple salt solutions supplemented with glucose, pyruvate, and lactate (such as those designed for the mouse embryo) were utilized. Typically, both types of media were supplemented with human serum

[8]. Significantly, in either tissue culture media or simple salt solutions, the human embryo would only readily develop through to the 4- to eight-cell stage. Consequently, the transfer of cleavage stages back to the uterus on Day 2 or Day 3 became the standard of care out of necessity (as embryos did not develop well to the blastocysts stage in such conditions). It is important to recognize that the cleavage-stage embryo resides in the fallopian tube and not the uterus, and that in all animal models, the asynchronous transfer of cleavage stages to the uterus results in compromised pregnancy outcomes [3,19,20]. Therefore, it was important to establish a system to maintain the human embryo for five to six days up to the viable blastocyst stage, thereby ensuring synchronization of the embryonic stage to the uterus [3,21].

Extensive studies on embryo metabolism [22] combined with the analysis of oviduct and uterine fluids of naturally cycling patients [12] culminated in the creation of stage-specific sequential media capable of supporting human embryo development from the pronucleate oocyte to the viable blastocyst stage [23]. As well as providing gradients of carbohydrates which mirror those found in the female reproductive tract, such media also contain amino acids, the latter of which have been shown to fulfill many key roles in supporting the oocyte and embryo [24]. As stated earlier, it is important that all media used for gamete collection and embryo culture contain amino acids [25]. The beneficial effects of amino acids on early embryo development have been proposed to come from their use not solely as energy substrates, but rather as intracellular osmolytes [26–28] and regulators of intracellular pH [29,30]. The use of amino acids as intracellular regulators is common among unicellular organisms, and the utilization of such amino acids as glycine, alanine, serine, proline, and taurine by the precompacted embryo may well stem from the simplistic organization of individual cells within the embryo and hence unique requirements to regulate their homeostasis/pH [31]. Therefore, the use of media lacking amino acids, such as phosphate-buffered saline or human tubal fluid medium [32], in particular during oocyte retrieval and the initial handling and all culture steps, should be avoided.

Albumin is the most abundant protein in the oviduct and uterus, and consequently, human serum albumin is routinely added to embryo culture media. However, the composition of the fluids of the female reproductive tract is complex, containing mucins and glycosaminoglycans, the levels of which vary during the cycle. Hyaluronan, both a polysaccharide and glycosaminoglycan, is of particular relevance to embryo culture, as it works in synergy with albumin to promote embryo development [33]. Further, the concentration of hyaluronan increases in the uterus at the time the blastocyst resides there and at implantation [34,35]. Further, the addition of hyaluronate to the culture medium has also been shown to increase the cryotolerance of embryos [36,37]. Perhaps of greatest significance, however, is the finding that the addition of hyaluronan to embryo culture medium significantly increases mouse blastocyst implantation and fetal development after the transfer [33]. These data led to the development of a specific transfer medium characterized by high levels of hyaluronan. Subsequent clinical data confirm that hyaluronan-enriched transfer medium is associated with increased implantation and take-home baby rate [38,39].

Since the development of sequential media [40,41], other culture systems have focused on a continuous medium to support blastocyst development in the IVF clinic, the latter of relevance to time-lapse culture systems [42,43]. As a result, we now have a range of culture media types to use in human IVF. Although the data are insufficient to recommend either sequential or single-step media as the best for the culture of embryos to Days 5/6 [44], using an oocyte donation model, sequential media have been shown to support fetal heart rates of 65% per transferred blastocyst in an oocyte donation program [24,45]. Whichever culture media system is employed, it should ideally be maintained at a pH of 7.3–7.4. This pH can be achieved by ensuring the correct concentration of CO_2 in the incubator (typically ~6%, depending on altitude) and the embryology workstation. The medium pH does vary with composition, so care should be taken to follow each media manufacturer's guidelines. In addition, the use of physiological oxygen (~5%) is of extreme importance, given the well-documented negative impacts of atmospheric oxygen (20%) have on embryo gene expression, the proteome, and metabolism of the embryo [1].

Given that one cannot in totally eliminate the exposure of gametes and embryos to atmospheric oxygen during the IVF process (at collection, during gamete and embryo manipulation, cryopreservation and transfer, and fertilization check if not using a time-lapse system), the role of antioxidant supplementation to media has been reevaluated [46–49]. The effectiveness, of antioxidant supplementation to media, in improving fertilization and pregnancy rates awaits further clinical trials.

Culture dishes can be prepared in the afternoon before use. Typical incubation volume for the human embryo is 50 μL for up to 4 embryos for 48 h when using drop cultures (medium should be refreshed/replaced after 48 h), or 25 μL for single embryo culture for four to five days when using timelapse microscopy. All media require an overlay of paraffin oil to prevent medium evaporation and to assist in minimizing pH drifts outside the incubator. Whatever the source of oil utilized, it is paramount that is appropriately prescreened with a one-cell mouse embryo assay [50]. When embryos are to be moved/manipulated, it is paramount that gentle pipetting is employed to avoid inducing intracellular stress responses [51]. This is facilitated through the use of pipettes whose diameters are just slightly larger than the embryo and the transfer of minimum volumes with the embryos when moved.

Blastocyst transfer has now become the standard of care, with single embryo transfers made possible by the high implantation rates associated with blastocyst transfer. There are several important advantages associated with blastocyst culture and transfer including:

- The synchronization of the embryo with the female tract [20,52], leading to increased implantation rates [53], leading to higher implantation rates, and reducing the need for multiple embryo transfers.
- The ability to assess embryo development and viability overextended culture. This can be achieved by both the identification of those embryos with little developmental potential, as manifest by slow development or degeneration in culture, and by the introduction of noninvasive tests of developmental potential to select the most viable embryos from within a cohort for transfer.
- Minimize the exposure of the embryo to a hyperstimulated uterine environment [54].
- Assessment of true embryo viability, that is, assessing the embryo postgenome activation [55].
- Reduced uterine contractions on Day 5 of embryo development, minimizing the chance that an embryo will be expelled from the uterus [56].
- Increased ability to undergo cryopreservation [57–59].

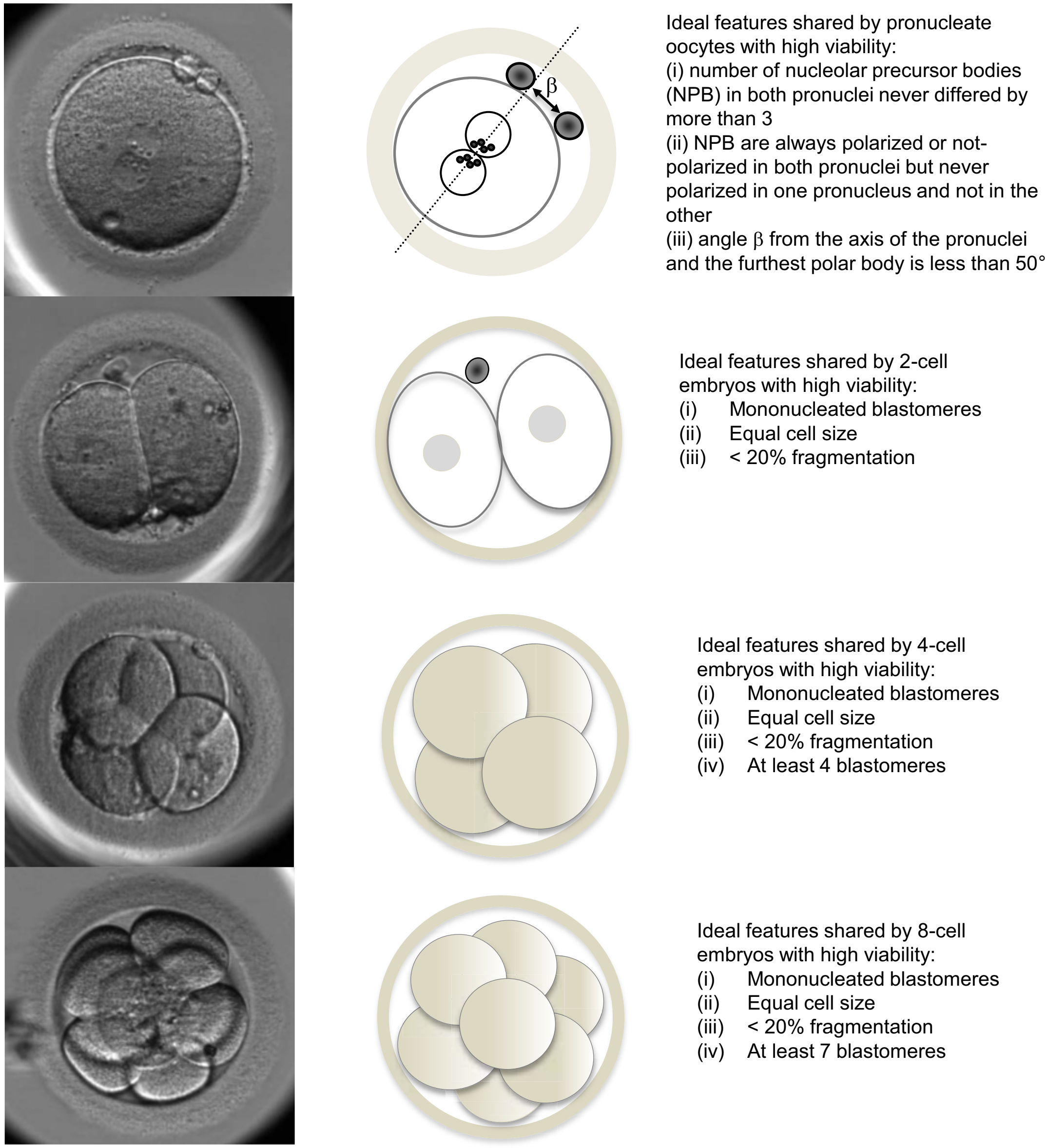

FIGURE 27.4 **Key morphological features of embryos with high viability.** *From Gardner DK, Balaban B. Assessment of human embryo development using morphological criteria in an era of time-lapse, algorithms and 'OMICS': is looking good still important? Mol Hum Reprod 2016;22:704-718 with permission.*

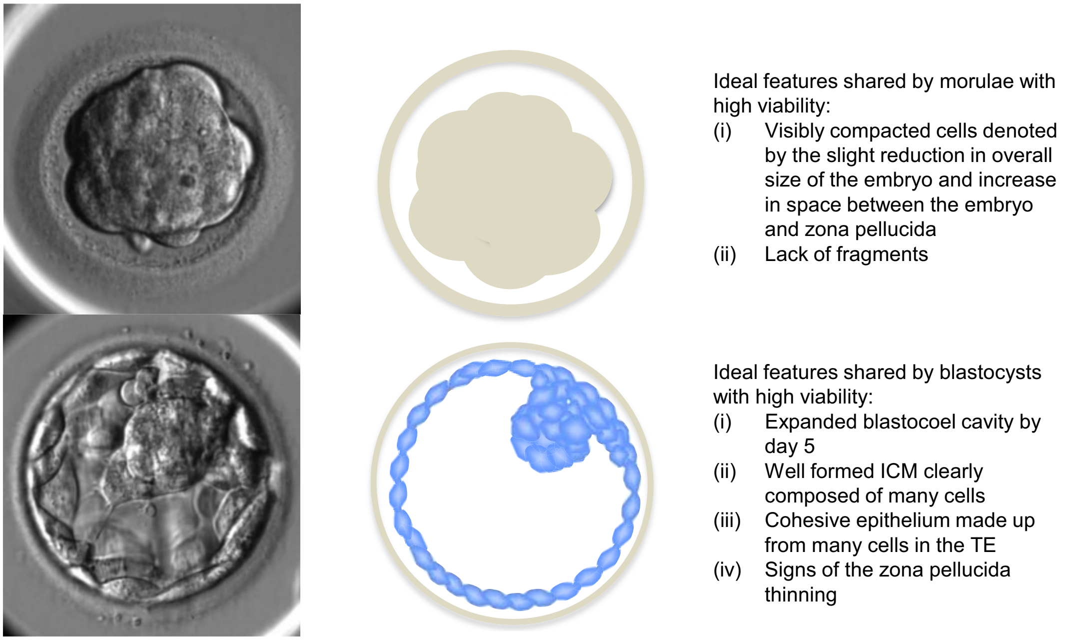

FIGURE 27.4 cont'd

- The generation of blastocysts facilitates trophectoderm biopsy for the screening of genetic diseases. Trophectoderm biopsy represents the earliest form of genetic diagnosis of nonembryonic material [60].
- Reduced pregnancy loss [61].

Embryo grading and selection

The embryologist needs to be able to characterize the quality of the embryo at each successive stage of development. Conversely, the characteristics and features associated with high-quality embryos are shown in Fig. 27.4 and the characteristics of poor quality embryos are shown in Fig. 27.5. When an embryo forms the blastocyst stage, it is possible to grade the two different cell types, the inner cell mass and the trophectoderm, and to give the blastocyst an alphanumeric score [62] (Fig. 27.6), which is highly correlated with implantation potential of the embryo [63,64]. With the advent of time-lapse incubation, one is able to assess the embryo every 7—10 min, culminating in a large amount of information, which has the potential

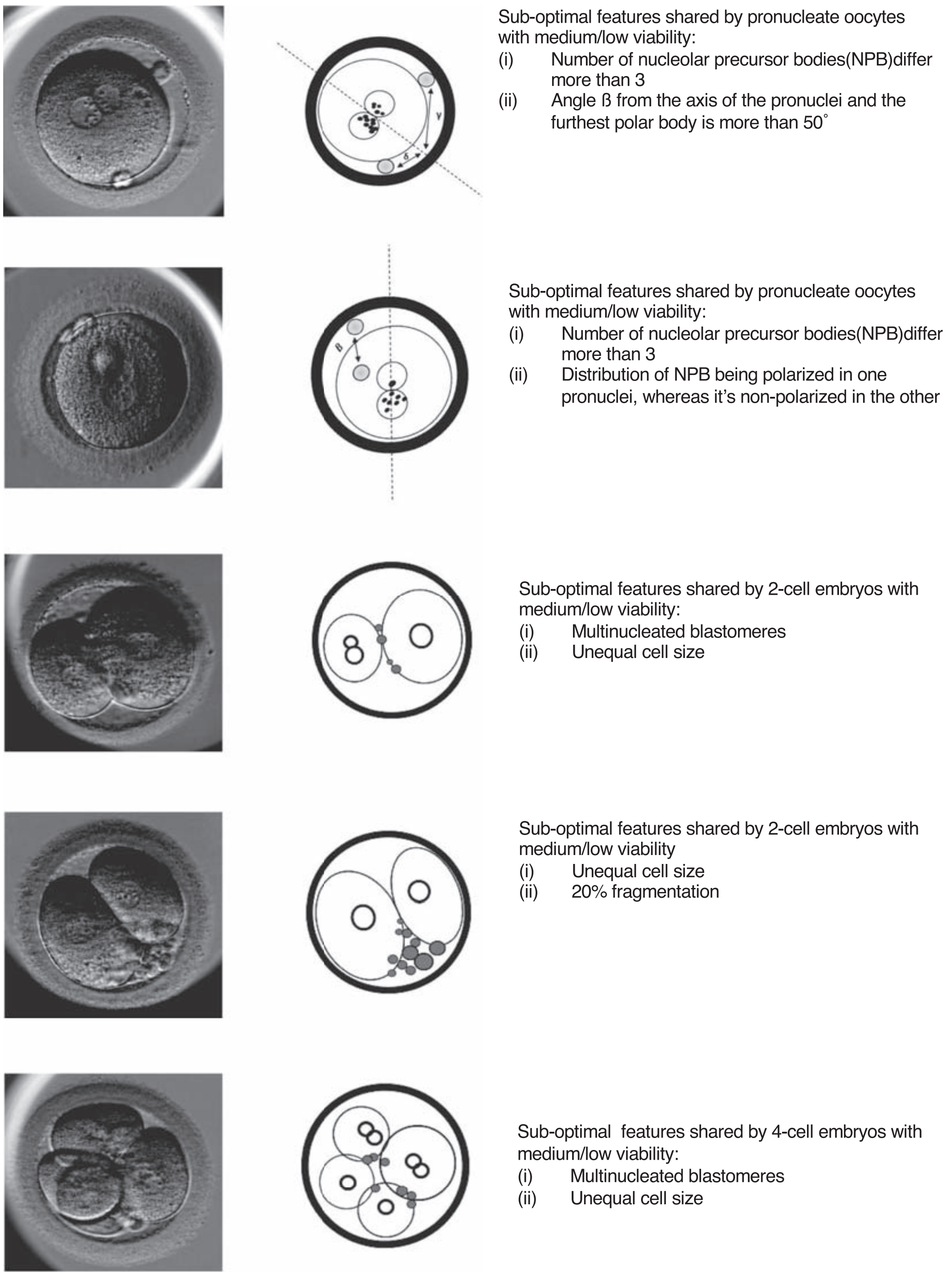

FIGURE 27.5 Key morphological features of embryos with medium to low viability. *From Gardner DK, Balaban B. Assessment of human embryo development using morphological criteria in an era of time-lapse, algorithms and 'OMICS': is looking good still important? Mol Hum Reprod 2016;22:704-718 with permission.*

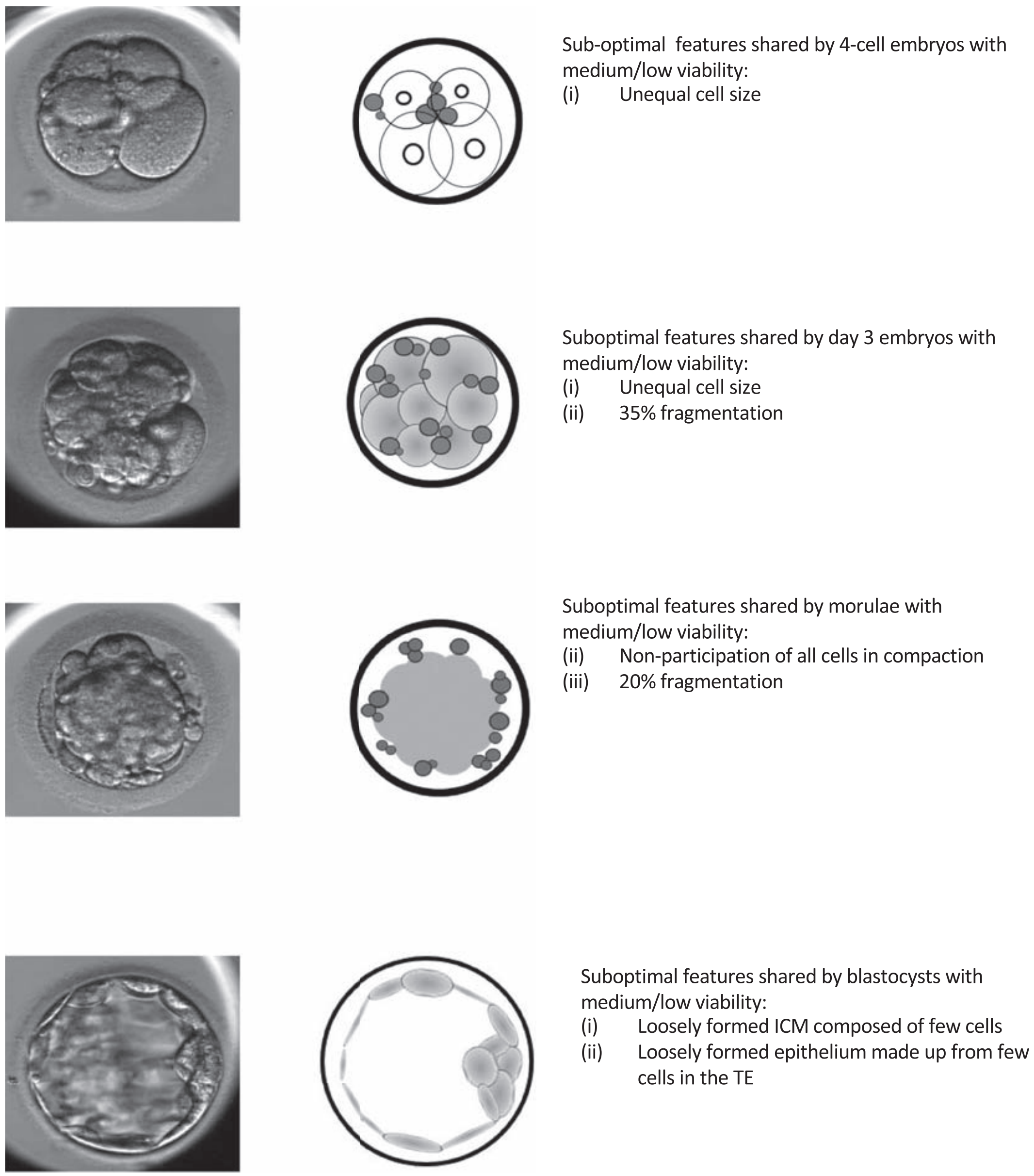

FIGURE 27.5 cont'd

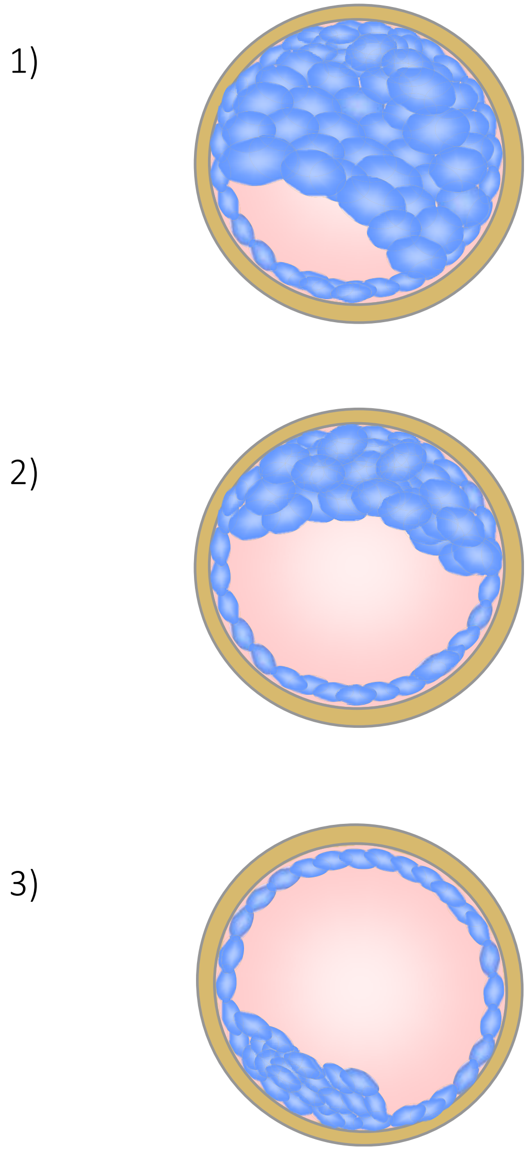

FIGURE 27.6 **Scoring system for human blastocysts.** Blastocysts are initially given a numerical score from 1 to 6 based upon their degree of expansion and hatching status:

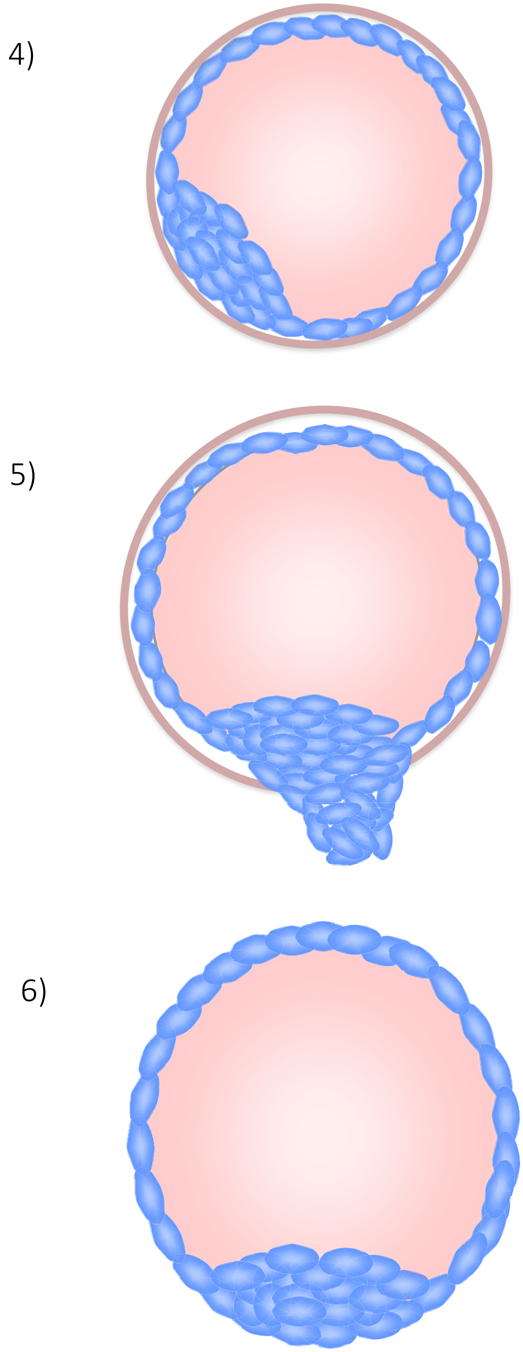

FIGURE 27.6 cont'd

to augment embryo selection, as algorithms [65–68] and artificial intelligence (AI) [69,70] can be applied to the morphokinetic data, creating aids to the embryologist for embryo selection and deselection for transfer.

Embryo transfer

It is essential that time and resources are spent in training the physician in embryo transfer. To put it bluntly, this is the step where pregnancies can be lost. Embryo transfer, just like embryo culture, should be perceived holistically and attention paid to all aspects of the patient preparation [71]. We advocate that embryos should be transferred in a medium with a high hyaluronan content [38] using ultrasound guidance.

Summary

With the development of physiologically based culture media [23] and a better understanding of the complexities of the embryo culture system [3], combined with oversight using a quality management system [72], and appropriate lab design and air handling [73], we can now readily culture the human embryo to the blastocyst stage and obtain high pregnancy rates following single embryo transfer. Maintaining the integrity of gamete and embryo physiology throughout the IVF process is fundamental to successful outcomes [1,2]. With the advent of predictive grading systems, selection algorithms, and the use of AI, we are in a strong position to be able to rank embryos for transfer, ensuring we reduce the time to

(1) Early blastocyst, the blastocoel occupies less than half the volume of the embryo.
(2) Blastocyst, the blastocoel occupies half of the volume of the embryo or more.
(3) Full blastocyst, the blastocoel completely fills the embryo, but the zona is not thinned.
(4) Expanded blastocyst, the blastocoel volume is now larger than that of the early embryo and the zona is thinning.
(5) Hatching blastocyst, the trophectoderm has started to herniate through the zona.
(6) Hatched blastocyst, the blastocyst has completely escaped from the zona.

The initial phase of the assessment can be performed on a dissection microscope. The second step in scoring the blastocysts should be performed on an inverted microscope. For blastocysts graded as 3 to 6 (i.e., full blastocysts onwards), the development of the inner cell mass (ICM) and trophectoderm can then be assessed.

ICM Grading

A. Tightly packed, many cells
B. Loosely grouped, several cells
C. Very few cells

Trophectoderm Grading

A. Many cells forming a cohesive epithelium
B. Few cells forming a loose epithelium
C. Very few large cells

pregnancy. Furthermore, in the near future, we shall see an introduction of new technologies to support better gamete preparation and selection, to improve culture systems and ultimately the automation of key processes, thereby further increase both the success and consistency of the IVF laboratory [74,75].

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CHAPTER

28

Introduction to the IVM laboratory

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Oocyte maturation process

Oocyte in vitro maturation (IVM) is the maturation of immature oocytes retrieved from small antral follicles. IVM of human oocytes is one of the reproductive techniques, which has been practiced for almost 30 years and is gaining popularity. The IVM has been applied for infertile women who have polycystic ovaries to avoid ovarian hyper stimulation syndrome, for poor responders [1,2], patients with repeated assisted reproduction failure due to resistant ovary syndrome [3], and for fertility preservation [4]. The research for improvements in IVM is continuing, and improved clinical outcomes have been established by several centers [5–7]. Nonetheless, the complexity of performing IVM and its lower subsequent clinical outcomes compared with conventional IVF represents a major obstacle that needs to be overcome for widespread uptake of IVM [7]. Therefore, it is important to understand the different mechanisms of oocyte maturation between in vivo and in vitro, in order to obtain viable oocytes in vitro.

Basically, oocyte maturation can be divided into nuclear maturation and cytoplasmic maturation [8]. Oocyte nuclear maturation implies the reinitiation of the first meiotic division from Prophase-I (germinal vesicle (GV)-stage) and progression to Metaphase II (MII) with 1st polar body (PB). This process is divided into several parts including meiotic resumption/ germinal vesicle breakdown (GVBD), chromatin condensation, formation of the meiotic spindle, separation of chromosome with extrusion of first PB, and meiotic rearrest before fertilization [8]. Cytoplasmic maturation refers to an accumulation of factors and involves metabolic structural changes in the organelle [8]. Synchronization of nuclear and cytoplasmic maturation must occur to undergo successful fertilization and subsequent development for a mature oocyte [9].

Oocyte maturation in vivo is regulated through hormonal signals, interactions with surrounding somatic cells, and involvement of transcription factors regulating gene expression [10]. In a regular menstrual cycle, one follicle gets dominance from several growing antral follicles and grows to preovulatory follicle, but the oocyte inside of the follicle is still arrested

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at Prophase-I until ovulatory luteinizing hormone (LH) surge [8]. This meiotic arrest is controlled by high cyclic adenosine 3′, 5′-monophosphate (cAMP) levels within the oocyte. There are gap junction connections between oocyte and cumuli cells (CCs), and between CC themselves [10]. Three mechanisms are known to maintain the high cAMP level within the oocyte. (1) cAMP enters the oocyte from CC through the gap junction [11]. (2) cAMP is produced by the oocyte itself [12−14]. (3) Guanosine 3′,5′-cyclic monophosphate (cGMP) passes through gap junctions into the oocyte [15−19] and inhibits hydrolysis of cAMP by the phosphodiesterase 3A (PDE3A:oocyte specific phosphodiesterase) [20]. The cGMP is produced in the granulosa cells via the activity of the guanylate cyclase receptor natriuretic peptide receptor 2 (NPR2) by C-type natriuretic peptide (CNP) [19]. The high intraoocyte cAMP concentration maintains a high protein kinase A (PKA) activity which inactivates meiosis promoting factor (MPF) by inducing phosphorylation of cell cycle components and thus blocks meiotic progression [21−23].

After the LH surge, oocyte maturation is induced by cascade signaling pathways as well as physiological changes in preovulatory follicles in vivo. LH activation of mural granulosa cells induces the expression of the epidermal growth factor (EGF)-like growth factors such as betacellulin, amphiregulin, and epiregulin [24,25]. The EGFs bind to their receptors in CC and activate mitogen-activated protein kinase (MAPK). The MAPK induces synthesis of downstream meiosis resumption-inducing factor(s) as well as blocking the gap junction. In addition, the LH surge inactivates NPR2 and activates cGMP phosphodiesterase PDE5, and induces a rapid drop of cGMP levels inside of follicle resulting in decreasing cGMP supply to the oocyte [17,18,26,27]. Eventually, cAMP levels in an oocyte drop, inactivation of PKA, dephosphorylation of the key components, and start meiosis subsequently. Physiologically, LH surge induces increasing volume of a Graafian follicle and CC expansion which is important for oocyte maturation. Hyaluronan (HA) is mainly involved in the CC expansion and is synthesized by hyaluronan synthase 2 (HAS2) in the plasma membrane of CC [28−30]. The gap junctions in the cumulus−oocyte complexes (COCs) are disrupted by the CC expansion which stops the transport of cAMP and cGMP through the gap junction and leads to the activation of MPF and meiotic resumption of oocytes [31,32]. Soluble factors such as growth differentiation factor-9 (GDF-9), bone morphogenetic protein 15 (BMP-15), and BMP-6 produced by oocyte itself are also actively involved in HA synthesis related with CC expansion [33,34]. These factors induce HAS 2 gene expression and CC expansion in the presence of follicle-stimulating hormone (FSH) [33−35].

Compared to in vivo oocyte maturation process, since there are no signals to maintain oocyte arrest in Prophase-I in vitro, the immature oocytes retrieved from antral follicles start hormone-independent, spontaneous meiotic maturation [36]. This spontaneous maturation causes a premature breakdown of oocyte−cumulus cell gap junctions, which is supposed be sustain in vivo at those stages, leading to loss of beneficial CC metabolites, such as nucleotides and nutrients, which may be contributing to the composition of the oocyte mature cytoplasm (cytoplasm maturation).

The classic IVM refers to the maturation of immature COCs in culture from GV-stage (Prophase-I) to reach MII after their recovery from small antral follicles without stimulating with any gonadotropins [37]. In clinical human IVM program, however, in vivo stimulation with gonadotropins has been commonly used to improve the quality and quantity of oocytes either with a few days of gonadotropin (FSH)-, one ovulatory dose of human chorionic gonadotropin (hCG-), or combined FSH-hCG [38]. Therefore, the techniques used for clinical IVM differ substantially across clinics, and they result in extremely variable pregnancy rates, partially due to some of these differences in protocols.

Oocyte identification

Laboratory process of human IVM program is more time consuming and technically different than the traditional IVF cycles [39]. More labor is required from both clinician and embryologist to handle an IVM cycle than conventional IVF. Before starting an IVM program, the embryologists and clinicians as well should obtain adequate training and experience by a supervisor skilled in this technique. The diameter of needles used to puncture small follicles (2–13 mm) is between 16 and 21-gauge in most of IVF centers; some clinics use two needle systems to aspirate the follicles. The thin needle and multiple punctures of small follicles with lower pressure may block the needle [39]. Flushing the needle often with flushing media and/or adding 2 U/mL of heparin is needed to prevent blood clots inside the aspiration needle. Regarding direct aspiration of immature oocytes from ex vivo ovarian tissues or ovaries during the cesarean section (CS), the tissues/ovaries are held with one hand and direct aspiration of small follicles containing immature oocytes is performed using a simple 5 or 10 mL syringe (containing HEPES-buffered IVF media with proteins) and a 21 or 22-gauge needle.

There are two methods for oocyte identification from follicular aspirates, the direct and the filter method. In the direct identification method, the follicular aspirate is poured into a Petri dish and first examined under a stereomicroscope to identify COC similar to IVF cycles. Immature oocytes without an expanded cumulus mass are much more difficult to identify than oocytes from IVF cycles (Fig. 28.1). In non-hCG-primed IVM cycles, the COC aspirated have compacted or small amount of cells without CC expansion (Fig. 28.1A). It is best to pour a small volume of follicular aspirate into the Petri dish to facilitate direct identification of COC. However, it is still difficult for embryologists to identify, and the time of oocyte identification is increased. The other simpler method is to use a cell strainer device made of a nylon mesh of 70 μm pores (Falcon, 70 μm mesh size, BD Biosciences) (Fig. 28.2) [39]. The

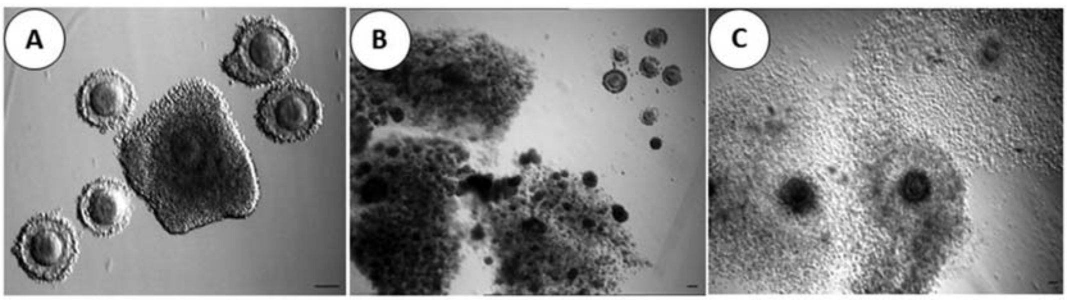

FIGURE 28.1 Cumulus–oocyte complex (COC) retrieved from non-, FSH (hMG)-, or HCG-stimulated IVM cycles. (**A**) Oocytes just after oocyte retrieval in non- or FSH (hMG)-primed IVM cycles. (**B**) Oocytes just after oocyte retrieval in HCG-primed IVM cycles. (**C**) Oocytes just after oocyte retrieval in COH cycles. Bar 100 μm. Reproduced from Son WY, Tan SL. Laboratory and embryological aspects of hCG-primed in vitro maturation cycles for patients with polycystic ovaries. Hum Reprod update 2010;16:675–89.

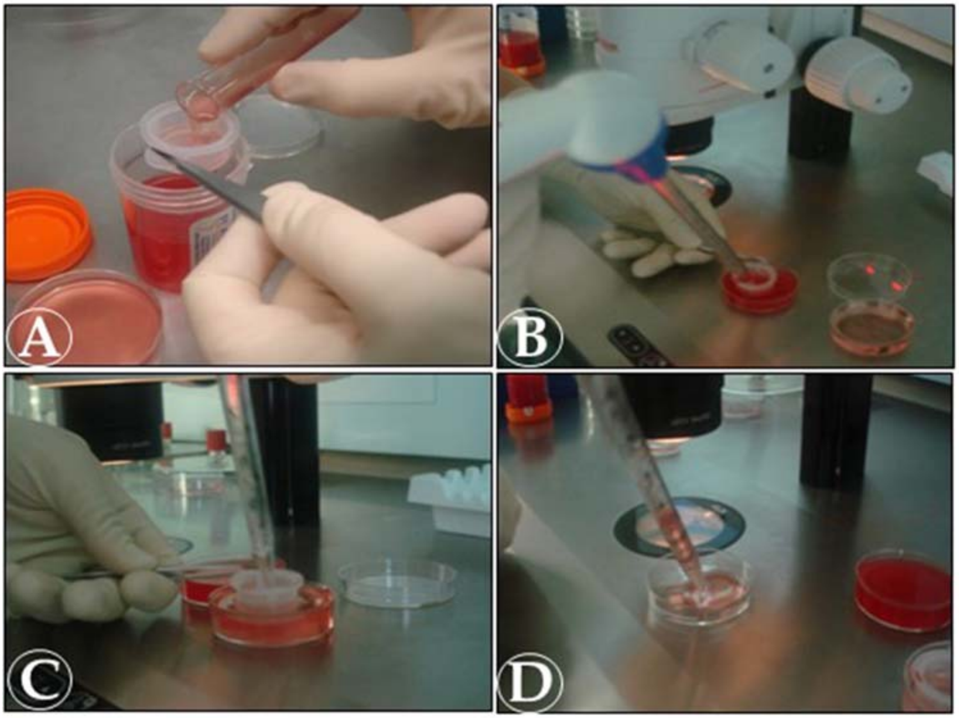

FIGURE 28.2 Filter method for identifying COCs from follicular aspirates. (**A**) Filter of follicular aspirates using a cell strainer device made of a nylon mesh with 70-μm pores. (**B**) Washing the collected aspirates with medium. (**C**) Collecting washed follicular aspirates. (**D**) Transferring the collected aspirates to a petri dish to search for COC under a microscope. *COC*, cumulus–oocyte complex. Reproduced from Son WY, Tan SL. Laboratory and embryological aspects of hCG-primed in vitro maturation cycles for patients with polycystic ovaries. Hum Reprod update 2010;16:675–89.

follicular aspirates in collection tubes are filtered with the cell strainer without identifying directly. After finishing follicular aspiration, the collected aspirates on the device are washed with fresh buffered medium to remove red blood cells and small cells (Fig. 28.2A–C). They are then transferred to a new Petri dish to search for COC under a stereomicroscope (Fig. 28.2D). This method is routinely used in the IVM cycles without hCG priming such as non-, FSH-primed IVM cycles and follicle aspirates from ovarian tissue as well. The filter method is easy and efficient, but there exists a delay in communication between clinician and embryologist regarding the number and morphology of oocytes retrieved during the process. Actually, most of clinicians want to know the number of oocytes obtained while doing the egg retrieval to see if their aspiration process is going well.

In hCG-primed IVM cycle, one of the big differences compared to non-hCG-primed cycles is to get COC with expanded CC even though the expansion is smaller than seen in IVF cycles (Fig. 28.1). The hCG induces the CC expansion and facilitates COC detachment from the follicles [39]. It makes oocyte retrieval and COC identification for both clinician and embryologist easier than without hCG. In conventional IVF cycles, for instance, the IVF cycles would sometimes be canceled because there are no eggs or small number of immature oocytes

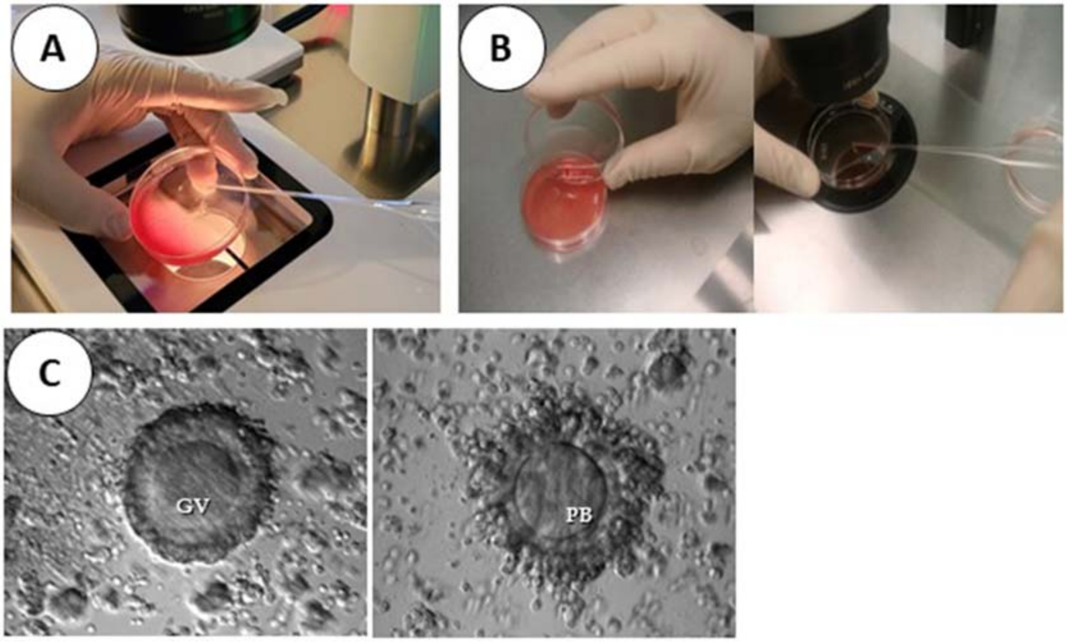

FIGURE 28.3 The process to evaluate oocyte maturity of COC with expanded cumulus cells. (**A**) Sliding method. (**B**) Spreading method. (**C**) COC with GV or 1 PB retrieved from hCG-primed IVM cycles. *COC*, cumulus–oocyte complex; *GV*, germinal vesicle; *PB*, 1st polar body. Reproduced from Son WY, Tan SL. Laboratory and embryological aspects of hCG-primed in vitro maturation cycles for patients with polycystic ovaries. Hum Reprod update 2010;16:675–89.

retrieved from bigger follicles if patients did not properly take hCG or took hCG at the wrong time.

In addition, some in vivo–matured oocytes could be obtained from the COC with expanded CC in IVM cycles with hCG priming. Two techniques can be used to evaluate oocyte maturity: the sliding method or the spreading method (Fig. 28.3). In the sliding method, the COC is allowed to slide slowly from one side to the other at the bottom of the Petri dish while being observed under the microscope (Fig. 28.3A). It is not easy for the embryologist to maintain focus in different frames under microscope while assessing the maturity. Spreading method can be used easily for any embryologist: after identifying expanded COC, they are removed from the Petri dish and follicular aspirates in the Petri dish are discarded while keeping a small amount of fluid in the dish (Fig. 28.3B). Then the identified COCs are put back into the Petri dish to allow them to spread so as to observe the oocyte cytoplasm under the dissecting microscope (Fig. 28.3C). This procedure is easier than the sliding method, but should be performed quickly before the follicular fluid dries out on the Petri dish. It would be difficult to identify oocytes with the first PB because of orientation of the PB and CC around, but simpler to separate oocytes with and without GV in cytoplasm with this method (Fig. 28.3C). If no GV is observed in the oocyte cytoplasm, CCs are removed until the presence of the PB is identified to observe maturity. However, if the embryologist overlooks MII oocytes retrieved, the oocyte will age for at least 24 h by the time of CC removal and their developmental competence might be compromised. Therefore, it is important to clearly identify the in vivo–matured oocytes in IVM cycles with hCG priming. Some IVF clinics are removing expanded CCs 6 h after egg retrieval to evaluate precise oocyte maturity [40], and other IVF clinics remove the CCs three hours after finishing egg retrieval [41]. Therefore, in the hCG-primed IVM cycles, COCs with expanded CCs are identified directly under microscope at first in order to assess oocyte maturity and the remaining aspirates are filtered through a cell strainer to facilitate the identification of those oocytes with a small amount or compact CCs. This method allows for more efficient and timely communication between clinician and embryologist regarding the number and morphology of oocytes retrieved during the process.

IVM of immature oocytes, culture medium for IVM and supplements

As mentioned before, immature oocytes retrieved start spontaneous oocyte maturation in vitro which may cause incomplete cytoplasm maturation. Based on the that, some studies tried to mimic in vivo systems called pre-IVM before culturing in IVM media by delaying or temporarily preventing spontaneous IVM for a short period of time (two hours) with chemicals such as cAMP analog, kinase, or PDE inhibitors [42,43]. It has been called "biphasic IVM" and includes pre-IVM and IVM phase. These strategies were designed to better support oocyte cytoplasm maturation and improve oocyte and embryo developmental competence after IVM. However, the effect of using those chemicals was not significant on the outcomes in humans.

Recently, another biphasic IVM method has been reported, and they use CNP to maintain GV arrest for 24 h in the pre-IVM phase. As mentioned previously, the CNP controls the production of cGMP levels in granulosa cells inside the follicle [19]. CNP-treated COCs sustain gap-junctional communication between the oocyte and cumulus cells longer [19,44]. This recent biphasic IVM system has been called "CAPA-IVM" [45], and they have reported very promising outcomes [7,45,46]. This culture system requires an extra day of culture and has only recently been clinically introduced in an IVM laboratory; therefore, its efficiency still needs to be confirmed by other IVF centers.

However, the majority of other clinics culture immature oocytes directly in IVM media without using the pre-IVM phase. There is no consensus on the medium formulation that is best suited for IVM [47]. The composition of most media used for human IVM is based on experiences with other mammalian species, composed of high levels of glucose and normal levels of pyruvic and lactic acids, along with essential and nonessential amino acids [47]. Two commercialized IVM media, with the advantage that they are certified as IVF quality controlled, such as SAGE (Cooper Surgical, USA) and Medi-Cult (Origio, Denmark) are available and have reported similar embryological outcomes [48]. Also, there were similar results on embryological and clinical outcomes between TCM 199-based and Medi-Cult commercial IVM media [49,50] or between SAGE commercial IVM media and blastocyst IVF media [48]. Therefore, although commercialized basic human IVM culture media are currently available, the developmental competency of immature oocytes was not different with complex culture medium or regular IVF media, and thus, further research remains to be done in order to define optimal culture medium for human immature oocytes. Regular IVF blastocyst culture medium which has high concentration of glucose may be a good choice for IVM of immature oocytes, especially for the clinic where IVM cycles are not routine [39].

Currently, most IVM protocols supplement serum, FSH, or other additives such as LH/hCG and/or EGF, EGF-like growth factor into IVM culture medium based on their physiological role in oocyte maturation in vivo. Since serum sources from other patients or from animals have potential sources of infectious agents, the patient's own serum, human serum albumin, or synthetic serum substitute has been used as protein supplementation in IVM [51,52].

FSH is most commonly added to the culture medium to help mediate subsequent oocyte maturation since FSH is involved in the development of preovulatory follicles in vivo [33], for induction of LH receptors and for inducing EGF-like growth factors. In the literature, the concentration of FSH added to the media ranges from 0.075 IU/mL to 0.1 IU/mL [5,49,53], and increasing concentration did not appear to increase maturation rates [54]. LH and hCG are necessary components of IVM media to mediate the resumption of meiosis and the final stages of maturation that would normally be completed following the LH surge and ovulation in vivo. Improvements in human oocyte maturation and embryo cleavage in the presence of FSH and LH have been reported [55,56]. Hreinsson et al. [57] reported that hCG and LH are equally effective in promoting oocyte maturation in vitro. In addition, matured oocytes from hCG-supplemented medium had a higher quality based on the measured glucose-6-phosphate dehydrogenase activity with immature oocytes obtained from controlled ovarian hyperstimulation (COH) cycles [58]. Concentration of LH or hCG ranges from 0.1 to 0.75 IU/mL [49,54]. However, a report suggests that hCG in the culture medium had no effect on IVM and embryonic development [59]. Therefore, further studies are needed to clarify the role of gonadotropin and their optimal concentration in culture medium. These gonadotropins may not play the same role in oocyte maturation in vitro since mural cells play a role in mediating LH/hCG signals involved for upregulating EGF-like growth factors in vivo but are not present in immature COC after collecting in vitro. The EGF itself could help immature oocytes to achieve nuclear and cytoplasmic maturity after culture [39,60]. Recently, some IVF groups have added either EGF-like growth factor (amphiregulin or/and epiregulin) [7,45,61,62], oocyte secreting factors (GDF-9 and BMP-15) [63,64], or antioxidant (melatonin or coenzyme Q10) [65–67] to the IVM media, and they reported better maturation and embryo development. However, more significant research is needed in this area to create optimal combination and concentrations of supplements.

Culture time for assessing in vitro maturation of immature oocytes

As mentioned previously, it is important to identify in vivo mature oocytes at the time of collection in the hCG-primed IVM cycles. Identification of MII stage oocytes with extruded first PB from the GVBD-stage oocytes is actually quite hard without removing CCs because of different orientation of the first PB on the Petri dish. The GVBD oocytes could become MII on the same day after removing CCs and in this case would be inseminated. On Day 1, oocyte maturity would be evaluated again early in the morning and ICSI would be performed immediately if there are any MII oocytes. By the end of Day 1, the presence of zygotes would be evaluated since aged oocytes could show earlier fertilization sign than usual and separated from the other oocytes that had been injected at the same time but show no signs of fertilization. For those oocytes where zygotes were not observed, signs of fertilization would be checked again the next morning. The oocyte maturity of GV-stage oocytes at the time of collection in hCG-primed IVM cycles does not need to be assessed again on retrieval day. In the non-hCG-primed IVM cycles, COCs retrieved from antral follicles usually have a compact or sparse mass of CCs with oocytes at GV-stage (Fig. 28.1A), and therefore oocyte maturity does not need be assessed on the collection day.

A number of MII stage oocytes could be obtained after 24 h of maturation from GV-stage oocytes at collection in both non- or hCG-primed IVM cycles [49,59,68]. Most of the IVM studies showed 40%−60% of IVM after Day 1 culture (24−30 h) in IVM cycles [59,68]. In some early studies, however, oocytes were inseminated 48 or 56 h after assessing maturation in nonprimed IVM cycles and did not even assess oocyte maturity on Day 1 [51,69]. Therefore, many oocytes must have been arrested at the MII stage for 24−30 h before insemination. This places them well past the optimal fertilization time and was one of the main reasons for poor clinical outcomes for the first several years after starting human IVM program.

It would not be suggested to use MII oocytes matured after Day 2 culture unless there is not enough mature oocytes on Day 1 since the embryos produced from late matured oocytes (Day 2) had poor developmental potential and had higher aneuploidy rate [70].

Since time-lapse incubator system (TLS) has been developed, the time period of oocyte IVM from GV-stage oocytes could be observed, although most of CCs at GV-stages are removed to see the first PB extrusion in the TLS which may cause them to act differently than oocytes with intact CC. The total duration of meiotic maturation in human, including the time to GVBD, during in vitro culture collected from COH cycles was estimated to be ∼ 20−22 h [71]. In the GV-stage oocytes retrieved from FSH-hCG-stimulated IVM cycles, the average time from GV- to GVBD-stage, was 3.3 h (± 2.3, range: 0.5−9.3 h), and GBVD- to MII-stage was 15.5 h (± 1.5, range: 10.7−18.6 h) after denudation of CCs (Fig. 28.4) [72]. Based on the two studies, the maturation timing between different sources of immature oocytes seems to be similar. Therefore, oocyte maturity should be assessed after Day 1 culture in the entire IVM program.

Insemination of mature oocytes produced from IVM cycles

Historically, ICSI has been used to increase the chances of fertilization whether or not a male factor has been detected as a result of the theoretical concern of zona pellucida

FIGURE 28.4 Time course of in vitro maturation from culture in IVM culture media of human immature GV oocyte obtained from FSH-hCG-primed IVM cycles. *GV*, germinal vesicle; *GVBD*, germinal vesicle breakdown; *M-II*, metaphase II. Reproduced from Son WY, Tan SL. Laboratory and embryological aspects of hCG-primed in vitro maturation cycles for patients with polycystic ovaries. Hum Reprod update 2010;16:675−89.

hardening during the in vitro culture of the immature oocytes [73]. ICSI resulted in a higher fertilization rate than IVF in several studies [49,74]. However, Walls et al. [75] reported similar fertilization rate between IVF and ICSI insemination in IVM oocytes. Therefore, it is still not clear whether ICSI is definitely beneficial or absolutely necessary to effectively inseminate IVM oocytes in the absence of sperm factors. Nevertheless, ICSI is the preferred method of insemination to increase fertilization or to avoid the risk of unexpected fertilization failure in the majority of IVM studies.

Optimal ICSI timing

One of the characteristics in the human IVM cycles is the asynchronous maturation of the immature oocytes observed following in vitro culture compared to COH cycles. In the human IVM program, the number of embryos fertilized is also an important factor to increase the chance of successful pregnancy [76,77]. Therefore, it is important to determine optimal ICSI timing to induce successful fertilization and embryo development when GBVD stage oocytes become MII on the same day.

Balakier et al. suggested [78] that the IVM oocytes obtained from COH cycles needed at least 3 h before insemination after extruding the 1st PB to obtain reasonable fertilization rates. Meanwhile, developmental potential would be negatively affected if IVM oocytes were aging postmaturation. Thus, defining the optimal interval between the first PB extrusion and ICSI is crucial to increase fertilization and embryo development rates, especially in human IVM program. Hyun et al. [77] reported that human oocytes matured in vitro needed at least 1 h after the first PB extrusion to complete nuclear maturation, and better-quality embryos were produced from the group injected three hours after 1st PB in hCG-primed IVM cycles. Recently, Gunasheela et al. also confirmed that the fertilization rate of the group who had ICSI 3–6 h after the 1st PB extrusion was higher than those of other groups using TLS in FSH-hCG-primed IVM cycles [79]. In this study, a lower number of good-quality morula/blastocyst embryos (0%) developed from the group where ICSI was performed < 3 h after the 1st PB extrusion compared to other groups.

It is important to note that the time to complete nuclear/cytoplasmic maturation after the first PB extrusion will depend on the source of immature oocytes (from COH cycles or IVM cycles) as well as on each IVM culture system. Practically, however, it would be easier to remove CCs 24 h after culturing and perform ICSI 3 h after observing the first PB in order to avoid several ICSI for an IVM cycle.

Culture of IVM embryos and embryo transfer

Once zygotes are formed, the remaining embryology work such as embryo culture and transfer is the same as IVF cycles.

The cleavage rate of IVM embryos was similar to IVF embryos, and blastocyst rate was also reasonable although lower than IVF embryos [80]. In the initial IVM program, cleavage stage transfer in IVM cycles was performed in most of the IVF centers to avoid the possibility of blastocyst development failure. Since the quality and quantity of embryos obtained from

IVM are not sufficient, there was a trend to increase number of embryos transferred compared to IVF cycles which may cause multiple pregnancies [81]. Therefore, some IVF centers tried to culture and transfer at blastocyst stage in order to reduce the number of embryos transferred and to synchronize between embryos and endometrium and have reported successful clinical outcomes [80,82]. Son et al. [82] applied a criterion to culture IVM embryos to blastocyst based on quantity and quality of embryos (more than seven zygotes and three good-quality cleavage embryos) and reported reasonable blastocyst rate (41.6%) and clinical outcomes (clinical pregnancy rate: 51.9%) in hCG-primed IVM cycles. Walls also reported about 40% useable blastocyst rate in FSH-primed IVM cycles [80]. Therefore, the stage at which embryo transfer (ET) ought to take place should be based on the number and quality of embryos. We would suggest adding one more good-quality cleavage embryo to decide to culture to blastocyst stage in IVM cycles compared to IVF cycles.

Cryopreservation of oocytes and embryos obtained from IVM cycles

Cryopreservation using vitrification method of oocytes and embryos retrieved from IVF cycles is successfully used in the regular and fertility preservation programs. In IVM program, the reasons to freeze embryos are either to store the remaining embryos after ET, inadequate endometrial thickness, or for fertility preservation [4,83,84]. In IVM cycles, the duration of estradiol exposure in the follicular phase is shorter than in COH cycles. Researchers think that one of the reasons that IVM cycles are less efficient than IVF cycles may be due to inadequate endometrium condition for fresh ET. In the hCG-primed cycles, however, it seemed that there is no difference in clinical outcomes between fresh and frozen ETs at both cleavage and blastocyst stage [83,85]. Junk and Yeap also reported a good clinical outcome after transferring at blastocyst stage generated from FSH-primed IVM cycles after starting estradiol 2 days before oocyte retrieval [5]. Nevertheless, in FSH-primed IVM cycles, clinical outcomes in frozen ET cycles were better than those of fresh ET cycles [86–88], and therefore a freeze-only strategy was proposed in the IVM cycles without hCG priming [88,89]. Good survival rate and reasonable clinical outcomes have also been reported using the vitrification method with either cleavage- or blastocyst-stage embryos produced from clinical IVM programs [83,84,87]. These results suggested that embryos produced from regular IVM cycles can be safely cryopreserved through vitrification.

Based on the improvement of oocyte and embryo cryopreservation method called “vitrification,” the American Society of Reproductive Medicine has endorsed oocyte cryopreservation as a fertility preservation strategy for women with cancer and other illnesses requiring treatments that pose a serious threat to their future fertility [90]. COH for IVF is contraindicated for patients with certain forms of cancer. For instance, the women with estrogen-sensitive cancers are usually being treated by methods minimizing serum estradiol concentration with an aromatase inhibitor or prescribing tamoxifen [91]. In these cases, however, immature oocyte collection can be an alternative as circulating estrogen concentrations remain low for the patients, and this has been proven to be safe and effective in patients with estrogen-sensitive breast cancer [92,93]. In addition, immature oocytes collected from the COH cycles could be a viable source of oocytes that can be used to improve the efficacy of fertility preservation treatments by increasing the number of mature oocytes available for freezing or fertilization [94].

Time to cancer treatment is a critical factor for many cancer patients. They do not have time to do an IVF cycle before starting chemo- or radiation-therapy. Since IVM treatment can be initiated immediately and at virtually any time of the cycle without stimulation, thus avoiding the need to delay chemo/radio therapy or surgery. IVM strategy can easily be combined with other fertility preservation strategies such as ovarian tissue preservation, CS, and follicle culture to improve the efficiency of fertility preservation.

Although very few live births have been reported after IVM oocyte cryopreservation in regular IVM program [95–100], there are a few reports of successful live birth using the cryopreserved oocytes obtained from ovarian tissue in IVM program for fertility preservation related to a medical reason [101]. In addition, some healthy live birth cases have been reported after using vitrified/warmed IVM embryos produced from immature oocytes retrieved from ex vivo ovarian tissue [102–104], transvaginal follicular puncture [105,106], and the clinical outcomes in a series of patients of technique IVM embryo freezing have been evaluated in the fertility preservation program [101,107].

Theoretically, there are two approaches for preserving immature oocytes: oocyte cryopreservation at the mature stage (after IVM) or at the GV-stage (before IVM). According to papers in the literature, it seems that vitrifying oocytes at MII stage after IVM is better than at the GV stage in terms of survival rate and embryo developmental potential after warming [108–110]. Cohen and his colleagues reported that vitrified-thawed IVM oocytes had a low survival rate and embryological and clinical outcomes compared with fresh IVM oocytes [100]. Similarly, another study showed very poor pregnancy and delivery outcomes from vitrified-warmed embryos derived from IVM oocytes for cancer patients [107]. Therefore, it seems that the reproductive potential of vitrified IVM oocytes/embryos obtained from IVM program is impaired owing to the present vitrification-warming procedure. Therefore, this procedure will require further advances in IVM- or/and cryo-technology. Different strategies may be adopted to improve the results: (1) improve the quality of in vitro–produced oocytes by optimizing the in vitro conditions and/or (2) adapt cryopreservation methods to the cellular properties of oocytes before/after IVM [110].

Most challenging aspects to introduce IVM into the routine laboratory

There are benefits of IVM such as it being less expensive, less risky, and can be repeated frequently [39]. However, it takes more time, skills, and energy to perform an IVM cycle compared to conventional IVF cycles [39]. Nevertheless, the overall efficiency of IVM is lower than IVF cycles. Oocyte retrieval for clinician in an IVM cycle is more challenging than an IVF cycle. It is much more difficult for clinician puncturing small antral follicles to retrieve oocytes, especially without triggering with an ovulatory dose of hCG or GnRH agonist. Only some skillful clinicians would have good oocyte recovery rate, and it would take more time to train the clinician to get the skill compared to that of IVF cycle. For the embryologist, it also requires more work to handle an IVM cycle than conventional IVF; (1) since there are no good commercial media and consensus protocol to prepare media, each laboratory would prepare their own culture medium. The composition of the culture medium varies between laboratories, and it is easy to make mistakes in the preparation of this culture medium [111]. (2) It also requires more time to identify COCs in follicular aspirates at collection time since the immature oocytes do not have typical expanded CC which is seen in IVF cycles. (3) In addition, multiple rounds of maturation assessment and ICSI are needed. This makes it complicated to operate, especially in hCG-primed IVM cycles.

Therefore, it is important to find more innovative and sophisticated ways to improve simple laboratory procedures to introduce IVM program into the routine. The reason the human IVM program is less efficient is because of quantity and quality of oocytes obtained in IVM cycles. First, the oocyte retrieval rate and in vitro oocyte maturation rate after IVM are lower than those after COH cycles. Second, the quality of oocytes derived from current IVM culture systems is generally lower than that of oocytes matured in vivo [47]. Recently, efforts have been made to try and improve human IVM culture system such as "CAPA-IVM" biphasic culture system using natural meiosis inhibitor "CNP" [7,45,46] or microvibration culture system [112]. In addition, some studies are adding EGF, EGF-like growth factors, oocyte-secreting factors, or antioxidants into the IVM culture media to mimic in vivo microenvironment of follicle according to results found in animal oocyte biology research [7,45,61−67]. If IVM culture system and commercialized media with high quality and safety incorporated any one or combinations of these factors, it would be easier for the embryology laboratory to introduce human IVM program as a routine program and for its widespread use. In authors' humble opinion, it is not only culture medium that is important but also the physical aspect of culture such as a 3-D culture system with proper pressure (artificial follicle) should also be considered [113−115].

IVM program will not completely replace the IVF program, but it would be very useful to help the people who are not able to get COH if the efficiency were improved as a routine procedure. In addition, it would contribute to increase the number of mature oocytes available even from COH cycles. IVM is a useful and attractive technology and may be widely used worldwide in the near future. In order to reach such goal, emphasis should be placed on open and collaborative efforts by clinicians, scientists, clinical embryologists, and industry.

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CHAPTER

29

Time-lapse imaging

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Introduction

In assisted reproduction treatments, embryo selection is one of the key procedures to ensure the success of the cycle. Dynamic observation of morphological events during human embryo development provides valuable microscopic information which contributes significantly to the success of human-assisted reproduction programs. It is essential to identify and select those viable embryos with the highest implantation potential. Nowadays, we have the technology at our fingertips to achieve these results.

Background of time-lapse systems

There are many embryo selection criteria; the most widely used are those based on morphology. Morphological evaluation is the standard method for embryo evaluation, although it is subjective and associated with considerable inter- and intraobserver variability. Therefore, the selection of the embryo to be transferred cannot be based solely on morphological evaluation at specific points in its development [37,43].

Automated time-lapse imaging systems provide a noninvasive alternative to static morphological assessment of embryos, allowing observation of the complete evolution of the embryo without the need to disturb culture conditions, maintaining stable pH, temperature, gas concentrations, and humidity [52]. In addition, a number of predictive parameters based on developmental kinetics have been detected and used to create predictive models for implantation, live birth (LB), and aneuploidy. These systems are able to observe cellular events such as direct cleavage from one to three cells, fragment resorption, and blastomere fusion that otherwise could not be detected by conventional methods [25].

Another use of time-lapse in the clinic is its implementation in vitrification processes. Several studies have shown that some factors such as cryoprotectants, changes in volume

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due to osmotic effect, or sudden changes in temperature can alter embryo integrity and compromise its viability. Therefore, many of the embryos classified with good morphological quality before vitrification do not maintain their characteristics after devitrification, making it difficult to select the optimal embryo [8,17,26]. However, thanks to the dynamic evaluation offered by time-lapse systems (TLSs), it has been possible to generate embryo classification algorithms after devitrification. The algorithm measures the maximum blastocyst area and the initial area after devitrification [17]. Furthermore, using time-lapse, the degree of expansion prior to vitrification has been found to correlate with lower survival to vitrification but higher implantation and LB rates [17,39]. In short, the introduction of time-lapse in assisted reproduction has been a breakthrough in the understanding of fertilization processes and human embryo development [51].

However, the first step for the introduction of TLSs in assisted reproduction practice was the clinical validation of time-lapse incubators. Meseguer et al. in 2012 published a retrospective study evaluating the effect of clinical outcomes using time-lapse incubators compared with traditional incubators. This study was based on an analysis of all intracytoplasmic sperm injection (ICSI) treatments performed in all intravenous injection (IVI) clinics in which TLS had been used clinically. Specifically, the study included 8414 cycles of ICSI treatments, and it was the first comprehensive evaluation of TLSs and morphokinetic assessment to improve clinical pregnancy rates. Specifically, the relative improvement between both incubators was up to 15.7% more per embryo transfer when time-lapse was used [52]. Other authors reached the same conclusions; TLSs were incubators which ensured the correct embryonic development. Additionally, TLSs increase the quantity and quality of information received by the embryologist [67].

Morphokinetics

The development of incubators and TLSs for human embryo culture has led to the acquisition of a large number of data related to embryonic development times, and these data are known as morphokinetic parameters. Different research groups have identified new parameters and have demonstrated their predictive capacity.

Ability of morphokinetics in the prediction of aneuploidy

Aneuploidy is characterized by the presence of an abnormal number of chromosomes, and it is a major cause of reduced fertility and pregnancy loss in humans [36,76].

Embryonic aneuploidies are caused by meiotic and mitotic chromosome segregation errors, although the underlying cellular mechanisms are not yet known exactly [36]. There are different risk factors which predispose to the appearance of these chromosomal aberrations, but the most established is advanced maternal age. Specifically, aneuploidy rate increases from an initial value of 30% in women aged 30 years to 90% in mothers older than 40 years, before the onset of menopause [71]. Several studies relate this dramatic increase to defects in the molecular and cellular processes experienced by aging oocytes, such as mitochondrial dysfunction, telomere shortening, cohesin dysfunction, and meiotic spindle abnormalities [16,31].

Most embryos with aneuploidies are not viable. However, some manifestations of these chromosomal defects are compatible with the development of the embryo and the origin of LB. Nevertheless, these chromosomal aneuploidies cause syndromes that severely affect the health of children and their quality of life. Therefore, one of the most important tasks is to avoid the selection of aneuploid embryos for transfer [18,74].

The only method currently available in fertility clinics to assess embryo ploidy is the preimplantation genetic testing for aneuploidies (PGT-A) that provides an accurate number of complete chromosomal copies [11]. PGT-A has been established worldwide as a clinical service. In patients with more than one embryo, IVF together with PGT-A reduces both the overall treatment time and embryo transfer failure, as well as the chance of clinical miscarriage compared to performing conventional IVF without PGT-A [22,57,68].

Despite the advantages that the clinical practice of preimplantation genetic testing has introduced, there are still several drawbacks to be overcome that condition its use. PGT-A is a highly invasive technique that can condition embryo development (if done in early development stage). In addition, it is a high-cost test which makes the assisted reproduction treatment more expensive. It is also important to take into account the need to have highly qualified personnel in the clinic to perform embryo biopsies; otherwise, the viability of the embryo could be compromised [61].

Compared to preimplantation genetic testing, morphokinetics offers a noninvasive alternative that is completely harmless to the embryo. In addition, it is a cheaper option and does not require specifically qualified personnel for its evaluation [13]. Hence, TLSs can overcome the disadvantages of PGT by using morphokinetic parameters to predict embryonic ploidy. Among the different studies that focus their research on finding the relationship between embryo morphokinetics and embryo ploidy, the study by Chavez et al. is worth mentioning. In this study, they observed that euploid embryos presented precise division times from the first cell division to the four-cell stage. However, 30% of the embryos classified as aneuploid presented "normal" division times. Many studies have shown that most embryos with abnormal cell division times have chromosomal abnormalities [4,19,23,51].

At present, different studies focus their efforts on analyzing the predictive capacity of different morphokinetic parameters for embryonic aneuploidy. One of the first morphokinetic events observed during the evaluation of embryo development is the appearance and disappearance of pronuclei (tPNa and tPNf, respectively). Different authors have demonstrated the usefulness of tPNa and tPNf in their studies. Refs. [14,10] were able to relate these morphokinetic parameters to embryonic ploidy, demonstrating that euploid embryos took less time to develop pronuclei and extrude the second polar body than chromosomally abnormal embryos [10,14]. Other authors found a stronger relationship between embryonic ploidy and pronuclei duration (dPN) defined as the time to pronuclei fading minus the time to pronuclei appearance (dPN = tPNf-tPNa). In the studies of Refs. [72,60], it could be observed that pronucleus duration in aneuploid embryos was longer than in euploid embryos [60,72].

The aneuploidy rate has been correlated with different early morphokinetic parameters such as the time course between the first and second mitosis (cc2 = t3-t2) or the prolongation of the duration of the first cytokinesis (cc1 = t2-tpb2) [13,14,33]. In another study, they saw that the morphokinetic parameters of early embryonic developmental stages that had the most value in predicting ploidy were t5-t2 and cc3 [6]. Subsequently, Minasi et al. observed that the four-cell stage (t4) and the time to second synchrony (S2) were reached significantly earlier in euploid embryos compared with aneuploid embryos [54].

However, other authors focused on morphokinetic parameters of late stage of embryonic development. While some studies observed that morphokinetic parameters of late stages of embryonic development could not predict aneuploidy [44,65,75] other studies show the opposite. Melzer et al. in 2012 observed that the compactation start time (tSC) was faster in aneuploid embryos compared with euploid embryos [50]. In contrast, these results are at odds with those of Campbell et al. which showed that aneuploid embryos initiated compaction later than euploid embryos [10]. Huang and colleagues showed that blastocyst expansion time was directly related to embryo ploidy. The average euploid blastocysts observed were significantly faster than in aneuploid blastocysts [39]. Furthermore, the parameters observed by TLSs in the study of Minasi such as time to onset of blastulation (tSB), time to expanded blastocyst (tEB), and time to blastocyst hatching (tHB) were different in euploid and aneuploid embryos [54]. Campbell et al. in their 2013 study reached the same conclusions, using late morphokinetic parameters (tSC, tSB, and tB) increased the predictive power of their euploid embryo selection models [10].

Despite the evidence of the different studies presented above, currently aneuploidy prediction models based solely on morphokinetic parameters are not sufficiently accurate to replace PGT-A. Swain et al. in 2013 conducted a study whose purpose was to review the most relevant research that examined the relationship between embryonic morphokinetics and aneuploidy. That study concluded that morphokinetic parameters along with morphology can help in the selection of euploid embryos. However, the accuracy of these parameters requires further research [70]. Therefore, the latest trend in the current clinic is to introduce artificial intelligence (AI) for the prediction of embryonic aneuploidy as will be discussed below.

Ability of morphokinetics to predict clinical outcomes

With the advent of time-lapse, it has been possible to determine many morphokinetic parameters that have been used to create different models capable of making various predictions such as blastocyst formation, implantation, and LB. First, priority was given to the study of morphokinetic parameters of the early stages. This is because visualization of late divisional events is more difficult to determine because tracking cells at this stage is a difficult task due to the fact that cells tend to cover each other and, therefore, it becomes necessary to track them in different focal planes. In addition, the more fragments observed in the image, the more difficult it is to count and determine whether division occurred [51].

Some clinics continue transferring embryos at Day 3 of development, although success rates have been seen to increase at Day 5 [48,63,75,76]. To reduce the subjectivity of embryo selection based on morphology, it is necessary to have as much information as possible. In addition, there are embryos that are morphologically good but are blocked and do not reach blastocyst [15]. In these cases, morphokinetics plays a very important role in predicting blastocyst formation. These models use only the morphokinetic parameters of the early stages of embryonic development. Among the parameters with the highest predictive capacity, we find the third cell division time (t4-t2), time of transition of 5-blastomere embryos to 8-blastomere embryos (t8-t5), and the time of morula formation (tM) [47,56]. To these parameters were added t5 and cc3 as parameters able to predict blastocyst quality, with slower blastocysts being of better quality [15]. Although, another earlier study demonstrated that embryos that divided 2 to 8 cells faster were those most likely to reach blastocyst, expand, and implant [21].

In parallel, other authors, studying the early stages of development, have found morphokinetic parameters that are linked to embryo implantation. Some of them are the duration of the second cell cycle (cc2), the time of appearance and fading of pronuclei (tPNa and tPNf) [3,12,24,46,62] any of the derived combinations of division times in 2, 3, 4, and up to 8 cells (t2, t3, t4, and t8, respectively) [5,35,46,51,62,73] followed by the second synchrony or 4 cell division time minus 3 cell division time (S2 = t4−t3) [35,51,73]. Many of these parameters have been used for the creation of algorithms capable of predicting implantation as can be seen in Fig. 29.1 [5,51,64].

Despite having highlighted the usefulness of early-stage morphokinetic parameters and their greater ease of study, these early-stage parameters do not have the ability to predict embryonic ploidy status with good accuracy. However, later morphological events and later morphokinetic parameters provide more information. This is the reason why late-stage morphokinetic parameters have been incorporated into algorithms that are capable of predicting LBs. An example of this is Fishel's algorithm—it takes into account morphokinetic parameters of early developmental stages such as t2, cc2, and t2/t5−t2 and late stages such as

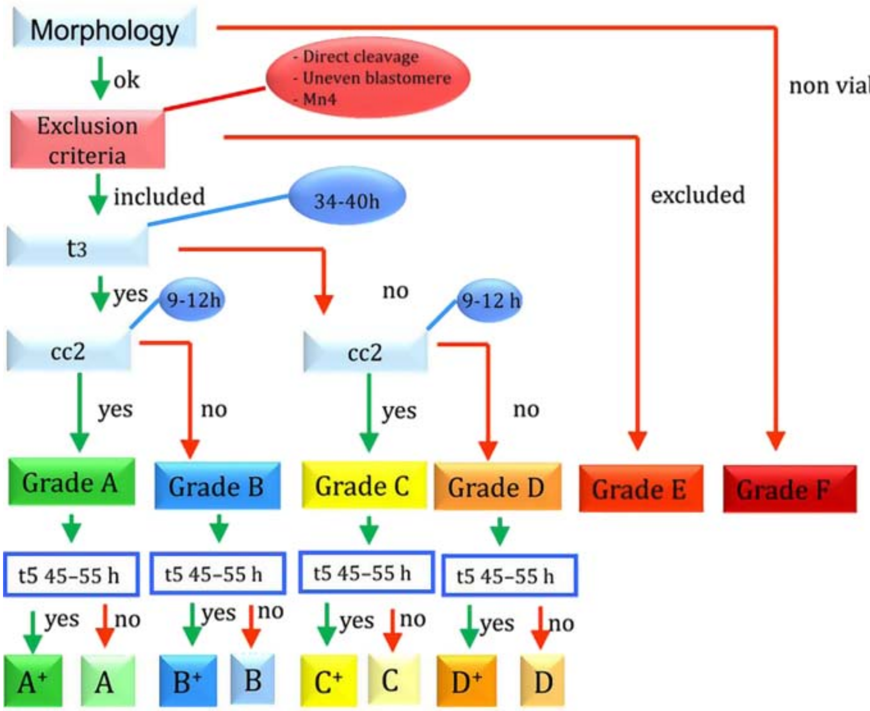

FIGURE 29.1 Algorithm that classifies embryos by groups according to their implantation capacity. It was performed with a sample size of 1664 cycles from 4 different clinics. The first screening were morphology and exclusion factors. The last screening were the morphokinetic parameters t3, cc2, and t5 [5].

time to start of blastulation (tSB) and duration of blastulation (dB = tB[time to blastocyst arrival]-tSB) [28–30]. Other morphokinetic parameters predictive of LB are two-cell disappearance time and time to arrival at expanded blastocyst diameter larger than 160 μm (texpB2). Time to morula stage (tM) and trophectoderm quality were added to these parameters [7,66].

Notwithstanding the immense number of publications in the field of morphokinetics, no universal algorithm has been developed which can be used by any clinic without prior modification and it is able to predict clinical outcomes. This is because morphokinetics is highly dependent on the clinical environment such as different culture conditions, pH, temperature, humidity, O_2 concentration, or type of culture medium [32,40]. Nonetheless, the commercial house Vitrolife created an algorithm called KIDScoreD5TM. This algorithm was elaborated with more than 5000 known implantation data (KID) embryos from different assisted reproduction clinics. In this way, interclinic variability is reduced [34]. Recently, the ability of the KIDScoreD5 algorithm to predict the probability of implantation, ongoing pregnancy, and LB has been validated [34,40]. However, further studies should be done in other clinics to confirm its universal use.

Implementation of artificial intelligence in assisted reproduction laboratories

AI is the area of computer technology concerned with the simulation of intelligent behavior and the ability of machines to mimic cognitive thinking, learning, and problem solving usually associated with humans [42]. Some authors define AI as the ability of a computational system to learn and exercise intelligent behavior [69]. AI excels at recognizing complex patterns from generated images and produces quantitative data from the received information using a learning system based on selection algorithms [38].

Assisted reproduction clinics are currently betting on the benefits of AI. The emerging role of AI in this area includes diverse applications such as sperm selection, follicle differentiation for stimulation, embryo development monitoring, and prediction of implantation potential from time-lapse images of blastocysts as discussed above [25]. Currently, in assisted reproduction laboratories, the most commonly used AI modalities are the techniques known as *machine learning* (ML) and *deep learning* (DL). ML is a supervised computational method that uses algorithms to make a detailed determination or prediction through automated learning from the data provided to it [53]. Among the most widely used ML techniques, we can highlight artificial neural networks (ANNs) such as convolutional neural networks [49]. ANNs are intelligent systems that mimic the behavior of the human brain by combining different hardware (e.g., graphics processing units) and software (such as big data) modalities. In this way, this ML modality manages to create algorithms that simulate processes to solve complex problems through supervised learning [20].

DL is a variable of ML whose main distinguishing feature is that its learning is unsupervised so that it recognizes a piece of data from the multitude of data provided by making changes in its parameters until it gets close to the real value [45]. The learning of this system is based on the repetition of the action in the same way as the human nervous system. Therefore, the DL needs a lot of data to achieve automated learning [41] (Fig. 29.2).

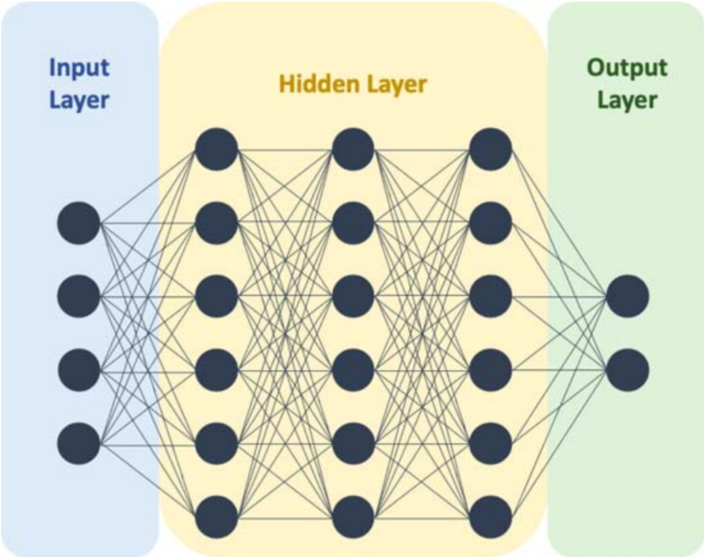

FIGURE 29.2 An artificial neural network is composed of artificial neurons or nodes that receive information from the outside (input layer) or from other neurons (hidden layer), similar to the nerve impulses received by neurons in the human brain. Information enters a neuron; it is processed and generates an output value that feeds other neurons in the network, or it is the output to the outside of the network (output layer) generating an output value.

AI offers us the opportunity to follow the dynamic patterns of embryo development by creating algorithms that reduce intra- and interobserver subjectivity and consequently the possibility of human error [27]. In addition, AI algorithms can offer a valuable tool to assisted reproduction professionals to standardize, automate, and develop different processes to improve patient care and treatment outcomes, thereby also reducing the workload in laboratories as it reduces the time spent on activities allowing them to be automated [77].

Applications of artificial intelligence in the in vitro fertilization laboratory

The great potential of AI platforms is reflected in the solutions offered, among others, to improve the diagnosis and selection of the most appropriate treatment for each patient, use algorithms to clarify success rates, provide personalized treatment plans, use computer vision for the evaluation of sperm, eggs, and embryos, and maximize the endometrial factor among many other applications within assisted reproductive techniques [27].

Regarding the usefulness of AI in the prediction of clinical outcomes, it is worth highlighting the efforts made by different researchers to create predictive algorithms coupled to the TLSs mentioned above. The first commercialized predictive AI approach was the Eeva system from the research team of Wong et al. in 2010. This system integrated automatic software along with two predictive parameters based on the duration of cytokinesis and the duration of times between mitoses to reach the four-cell stage [59]. This platform can be attached to any conventional incubator, although it is currently incorporated in the Geri incubator. The introduction into the clinic of the Eeva system and other new automated annotation systems such as Geri Connect and Geri Assess 2.0 has meant reduced inter- and intraobserver variability and subjectivity in the selection of the optimal embryo [18]. In addition, success results for implantation, clinical pregnancy, and evolved pregnancy rates have also been increased with the use of these softwares [1,2,73] (Fig. 29.3).

Subsequently, Bori published a study in which certain embryonic characteristics were identified such as blastocyst expanded diameter and trophectoderm cell cycle length had statistically different values in implanted and nonimplanted embryos. The use of AI to analyze big data provided by TLSs showed that the most predictive model was made up of all of the variables described (conventional morphokinetics + novel morphodynamics) and not just those that were individually discriminatory [9]. The same research group developed an AI model based on ANNs using time-lapse blastocyst images and the embryo protein profile in 81 euploid embryos. The ANN achieved to detect euploid embryos capable of resulting in a LB with an excellent accuracy (100%), especially considering IL-6 and MMP-1 [9].

As previously discussed, the selection of a euploid embryo for transfer is the key to reproductive success. However, both PGT-A and morphokinetics have several disadvantages. For this reason, the current trend in assisted reproduction is to create models based on AI that are capable of identifying aneuploid embryos with considerable sensitivity and effectiveness. Following these premises, models have been created such as the one by Neocleous and collaborators in 2016; they performed a model of ANNs trained with clinical data from singleton pregnancies with markers collected from the mother and the fetus, capable of detecting fetal aneuploidies. Detection rates of 100% were achieved for trisomy 21 and a higher rate of 80% for other common trisomies such as 13, 18, Turner syndrome, and triploid embryos [58]. Subsequently, Miyagi Y et al. in 2019 presented a ML system applied to blastocyst images evaluated by conventional means. The AI model was trained by comparing 80 images of

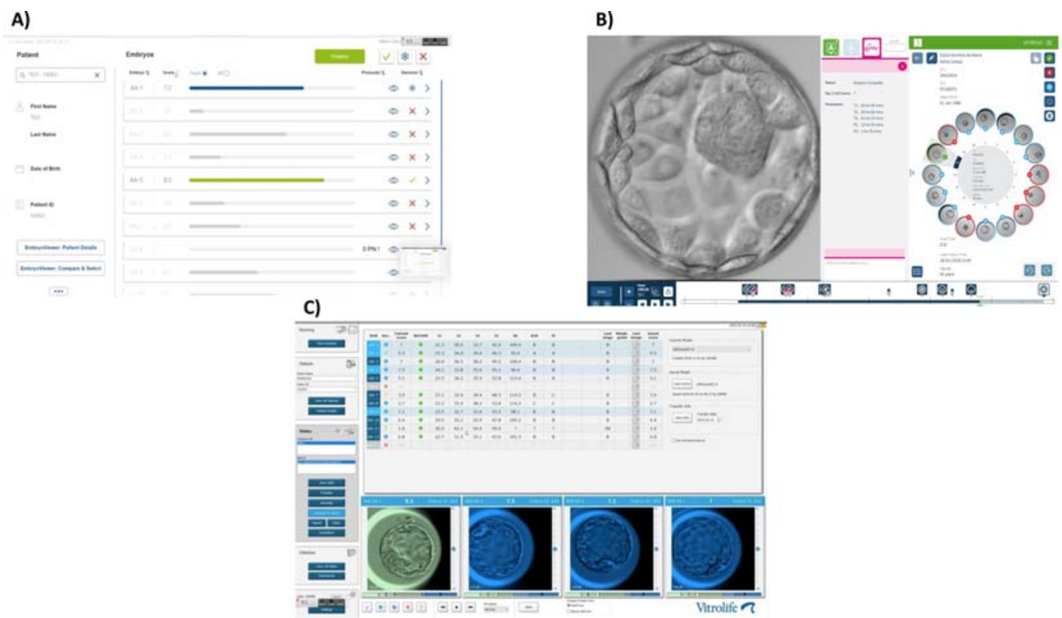

FIGURE 29.3 Commercially available image analysis platforms for embryo evaluation and selection. (A) IDA-Score, (B) Eeva, (C) KIDScoreD5.

blastocysts that resulted in LBs and 80 images of blastocysts that resulted in abortions. This created a program that retrospectively classified embryos according to the probability of LB without aneuploidy with an accuracy of 0.65 [55].

AI is therefore a very powerful tool, with great applicability and at the same time novel in our sector. There are more and more studies that reaffirm its usefulness in many clinical aspects; however, we have not yet been able to get the full potential of this tool, and as research becomes more focused and familiar with AI, very interesting results will be obtained.

Conclusions

TLSs are presented as incubators capable of maintaining optimal conditions for embryo development. In addition, they allow us to perform a dynamic evaluation of the embryo, being able to see certain important events that would be lost with static evaluation. Likewise, thanks to time-lapse, morphokinetic parameters have been discovered which have been used for the creation of algorithms or models that have been able to predict aneuploidy, blastocyst rate, implantation rate, and LB rate, among others. However, the algorithms presented so far, with the exception of KIDScore D5, require some modifications before they can be used in daily clinical practice. Added to this is the need for manual and subjective annotation of morphokinetic parameters. Using these drawbacks, AI in a fully automated and objective manner can predict clinical outcomes more accurately than existing algorithms.

Technology is advancing and with it assisted reproduction clinics. We have more and more tools at our disposal, and our duty as professionals in the sector is to introduce them into our daily practice and seek their usefulness in order to benefit from each technological advance. The future lies in their implementation in our sector with the aim of maximizing success rates.

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CHAPTER

30

Troubleshooting the in vitro fertilization laboratory

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Introduction

While all in vitro fertilization (IVF) labs believe that they are doing everything right, and hope that things will (continue to) run smoothly with consistently high success rates, we all know that from time-to-time, there will be problems. Problems might be caused by factors outside the lab's control, but sometimes they arise because there was insufficient attention to detail, or the lab continued to use an outdated technique, or because a product had changed and either the lab was unaware of this or did not recognize the significance of the change. Regardless of the origin of the problem, sometimes someone has to troubleshoot a part of the lab system. Of course, the more proactive the Laboratory Director is in managing systems the less likely this will be, the less serious it is likely to be, and hopefully the easier the problem will be to solve (see "Proactive Troubleshooting," below). But sometimes one just has to be reactive, and in such circumstances, *root cause analysis* (RCA) is the conceptual basis for troubleshooting [1].

Fig. 30.1 and Table 30.1 provide an overview of the troubleshooting procedure and illustrate how it is based on the application of scientific method. While as scientists we might ask "Isn't this all just common sense?", that might not be apparent to someone who has not had formal scientific training. For a trained scientist, the structured approach to problem-solving should be intuitive—and they will subconsciously go through the steps of an RCA automatically when confronted by a problem. Formalizing the procedure enables less well-trained or less-experienced people to employ this approach, and also provides a framework for documenting the task, which is a crucial aspect of accreditation [1,2].

An effective troubleshooter is essentially a scientific detective: using the same rigorous "follow the evidence" approach, examining every detail, and deciding whether it is relevant to the mystery at hand (whether it is "contributory" in RCA terminology). Recognizing more obscure factors might require specialist knowledge, the lack of which might be a great hindrance to solving the mystery. This is why it is vital that when teaching someone a method,

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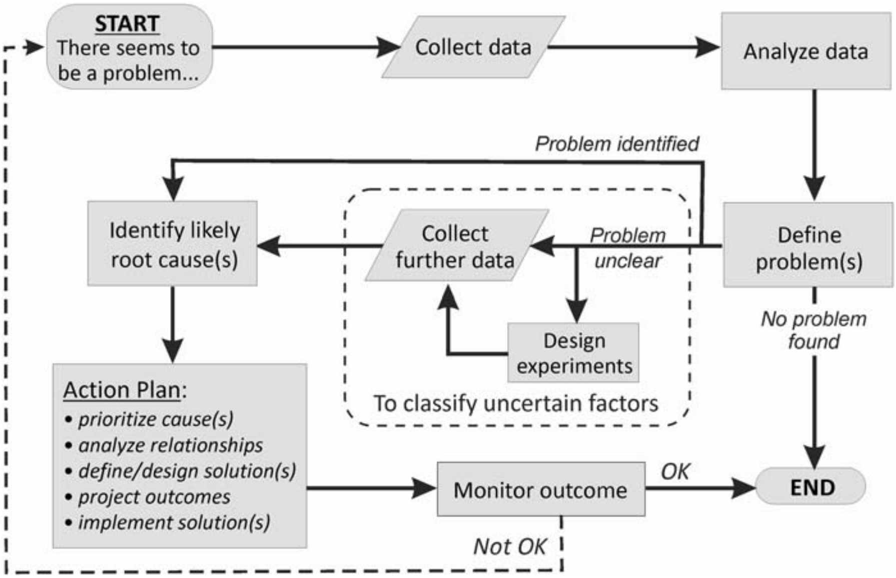

FIGURE 30.1 Flow chart of the root cause analysis procedure.

the trainer/mentor takes the time to explain "why" things are done the way they are (and perhaps why they are not done certain other ways): just describing the "how" will not produce a competent scientist [3].

Process mapping and process control

It should be evident that effective troubleshooting depends on being able to visualize anything that is done in the IVF lab as a process—ideally from the outset—with all the lab activities being structured and organized for efficiency and efficacy. ***Process management*** is simply how one visualizes (e.g., using process mapping), understands, and analyzes how something is done, which encompasses both how it was *intended* to be done and how it is *actually* done.

Identifying and measuring extrinsic factors that affect a process requires a thorough understanding of the underlying basic science, not just the biology but also the chemistry, the physics, and even the engineering that together affect all lab processes [4]. Without such a comprehensive understanding, the ability to minimize or eliminate the problems—achieve ***process control***—will be greatly compromised.

Process control provides essential information in terms of:

(a) How the process operates normally:
- Its internal control factors
- Its historical performance (as shown by *key performance indicators*, KPIs)
- Its inherent variability (from KPI control charts)

TABLE 30.1 Overview of the seven stages of the troubleshooting process.

Stage	Action	Specifics
Perception	There is perceived to be a problem	Someone raises a concern that "there seems to be a problem" or "something seems to be wrong"
Inspection	Define the possible problems	Define the review specification: • Identify the particular issue • Identify pertinent KPIs • Establish appropriate benchmarks
Verification	Establish that there really is a problem	Review and analyze the KPIs
Investigation	Perform a root cause analysis	Inspect all aspects of the clinic's operations (not just the IVF lab) that could impact on the underlying process(es)
		Audit results, manuals, policies, and the actual performed processes
		Identify possible contributory factors; if it is unclear whether something is contributory, then investigate further via prospective collection of additional data
		Develop a risk matrix of the possible contributory factors
Remediation	Develop an action plan	Plan the likely effective change(s)
		Educate and retrain staff as necessary
		Project the improvement(s) that will be achieved = Target(s)
		Identify appropriate KPIs to monitor the change(s)
		Define a timeline for the change(s) and follow-up date(s)
	Effect change(s)	Implement the change(s)
Confirmation	Follow-up on the action(s) taken and review their effectiveness	Monitor the effectiveness of the change(s) using the KPI(s) and appropriate statistical testing
		Compare the achieved improvement(s) against the target(s): Target(s) achieved—troubleshooting complete, proceed to documentation stage Target(s) not (all) achieved—troubleshooting is not complete, return to inspection stage and repeat
Documentation	Prepare a formal written report	Essential content: • document all the actions taken at each stage • tabulate all data and include statistical analyses • State final conclusion(s)
	Report authorization	Review and acceptance by the quality committee
		Report entered into the organization's quality framework

KPI = Key Performance Indicator.

- Extrinsic factors that can affect it

(b) What the process is capable of achieving:
- Minimum performance standards or "competence" (based on benchmarks)
- Aspirational performance benchmarks ("best practice"), reflecting the proof of the possible

A minimum set of IVF lab KPIs, including both competency and aspirational benchmark values, has been published [5], as has general guidance on deriving KPIs and using then to create Shewhart control charts [1]. Users also need to understand the *measurement uncertainty* of KPIs to understand the strength of the evidence provided by a control chart.

Using root cause analysis

The RCA is the standard retrospective risk management tool used as a framework for troubleshooting and depends on (a) being able to visualize the problem within the context of one or more processes and having information available on the process's controls and (b) performance (process outcomes) [1]. Table 30.2 lists the general areas of concern that must always be considered during troubleshooting. During the *investigation* stage of an RCA, consider the process in terms of what goes in, what happens to it, and what comes out.

When following the RCA procedure as illustrated in Fig. 30.1, after all the possible root causes have been identified, they need to be classified as either contributory or noncontributory; for those that cannot be classified, seek additional data to permit their classification (this might require the collation or even collection of extra data). Then for each potential root cause, assign three scores: (1) the likelihood of it occurring; (2) the consequence or severity if it happens; and (3) how likely it is that it might be detected. While many people employ a 1–5 scale for each of these three scores, this often ends up with a rather narrow dynamic range of actual values assigned (since the process design should have removed the possibility of ever reaching the highest score for any aspect) which can confound prioritization. Consequently, we prefer to use 10-point grading scales, as shown in Table 30.3.

The three assigned values are multiplied together to give a *risk priority number* (RPN), and obviously the highest-scoring factors are investigated as the most likely root causes. With 5-point scales, the maximum RPN should not exceed 64, with many real-world RPNs being clustered in the 8–27 range, but with a 10-point scale, the dynamic range is much wider, theoretically extending to 729, with many real-world RPNs ranging between 30 and 280. When there are several possible contributory factors, a *risk matrix* can be constructed graphically to facilitate appreciation of the situation by visual learners.

Once the likely contributory factors have been identified, an *action plan* is created to effect remediation. Keeping an open mind is vital—do not make assumptions or jump to conclusions. When preparing the Action Plan and writing the Report, discuss the possible "contributory factors," but avoid using the words "cause" or "fault" for psychological and legal reasons. Also, do not name individuals in Reports, identify roles generically.

We have published elsewhere examples of troubleshooting a biological process, an adverse incident, and a maintenance process, as well as an illustrative description of a well-controlled IVF lab to show the control mechanisms that might be put in place to ensure that each process

TABLE 30.2 The five key areas that must always be considered in risk management and when troubleshooting.

Aspect	Focus	Particulars
Biological	Patient factors	Atypical age distribution Aberrant casemix or change in referral pattern Unusual pathophysiology
	Treatment factors	Stimulation responses Inadequate luteal phase support Undiagnosed sperm dysfunction
Organizational	Management choices	Treatment protocol choice Poor criteria for deciding IVF versus ICSI
	Procedure timelines	Trigger timing Delays in sperm preparation ICSI stripping/injection times assessments
Technical	Medical procedural aspects	Allowing follicular fluid to cool during oocyte retrieval Trainees with poor embryo transfer technique
	Laboratory procedural aspects	Temperature or pH shifts during gamete or embryo handling and/or assessment
	Contact materials	Needles and catheters Plasticware Handling devices Culture media
	Failures or errors	Suboptimal methodology Poor SOPs Unauthorized method change in laboratory SOP
Extrinsic	Macro environmental factors	Lab design/construction Air quality (especially volatile organic compounds, VOCs(s))
	Microenvironmental factors	Workstation and incubator calibration and operation
Random chance		

SOPs = standard operating procedures; VOC = volatile organic compound.

or system continues to function within its expected parameters [1]. Others have published more exhaustive examples of troubleshooting and problem-solving in the IVF lab [6].

Is the origin of a problem always found?

Some experts say that troubleshooting will always identify the root cause of a problem. Certainly this would be true in a perfect world, but from our extensive experience auditing IVF labs, this is simply not the case: far too often there are insufficient—and often no—data available from the routine lab records to establish whether a possible root cause might even

TABLE 30.3 A ratings scheme for performing Root Cause Analysis (RCA) and Failure Modes and Effects Analysis (FMEAs) using wider dynamic range values for likelihood, consequence, and detectability.

Rating	Likelihood of occurrence	Consequence or severity (examples)	Detectability
0	Will not occur—so it is not a real risk	None, so it cannot be considered as a real risk	Certain, so there should be no risk of it occurring
1	Very unlikely (a well-controlled or minimized risk, but cannot be eliminated completely)	Trivial (no measurable adverse risk)	Probable
2	Unlikely (circumstances for occurrence are all controlled as far as possible, but external factors might remain outside the organization's control)	Minimal (more of an inconvenience, with no identifiable impact on patient care)	Very likely
3	Somewhat possible	Quite minor (impacts organization internal systems only)	Likely
4	Possible	Minor (adversely affects efficiency but with no measurable impact on treatment)	Quite likely
5	Very possible	Quite serious (definite risk of diminished treatment outcome)	Very possible
6	Quite likely	Serious (definite adverse effect on patient management or treatment outcome)	Possible
7	Likely	Very serious (definite adverse impact on multiple patients' management or treatment outcomes, or risk of injury to patients or staff)	Somewhat possible
8	Very likely	Major (loss of embryos, OHSS, infection of patients or staff, actual injury to patients or staff)	Unlikely
9	Probable	Extreme (loss of life, damage to facility)—risk of occurrence must be at least "very unlikely"	Very unlikely
10	Certain—such a situation should not exist in the everyday world	Catastrophic (loss of multiple lives, destruction of facility) "acts of god," war or terrorism	Impossible

OHSS = ovarian hyperstimulation syndrome.

be considered contributory or not, much less establish the actual explanation. When confronted with such situations, we describe them as trying to "troubleshoot in the dark."

Common issues of information inadequacy or nonexistence include:

- absence of calibration or QC data for critical equipment;
- insufficiently detailed, or missing, observations at fertilization check, or during embryo development;

- inadequate or not up-to-date KPIs, or missing KPIs; and
- inability to link clinical information (e.g., stimulation response/trigger timing) with laboratory observations and/or clinical outcomes without having to resort to manual audits of patient charts.

This is why the information collected as part of the routine lab work needs to be comprehensive, rather than just the minimum needed to maintain a basic set of KPIs to "get through" accreditation, or only what the Medical Director sees as important in the short term. IVF scientists need to think ahead and anticipate what information might be needed in order to effectively troubleshoot possible problems (not just the perceived likely ones). We have always preferred to take the approach of collecting more detailed information than that needed to monitor routine lab operations so that should we need to troubleshoot an unexpected issue we'll likely have the necessary data readily available. The alternative circumstance of identifying inadequate or missing data during an RCA and then having to either dig it out of paper records (assuming it was ever recorded) or collect it prospectively means that additional patients' care will probably be compromised while the data are being compiled or collected and analyzed—which is clearly a failure of professional standards and a breach of patient trust. Hence, routine data collection directly into a database, ideally with integral analytics capability, is key, and clearly illustrates the need for IVF labs to work smarter not harder [1].

Proactive troubleshooting

But would not life be easier, and our care of the patients better, if we were able to reduce the likelihood of needing to troubleshoot problems? For the three decades, we have embraced the principle of "proactive troubleshooting," which means that all our processes and systems have been subjected (repeatedly) to the other core risk management tool, *failure modes and effects analysis* or FMEA [1]. Performing an FMEA employs the same general principles as the RCA except that we start by considering what might go wrong with a process (Gedanken experiments)—anticipating all the possible failure modes prospectively and assigning *likelihood, consequence, and detectability* scores using the same scales as for an RCA. In the *action plan* phase, design and apply controls to reduce the possibility of the problem occurring, to increase the likelihood of detecting the problem early, and to reduce the possible consequence of the problem should it arise. We have called this general approach "trouble-proofing."

Fundamentally, in applying the FMEA tool, we are analyzing not just how a process is performing compared to aspirational benchmarks but also taking it apart in order to analyze:

(a) how well we are supporting the process in terms of the materials used (e.g., choice of plasticware, culture medium formulation, culture dish configurations);
(b) how well the controlling factors are being regulated (e.g., workstation and incubator temperature, pCO_2/pH, humidity); and
(c) whether adverse extrinsic factors (e.g., lab environment, operator variability) might be affecting the process.

After refining the control and support mechanisms for the process steps, and perhaps adding extra ways of monitoring the process, the process is reassembled—and, of course, the standard operating procedure (SOP) updated and staff retrained as necessary. The performance of the improved process is monitored via control charts of one or more pertinent KPIs using their existing control mean and ranges. Evidence of an improved process is usually seen as an improved control mean and tighter control ranges. Once change has been established, the control mean and ranges will need to be updated using, typically, the first six periods of KPI data following the introduction of the improved process.

It is important to remember that improvements to a process might not necessarily be reflected in an outcome KPI such as better embryo quality or a higher implantation rate but could be achieved in other quality characteristics of the process such as reduced operator time, better ability to detect an adverse effect, or reduced likelihood of an error being made. Hence, processes that have been subjected to FMEA optimization will not only provide better and more stable outcomes but will themselves also be more robust: less subject to fluctuations in lab environmental/equipment conditions or operator behavior, and capable of performing identically in different locations. Our basic IVF lab systems have been FMEA-optimized over the past 25 years and have been introduced into many new labs or satellite labs, where they have been proven capable of achieving excellent results regardless of geographic location, subject only to the developmental competence of the oocytes.

The issue of oocyte competence is often not recognized. Far too often, a decrease in outcomes is presumed to be caused by a lab problem: How often have we all heard the cry of "What's gone wrong in the lab?" But in many situations, we have seen that perceived decreases in lab performance were attributable to changes in, or poor control of, critical clinical factors such as ensuring application of correct trigger criteria. Variable delays in triggering increase the prevalence of postmature oocytes, which have impaired developmental potential. Fig. 30.2 shows the prevalence of "late triggers" that came out of a patient chart audit performed during an investigation of a perceived fall in cycle outcomes at a leading IVF center. Unfortunately, in the absence of oocyte assessments at or shortly after retrieval (e.g., using criteria such as those summarized in Table 30.4), an IVF lab has no relevant data to establish the risk of decreased embryo competence resulting from an increase in the proportion, and/or severity, of oocyte postmaturity. A classic case of trying to troubleshoot in the dark.

A key part of these optimization exercises is that the limitations of available *indicators* are revealed before anything has gone wrong and patient outcomes have been compromised. If the optimization process requires information that is not being recorded then one or more *failure modes* could be unclassifiable, or classified as potentially contributory but not able to be investigated or targeted for remediation: such factors would then require prospective data collection that could extend patient compromise, as discussed earlier in this chapter.

The value of quality

Improving success rates

Assisted conception treatment is an excellent example of complex patient care provided by a multidisciplinary team. It has long been recognized that close collaboration between especially the physicians and embryologists is key to achieving the highest success rates. Fig. 30.3

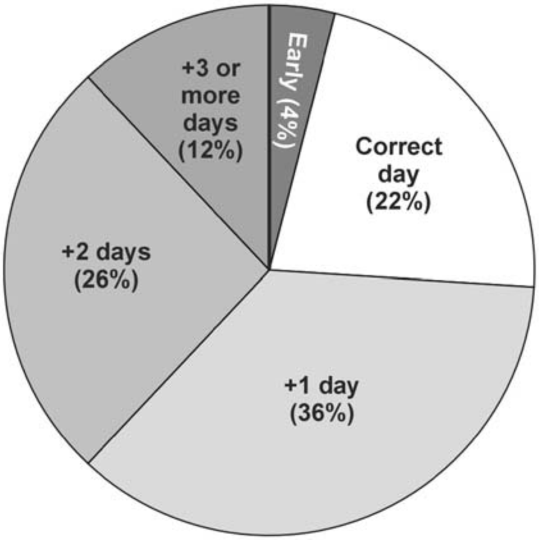

FIGURE 30.2 Prevalence of delays in ovulation trigger timing found in a retrospective review of 100 consecutive stimulation cycles (recorded data on ultrasound-based follicular counts and measurements and estradiol measurements) during a troubleshooting audit. The correct day of trigger was based on the center's own stated procedural criteria.

TABLE 30.4 Criteria that can be used for differentiating oocyte immaturity, maturity, and postmaturity at or shortly after oocyte retrieval.

Aspect	Immature	Mature	Postmature
Cumulus amount	Scanty	Plentiful	Minimal
Appearance	Dense	Expanded	Dispersed
Corona radiata	Tight	Radiating	Radiating
First polar body	Absent	Present	Fragmented
Ooplasm	Pale/dark	Pale	Dark
Oolemma	?	Elastic	Rubbery

shows the Day 5 embryo development rate (D5 EDR) in a center where every treatment cycle—regardless of whether a pregnancy was achieved or not—was reviewed *post hoc* by a multidisciplinary group. Stimulation information (including choice of regimen, gonadotrophin starting doses and dosage changes, follicular and estradiol monitoring results, and trigger decisions) and laboratory information (including oocyte maturity and grades, fertilization and embryo development data, and embryo grades) were considered objectively in light of the success of the stimulation protocol (functional competency of the oocyte cohort

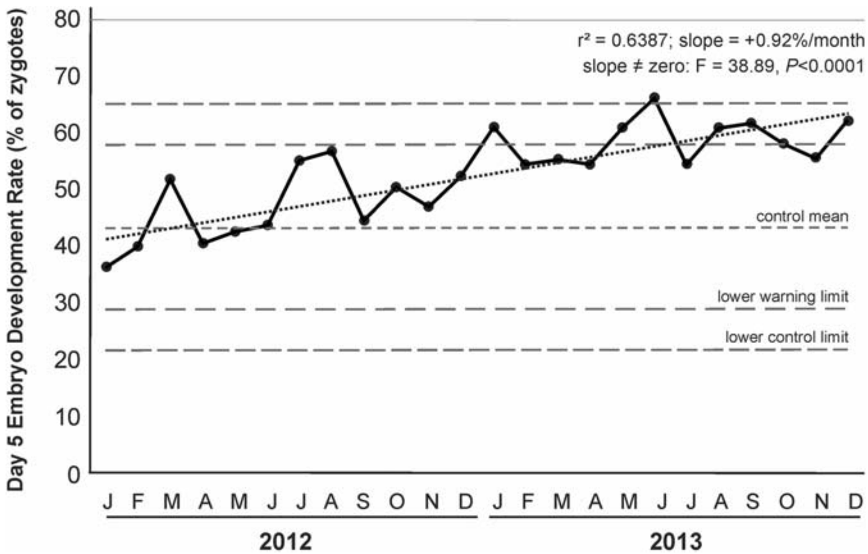

FIGURE 30.3 Illustration of the beneficial impact of multidisciplinary post hoc cycle reviews on day 5 embryo development against a background on consistent laboratory protocols. See text for further details.

obtained) and treatment cycle outcome, as well as used to plan possible future treatment cycles for each patient. Clinical and laboratory staff were consistent throughout the period, as were all laboratory protocols. The highly significant increase in D5 EDR over the two-year period clearly illustrates the importance of improved stimulation and triggering decision-making in managing future cycles—learning from the past within a multidisciplinary team—to improve treatment success [5,7,8].

Stability and calm

Another important feature of working in a process-optimized lab is that the likelihood of needing to troubleshoot problems is greatly reduced. Minimizing the need to keep "putting out fires," to struggle through the recovery from incidents, and to troubleshoot and fix problems, greatly improves not just the efficiency, efficacy, and morale of a lab (indeed, the reputation of the entire clinic) but also the Lab Director's quality of life [1].

Take home messages

- Collect as much data as possible; not just that needed to maintain a minimal set of KPIs but also sufficient to support a comprehensive ability to troubleshoot problems promptly using the *root cause analysis* approach.
- Data collection needs to be routine, and data entered into a database that allows its easy use for both generating KPI control charts and troubleshooting. Repeat handling of

data needs to be avoided; it is both wasteful of time and a major source of errors. The ability to integrate clinical and lab data is crucial.

- Employ extensive, and repeated, process optimization using *failure modes and effects analysis*: trouble-proofing is better for everyone than having to troubleshoot. Indeed, not trouble-proofing can be considered both unprofessional and a breach of the patients' trust.

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Note: 'Page numbers followed by "*f*" indicate figures and "*t*" indicate tables.'

Handbook of Current and Novel Protocols for the Treatment of Infertility

Edited by
Michael H. Dahan, Human M. Fatemi, Nikolaos P. Polyzos, Juan A. García-Velasco

The field of Assisted Reproductive Technology (ART) is rapidly evolving, and stimulation protocols, fertility strategies, and aspects of infertility treatments are constantly being updated as advances and new discoveries are made.

Handbook of Current and Novel Protocols for the Treatment of Infertility is a valuable resource of well-organized, comprehensive scientific data with practical guides and step-by-step protocol for infertility management. Written by worldwide experts, this book discusses different practice patterns and approaches used internationally and presents innovative topics including preimplantation genetic testing, time lapse imaging, and the role of artificial intelligence in ART.

This book provides up-to-date evidence-based guidance on daily practice and is an important resource for infertility providers, including trainees or experienced clinicians, embryologist, and is geared for both specialist in reproductive medicine and gynecologists interested in increasing their knowledge in the field of reproductive endocrinology and infertility.

Key Features

- Presents protocols for infertility management and new developments in practical techniques and understanding, including discussions on in vitro maturation, in vitro fertilization, and ovarian stimulation.
- Discusses innovative topics such as the role of artificial intelligence in infertility management, protocols using progesterone to prevent ovulation, dual-stim protocols, random start protocols, how to set up an in vitro fertilization (IVF) lab, and complications in IVF and the management of these complications.
- Chapters written by well-known experts on infertility management from different parts of the world, providing a worldwide perspective.

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